

Lean Proof Strategy Proposal for the Scheduling Rules Document

Stuart Swan, Platform Architect, Siemens EDA

27 June 2026

0. Goal and scope

The goal is to formalize the scheduling rules document's mathematical proof core in Lean: trace compatibility, witness construction, cutpoint correspondence, elaboration, and one-way liveness/deadlock preservation. The Lean proof would certify the formalized theorem stack under the stated assumptions, not every informal prose statement in the document.

The scheduling rules document's own framing is conditional: the formal results depend on assumptions, observation-boundary choices, witness-selection conditions, and liveness/progress premises.

The top-level safety conclusion is the document's witness-relative ordered trace-compatibility predicate, not ordinary trace inclusion or trace equivalence. It is defined over complete observable units, execution-specific ordering and attribution data, and selected witness annotations; it permits independent units to commute or to use different cycle groupings while preserving the required $<_J$ constraints. Generic simulation-to-trace-inclusion results may supply infrastructure, but they do not discharge this scheduling-specific conclusion.

The Lean development should therefore be organized around explicit theorem instances and assumption bundles, rather than hidden global axioms.

Notation and signature status. The Lean-like declarations in this strategy are intended to fix proof architecture, theorem dependencies, and object domains. Unless a declaration is explicitly identified as a final Lean signature, it should be read as schematic pseudo-Lean. In particular, channel, process, side, prefix, and bucket types must be unified in the concrete development through the selected ComparisonInstance, the side-parametric SystemOfSide map, and the shared proof-internal alphabet C.Sigma. This note is not a relaxation of the proof obligations; it prevents schematic notation such as Channel, C.Sys_ref.Channel, Prefix, and InfiniteExecutionBucketed from being mistaken for already type-checking Lean code. In the final Lean files, foundational declarations such as ObservableUnitWF, PrefixAnnotations, and AnnotatedPrefix must be placed before any

structures that consume them; the topical order of this strategy document is not a proposed Lean file order.

Version alignment note. This revision updates the Lean proof strategy to align with the scheduling-rules v45 Appendix K changes, while retaining the earlier v42-aligned strategy as the base. It retains the v42 alignment points: (i) Appendix G/J separates the informational causal-dependency graph from the strict Appendix J required-order relation $\prec_J$; (ii) Theorem J.1 produces an explicitly selected Sys_ref witness prefix and is organized along the fixed RTL-time ideal chain rather than by arbitrary $\prec_J$ -minimal extension; (iii) Corollary J.1 and Appendix K/R cutpoint transfers use an explicit RTL_B-to-source abstraction relation J.Abs and a corresponding implementation obligation J.AbsConf; and (iv) Appendix K A5 is now the conservative serial-blocking source-liveness assumption, while Lemma K.2b supplies the serial-source dominance bridge from an RTL-matched / overlapped Sys_ref witness deadlock to a serial-source deadlock.

The retained v42-specific alignment points are: (v) R2C combinational signal IO is now modeled as component-local, stutter-canonical fixed-point signal-observation units with selected observable slices Obs(C_comb), not as an unconditional per-cycle observable bucket; R2C matching is by component identity and local R2C occurrence ordinal, with cycle-locked components handled as an explicit stronger observation mode. (vi) ChannelEnabled is eligibility/availability for scheduler or arbiter selection, not same-cycle commit by itself; E4 is the safety/legality rule for committed transfers, while B2 supplies eventual selection and commit for continuously ChannelEnabled actions. (vii) hidden arbitration, grant, or back-pressure may not be smuggled in as an extra E4 availability conjunct at an ordinary boundary; it must be represented as fair selection among ChannelEnabled actions or as explicit Sys_B^elab boundary structure. (viii) K.0/K.0a frontier reasoning now explicitly proves full NextFront minimality over synchronization, anchored signal/R2C, and message items, and rendezvous endpoint-request transfer relies on J.Abs/J.AbsConf raw endpoint-request facts. (ix) K.1 uses fair drain-normalization, rather than pure idle-stutter, to justify Exec_post applicability at the relevant deadlock cutpoint.

Revision update note (v42 base plus v45 Appendix K refinement pass). This revision is a refinement pass against the v42-aligned strategy plus the scheduling-rules v45 Appendix K update. It retains the earlier v42 items and repairs several internal inconsistencies: (x) the Common Formal Definitions side-condition discharge record is given an explicit Lean-side audit structure whose WC field proves the actual WC predicate, with NB-CF, Appendix L snooping, direct construction, and mixed per-site coverage recorded as proof-producing provenance routes; Sys_B^elab(E), B-faithfulness, J.AbsConf, K.Drain, and Kε retain their theorem-scoped discharge evidence (Section 1.1); (xi) B-faithfulness of each back-

annotated capacity $B(c)$ is recorded as a named tool/RTL-flow compliance hypothesis alongside E4 channel-realization conformance and $J.AbsConf$, with its formal structure and assumption threading in Section 7, its audit discharge in Section 1.1, and its concrete-flow checklist entry in Section 20; (xii) the v42 B3 modularity note is reflected: for a single explicit abstract boundary with ordinary E4 buffered or rendezvous semantics, the B3 P_push/P_pop discharge is derivable from B2 plus E4, and Lean should prove that derivation while retaining B3 as the modular per-channel obligation checked against channel realizations and elaborated Sys_B^{elab} boundary chains (Section 6); (xiii) the Appendix K terminated-process convention is made an explicit derived theorem together with its scope warning: waiting on a terminated peer is not, by itself, an internal message-passing deadlock in the K.1 sense, and intended termination/EOF/done protocols must be modeled explicitly or verified outside Appendix K (Section 18); (xiv) initial channel occupancy is aligned with A10: document-aligned theorem instances use reset-empty occupancy ($initialOcc = 0$), and accounted nonzero initial occupancy is flagged as an extension outside the main document’s empty-FIFO model (Section 7); (xv) RULE 1 / RULE 2 source well-formedness is recorded as part of the R2 obligation for proof-backed flows (Section 9); (xvi) $ObsSigma$ gains the R2C component-local observation-unit slice already required by the document’s Definition I.Obs (Section 15); (xvii) $JTraceAssumptions$ gains the elaboration-projection well-formedness field that the J.1 proof sketch already consumed, with corresponding K-to-J trace and cutpoint projection constructors (Sections 17 and 18); (xviii) $ProcessStalledAtFront$ is unified with $FrontierBlocked$ so the Appendix K layer uses the Section 13 conjunction form of “blocked on message passing” (Section 18); (xix) $RequiredJOrder$ acyclicity is proved exactly as in the v42 Theorem J.1 argument, by the per-execution commit-cycle measure over the fixed legal prefix (Section 17); (xx) A5 is stated over finite conservative-serial source prefixes, and Theorem K.1 uses Lemma K.2b to convert the selected finite liberal Sys_ref prefix into a finite conservative-serial deadlocking prefix before A5 is invoked (Section 18); (xxi) $START_P$ / $FINISH_P$ anchors are read as per-reset-epoch dynamic occurrences, matching the v42 dynamic-reset treatment of direct inputs (Section 3); (xxii) Corollary L.4 (cadence-sensitive polling handled by isolation) is given an explicit named theorem alias (Section 14); (xxiii) a scope note records that the formalization targets the single-global-clock semantics of Appendices G–K (below); and (xxiv) the v45 Lemma K.2b proof spine is formalized as an eager-count reduction: serial reference executions are explicitly liberal/well-formed reference executions plus the no-eager restriction, and the one-step reduction is supported by endpoint-source-order prefix suppression, joint value/control consistency, occupancy/disabledness domination, no-third-party-enable monotonicity, and terminal-SCC persistence.

M5 alignment note (standing B1 in the main theorem stack). The latest scheduling-rules

Appendix G/J wording treats B1 as a standing RTL_B determinism assumption for the main Appendix G, Theorem J.1, Corollary J.1, and Appendix K theorem instances under the fixed stimulus. Lean therefore treats PostDeterministicSigmaTrace C as carried by the B-rule bundle used by JTraceAssumptions and KAssumptions, not as an optional premise introduced only when uniqueness is invoked. B4/B5 remain separate additional premises used for ε -normalization, ε -quiescent endpoint reasoning, and uniqueness up to finite ε -stutter.

22 June 2026 correction pass. This revision additionally: (xxv) makes Q_Ω a fixed witness-indexed dynamic universe rather than a side/state-dependent object; (xxvi) defines BFaithful and carries per-channel B-faithfulness through ImplementationAssumptions and the side-condition discharge record; (xxvii) states A5 directly over admissible finite conservative-serial source prefixes and makes K.2b return such a prefix, avoiding an unproved fair-extension step; (xxviii) separates the deadlock shape predicate from its reachability wrapper; (xxix) removes the duplicate placeholder K.0 interface in favor of K0_front_nextfront_classification; and (xxx) derives per-process post-side trace-projection determinism from system-level B1 without asserting isolated process determinism.

Second 22 June 2026 correction pass. This revision additionally: (xxxi) removes the residual finite-prefix/infinite-execution type split from the J.1-to-K.2b path; (xxxii) gives J1_selected_ref_prefix_is_liberal the complete selected post-prefix and reference-prefix signature; (xxxiii) places each process dynamic universe directly in WitnessBundle and separates witness-dependent universe well-formedness from universe data; (xxxiv) states the terminated-peer result against the canonical closed, drain-closed WFG-component conjunction; (xxxv) packages the exact selected post prefix, reference prefix, cutpoints, compatibility, normalization, and J.Abs correspondence in SelectedMatchedCutpoint; (xxxvi) strengthens B-faithfulness from a capacity upper bound to exact E4 boundary-behavior refinement; and (xxxvii) replaces free-standing status labels with method-indexed discharge objects whose constructors carry the corresponding syntactic, tool/RTL, or witness evidence.

Third 22 June 2026 correction pass. This revision additionally: (xxxviii) makes ClosedDrainClosedWFGComponent the single canonical component used by ExistsClosedDrainClosedWFGCycle, InternalMPDeadlockShapeAt, process-participation, and the terminated-peer corollaries, with ProcessStalledAtFront defined literally as FrontierBlocked; (xxxix) performs the K.2b eager-count reduction over semantic Prefix C.Sys_ref objects throughout, using RefPrefixBuckets only as a derived bucket projection and requiring each one-step reduction to construct an actual admissible reference prefix;

and (xl) updates the earlier B-faithfulness revision note to identify its Section 1.1 audit, Section 7 formal structure/assumption threading, and Section 20 concrete-flow checklist locations.

Fourth 22 June 2026 correction pass. This revision additionally: (xli) makes the side-condition audit theorem-scoped and requires its WC field to discharge the actual predicate WC C W, with NB-CF, snooping, and mixed per-site coverage represented as proof-producing provenance routes rather than as alternatives inside the audited proposition; (xlii) replaces process-only START_P / FINISH_P anchors with explicit reset-epoch-indexed dynamic synchronization occurrences, bounds terminated-process idle padding by the next reset affecting the process, and states the terminated-peer corollaries only for peers terminated in the current reset epoch at the examined cutpoint; and (xliii) removes duplicate A2–A4 fields from DockStructuralAssumptions and exposes those document-numbered premises as projections of the B4, B2, and B3 fields already carried by A1’s BRules component.

23 June 2026 alignment and proof-architecture pass. This revision additionally: (xliv) separates trace/admissibility assumptions from B4/B5 normalization by introducing TraceBRules and NormalizationBRules, using them in Appendix I, and then introducing JTraceAssumptions and JCutpointAssumptions; (xlv) makes preservation of Exec_ref under each K.2b eager deferral an explicit construction theorem, including the obligation to build an admissible fair/progress-satisfying reference extension for the edited finite prefix rather than treating semantic-prefix realizability as automatic; (xlvi) restructures the former monolithic Milestone B into trace, cutpoint-normalization, and WFG/liveness submilestones; (xlvii) aligns the audit summaries with the scheduling-rules treatment of NB-CF and snooping as provenance routes for WC; (xlviii) targets the attached 23 June 2026 Scheduling Rules document, including its B5 environmental-stability and strict earlier-cycle A5 clarifications; and (xlix) makes Appendix I deterministic replay operate over complete P-local ObservableUnit steps rather than individual Σ labels, so atomic rendezvous, synchronization, and stutter-canonical R2C units are never split; the R2C congruence case now derives canonical emission and fixed-point agreement from the prior r2c_unit_slice, R2C_WF, the fixed stimulus, and the applicable first-observation/change/observer/cycle-locked condition rather than from a common label alone.

24 June 2026 scheduling-alignment pass. This revision additionally: (l) aligns Appendix I/J with the corrected 24 June 2026 Scheduling Rules treatment of deterministic replay by separating canonical immediate post-unit replay from explicitly ε -normalized replay; (li) introduces IDR_immediate_post_unit_deterministic_replay for the plain J.1 ideal-chain

construction and retains `IDR_deterministic_replay` for ε -quiescent cutpoint theorems; (lii) makes the finite-P-local-event branch of I.5 explicit: B6 may be used only after a matched global `Sys_ref` prefix reaches a B5/global fixed point, while the non-fixed-point branch is obtained by projecting an admissible infinite global `Sys_ref` continuation supplied through `Exec_ref`; (liii) records the immediate replay-state invariant in `ReferenceldealChain.invariant` through `JConstructionInvariant` so J.1 imports no B4/B5 normalization premise; (liv) names the side-parametric finite fair drain-normalization helper consumed by the K.2b eager-deferral reduction and by the Theorem K.1 `Exec_post` bridge; and (lv) preserves the document result as Theorem R.2 while using the typed Lean identifier `R_theorem2_closed_cycle_and_one_way_drain_transfer`.

25 June 2026 Appendix T and theorem-carrier correction pass. This revision additionally: (lvi) retypes `CompatibleP`, `CompatibleSys`, and their projected variants over witness-relative `AnnotatedPrefix` carriers that include the selected dynamic universes, `Issue/Commit` annotations, occurrence matching, and request attribution rather than over bare `Prefix` pairs; (lvii) introduces `InitialComparisonCorrespondence`, `ResetEpochCorrespondence`, and `InitialAndResetCorrespondence` as explicit theorem premises and base-case obligations; (lviii) replaces the former frontier-or-pending `NextJUnitRealizable` predicate with `FiniteMatchedUnitExtension`, which proves correct dynamic-occurrence selection, source enablement, label/value agreement, atomic-counterpart availability, and an actual finite ε^* ;unit source extension; (lix) replaces independent existential replay prefixes with `ReferenceldealChain`, one coherent nested chain whose successor starts at the prior terminal state and whose terminal element is the selected source witness; (lx) narrows Proposition J.0 and its projected form to `TraceBRules` so the plain J.1 dependency closure contains no B4/B5 normalization premise; (lxi) adds explicit assumption-projection constructors among `JTraceAssumptions`, `IConstructionAssumptions`, `JCutpointAssumptions`, and the Appendix K bundles; and (lxii) aligns the strategy with the scheduling-rules document's 25 June 2026 Appendix T through a coverage matrix and revised development milestones.

25 June 2026 milestone-restoration and cleanup pass. This revision additionally: (lxiii) restores Milestones B2 and B3, including their cutpoint-normalization and WFG/liveness target lists, so the Milestone B hierarchy and Appendix T coverage matrix again name existing development stages; (lxiv) restores the projected-safety and restricted-reverse guardrails adjacent to the J.1 theorem declarations; (lxv) rewrites `PrefixAnnotations.annotationWF` so its dependent arguments are visibly part of the field type rather than an orphaned fragment; and (lxvi) updates revision note (liii) to locate the immediate replay-state invariant in `ReferenceldealChain.invariant / JConstructionInvariant`.

25 June 2026 theorem-dependency, reset-threading, and proof-plan correction pass. This revision additionally: (lxvii) threads the Appendix B AutomaticFlush object through KStructuralAssumptions and exposes a projection consumed by fair drain normalization, K.2b, and K.1; (lxviii) makes ResetEpochCorrespondence identify matched dynamic reset occurrences and prove reset-empty occupancy on both sides, then threads a MatchedEpochPreparation through every FiniteMatchedUnitExtension so the ReferenceldealChain cannot cross a reset implicitly; (lxix) adds a blocking-core direct WC construction and schedules it before the first B1 theorem that consumes WC; (lxx) replaces the ambiguous prefix-based ReactiveEnvironment.causal field with a cycle-indexed semantic carrier over the complete prior global bucket history, including elapsed empty cycles, while adding ReactiveEnvironmentWF so ordinary Appendix I/J theorem instances remain cross-cycle Σ -causal: equal prior global Σ -visible histories produce equal environment responses independently of absolute cycle or elapsed empty buckets, and genuinely cadence/time-driven behavior is routed to the existing Appendix D/L latency-sensitive mechanisms; (lxxi) adds ordinary-boundary B-faithfulness construction and discharge targets to Milestone A; (lxxii) makes DirectInputBucketWF carry DirectInputContract and derives late-sample value agreement from NoPermittedSyncControlledUpdateInAnchorWindow rather than accepting raw no-change evidence; (lxxiii) scopes JAbsConf to reachable selected prefixes by requiring the same Exec_post and Exec_ref evidence used by the cutpoint theorem; and (lxxiv) separates the B2/B3 admissibility dependency of plain J.1 from the B5 normalization dependency of cutpoint and Appendix K theorems in the Appendix T matrix.

25 June 2026 cross-document consistency and proof-carrier correction pass. This revision additionally: (lxxv) replaces the pipelined-only K-layer automatic-flush projection with a generalized DrainOnlyBlockedFrontierDiscipline that is constructed either from Appendix B AutomaticFlush for pipelined blocking intervals or from ordinary non-overlapped control for other liberal same-interval issue patterns, and threads that discipline through fair drain normalization and eager-deferral-edited prefixes; (lxxvi) gives J0_pairwise_composition and its projected variant explicit pair-specific ElaborationProjectionWF premises, so their ElaborationProjectionLegality conclusions are derivable from their signatures; (lxxvii) states initial reset-empty occupancy separately on the selected Sys_ref and RTL_B initial states rather than through an undefined shared InitialOcc carrier; (lxxviii) adds named J.1 observable-projection uniqueness and normalized finite- ϵ -stutter uniqueness theorems while preserving non-uniqueness of the stateful source witness; (lxxix) unifies the external-array carrier on SourceArrayAccess and ArrayAddress, integrates MemoryEvent through the core IO mapping, and expands the array milestone to include transformation, conflict-free reordering, shared-storage coherence, and synchronization-interval bridge obligations;

(lxxx) expands the elaboration milestone to expose projection, occupancy/accounting, atomicity, required-order, E4, sync-epoch-gate, K-prerequisite, and initial/reset intermediate proofs; and (lxxxi) updates the Appendix T coverage matrix and corrects the residual missing-space typo in the Appendix L milestone.

25 June 2026 primitive-declaration and milestone-completeness pass. This revision additionally: (lxxxii) introduces the previously implicit K-layer predicates `PipelinedControlCanContainOverlappedDynamicWork` and `NoNewDynamicOperationLaunchAfterBlockedPoint`, the boundary-accounting carrier `StoredOrCreditedBoundaryItem`, and the implementation-derived theorem `finite_explicit_boundary_storage_of_implementation` before their first consumers; (lxxxiii) introduces `OrderedObservableTrace`, `OrderedObservableProjectionOfPrefix`, `SamePostObservationCut`, `SameOrderedSigmaProjection`, and `FiniteEpsilonStutterDifference` before the J.1 uniqueness theorems; and (lxxxiv) adds corresponding Milestone A, B1, B2, B3, and H obligations so every new primitive is either explicitly declared or scheduled as a named construction target.

26 June 2026 K.Drain and R.16 alignment pass. This revision additionally: (lxxxv) adds K.Drain as a theorem-scoped Appendix K audit condition, while retaining NB-CF and Appendix L snooping solely as proof-producing provenance routes for WC; (lxxxvi) makes `DrainOnlyBlockedFrontierDiscipline` the complete seven-clause Lean carrier for Definition R.16, including the blocked point, no new source-later dynamic-operation launch, no withdrawal, non-frontier requests excluded as WFG sources, drain-only committed progress, finite non-replenishing drain, and explicit sync-boundary policy; (lxxxvii) separates K.Drain from $K\varepsilon$ so WFG-inert ε -only behavior remains the independent A11/ $K\varepsilon$ premise rather than a conjunct hidden inside the drain discipline; (lxxxviii) removes full `Remain/NextFront` frontier minimality from the implementation-side drain assumption and leaves that stronger fact to `K0_front_nextfront_classification`; and (lxxxix) updates the Section 1.1 audit, Section 13 declarations, Section 20 checklist, Section 21 proof inventory, Appendix T matrix, and Milestone B3 targets to the attached 26 June 2026 Scheduling Rules document.

26 June 2026 cutpoint-carrier, admissible-extension, and theorem-scoped K.Drain correction pass. This revision additionally: (xc) separates the exact J.1-selected prefixes from their later ε -normalized extensions through `PostEpsilonNormalizationExtension`, `RefEpsilonNormalizationExtension`, and `SelectedNormalizedPrefixPair`; (xci) makes `J1_cutpoint_correspondence` consume one reachable selected-normalization carrier containing explicit `Exec_post` and `Exec_ref` evidence; (xcii) redesigns `SelectedMatchedCutpoint` so τ_{post} remains the original requested post prefix while

τ_{post} and τ_{ref} name the normalized representatives; (xciii) removes `terminal_exec_ref` from the core `ReferenceIdealChain` and adds `reference_ideal_chain_terminal_has_admissible_extension` as a named B1 theorem; (xciv) scopes `KDrain` to `reachable`, selected proof-relevant intervals and retains `UniversalDrainConformance` as a sufficient stronger implementation profile; (xcv) states that the Lean `K.Drain` audit discharges all seven clauses of Definition R.16 even if an external prose summary is abbreviated; (xcvi) updates Section 21, the Appendix T coverage matrix, and Milestones A0/B1/B2/B3; and (xcvii) adds an early signature-skeleton target for the J.1, cutpoint, and K.1 interfaces.

26 June 2026 carrier-architecture, selected-elaboration, and sync-gate correction pass. This revision additionally: (xcviii) removes the circular `ComparisonInstance` / `ModelingContracts` / `AssumptionBundle` declaration pattern by keeping `ComparisonInstance` as the semantic comparison data, deriving `Sys_ref` from a dependent reference choice, and supplying modeling contracts and theorem assumptions through later evidence bundles rather than fields of the comparison object; (xcix) replaces the loose `SysRefChoice.Sys_ref` plus `sys_ref_matches_kind` proposition with an ordinary/elaborated dependent selection whose elaborated constructor carries the exact selected package `E` and whose `Sys_ref` is definitionally `E.Sys_elab`; (c) makes the elaboration audit branch depend on that selected reference choice, so an elaborated theorem instance cannot mark elaboration inapplicable and an ordinary instance must prove that no proof-relevant elaboration is required; (ci) repairs sync-boundary withholding by separating the future suffix operation whose work consumes post-sync capacity from the possibly different upstream producer request that is actually refused at the explicit gate; (cii) adds Appendix K A6, unique complementary ownership, and the shared-resource elaboration prerequisite to the normative Appendix T.3.4 row and Milestone B3; (ciii) removes the duplicate `ModelingContracts` declaration, fixes the missing `ComparisonInstance` argument in `ClockOnlyIdleStep`, and derives the sync-controlled no-update window from the anchor/window premises rather than carrying an unused anchor hypothesis; and (civ) records the seven-clause numbering convention explicitly: finite non-replenishing drain is clause (6), and sync-boundary/epoch-gate withdrawal is clause (7).

Terminology note on v42/v45 labels. References to “v42” and “v45” in this strategy identify alignment decisions inherited from earlier review passes. Unless explicitly stated otherwise, the theorem instances, assumptions, and appendix references in this revision target the attached 26 June 2026 Scheduling Rules document, including Appendix T’s scheduling-specific proof-obligation inventory.

Scope note (single clock). The formal core targeted by this strategy is the single-global-clock semantics of Appendices G–K of the scheduling-rules document. Appendix O of that document is an informative multi-clock sketch only. A multi-clock Lean development would additionally require the global event order or partial order over local-domain cycles, precise CDC commit semantics, cross-domain occupancy sampling rules, synchronizer fairness/metastability abstractions, and CDC progress obligations that Appendix O explicitly defers; those layers are out of scope for this strategy unless separately supplied.

The primary scheduling rules document on which this doc is based is available here:

https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/catapult_user_view_scheduling_rules.pdf

The latest version of this Lean proof strategy document is available here:

https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/Lean_proof_strategy.pdf

The Matchlib examples kit that contains training slides, examples, etc., is available here:

<https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/tree/master>

1. Central comparison object

The proof should be parameterized by a selected comparison instance:

```
-- Actual Lean file order places ElaborationPackage and SysRefChoice before
-- ComparisonInstance. The topical order of this strategy is not the file order.
structure ComparisonInstance where
  RTL_B : System
  -- Shared proof-internal observable alphabet for the Sys_ref / RTL_B comparison.
  -- Both sides emit events in this same Sigma schema.
  Sigma : Alphabet
  -- Optional outer DUT/testbench-visible projection alphabet.
  Sigma_DUT : Alphabet
  sigmaDUTProjection : ProjectionMap Sigma Sigma_DUT
  -- The ordinary outer bounded-capacity reference and its channel type are fixed
  -- before the reference choice is made.
  outerSysB : System
  Chan_outer : Type
```

-- The dependent choice is either ordinary outerSysB or one exact selected
-- elaboration package. It definitionally determines Sys_ref.

refChoice :

SysRefChoice outerSysB Chan_outer Sigma Sigma_DUT

env : ReactiveEnvironment Sigma

-- Optional modular combinational model used by R2C.

combModel? : Option CombComposition

namespace ComparisonInstance

-- Sys_ref is derived, not stored independently. Therefore no separate
-- sys_ref_consistent proof can be omitted, weakened, or instantiated with an
-- unrelated proposition.

abbrev Sys_ref (C : ComparisonInstance) : System :=

C.refChoice.Sys_ref

abbrev selectedElaboration? (C : ComparisonInstance) :=

C.refChoice.elab?

end ComparisonInstance

-- ModelingContracts C and theorem-specific assumption bundles are declared
-- after their component structures. They are supplied as explicit fields of
-- JTraceAssumptions, JCutpointAssumptions, KStructuralAssumptions, and related
-- proof contexts; they are not recursively embedded in ComparisonInstance.
-- Section 4.2 contains the single HasModelingContracts declaration that supplies
-- the C.contracts notation used by lower-level schematic declarations.

The comparison relation between C.Sys_ref and C.RTL_B is defined over the single shared proof-internal alphabet C.Sigma. The reference and post-HLS executions may have different internal epsilon-structure, bucket decomposition, or implementation state, but their observable events are encoded in the same Sigma event schema before compatibility is applied. Compatibility itself is witness-relative and is not typed over bare Prefix pairs: each compared prefix is carried together with the selected dynamic universes, Issue/Commit annotations, occurrence matching, request attribution, and observable-unit well-formedness needed by E3/E5, mu_Q, BoundaryReq, and Appendix J.

Sigma_DUT is not a second post-side alphabet. It is only the chosen outer DUT/testbench-

visible projection of $C.\text{Sigma}$. In elaborated cases, introduced proof-internal staged/bundle/join events may be hidden or accounted for by $\text{sigmaDUTProjection} / \text{Pi_elab}$ when projecting to Sigma_DUT , while the proof-internal compatibility theorem still ranges over $C.\text{Sigma}$.

`ComparisonInstance` intentionally contains only semantic comparison data. `ModelingContracts C` and the theorem-specific assumption bundles are later evidence objects. This avoids a recursive declaration in which `ComparisonInstance` contains `ModelingContracts C` while `ModelingContracts` itself is defined from `C`. Any schematic use of `C.contracts` below is read through `HasModelingContracts C`, installed locally from the applicable theorem bundle.

`C.Sys_ref` is the chosen source reference model for the theorem instance:
ordinary bounded-FIFO case: `C.refChoice = .ordinary` and `C.Sys_ref` is definitionally `C.outerSysB`;
elaborated case: `C.refChoice = .elaborated` selected and `C.Sys_ref` is definitionally `selected.E.Sys_elab`.

There is no separately stored `Sys_ref` field and no free-standing `sys_ref_matches_kind` : `Prop`. The dependent constructor records the exact selected elaboration package, and the derived `Sys_ref` definition cannot diverge from it. The formal theorems still do not require a unique algorithm that constructs $\text{Sys_B}^{\text{elab}}$; the selected package is theorem-instance data. They do require that every elaboration-dependent proof, audit entry, projection, B-faithfulness argument, and sync-boundary gate refer to that same selected package.

1.1 Side-condition discharge record

The scheduling-rules Common Formal Definitions require an explicit side-condition discharge record: before Theorem J.1, Corollary J.1 cutpoint transfer, or the Appendix K liveness-preservation theorem is applied to a concrete design, the verification flow shall state, for each named semantic side condition, whether the condition is inapplicable, satisfied by a simple syntactic/design restriction, discharged by tool/RTL evidence, or discharged by a source/RTL witness construction. The named semantic conditions are `WC`, $\text{Sys_B}^{\text{elab}}$, per-channel B-faithfulness, `J.AbsConf`, `K.Drain`, and $K\varepsilon$. NB-CF, Appendix L snooping, direct construction, and mixed per-site coverage are proof-producing provenance routes for `WC`; they are not separate propositions that may replace `WC` in the audited theorem premises. Because the required subset depends on the theorem being applied, the Lean audit object is indexed by theorem scope. The record is not an additional semantic assumption beyond the premises already used by Appendices I–K; its purpose is

to make theorem application auditable and to ensure that a condition required by the selected theorem cannot be marked inapplicable.

The Lean encoding should be a bookkeeping structure over the same evidence objects the strategy already defines, so that an instantiated theorem instance carries its own audit trail:

```
structure NotRequired (P : Prop) : Prop where
  reason : SideConditionNotApplicable P
```

```
structure SyntacticEvidence (P : Prop) : Prop where
  proof : P
  restriction : NamedSyntacticOrDesignRestriction P
```

```
structure ToolOrRTLEvidence (P : Prop) : Prop where
  proof : P
  certificate : NamedToolRTLCertificateOrCheck P
```

```
structure WitnessConstructionEvidence (P : Prop) : Prop where
  proof : P
  construction : NamedSourceRTLWitnessConstruction P
```

```
inductive Discharge (P : Prop) : Type
| inapplicable (evidence : NotRequired P)
| syntacticRestriction (evidence : SyntacticEvidence P)
| toolOrRTLEvidence (evidence : ToolOrRTLEvidence P)
| witnessConstruction (evidence : WitnessConstructionEvidence P)
```

-- RequiredDischarge has no inapplicable constructor.

```
inductive RequiredDischarge (P : Prop) : Type
| syntacticRestriction (evidence : SyntacticEvidence P)
| toolOrRTLEvidence (evidence : ToolOrRTLEvidence P)
| witnessConstruction (evidence : WitnessConstructionEvidence P)
```

```
inductive TheoremScope
| j1Trace
| j1Cutpoint
| k1Liveness
```

```
namespace TheoremScope
```

```
def requiresCutpointTransfer : TheoremScope → Bool
```

```
| .j1Trace => false
```

```
| .j1Cutpoint => true
```

```
| .k1Liveness => true
```

```
def requiresKDrain : TheoremScope → Bool
```

```
| .j1Trace => false
```

```
| .j1Cutpoint => false
```

```
| .k1Liveness => true
```

```
def requiresKepsilon : TheoremScope → Bool
```

```
| .j1Trace => false
```

```
| .j1Cutpoint => false
```

```
| .k1Liveness => true
```

```
end TheoremScope
```

```
def ScopeDischarge (required : Bool) (P : Prop) : Type :=
```

```
if required then RequiredDischarge P else Discharge P
```

```
-- Elaboration is indexed by the actual selected reference choice, not only by
```

```
-- theorem scope. An ordinary instance must prove that no proof-relevant
```

```
-- elaboration is needed. An elaborated instance must prove well-formedness of
```

```
-- the exact package carried by C.refChoice; no inapplicable constructor exists.
```

```
def ElaborationAuditObligation
```

```
(C : ComparisonInstance) : Type :=
```

```
match C.refChoice with
```

```
| .ordinary => NoProofRelevantElaborationRequired C
```

```
| .elaborated selected => WellFormedElaborationPackage selected.E
```

```
-- Each WC route carries a proof of the actual theorem premise WC C W.
```

```
structure SnoopingWCDischarge
```

```
(C : ComparisonInstance)
```

```
(W : WitnessBundle C) where
```

```
assumptions : SnoopingWrapperAssumptions C W
```

```
wc_proof : WC C W
```

construction : NamedSourceRTLWitnessConstruction (WC C W)

structure NBCFWCDischarge
(C : ComparisonInstance)
[HasModelingContracts C]
(W : WitnessBundle C) where
witness_is_identity_or_arbitrary :
W = IdentityOrArbitraryNBWitnessBundle C
nb_cf : $\forall P, \text{NBCF } C P$
local_protocols :
LatencySensitiveLocalProtocolWC
C W C.contracts.latencySensitiveLocalProtocols
wc_proof : WC C W

structure MixedWCDischarge
(C : ComparisonInstance)
(W : WitnessBundle C) where
partition : NBSiteDischargePartition C W
complete : AllRelevantNBSitesCovered C W partition
wc_proof : WC C W
construction : NamedSourceRTLWitnessConstruction (WC C W)

inductive WCDischarge
(C : ComparisonInstance)
(W : WitnessBundle C) : Type
| direct (evidence : WitnessConstructionEvidence (WC C W))
| fromSnooping (evidence : SnoopingWCDischarge C W)
| fromNBCF (evidence : NBCFWCDischarge C W)
| mixed (evidence : MixedWCDischarge C W)

def WCDischarge.proof
(C : ComparisonInstance)
(W : WitnessBundle C) :
WCDischarge C W $\rightarrow$ WC C W
| .direct evidence => evidence.proof
| .fromSnooping evidence => evidence.wc_proof
| .fromNBCF evidence => evidence.wc_proof
| .mixed evidence => evidence.wc_proof

structure SideConditionDischargeRecord

(C : ComparisonInstance)

(W : WitnessBundle C)

(scope : TheoremScope) where

wc : WCDischarge C W

-- This field is definitionally specialized by C.refChoice. It cannot certify an

-- elaborated instance with an unrelated package, and it cannot mark an

-- elaborated instance inapplicable.

elaboration : ElaborationAuditObligation C

bfaithfulness :

$\forall c,$

BackAnnotatedExplicitAbstractChannel C c $\rightarrow$

RequiredDischarge (BFaithful C c)

jabsconf : ScopeDischarge

(TheoremScope.requiresCutpointTransfer scope)

(Nonempty (JAbsConf C W))

kdrain : ScopeDischarge

(TheoremScope.requiresKDrain scope)

(KDrain C W)

kepsilon : ScopeDischarge

(TheoremScope.requiresKepsilon scope)

(A11_Kepsilon_WFGInert C)

The discharge record duplicates no semantics: every evidence route carries a proof of the same predicate consumed by the theorem stack. The wc field proves WC C W itself; NB-CF, snooping, and mixed per-site coverage are provenance for that proof rather than substitute propositions. The global NB-CF route also records identity/arbitrary witness selection and the latency-sensitive local-protocol premise required by Corollary L.3. A mixed route carries a complete partition of all relevant NB sites.

The elaboration field is stronger than an ordinary ScopeDischarge. Its type is computed from C.refChoice: an ordinary instance supplies NoProofRelevantElaborationRequired C, while an elaborated instance supplies WellFormedElaborationPackage for the exact selected package. Therefore an elaborated comparison cannot be labeled inapplicable,

and evidence about an unrelated package cannot satisfy the field. The theorem-scope index separately makes J.AbsConf mandatory for cutpoint-transfer and Appendix K applications and makes K.Drain and Kepsilon independently mandatory for Appendix K. Per-channel B-faithfulness continues to use RequiredDischarge.

Cross-document K.Drain audit interpretation. The Lean kdrain field always discharges the complete seven-clause Definition R.16 property for the theorem-scoped proof-relevant intervals. A shortened external audit summary cannot certify only a subset: no new source-later dynamic-operation launch, non-frontier request exclusion from WFG sources, finite non-replenishing drain, and universal explicit sync-boundary/epoch-gate coverage remain part of the required proposition. The clause numbering used by Lean is: (6) finite non-replenishing drain and (7) explicit sync-boundary policy. If an Appendix B display omits the number on the finite-storage/drain item, that is an editorial numbering defect, not a six-clause semantic variant. A flow may instead prove the stronger UniversalDrainConformance profile and use the implication theorem below.

2. Foundation layer: bucketed executions, not flat traces

The primary execution model should be cycle-bucketed, not a flat sequence of $\text{eps} \mid \text{tick} \mid \text{sigma } e$. The reason is that the document allows multiple same-cycle Σ -events without imposing artificial order among independent same-cycle events. The earlier review correctly noted that a flat trace would either impose a fake linearization or require an added bucket abstraction anyway.

Use:

structure State where

cycle : Nat

epsPos : Nat

store : Store

with explicit advancement rules:

- Intra-cycle ε -normalization steps advance epsPos only.
- Same-cycle Σ -events advance epsPos only.
- Clock ticks advance cycle and reset epsPos.
- Cycle-advancing silent progress is represented at the bucket level: a
- CycleBucket with $\text{sigmaEvents} = \emptyset$ may still consume a real clock cycle and
- may represent HLS-inserted FSM latency, arbitration/bookkeeping latency,
- failed non-blocking polls, or pipeline advances that emit no Σ -event.

Define buckets:

```
structure CycleBucket ( $\Sigma$  : Type) where
  cycle : Nat
  startState : State
  epsPrefix : List EpsStep
  -- The  $\Sigma$ -events committed in this clock cycle.
  -- This is an unordered finite set. A CycleBucket does not carry an intrinsic
  -- sequence or bucket-local partial order among same-cycle events.
  sigmaEvents : Finset  $\Sigma$ 
  epsSuffix : List EpsStep
  endState : State
  endQuiescent : Prop
  def ExecutionBucketed ( $\Sigma$  : Type) :=
  List (CycleBucket  $\Sigma$ )
  def InfiniteExecutionBucketed ( $\Sigma$  : Type) :=
  Nat  $\rightarrow$  CycleBucket  $\Sigma$ 
```

Bucket well-formedness predicates, not the Finset representation alone, enforce per-cycle scalar constraints such as at most one Push_c and at most one Pop_c per channel in a cycle. Required ordering is not stored in CycleBucket. It is derived globally from observable units and from the Appendix J required-order relation $\prec_J$, not from bucket membership or bucket-local sequencing.

The scheduling-rules v42 text intentionally separates two relations:

causal-dependency graph ($\rightarrow$), an information-flow graph over observable IO actions that is useful for explaining functional dependencies but need not be acyclic in every legal execution; and
RequiredJOrder / $\prec_J$, the well-founded strict order over observable units used by the Appendix J finite-ideal induction and by CompatibleP / CompatibleSys.

This provides the right foundation for:

ε -quiescent cutpoints;
rendezvous same-cycle commits;
R2C component-local fixed-point signal-observation units;
J.1 ideal induction over observable units and the mandated $\prec_J$ relation;
B2/B3 temporal reasoning over infinite bucketed executions.

Important separation: ExecutionBucketed is the semantic carrier for state advancement, ε -

quiescence, occupancy updates, R2C fixed points, R2C stutter-canonical unit emission, and fairness/progress reasoning. It is not itself the cross-model trace-compatibility predicate.

In particular, system-level compatibility $\tau_{\text{ref},\text{system}} \approx_{\text{sys}} \tau_{\text{post},\text{system}}$ must not be defined as equality of CycleBucket lists, equality of bucket cycle numbers, or bucket-by-bucket equality of sigmaEvents. The document's Appendix J compatibility relation is an ordered cross-model observable-compatibility predicate. It preserves the explicitly required $\leq_J$ constraints, atomic multi-endpoint units, E1–E5 obligations, FIFO/synchronization/signal-anchor obligations, and matched value labels, while allowing independent observable units to commute or to be grouped into different cycle buckets in Sys_ref and RTL_B. Same-cycle membership in a CycleBucket is therefore a timing fact for one execution, not a source of intrinsic order.

Thus:

bucket equality may be required inside explicitly atomic units, such as rendezvous Push/Pop pairs and SyncChannel/ac_sync handshakes;

bucket decomposition may be used to prove local state-transition and enablement facts;

but CompatibleSys / $\approx_{\text{sys}}$ is defined over observable units and the mandated $\leq_J$ relation, not over identical bucket decompositions.

3. Events, operation instances, and source order

Define dynamic operation universes:

inductive MsgKind

| Push

| Pop

| PushNB

| PopNB

def MsgKind.isBlocking : MsgKind → Bool

| MsgKind.Push => true

| MsgKind.Pop => true

| MsgKind.PushNB => false

| MsgKind.PopNB => false

structure DynMsgInstance where

```

proc : Process
channel : Channel
kind : MsgKind -- Push, Pop, PushNB, PopNB
sourceLoc : SourceLoc
instanceId : Nat

def Blocking (q : DynMsgInstance) : Prop :=
q.kind.isBlocking = true

def NonBlocking (q : DynMsgInstance) : Prop :=
q.kind.isBlocking = false

```

```

structure FrontierItem where
  proc      : Process
  itemKind  : FrontierItemKind
  sourceLoc : SourceLoc
  instanceId : Nat

```

Core source-order objects are witness-specific. They range only over dynamic operation instances that actually occur in the compared execution / witness. Untaken branches and unexecuted loop iterations are not members of this dynamic universe.

```

Qexec : Process → Set DynMsgInstance
QOmega : Process → Set DynMsgInstance
Omega : Process → Set FrontierItem

```

$Qexec, P$ is the execution-level dynamic-instance universe. It includes executed failed non-blocking polls that may produce no Σ -event.

Ω, P is the frontier/order universe used by `Front`, `NextFront`, `WFG`, `E3/E5`, `BoundaryReq` attribution, and source-order reasoning. Failed NB-only executed polls are not included in Ω, P merely because they executed.

$Q\Omega, P$ denotes the executed dynamic message-operation-instance slice induced by Ω, P . Since $Qexec, P$ is a set of `DynMsgInstance` and Ω, P is a set of `FrontierItem`, this is not a direct set intersection. It is defined through the message-instance projection from frontier items:

```

QOmega,P :=
{ q : DynMsgInstance |
  q ∈ Qexec,P ∧
  ∃ x : FrontierItem,
  x ∈ Omega,P ∧
  MsgInstanceOfFrontierItem C P x = some q }

```

```

def MsgInstanceOfFrontierItem
(C : ComparisonInstance)
(P : Process)
(x : FrontierItem) : Option DynMsgInstance :=
...

```

```

-- R2/E2 signal anchors include explicit synchronization calls and the
-- process-boundary START_P / FINISH_P events. Every anchor is a dynamic
-- occurrence in a particular reset epoch; process identity alone is not enough
-- to identify an anchor when a process can be reset and restarted.

```

```

structure ResetEpochId where
index : Nat
deriving DecidableEq

```

```

inductive SyncOccurrenceKind
| start
| call (s : SyncCall)
| finish

```

```

structure SyncOccurrence where
proc : Process
epoch : ResetEpochId
kind : SyncOccurrenceKind
instanceId : Nat

```

```

abbrev SyncAnchor := SyncOccurrence

```

```

def SyncAnchorClock
(C : ComparisonInstance)
(W : WitnessBundle C)
(a : SyncAnchor) : Cycle :=

```

SyncOccurrenceClock C W a

```
def SyncAnchorRealizedInUniverse
(C : ComparisonInstance)
(P : Process)
(W : WitnessBundle C)
( $\Omega$  : Set FrontierItem)
(a : SyncAnchor) : Prop :=
a.proc = P  $\wedge$ 
SyncOccurrenceWellFormed C W a  $\wedge$ 
 $\exists x$ ,
 $x \in \Omega$   $\wedge$ 
SyncOccurrenceFrontierItem C P a x  $\wedge$ 
ExecutedInWitness C W P x
```

-- Pure universe data. It contains no current state and no side parameter.

```
structure DynamicUniverseData
(C : ComparisonInstance)
(P : Process) where
Q_exec : Set DynMsgInstance
 $\Omega$  : Set FrontierItem
Q_ $\Omega$  : Set DynMsgInstance
qomega_characterization :
 $\forall q$ ,
 $q \in Q_{\Omega} \leftrightarrow$ 
 $q \in Q_{\text{exec}} \wedge$ 
 $\exists x, x \in \Omega \wedge \text{MsgInstanceOfFrontierItem C P } x = \text{some } q$ 
message_frontier_items_have_exec_instance :
 $\forall x q$ ,
 $x \in \Omega \rightarrow$ 
MsgInstanceOfFrontierItem C P x = some q  $\rightarrow$ 
 $q \in Q_{\text{exec}} \wedge q \in Q_{\Omega}$ 
qomega_items_have_frontier_item :
 $\forall q$ ,
 $q \in Q_{\Omega} \rightarrow$ 
 $\exists x, x \in \Omega \wedge \text{MsgInstanceOfFrontierItem C P } x = \text{some } q$ 
leP :
{x : FrontierItem //  $x \in \Omega$ }  $\rightarrow$ 
```

```

{y : FrontierItem // y ∈ Ω} → Prop
leP_partial_order : PartialOrder leP

```

```

prefS :
SyncCall → Set {x : FrontierItem // x ∈ Ω}
suffS :
SyncCall → Set {x : FrontierItem // x ∈ Ω}
pred_S :
{x : FrontierItem // x ∈ Ω} → Option SyncAnchor
succ_S :
{x : FrontierItem // x ∈ Ω} → Option SyncAnchor
read_anchor_defined :
∀ x,
SequentialSignalReadItem x.val → pred_S x ≠ none ∧ succ_S x = none
write_anchor_defined :
∀ x,
SequentialSignalWriteItem x.val → succ_S x ≠ none ∧ pred_S x = none

```

```

-- Witness-dependent facts are kept outside DynamicUniverseData. This breaks the
-- former circularity while still proving that W.universe P is exactly the
-- executed dynamic universe selected by W.

```

```

structure WitnessUniverseWF
(C : ComparisonInstance)
(W : WitnessBundle C)
(P : Process) where
executed_only :
∀ x, x ∈ (W.universe P).Ω → ExecutedInWitness C W P x
pred_S_anchor_realized :
∀ x a,
SequentialSignalReadItem x.val →
(W.universe P).pred_S x = some a →
SyncAnchorRealizedInUniverse C P W (W.universe P).Ω a
succ_S_anchor_realized :
∀ x a,
SequentialSignalWriteItem x.val →
(W.universe P).succ_S x = some a →
SyncAnchorRealizedInUniverse C P W (W.universe P).Ω a

```

```

pred_S_is_closest_preceding_anchor :
  ∀ x a,
  SequentialSignalReadItem x.val →
  (W.universe P).pred_S x = some a →
  ClosestPrecedingDynamicSyncAnchor C P W a x.val
succ_S_is_closest_succeeding_anchor :
  ∀ x a,
  SequentialSignalWriteItem x.val →
  (W.universe P).succ_S x = some a →
  ClosestSucceedingDynamicSyncAnchor C P W a x.val
signal_read_ordered_at_pred_S :
  ∀ x a,
  SequentialSignalReadItem x.val →
  (W.universe P).pred_S x = some a →
  SignalItemClock C W x.val = SyncAnchorClock C W a
signal_write_ordered_at_succ_S :
  ∀ x a,
  SequentialSignalWriteItem x.val →
  (W.universe P).succ_S x = some a →
  SignalItemClock C W x.val = SyncAnchorClock C W a

```

```

def QOmegaOfProcess
(C : ComparisonInstance)
(W : WitnessBundle C)
(P : Process) : Set DynMsgInstance :=
(W.universe P).Q_Ω

```

MsgInstanceOfFrontierItem is defined only for message frontier items. Pure synchronization and signal-anchor items map to none. The pred_S and succ_S maps in W.universe P place ordinary sequential signal reads and writes at their realized dynamic synchronization anchors. START_P and FINISH_P are explicitly represented here as distinct reset-epoch-indexed dynamic occurrences; a later reset creates a new epoch and new start/finish occurrence identifiers.

The Q_Ω field is therefore fixed by W.universe P. It is not recomputed from a state, and the reference and post sides do not choose independent universes. State-dependent predicates such as NextFront, BoundaryReq, ChannelEnabled, and ChannelDisabled range over this same fixed witness universe.

Non-blocking calls need explicit modeling:

- failed PushNB / PopNB polls are executed dynamic instances in Qexec, but produce no Σ -event;
- successful non-blocking calls produce the corresponding committed Push_c(v) / Pop_c(v) event;
- repeated NB calls are distinct dynamic source instances even if a request signal remains physically asserted.

This matches the document's issue/commit treatment of NB calls.

structure NonBlockingInstanceWF
(C : ComparisonInstance)
(P : Process)
(W : WitnessBundle C)
(U : DynamicUniverseData C P) where

nb_instances_are_nonpersistent :

$\forall q,$
 $q \in U.Q_exec \rightarrow$
NonBlocking q $\rightarrow$
 $\neg \text{PersistentBlockingRequestInstance } C \ P \ q$

nb_reexecution_distinct_instances :

$\forall q_1 \ q_2 \ t_1 \ t_2,$
 $q_1 \in U.Q_exec \rightarrow$
 $q_2 \in U.Q_exec \rightarrow$
NonBlocking $q_1 \rightarrow$
NonBlocking $q_2 \rightarrow$
SameSourceNBPollSite $q_1 \ q_2 \rightarrow$
ExecutedAtCycle C W $q_1 \ t_1 \rightarrow$
ExecutedAtCycle C W $q_2 \ t_2 \rightarrow$
 $t_1 \neq t_2 \rightarrow$
 $q_1 \neq q_2$

nb_physical_assertion_does_not_identify_instances :

$\forall q_1 \ q_2 \ t_1 \ t_2,$
 $q_1 \in U.Q_exec \rightarrow$
 $q_2 \in U.Q_exec \rightarrow$
NonBlocking $q_1 \rightarrow$

NonBlocking $q_2 \rightarrow$
 SameEndpoint $q_1 q_2 \rightarrow$
 EndpointRequestContinuouslyAsserted $C W q_1.channel t_1 t_2 \rightarrow$
 ExecutedAtCycle $C W q_1 t_1 \rightarrow$
 ExecutedAtCycle $C W q_2 t_2 \rightarrow$
 $t_1 \neq t_2 \rightarrow$
 $q_1 \neq q_2$

nb_not_fair_action :
 $\forall q,$
 $q \in U.Q_exec \rightarrow$
 NonBlocking $q \rightarrow$
 $\neg \exists h, FairAction.push q h \vee FairAction.pop q h$

The last field is only a type-level sanity condition for the fairness layer:
 B2 fairness ranges over persistent blocking Push/Pop requests and real
 synchronization calls. A successful NB call may still contribute the ordinary
 committed Push_c / Pop_c Σ -event, but the NB call instance itself does not become
 a persistent FairAction.

3.1 Initial state and reset-epoch correspondence

Appendix T.2.1 is a named base-case obligation, not an implicit consequence of
 C.WellFormed. The initial post-reset RTL_B state and the prescribed Sys_ref initial state
 must agree on process control, source-visible values, explicit abstract-channel state,
 request/synchronization state, selected signal anchors, and the proof's abstraction slices.
 Reset-empty occupancy is required separately in the selected Sys_ref and RTL_B initial
 states for the document-aligned A10 instances; it is not represented by an undefined
 shared InitialOcc value. No hidden accepted message, stale endpoint request, or stale
 synchronization state may survive reset without a corresponding source object.

```

structure InitialComparisonCorrespondence
(C : ComparisonInstance)
(W : WitnessBundle C) where
  ref_initial : State
  post_initial : State
  ref_is_prescribed_initial : PrescribedInitialState C .ref ref_initial
  post_is_postreset_initial : PostResetInitialState C .post post_initial
  
```

process_control : InitialProcessControlAgreement C W ref_initial post_initial
 source_values : InitialSourceVisibleValueAgreement C W ref_initial post_initial
 abstract_channels : InitialAbstractChannelAgreement C W ref_initial post_initial
 reset_empty :
 $\forall c,$
 DocumentAlignedChannel C $c \rightarrow$
 $B \ C \ c > 0 \rightarrow$
 OccupancyAt C .ref ref_initial $c = 0 \wedge$
 OccupancyAt C .post post_initial $c = 0$
 no_hidden_messages : NoUnaccountedAcceptedMessage C W post_initial
 no_stale_requests : NoUnaccountedPendingEndpointRequest C W post_initial
 no_stale_sync : NoUnaccountedSynchronizationState C W post_initial
 anchors : InitialSignalAndDirectInputAnchorAgreement C W ref_initial post_initial
 initial_abs : InitialAbstractionSliceAgreement C W ref_initial post_initial

structure ResetEpochCorrespondence
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (epoch : ResetEpochId) where
 relevant : RelevantResetEpoch C epoch
 ref_reset_occurrence : DynamicResetOccurrence C .ref epoch
 post_reset_occurrence : DynamicResetOccurrence C .post epoch
 occurrences_correspond :
 MatchedDynamicResetOccurrences C W epoch
 ref_reset_occurrence post_reset_occurrence
 ref_restart : State
 post_restart : State
 processes : ResetAffectedProcessSet C epoch
 restart_states : ResetRestartStateAgreement C W epoch ref_restart post_restart
 channel_state : ResetEpochChannelAgreement C W epoch ref_restart post_restart
 reset_empty :
 $\forall c,$
 ResetAffectedChannel C epoch $c \rightarrow$
 OccupancyAt C .ref ref_restart $c = 0 \wedge$
 OccupancyAt C .post post_restart $c = 0$
 request_and_sync_state : ResetEpochRequestSyncAgreement C W epoch ref_restart
 post_restart
 anchors_and_direct_inputs : ResetEpochAnchorDirectInputAgreement C W epoch

ref_restart post_restart
 no_stale_state : NoCrossEpochHiddenMessageRequestOrSyncState C W epoch
 post_restart
 fresh_dynamic_identity :
 NoPreResetDynamicOccurrenceIdentityLeaksAcrossEpoch C W epoch
 restart_abs : ResetEpochAbstractionSliceAgreement C W epoch ref_restart post_restart

structure InitialAndResetCorrespondence
 (C : ComparisonInstance)
 (W : WitnessBundle C) where
 initial : InitialComparisonCorrespondence C W
 each_relevant_reset :
 $\forall$ epoch, RelevantResetEpoch C epoch $\rightarrow$ ResetEpochCorrespondence C W epoch

A J.1 successor may remain in the current reset epoch or may first cross one relevant dynamic reset. The reset crossing must be represented by an actual finite matched execution splice; it cannot be hidden inside the ordinary epsilon-star replay segment.

structure MatchedResetEpochSplice
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (α _post : PostAnnotatedPrefix C W)
 (α _ref_before : RefAnnotatedPrefix C W)
 (u : ObservableUnit C.Sigma) where
 epoch : ResetEpochId
 correspondence : ResetEpochCorrespondence C W epoch
 post_reset_precedes_unit :
 SelectedPostResetOccurrencePrecedesUnit
 C W α _post correspondence.post_reset_occurrence u
 ref_after_reset : RefAnnotatedPrefix C W
 finite_ref_reset_extension :
 FiniteMatchedResetTransition
 C W α _ref_before ref_after_reset correspondence.ref_reset_occurrence
 prefix_extends : PrefixExtends ref_after_reset.prefix α _ref_before.prefix
 restart_state_selected :
 TerminalState ref_after_reset.prefix = correspondence.ref_restart
 post_restart_selected :
 PostStateAfterResetOccurrence

$C \ W \ \alpha_{\text{post}}$ correspondence.post_reset_occurrence correspondence.post_restart
restart_invariant :

JConstructionInvariantAtResetRestart

$C \ W \ \alpha_{\text{post}}$ ref_after_reset epoch

no_cross_epoch_identity_reuse :

FreshDynamicUniverseAfterReset $C \ W \ \alpha_{\text{ref_before}}$ ref_after_reset epoch

structure MatchedEpochPreparation

(C : ComparisonInstance)

(W : WitnessBundle C)

(α_{post} : PostAnnotatedPrefix C W)

($\alpha_{\text{ref_before}}$: RefAnnotatedPrefix C W)

(u : ObservableUnit C.Sigma) where

prepared_ref : RefAnnotatedPrefix C W

complete :

(prepared_ref = $\alpha_{\text{ref_before}}$ $\wedge$

NoRelevantResetBetweenPriorBoundaryAndUnit C W α_{post} $\alpha_{\text{ref_before}}$ u) $\vee$

($\exists$ splice : MatchedResetEpochSplice C W α_{post} $\alpha_{\text{ref_before}}$ u,

prepared_ref = splice.ref_after_reset)

accounts_for_all_prior_resets :

EveryRelevantPostResetBeforeUnitAccountedFor

$C \ W \ \alpha_{\text{post}}$ $\alpha_{\text{ref_before}}$ prepared_ref u

The immediate-replay base case SameInitialSourceState is derived from InitialComparisonCorrespondence. At every later dynamic reset, MatchedEpochPreparation must select the corresponding ResetEpochCorrespondence, construct the finite reset splice, re-establish the J-construction invariant, and restart with reset-empty occupancy and fresh dynamic-operation identities.

4. Reactive environment and modeling contracts

4.1 Reactive environment

“Fixed stimulus” should be represented by a causal global strategy over the complete prior bucket history. The semantic carrier retains the current cycle and elapsed empty buckets so that an explicitly cycle-accurate or time-driven environment can be represented when such behavior is in scope. However, the ordinary latency-insensitive Appendix I/J theorem instances require the stronger ReactiveEnvironmentWF condition: at corresponding

environment decision points, equal prior global Σ -visible histories must produce equal externally supplied inputs and peer/environment responses, independently of differences in absolute cycle count or in the number of elapsed empty buckets. Thus the cycle index is representational infrastructure; it is not an additional source of stimulus variation in the ordinary source core.

structure ReactiveEnvironment (Σ : Type) where

inputAt :

(t : Cycle) →

SigmaBucketHistoryBefore Σ t →

ExternallInputs

-- Same-cycle extensionality over the complete semantic bucket history.

causal_same_cycle :

$\forall$ t h₁ h₂,

SameSigmaBucketHistoryBefore t h₁ h₂ →

inputAt t h₁ = inputAt t h₂

-- Retained by the semantic carrier even though ordinary Appendix I/J

-- instances may not branch on empty-bucket timing alone.

history_preserves_elapsed_empty_cycles :

SigmaBucketHistoryPreservesElapsedCycles Σ

-- SamePriorSigmaVisibleHistory compares the complete prior global Σ -visible

-- history while forgetting only ε timing and elapsed empty buckets that are not

-- themselves represented by a Σ event. An explicit timer, deadline, or cycle

-- marker placed in Σ remains visible and is therefore not erased.

def SamePriorSigmaVisibleHistory

{ Σ : Type}

{t₁ t₂ : Cycle}

(h₁ : SigmaBucketHistoryBefore Σ t₁)

(h₂ : SigmaBucketHistoryBefore Σ t₂) : Prop :=

SigmaVisibleProjection h₁ = SigmaVisibleProjection h₂

structure ReactiveEnvironmentWF

{ Σ : Type}

(env : ReactiveEnvironment Σ) : Prop where

sigma_causal_cross_cycle :

$$\begin{aligned} &\forall t_1 t_2 h_1 h_2, \\ &\text{SamePriorSigmaVisibleHistory } h_1 h_2 \rightarrow \\ &\text{env.inputAt } t_1 h_1 = \text{env.inputAt } t_2 h_2 \end{aligned}$$

theorem reactive_environment_replay_from_same_sigma_history

```
{Σ : Type}
(env : ReactiveEnvironment Σ)
(hEnv : ReactiveEnvironmentWF env)
{t_ref t_post : Cycle}
(h_ref : SigmaBucketHistoryBefore Σ t_ref)
(h_post : SigmaBucketHistoryBefore Σ t_post)
(hSame : SamePriorSigmaVisibleHistory h_ref h_post) :
env.inputAt t_ref h_ref = env.inputAt t_post h_post :=
by
  exact hEnv.sigma_causal_cross_cycle t_ref t_post h_ref h_post hSame
```

This restores the scheduling document’s global Σ -causal fixed-stimulus premise while preserving the bucketed execution model. InputValue C.env sig t is derived from inputAt t and the history before t, but the Appendix I/J replay step uses reactive_environment_replay_from_same_sigma_history rather than equality at a common absolute cycle. An environment whose externally relevant response depends on raw cycle count, timeout or retry cadence, or the number of elapsed empty buckets without that timing information being represented in Σ does not satisfy the ordinary ReactiveEnvironmentWF condition. Such behavior is latency-sensitive and must be isolated at an Appendix D boundary, snoop- or witness-aligned under Appendix L, supplied through an explicit cycle-accurate local protocol model, or made explicit in the chosen observable alphabet. This is the same classification used by Definition L.1 and Corollary L.4 for cadence-sensitive behavior.

4.2 Modeling contracts

Direct-input contracts, array-access contracts, and latency-sensitive local-protocol isolation contracts should be parallel top-level modeling contracts, not partly folded into RRules and partly left separate.

The scheduling-rules document treats cadence-sensitive retry/timeout polling as latency-sensitive local protocol logic. In the Lean development this should be an explicit modeling contract, not a hidden theorem-side convention. Failed non-blocking polls still remain ε -steps with no committed Push_c / Pop_c Σ -event; the contract instead states that a polling site whose externally visible behavior depends on the number or cadence of failed polls is

outside the ordinary latency-insensitive Appendix I/J source core unless it has been isolated, snoop-aligned, or supplied with an equivalent explicit cycle-accurate local protocol model at the chosen boundary.

inductive LocalProtocolHandling
| ordinaryLatencyInsensitiveCore
| isolatedBoundary
| snoopAligned
| explicitCycleAccurateModel

structure LatencySensitiveLocalProtocolContract

(C : ComparisonInstance) where

handlingOf :

Process → SourceLoc → LocalProtocolHandling

cadence_sensitive_nb_sites_handled :

∀ P loc,

CadenceSensitiveNBPollingSite C P loc →

handlingOf P loc = LocalProtocolHandling.isolatedBoundary ∨

handlingOf P loc = LocalProtocolHandling.snoopAligned ∨

handlingOf P loc = LocalProtocolHandling.explicitCycleAccurateModel

ordinary_core_excludes_unhandled_cadence :

∀ P loc,

handlingOf P loc = LocalProtocolHandling.ordinaryLatencyInsensitiveCore →

¬ CadenceSensitiveNBPollingSite C P loc

isolated_boundary_has_stable_observable_interface :

∀ P loc,

handlingOf P loc = LocalProtocolHandling.isolatedBoundary →

LatencyInsensitiveOrExplicitlyModeledBoundary C P loc

snoop_aligned_sites_have_admissible_witness :

∀ P loc,

handlingOf P loc = LocalProtocolHandling.snoopAligned →

SnoopAlignedLocalProtocolSite C P loc

cycle_accurate_sites_have_explicit_model :

∀ P loc,

handlingOf P loc = LocalProtocolHandling.explicitCycleAccurateModel →

ExplicitCycleAccurateLocalProtocolModel C P loc


```

structure ModelingContracts (C : ComparisonInstance) where
directInputs : DirectInputContract C
arrays : ArrayAccessMappingRules C
latencySensitiveLocalProtocols : LatencySensitiveLocalProtocolContract C

```

```

-- This class is declared after ModelingContracts. It provides the C.contracts
-- notation used in lower-level schematic declarations without recursively
-- embedding ModelingContracts C in ComparisonInstance.
class HasModelingContracts (C : ComparisonInstance) where
value : ModelingContracts C

```

```

namespace ComparisonInstance

```

```

def contracts (C : ComparisonInstance)
[H : HasModelingContracts C] : ModelingContracts C := H.value

```

```

end ComparisonInstance

```

This keeps RRules close to the document's actual R-rule set and treats direct-input, external-array, and latency-sensitive local-protocol obligations as environment/modeling contracts. In particular, timeout/retry/poll-arbiter cadence is not added to the ordinary Σ alphabet; it is either local to an isolated block, selected by a snooping witness for that block, or covered by an explicit cycle-accurate local protocol model.

5. Direct-input contract and late sampling

The direct-input contract should contain only primitive environmental stability/update obligations. The fact that late sampling preserves the logical anchor should be a derived theorem, not a separate hypothesis.

The document says `hls_direct_input` permits late physical sampling because the environment holds the signal stable after reset, while `hls_direct_input_sync` permits controlled updates only at the designated synchronization handshake. The document also states that logical signal Read/Write events are timestamped at the E2 anchor, not at any later physical RTL sampling cycle; late sampling is ε -internal.

Define:

The direct-input stability rule applies only within a single reset epoch, and only to the effective static direct-input discipline after pragma precedence has

been resolved. The scheduling document explicitly permits dynamic resetting during normal HW operation, including resetting of direct-input-controlled regions. It also permits a signal covered by `hls_direct_input` to be governed instead by an `hls_direct_input_sync` update contract. In that case the signal remains a direct input for late-sampling purposes, but it is not required to be stable for the entire reset epoch; it is required to change only at the designated sync-controlled update point and to remain stable between such permitted updates.

```
def NoRelevantResetBetween
(C : ComparisonInstance)
(sig : Signal)
(t1 t2 : Cycle) : Prop :=
  ¬ ∃ r,
  t1 < r ∧
  r ≤ t2 ∧
  ReceiverResetAffectsSignal C sig r
```

-- DirectInput is the umbrella property: the signal may be physically sampled
 -- late by the implementation. EffectiveStaticDirectInput is the ordinary
 -- `hls_direct_input` case after `hls_direct_input_sync` precedence has been
 -- resolved.

```
def EffectiveStaticDirectInput
(C : ComparisonInstance)
(sig : Signal) : Prop :=
  DirectInput sig ∧ ¬ ∃ s, DirectInputSyncControlled sig s
```

-- A sync-controlled direct input is still a DirectInput, but its stability
 -- obligation is the `hls_direct_input_sync` discipline rather than whole-epoch
 -- stability. The indexed form records the controlling synchronization call;
 -- the existential form is a convenient umbrella predicate.

```
def EffectiveSyncControlledDirectInputBy
(C : ComparisonInstance)
(sig : Signal)
(s : SyncCall) : Prop :=
  DirectInput sig ∧ DirectInputSyncControlled sig s
```

```
def EffectiveSyncControlledDirectInput
(C : ComparisonInstance)
(sig : Signal) : Prop :=
  ∃ s, EffectiveSyncControlledDirectInputBy C sig s
-- The field syncControlledUpdateOnlyAtSync records the global discipline that
-- any value change of a sync-controlled direct input must occur at its
-- controlling synchronization handshake.
```

```

def NoSyncControlledValueChangeBetween
(C : ComparisonInstance)
(sig : Signal)
(t1 t2 : Cycle) : Prop :=
  ¬ ∃ t,
  t1 < t ∧
  t ≤ t2 ∧
  ValueChangesAt C.env sig t

```

```

def NoControllingSyncHandshakeBetween
(C : ComparisonInstance)
(s : SyncCall)
(t1 t2 : Cycle) : Prop :=
  ¬ ∃ t,
  t1 < t ∧
  t ≤ t2 ∧
  SyncHandshakeOccursAt C s t

```

```

def NoPermittedSyncControlledUpdateInAnchorWindow
(C : ComparisonInstance)
(sig : Signal)
(s : SyncCall)
(anchor sampleT : Cycle) : Prop :=
  NoControllingSyncHandshakeBetween C s anchor sampleT

```

-- DirectInputAnchorControlledBySync identifies the controlling sync, and the
 -- selected DirectInputAnchorWindow excludes a new controlling-sync handshake
 -- before the physical late sample. Thus the no-permitted-update fact is derived
 -- from the anchor/window premises rather than supplied independently.

```

theorem no_permitted_sync_controlled_update_from_anchor_window
(C : ComparisonInstance)
(sig : Signal)
(s : SyncCall)
(anchor sampleT : Cycle)
(hAnchor : DirectInputAnchorControlledBySync C sig anchor s)
(hWindow : DirectInputAnchorWindow C sig anchor sampleT) :
  NoPermittedSyncControlledUpdateInAnchorWindow C sig s anchor sampleT :=
  by
    exact DirectInputAnchorWindow.no_controlling_sync_handshake
      C sig s anchor sampleT hAnchor hWindow
structure DirectInputContract (C : ComparisonInstance) where
  staticStableWithinResetEpoch :
    ∀ sig t1 t2,
    EffectiveStaticDirectInput C sig →

```

AfterAllReceiversOutOfReset C sig $t_1 \rightarrow$
 $t_1 \leq t_2 \rightarrow$
 NoRelevantResetBetween C sig $t_1 t_2 \rightarrow$
 InputValue C.env sig $t_1 = \text{InputValue C.env sig } t_2$

syncControlledUpdateOnlyAtSync :
 $\forall$ sig s t,
 DirectInputSyncControlled sig s $\rightarrow$
 ValueChangesAt C.env sig t $\rightarrow$
 SyncHandshakeOccursAt C s t

syncControlledStableBetweenUpdates :
 $\forall$ sig s $t_1 t_2$,
 EffectiveSyncControlledDirectInputBy C sig s $\rightarrow$
 AfterAllReceiversOutOfReset C sig $t_1 \rightarrow$
 $t_1 \leq t_2 \rightarrow$
 NoRelevantResetBetween C sig $t_1 t_2 \rightarrow$
 NoSyncControlledValueChangeBetween C sig $t_1 t_2 \rightarrow$
 InputValue C.env sig $t_1 = \text{InputValue C.env sig } t_2$

theorem no_sync_controlled_value_change_between_from_contract

(C : ComparisonInstance)

(hDI : DirectInputContract C)

(sig : Signal)

(s : SyncCall)

($t_1 t_2$: Cycle)

(hSync : EffectiveSyncControlledDirectInputBy C sig s)

(hNoPermitted :

NoPermittedSyncControlledUpdateInAnchorWindow C sig s $t_1 t_2$) :

NoSyncControlledValueChangeBetween C sig $t_1 t_2$:=

by

-- If the value changed in the window, syncControlledUpdateOnlyAtSync would place

-- that change at the controlling sync handshake, contradicting hNoPermitted.

intro hChange

rcases hChange with $\langle t, hLt, hLe, hValChange \rangle$

have hCtrl : DirectInputSyncControlled sig s := hSync.2

have hHandshake : SyncHandshakeOccursAt C s t :=

hDI.syncControlledUpdateOnlyAtSync sig s t hCtrl hValChange

exact hNoPermitted $\langle t, hLt, hLe, hHandshake \rangle$

Then prove the late-sampling preservation theorem by cases on the effective direct-input discipline. The ordinary static case uses reset-epoch stability. The sync-controlled case first derives the local no-value-change window from the controlling-sync update discipline and the absence of a permitted controlling

sync update in the anchor-to-sample window; it does not take raw value stability as an independent side condition.

theorem lateSamplingPreservesLogicalAnchor_static

(C : ComparisonInstance)

(hDI : DirectInputContract C)

(sig : Signal)

(anchor sampleT : Cycle)

(hStatic : EffectiveStaticDirectInput C sig)

(hOutOfReset : AfterAllReceiversOutOfReset C sig anchor)

(hLe : anchor $\leq$ sampleT)

(hNoReset : NoRelevantResetBetween C sig anchor sampleT) :

InputValue C.env sig anchor = InputValue C.env sig sampleT :=

hDI.staticStableWithinResetEpoch

sig anchor sampleT hStatic hOutOfReset hLe hNoReset

theorem lateSamplingPreservesLogicalAnchor_syncControlled

(C : ComparisonInstance)

(hDI : DirectInputContract C)

(sig : Signal)

(s : SyncCall)

(anchor sampleT : Cycle)

(hSync : EffectiveSyncControlledDirectInputBy C sig s)

(hOutOfReset : AfterAllReceiversOutOfReset C sig anchor)

(hAnchor : DirectInputAnchorControlledBySync C sig anchor s)

(hWindow : DirectInputAnchorWindow C sig anchor sampleT)

(hLe : anchor $\leq$ sampleT)

(hNoReset : NoRelevantResetBetween C sig anchor sampleT) :

InputValue C.env sig anchor = InputValue C.env sig sampleT :=

by

have hNoPermitted :

NoPermittedSyncControlledUpdateInAnchorWindow C sig s anchor sampleT :=

no_permitted_sync_controlled_update_from_anchor_window

C sig s anchor sampleT hAnchor hWindow

have hNoChange :

NoSyncControlledValueChangeBetween C sig anchor sampleT :=

no_sync_controlled_value_change_between_from_contract

C hDI sig s anchor sampleT hSync hNoPermitted

exact hDI.syncControlledStableBetweenUpdates

sig s anchor sampleT hSync hOutOfReset hLe hNoReset hNoChange

theorem lateSamplingPreservesLogicalAnchor

(C : ComparisonInstance)

(hDI : DirectInputContract C)

(sig : Signal)

(anchor sampleT : Cycle)

(hIsDI : DirectInput sig)

```

(hOutOfReset : AfterAllReceiversOutOfReset C sig anchor)
(hWindow : DirectInputAnchorWindow C sig anchor sampleT)
(hLe : anchor ≤ sampleT)
(hNoReset : NoRelevantResetBetween C sig anchor sampleT)
(hEffective :
EffectiveStaticDirectInput C sig V
∃ s,
  EffectiveSyncControlledDirectInputBy C sig s ∧
  DirectInputAnchorControlledBySync C sig anchor s) :
InputValue C.env sig anchor = InputValue C.env sig sampleT :=
by
cases hEffective with
| inl hStatic =>
  exact lateSamplingPreservesLogicalAnchor_static
  C hDI sig anchor sampleT hStatic hOutOfReset hLe hNoReset
| inr hSyncCase =>
  rcases hSyncCase with ⟨s, hSync, hAnchor⟩
  exact lateSamplingPreservesLogicalAnchor_syncControlled
  C hDI sig s anchor sampleT hSync hOutOfReset hAnchor hWindow hLe hNoReset

```

Bucket-location rule:

For sequential processes, the logical direct-input $\text{Read}(\text{sig}, \text{val}) / \text{Write}(\text{sig}, \text{val})$ event belongs only to the E2 anchor bucket β_{anchor} . A physical late sample, if modeled, is an ε -step in a later bucket. The late ε -step creates no new Σ -event. $\text{lateSamplingPreservesLogicalAnchor}$ is used only to justify the value recorded in the logical sequential anchor bucket.

This sequential anchoring rule is not applied to combinational-process signal IO. Combinational processes do not have $\text{pred_S} / \text{succ_S}$ synchronization anchors. Their observable direct-input Read/Write labels, when any direct input participates in combinational signal IO, are included only inside an emitted R2C fixed-point signal-observation unit for the relevant combinational component. The unit is emitted according to the v42 R2C unit-generation convention: first defined observation, later observable-slice value changes, or an explicitly modeled observer/sampler requirement. Intermediate physical samples, delta-cycle reads/writes, redundant re-evaluations, transient values before the fixed point, and R2C stutter cycles that emit no unit are ε /stutter and are not separately Σ -visible.

-- These anchor predicates are derived from the per-process DynamicUniverse
-- fields pred_S and succ_S . Thus the anchor used below is the event's actual

-- E2/R2 anchor, not an unconstrained synchronization point.

structure DirectInputBucketWF (C : ComparisonInstance) where
contract : DirectInputContract C

sequential_read_recorded_only_at_pred_anchor :

$\forall$ sig val anchor sampleT β ,

DirectInput sig $\rightarrow$

SequentialSignalReadEvent (LogicalReadEvent sig val) $\rightarrow$

LogicalReadEvent sig val $\in$ BucketSigmaEvents C $\beta \rightarrow$

SequentialSignalReadAnchoredAt C (LogicalReadEvent sig val) anchor $\rightarrow$

DirectInputAnchorWindow C sig anchor sampleT $\rightarrow$

$\beta = \text{AnchorBucket C anchor}$

sequential_write_recorded_only_at_succ_anchor :

$\forall$ sig val anchor β ,

DirectInput sig $\rightarrow$

SequentialSignalWriteEvent (LogicalWriteEvent sig val) $\rightarrow$

LogicalWriteEvent sig val $\in$ BucketSigmaEvents C $\beta \rightarrow$

SequentialSignalWriteAnchoredAt C (LogicalWriteEvent sig val) anchor $\rightarrow$

$\beta = \text{AnchorBucket C anchor}$

late_sample_is_epsilon_only :

$\forall$ sig anchor sampleT,

DirectInputAnchorWindow C sig anchor sampleT $\rightarrow$

PhysicalLateSample C sig sampleT $\rightarrow$

EpsilonStepOnly C sig sampleT $\wedge$

$\neg \exists$ val β ,

PhysicalSampleEvent sig val sampleT $\in$ BucketSigmaEvents C β

The bucket record does not contain a primitive late_sample_value_matches_anchor assumption. Value agreement is derived from its DirectInputContract field and, for the sync-controlled branch, from the absence of a permitted update through the controlling synchronization handshake:

theorem DirectInputBucketWF.late_sample_value_matches_anchor

(C : ComparisonInstance)

(hWF : DirectInputBucketWF C)

(sig : Signal)
 (anchor sampleT : Cycle)
 (hDI : DirectInput sig)
 (hOutOfReset : AfterAllReceiversOutOfReset C sig anchor)
 (hWindow : DirectInputAnchorWindow C sig anchor sampleT)
 (hLe : anchor $\leq$ sampleT)
 (hNoReset : NoRelevantResetBetween C sig anchor sampleT)
 (hEffective :
 EffectiveStaticDirectInput C sig $\vee$
 $\exists$ s,
 EffectiveSyncControlledDirectInputBy C sig s $\wedge$
 DirectInputAnchorControlledBySync C sig anchor s $\wedge$
 NoPermittedSyncControlledUpdateInAnchorWindow
 C sig s anchor sampleT) :
 InputValue C.env sig anchor = InputValue C.env sig sampleT :=
 lateSamplingPreservesLogicalAnchor
 C hWF.contract sig anchor sampleT hDI hOutOfReset hWindow hLe hNoReset hEffective

This explicitly reconciles the bucket model with direct-input late sampling:
 CompatibleP and CompatibleSys compare the logical Σ -visible direct-input
 Read/Write events at the appropriate logical unit — the E2 anchor bucket for
 sequential-process signal IO, or the emitted component-local R2C fixed-point
 signal-observation unit for combinational-process signal IO — never the later
 physical sample micro-step.

Combinational-process signal IO uses R2C, not the direct-input late-sampling contract.
 The R2C rule is therefore a general combinational-process well-formedness
 obligation. It applies to every combinational-process signal Read/Write event in
 the compared Σ trace, regardless of whether the signal is marked DirectInput.

structure R2CObservationUnitWF (C : ComparisonInstance) where
 selected_observable_slice_defined :
 $\forall$ comp,
 R2CComponent C comp $\rightarrow$
 $\exists$ obs, ObservableSlice C comp obs

 unit_emitted_only_at_canonical_occurrence :
 $\forall$ comp unit t,
 R2CUnitOfComponent C comp unit $\rightarrow$

$R2UnitEmittedAt\ C\ unit\ t \rightarrow$
 $FirstDefinedR2CObservation\ C\ comp\ t\ V$
 $ObservableSliceChangedSincePreviousUnit\ C\ comp\ t\ V$
 $ExplicitObserverRequiresFreshR2CUnit\ C\ comp\ t\ V$
 $CycleLockedR2CComponent\ C\ comp$

$unit_uses_unique_fixed_point :$
 $\forall\ comp\ unit\ t\ \mu,$
 $R2UnitOfComponent\ C\ comp\ unit \rightarrow$
 $R2UnitEmittedAt\ C\ unit\ t \rightarrow$
 $CombFixedPointAtComponentCycle\ C\ comp\ t\ \mu \rightarrow$
 $\forall\ e,$
 $e \in unit.events \rightarrow$
 $EventReadsWritesStore\ e\ \mu$

$unit_not_split :$
 $\forall\ comp\ unit\ e_1\ e_2,$
 $R2UnitOfComponent\ C\ comp\ unit \rightarrow$
 $e_1 \in unit.events \rightarrow$
 $e_2 \in unit.events \rightarrow$
 $SameObservableUnit\ C\ .ref\ unit\ e_1\ e_2 \wedge SameObservableUnit\ C\ .post\ unit\ e_1\ e_2$

$stutter_cycles_emit_no_unit :$
 $\forall\ comp\ t,$
 $R2CStutterCycle\ C\ comp\ t \rightarrow$
 $\neg \exists\ unit,\ R2UnitOfComponent\ C\ comp\ unit \wedge R2UnitEmittedAt\ C\ unit\ t$

$matched_by_component_local_ordinal :$
 $\forall\ comp\ k\ u_ref\ u_post,$
 $KthR2CUnit\ C\ .ref\ comp\ k\ u_ref \rightarrow$
 $KthR2CUnit\ C\ .post\ comp\ k\ u_post \rightarrow$
 $R2UnitValuesAgree\ C\ comp\ u_ref\ u_post$

$multi_input_component_has_anchor_or_confluence_proof :$
 $\forall\ comp,$
 $MultiInputR2CComponent\ C\ comp \rightarrow$
 $SingleExplicitAnchoringDomain\ C\ comp\ V$

AllAdmissibleInterleavingsInduceSameR2CUnitSequence C comp V
R2CComponentHiddenBehindExplicitSynchronizationOrTransactor C comp

6. Temporal logic layer for B2/B3/B4/B5/B6

B2 and B3 should not be opaque Prop fields. They are temporal assumptions over infinite executions.

The document defines B2 as weak fairness over enabled actions, but v42 clarifies the enabled/selected/committed split. For message operations, ChannelEnabled means that the abstract boundary permits the operation to be selected; it does not by itself mean the transfer has already committed in that cycle. B2 therefore says: if an action remains continuously ChannelEnabled / eligible from some cycle onward, the scheduler or arbiter eventually selects it, and the selected action commits in the selected cycle. B3 remains the P_push(c) / P_pop(c) channel-progress obligation; it may be needed to discharge full/empty/rendezvous-blocked conditions before a pending transfer becomes ChannelEnabled.

Define temporal operators:

```
def AlwaysFrom
  (τ : InfiniteExecutionBucketed Σ)
  (t : Nat)
  (P : Nat → Prop) : Prop :=
  ∀ n, t ≤ n → P n
```

```
def EventuallyFrom
  (τ : InfiniteExecutionBucketed Σ)
  (t : Nat)
  (P : Nat → Prop) : Prop :=
  ∃ n, t ≤ n ∧ P n
```

Concrete fairness actions:

- B2 fairness ranges only over scheduler-selectable operations.
- For message operations, this means persistent blocking Push and Pop boundary
- requests. Non-blocking PushNB/PopNB polls are not FairAction cases: failed NB
- polls are ε -steps, and successful NB polls contribute the ordinary committed
- Push/Pop Σ -event without creating a persistent scheduler-fairness obligation.
- For synchronization, fairness ranges only over real synchronization calls

-- such as wait(), ac_wait, ac_sync, or SyncChannel calls. It does not range over
 -- the auxiliary process-boundary anchors START_P and FINISH_P.

inductive FairAction

| push (q : DynMsgInstance) (h : q.kind = MsgKind.Push)

| pop (q : DynMsgInstance) (h : q.kind = MsgKind.Pop)

| sync (s : SyncCall)

Weak fairness:

Weak fairness is a side-parametric temporal assumption over the FairAction type above. For message operations, FairAction contains only blocking Push and blocking Pop dynamic instances. For synchronization operations, FairAction contains only real SyncCall instances, not arbitrary SyncAnchor values. Thus START_P and FINISH_P may still be used as formal R2/E2 signal anchors when they are realized in the dynamic universe, but they are not scheduler-fairness actions. Non-blocking PushNB/PopNB instances are excluded at the type level; they are handled through the NB execution/witness machinery and through ordinary committed Push/Pop Σ -events when successful.

Weak fairness applies to any valid infinite execution of the selected comparison side. In particular, it is not a post-only premise: the reference-side execution used by Exec_ref and the post-side execution used by RTL_B both carry the same WF/B2 obligation whenever that side is being used in a liveness/progress theorem.

For message actions, FairActionEnabled delegates to ChannelEnabledSide and therefore denotes abstract-boundary eligibility for selection, not an already committed transfer. Selection is a separate scheduler/arbiter fact, and the B2 commit-soundness clause below states that a selected enabled message action, or selected enabled rendezvous pair, commits in that selected cycle.

def FairActionCommitsAt

(C : ComparisonInstance)

(side : Side)

(τ : InfiniteExecutionBucketed C.Sigma)

(a : FairAction)

(n : Cycle) : Prop :=

FairActionSelected C side τ a n $\wedge$

SelectedFairActionCommitsInCycle C side τ a n

```

def WeakFairness
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma) : Prop :=
 $\forall a t$ ,
AlwaysFrom  $\tau t$  (fun n => FairActionEnabled C side  $\tau a n$ )  $\rightarrow$ 
EventuallyFrom  $\tau t$  (fun n => FairActionCommitsAt C side  $\tau a n$ )

```

B3 channel-progress predicates should be defined in terms of continuous endpoint-level ready/valid requests, with operation attribution supplied separately. They should not require that the complementary endpoint's continuous request be witnessed by one persistent dynamic message-operation instance.

A blocking request is persistent from cycle t through its commit cycle inclusive when the same dynamic operation instance q keeps its BoundaryReq asserted throughout that interval. For Push operations, persistence also includes payload stability while the request is outstanding, including the commit cycle itself:

```

def PersistentRequestFrom
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(q : DynMsgInstance)
(t : Nat) : Prop :=
Blocking q  $\wedge$ 
 $\forall n$ ,
 $t \leq n \rightarrow$ 
 $\neg$  CommitsBefore C side  $\tau q n \rightarrow$ 
BoundaryReqAt C side  $\tau q n \wedge$ 
(q.kind = MsgKind.Push  $\rightarrow$ 
PayloadAt C side  $\tau q n$  = PayloadAt C side  $\tau q t$ )

```

PersistentRequestFrom remains the right predicate for a single outstanding blocking Push or Pop. It is deliberately not used to collapse a sequence of executed non-blocking polls into one dynamic instance. Repeated PushNB/PopNB polls are distinct source-level dynamic instances, even if the concrete endpoint request signal remains asserted across adjacent cycles.

```

inductive EndpointDirection
| push
| pop

```

```

def DirectionOf
(q : DynMsgInstance) : EndpointDirection :=
match q.kind with
| MsgKind.Push => EndpointDirection.push
| MsgKind.PushNB => EndpointDirection.push
| MsgKind.Pop => EndpointDirection.pop
| MsgKind.PopNB => EndpointDirection.pop

def IsPushKind
(k : MsgKind) : Prop :=
k = MsgKind.Push ∨ k = MsgKind.PushNB

def IsPopKind
(k : MsgKind) : Prop :=
k = MsgKind.Pop ∨ k = MsgKind.PopNB

def EndpointRequestAssertedAt
(C : ComparisonInstance)
(side : Side)
(τ : InfiniteExecutionBucketed C.Sigma)
(c : Channel)
(dir : EndpointDirection)
(n : Nat) : Prop :=
match dir with
| EndpointDirection.push => EndpointValidAssertedAt C side τ c n
| EndpointDirection.pop => EndpointReadyAssertedAt C side τ c n

structure EndpointRequestWitness
(C : ComparisonInstance)
(side : Side)
(τ : InfiniteExecutionBucketed C.Sigma)
(c : Channel)
(dir : EndpointDirection)
(n : Nat) where
endpoint_asserted :
EndpointRequestAssertedAt C side τ c dir n
source_attributed :
∃ q,
q.channel = c ∧
DirectionOf q = dir ∧
BoundaryReqAt C side τ q n
payload_present_if_push :
dir = EndpointDirection.push →

```

$\exists v, \text{EndpointPayloadAt } C \text{ side } \tau \text{ c } n = \text{some } v$

-- Endpoint request continuity is not a forever-after-t predicate. It is scoped
-- to the interval before the relevant discharge event D. After D occurs, the
-- endpoint may drop the request or change payload for the next operation.

def EndpointRequestHoldsUntilDischarged

(C : ComparisonInstance)

(side : Side)

(τ : InfiniteExecutionBucketed C.Sigma)

(c : Channel)

(dir : EndpointDirection)

(t : Nat)

(D : Nat $\rightarrow$ Prop) : Prop :=

HoldsUntilDischarged τ t

(fun n =>

EndpointRequestWitness C side τ c dir n)

D

-- Push payload stability is also scoped only to the outstanding interval. A
-- successful transfer may be followed by a deasserted request or a different
-- payload for a later operation.

def PushPayloadStableUntilDischarged

(C : ComparisonInstance)

(side : Side)

(τ : InfiniteExecutionBucketed C.Sigma)

(c : Channel)

(t : Nat)

(D : Nat $\rightarrow$ Prop) : Prop :=

$\exists v,$

HoldsUntilDischarged τ t

(fun n =>

EndpointPayloadAt C side τ c n = some v)

D

def ContinuousPushRequestUntilDischarged

(C : ComparisonInstance)

(side : Side)

(τ : InfiniteExecutionBucketed C.Sigma)

(c : Channel)

(t : Nat)

(D : Nat $\rightarrow$ Prop) : Prop :=

EndpointRequestHoldsUntilDischarged C side τ c EndpointDirection.push t D $\wedge$

PushPayloadStableUntilDischarged C side τ c t D

```

def ContinuousPopRequestUntilDischarged
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(c : Channel)
(t : Nat)
(D : Nat  $\rightarrow$  Prop) : Prop :=
EndpointRequestHoldsUntilDischarged C side  $\tau$  c EndpointDirection.pop t D

```

```

def BufferedChannel
(C : ComparisonInstance)
(c : Channel) : Prop :=
0 < capacity C c

```

```

def RendezvousChannel
(C : ComparisonInstance)
(c : Channel) : Prop :=
capacity C c = 0

```

```

def SomePopCommitOn
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(c : Channel)
(n : Nat) : Prop :=
 $\exists$  qPop,
qPop.channel = c  $\wedge$ 
IsPopKind qPop.kind  $\wedge$ 
CommitAtCycle C side  $\tau$  qPop n

```

```

def SomePushCommitOn
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(c : Channel)
(n : Nat) : Prop :=
 $\exists$  qPush,
qPush.channel = c  $\wedge$ 
IsPushKind qPush.kind  $\wedge$ 
CommitAtCycle C side  $\tau$  qPush n

```

```

def MatchingTransferCommitOnChannel
(C : ComparisonInstance)
(side : Side)

```

```

(τ : InfiniteExecutionBucketed C.Sigma)
(c : Channel)
(n : Nat) : Prop :=
  ∃ qPush qPop,
  qPush.channel = c ∧
  qPop.channel = c ∧
  IsPushKind qPush.kind ∧
  IsPopKind qPop.kind ∧
  MatchingTransferCommitOn C side τ qPush qPop n

```

```

def BufferedFullBothEndpointsRequesting
(C : ComparisonInstance)
(side : Side)
(τ : InfiniteExecutionBucketed C.Sigma)
(c : Channel)
(n : Nat) : Prop :=
  BufferedChannel C c ∧
  occAt C side τ c n = capacity C c ∧
  EndpointRequestAssertedAt C side τ c EndpointDirection.push n ∧
  EndpointRequestAssertedAt C side τ c EndpointDirection.pop n

```

```

def BufferedEmptyBothEndpointsRequesting
(C : ComparisonInstance)
(side : Side)
(τ : InfiniteExecutionBucketed C.Sigma)
(c : Channel)
(n : Nat) : Prop :=
  BufferedChannel C c ∧
  occAt C side τ c n = 0 ∧
  EndpointRequestAssertedAt C side τ c EndpointDirection.pop n ∧
  EndpointRequestAssertedAt C side τ c EndpointDirection.push n

```

```

def RendezvousBothEndpointsRequesting
(C : ComparisonInstance)
(side : Side)
(τ : InfiniteExecutionBucketed C.Sigma)
(c : Channel)
(n : Nat) : Prop :=
  RendezvousChannel C c ∧
  EndpointRequestAssertedAt C side τ c EndpointDirection.push n ∧
  EndpointRequestAssertedAt C side τ c EndpointDirection.pop n

```

-- P holds continuously from t until the first discharge cycle.
-- This avoids the vacuity that would result from requiring the stalled condition

-- to remain true forever after the discharge has already occurred.

```
def HoldsUntilDischarged
  (τ : InfiniteExecutionBucketed Σ)
  (t : Nat)
  (P : Nat → Prop)
  (D : Nat → Prop) : Prop :=
  ∀ n, t ≤ n → (∀ k, t ≤ k → k < n → ¬ D k) → P n
```

```
def EventuallyDischargedFrom
  (τ : InfiniteExecutionBucketed Σ)
  (t : Nat)
  (D : Nat → Prop) : Prop :=
  ∃ n, t ≤ n ∧ D n
```

Then:

```
def P_push
  (C : ComparisonInstance)
  (side : Side)
  (τ : InfiniteExecutionBucketed C.Sigma)
  (c : Channel) : Prop :=
  (BufferedChannel C c →
  ∀ t,
  BufferedFullBothEndpointsRequesting C side τ c t →
  ContinuousPushRequestUntilDischarged C side τ c t
  (fun n =>
  SomePopCommitOn C side τ c n) →
  ContinuousPopRequestUntilDischarged C side τ c t
  (fun n =>
  SomePopCommitOn C side τ c n) →
  EventuallyDischargedFrom τ t
  (fun n =>
  SomePopCommitOn C side τ c n))
  ∧
  (RendezvousChannel C c →
  ∀ t,
  RendezvousBothEndpointsRequesting C side τ c t →
  ContinuousPushRequestUntilDischarged C side τ c t
  (fun n =>
  MatchingTransferCommitOnChannel C side τ c n) →
  ContinuousPopRequestUntilDischarged C side τ c t
  (fun n =>
  MatchingTransferCommitOnChannel C side τ c n) →
  EventuallyDischargedFrom τ t
  (fun n =>
```

MatchingTransferCommitOnChannel C side τ c n))

```

def P_pop
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(c : Channel) : Prop :=
(BufferedChannel C c  $\rightarrow$ 
 $\forall t$ ,
BufferedEmptyBothEndpointsRequesting C side  $\tau$  c t  $\rightarrow$ 
ContinuousPopRequestUntilDischarged C side  $\tau$  c t
(fun n =>
SomePushCommitOn C side  $\tau$  c n)  $\rightarrow$ 
ContinuousPushRequestUntilDischarged C side  $\tau$  c t
(fun n =>
SomePushCommitOn C side  $\tau$  c n)  $\rightarrow$ 
EventuallyDischargedFrom  $\tau$  t
(fun n =>
SomePushCommitOn C side  $\tau$  c n))
 $\wedge$ 
(RendezvousChannel C c  $\rightarrow$ 
 $\forall t$ ,
RendezvousBothEndpointsRequesting C side  $\tau$  c t  $\rightarrow$ 
ContinuousPopRequestUntilDischarged C side  $\tau$  c t
(fun n =>
MatchingTransferCommitOnChannel C side  $\tau$  c n)  $\rightarrow$ 
ContinuousPushRequestUntilDischarged C side  $\tau$  c t
(fun n =>
MatchingTransferCommitOnChannel C side  $\tau$  c n)  $\rightarrow$ 
EventuallyDischargedFrom  $\tau$  t
(fun n =>
MatchingTransferCommitOnChannel C side  $\tau$  c n))

```

The removed HoldsUntilDischarged premises are not independent B3 assumptions. For bounded FIFOs, the needed full/empty persistence facts are derived from E4: if $\text{occAt } C \text{ side } \tau \text{ c } t = \text{capacity } C \text{ c}$ and no Pop_c commits before n , then the channel remains full at n ; dually, if $\text{occAt } C \text{ side } \tau \text{ c } t = 0$ and no Push_c commits before n , then the channel remains empty at n . For rendezvous channels, the corresponding “both endpoints requesting until transfer” fact is exactly the conjunction of the two continuous endpoint-request premises together with $\text{RendezvousChannel } C \text{ c}$.

Thus P_{push} and P_{pop} state only the real B3 progress obligation: once the appropriate blocked condition holds at t and both endpoint requests continue until the matching discharge event, that discharge event must eventually occur.

The occupancy-stability facts needed inside proofs should be supplied as E4 derived lemmas rather than duplicated as extra hypotheses in B3.

This matches the document's conditional B3 reading: B3 forces progress only when both endpoints continuously request the matching transfer until the corresponding discharge event. A producer stalled on a full channel is not automatically a B3 violation if the consumer is not yet requesting the matching pop, and a consumer stalled on an empty channel is not automatically a B3 violation if the producer is not yet requesting the matching push.

The continuous-request premise is endpoint-level and interval-scoped. For blocking Push/Pop it may be witnessed by one persistent dynamic operation instance using PersistentRequestFrom through the commit cycle. For non-blocking polling loops it may instead be witnessed by a sequence of distinct executed PushNB/PopNB instances on the same endpoint, provided the endpoint request remains continuously asserted until discharge and each cycle's assertion has BoundaryReq attribution to the appropriate executed dynamic instance.

The payload-stability premise for Push is likewise scoped only until discharge. After the discharge event, the producer may deassert valid or present a different payload for a later operation. Thus B3 tracks the ready/valid progress premise without requiring an endpoint request or payload to remain stable forever and without misidentifying multiple NB polls as one persistent q.

B3 modularity (v42). For a single explicit abstract boundary with ordinary E4 buffered or rendezvous semantics, the corresponding B3 discharge is implied by B2 plus E4 once the complementary endpoint request in the B3 premise remains continuously asserted: in the buffered full case, $\text{occ_c} = B(c) > 0$ makes the complementary Pop_c continuously ChannelEnabled, so B2 forces eventual selection and commit of that Pop_c, freeing space; in the buffered empty case, $\text{occ_c} = 0 < B(c)$ makes the complementary Push_c continuously ChannelEnabled, so B2 forces eventual selection and commit of that Push_c, providing data; and in the rendezvous case, the two persistent endpoint requests make the rendezvous pair continuously ChannelEnabled, so B2 supplies eventual rendezvous selection and commit. Lean should prove this derivation explicitly:

```
theorem B3_ordinary_boundary_from_B2_E4
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(c : Channel)
(hExec : InfiniteExecutionOfSide C side  $\tau$ )
(hWF : WeakFairness C side  $\tau$ )
(hE4 : E4OccupancyTraceWF C side  $\tau$ )
(hOrd : OrdinaryE4Boundary C c) :
```

$P_push \text{ C side } \tau \text{ c} \wedge P_pop \text{ C side } \tau \text{ c}$

Here `OrdinaryE4Boundary C c` records that c is a single explicit abstract boundary whose `ChannelEnabled` predicate is exactly the ordinary E4 buffered or rendezvous availability predicate, with no elaborated staged/bundle/join/epoch-gate structure interposed. The proof uses the E4-derived occupancy-stability lemmas (full stays full until a Pop commits; empty stays empty until a Push commits), the continuous complementary endpoint-request premise of P_push / P_pop , and B2 selection-commit soundness for the continuously enabled complementary action or rendezvous pair.

B3 is nevertheless retained as the explicit named B-rule obligation in `BRules` and `DockStructuralAssumptions`, exactly as in the v42 document: Lemma I.2, Appendix K, and the channel-validation methodology consume B3 in that modular per-channel form, and it is the form checked per explicit abstract channel in elaborated $\text{Sys_B}^{\text{elab}}$ boundary chains, where the boundary in question may not satisfy `OrdinaryE4Boundary`. Thus B3 should be read as the modular channel-progress assumption used by the proof and checked against channel realizations, not as a claim that the ordinary single-boundary clauses are logically independent of B2 plus E4. For ordinary instances, `B3_ordinary_boundary_from_B2_E4` lets the B3 field of `BRules` be discharged inside Lean rather than assumed separately.

The latest scheduling-rules document treats B1 as a standing determinism assumption for the post-HLS `RTL_B` side in the main Appendix G/J/K theorem stack. B1 is not a source determinism assumption: the source-side models $\text{Sys} / \text{Sys_B} / \text{Sys_ref}$ may admit multiple ε - or latency-different executions having the same Σ -projection, and witness selection / snooping resolve the required source-side correspondence when needed.

For a fixed comparison instance — including the fixed initial state and fixed reactive environment / external stimulus — B1 requires uniqueness of the labeled post-side Σ -trace, not uniqueness of the full internal ε -execution. Internal ε -steps, ε -depths, and other non-observable scheduling details may differ between valid post-side executions.

```
def PostDeterministicSigmaTrace
(C : ComparisonInstance) : Prop :=
  ∀ τ1 τ2,
  InfinitePostExecution C τ1 →
  InfinitePostExecution C τ2 →
  SigmaTraceOf C .post τ1 =
  SigmaTraceOf C .post τ2
```

-- Process-owned or process-attributed projection of the post-side Σ -trace.

```

def ProcessProjectedPostSigmaTrace
(C : ComparisonInstance)
(P : Process)
( $\tau$  : InfiniteExecutionBucketed C.Sigma) : SigmaTrace C.Sigma :=
ProjectProcessOwnedOrAttributedEvents C P (SigmaTraceOf C .post  $\tau$ )

```

```

def DeterministicProjectedPostTrace
(C : ComparisonInstance)
(P : Process) : Prop :=
 $\forall \tau_1 \tau_2,$ 
InfinitePostExecution C  $\tau_1 \rightarrow$ 
InfinitePostExecution C  $\tau_2 \rightarrow$ 
ProcessProjectedPostSigmaTrace C P  $\tau_1 =$ 
ProcessProjectedPostSigmaTrace C P  $\tau_2$ 

```

```

theorem post_deterministic_process_projection
(C : ComparisonInstance)
(hB1 : PostDeterministicSigmaTrace C)
(P : Process) :
DeterministicProjectedPostTrace C P :=
by
intro  $\tau_1 \tau_2 h\tau_1 h\tau_2$ 
unfold ProcessProjectedPostSigmaTrace
rw [hB1  $\tau_1 \tau_2 h\tau_1 h\tau_2$ ]

```

This is a projection consequence of system-level B1 under the same fixed comparison instance. It does not assert that process P is deterministic in isolation under arbitrary peer behavior, environments, or stimuli. Appendix I should consume this derived projection lemma wherever it needs uniqueness of P-owned or P-attributed post-side observations, rather than adding a separate per-process determinism assumption.

BRules / AssumptionBundle handling of B1. In the concrete Lean development, the B-rule bundle used by Appendix J and Appendix K theorem instances must carry B1 unconditionally for RTL_B under the fixed comparison instance. Equivalently, JTraceAssumptions C W, JCutpointAssumptions C W, and KAssumptions C W must expose a field or projection such as:

```

postDeterministicSigmaTrace : PostDeterministicSigmaTrace C

```

This field is the Lean representative of the standing B1 assumption. It is not a separate source-side determinism premise, and it is not conditional only on the phrase ‘when uniqueness is invoked.’ By contrast, B4 and B5 remain separate assumptions or derived structures used when ε -normalization, ε -quiescent endpoint reasoning, or uniqueness up to finite ε -stutter is needed.

Constructive ε -normalization and finite-stutter obligations for B4/B5.

B4 and B5 should not be merely opaque Prop fields when used to mechanize Lemma I.1-style arguments. They are assumptions, but they must expose enough structure to prove that internal ε -evolution cannot hide an infinite internal tail before the next Σ -action or stable wait.

This Lean strategy distinguishes two implementation-level measurements that are both silent with respect to Σ :

- intra-cycle ε -normalization: ε -microsteps inside one CycleBucket, used to reach an ε -quiescent cutpoint within a cycle; and
- cycle-advancing silent progress: one or more real clock cycles whose buckets contain no Σ -event for the process, used to model HLS-inserted FSM latency, arbitration/bookkeeping latency, failed non-blocking polls, and pipeline advances.

This distinction is only a Lean-internal decomposition. It is not a redefinition of B4. The main-document B4 premise remains a single finite ε -stutter bound for each process P: along any consecutive ε -only evolution in which P is internally advancing and is not externally stalled, P must reach either a Σ -action or a stable waiting state within a finite bound D_P .

Accordingly, ProcessEpsilonBound and IntraCycleNormalizationBound may remain separate Lean-internal implementation parameters, but neither one is the main-document B4 bound D_P . FiniteEpsilonDepth must expose a derived unified B4 consequence with one bound per process, and that derived bound is the Lean representative of the main document’s D_P .

In this strategy, the identification is:

$D_P \equiv \text{MainB4Bound } C \ P.$

ProcessEpsilonBound C P is only an internal bound on cycle-advancing silent progress, and IntraCycleNormalizationBound C P is only an internal bound on intra-cycle ε -normalization. They are useful for proof engineering, but together they must imply the single main-document B4 finite-stutter obligation. Any B5 statement of the form “within max_P D_P” is therefore represented in Lean using max_P MainB4Bound C P, not max_P ProcessEpsilonBound C P.

```
def ProcessEpsilonBound
(C : ComparisonInstance)
(P : Process) : Nat :=
pipe_depth C P
```

```
def IntraCycleNormalizationBound
(C : ComparisonInstance)
(P : Process) : Nat :=
eps_norm_depth C P
-- Main-document B4 bound D_P.
-- MainB4Bound C P is the single D_P-style finite  $\varepsilon$ -stutter bound exposed to
-- the theorem stack. ProcessEpsilonBound and IntraCycleNormalizationBound are
-- Lean-internal components only; they are not separate candidates for D_P. The
-- exact arithmetic expression is not important, provided MainB4Bound dominates
-- the Lean-internal cycle-progress and bucket-normalization bounds strongly
-- enough to prove the unified B4 consequence.
```

```
def MainB4Bound
(C : ComparisonInstance)
(P : Process) : Nat :=
(ProcessEpsilonBound C P + 1) *
(2 * IntraCycleNormalizationBound C P + 1)
```

```
def IntraCycleEpsilonStep
(C : ComparisonInstance)
(side : Side)
( $\sigma$   $\sigma'$  : State) : Prop :=
InternalStep C side  $\sigma$   $\sigma'$   $\wedge$ 
SigmaEventsBetween C side  $\sigma$   $\sigma' = \emptyset \wedge$ 
 $\sigma'.cycle = \sigma.cycle$ 
```

```

def IntraCycleEpsilonSegment
(C : ComparisonInstance)
(side : Side)
( $\sigma_0 \sigma_1$  : State)
(steps : List EpsStep) : Prop :=
EpsilonStepsFromTo C side  $\sigma_0$  steps  $\sigma_1$   $\wedge$ 
 $\forall \sigma \sigma'$ ,
AdjacentInEpsilonSteps C side  $\sigma_0$  steps  $\sigma_1$   $\sigma \sigma' \rightarrow$ 
IntraCycleEpsilonStep C side  $\sigma \sigma'$ 

```

```

def StableWaitOrSigmaEnabled
(C : ComparisonInstance)
(side : Side)
( $\sigma$  : State) : Prop :=
StableWaitCutpoint C side  $\sigma \vee$ 
 $\exists e$ , SigmaEnabledAt C side  $e \sigma$ 

```

-- Event ownership is side-parametric. In simple cases the ownership predicate
-- may reduce to a side-agnostic schema-level owner map over C.Sigma, but the
-- side parameter is still carried here so bucket membership, execution
-- well-formedness, and process attribution are checked in the same side context.

```

def SigmaCommitsForProcessInBucket
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\beta$  : CycleBucket C.Sigma) : Prop :=
 $\exists e$ ,
 $e \in \beta.\text{sigmaEvents} \wedge$ 
EventOwnedByProcess C side P e

```

```

def SigmaCommitsForProcessAt
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(n : Cycle) : Prop :=

```


SigmaCommitsForProcessInBucket C side P (τ n)

```
def ProcessSilentCycle
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\beta$  : CycleBucket C.Sigma) : Prop :=
WellFormedCycleBucket C side  $\beta$   $\wedge$ 
ProcessActiveInBucket C side P  $\beta$   $\wedge$ 
 $\neg$  SigmaCommitsForProcessInBucket C side P  $\beta$   $\wedge$ 
InternalProgressForProcess C side P  $\beta$ 
```

```
def SilentCycleSegment
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
( $t_0$   $t_1$  : Cycle) : Prop :=
 $t_0 \leq t_1$   $\wedge$ 
 $\forall n$ ,
 $t_0 \leq n \rightarrow$ 
 $n < t_1 \rightarrow$ 
ProcessSilentCycle C side P ( $\tau$  n)
```

```
def StableWaitOrSigmaEnabledAtCycle
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(n : Cycle) : Prop :=
StableWaitOrSigmaEnabled C side (( $\tau$  n).endState)  $\vee$ 
 $\exists e$ ,
EventOwnedByProcess C side P e  $\wedge$ 
SigmaEnabledAt C side e (( $\tau$  n).endState)
```

structure FiniteEpsilonDepth (C : ComparisonInstance) where

-- MainB4Bound is the Lean representative of the main-document D_P. These two

-- fields record that the Lean-internal component bounds are dominated by that
 -- single exposed D_P bound.

main_B4_bound_dominates_cycle_bound :

$\forall P,$

$\text{ProcessEpsilonBound } C \ P \leq \text{MainB4Bound } C \ P$

main_B4_bound_dominates_normalization_bound :

$\forall P,$

$\text{IntraCycleNormalizationBound } C \ P \leq \text{MainB4Bound } C \ P$

-- Cycle-level internal consequence used by Lemma I.1-style arguments.

-- This bound is intentionally stated using the Lean-internal

-- ProcessEpsilonBound. It is not the main-document D_P; it is one component

-- used to prove the unified D_P bound below.

silent_cycle_bound :

$\forall \text{ side } \tau \ P \ t_0 \ t_1,$

$\text{InfiniteExecutionOfSide } C \ \text{side } \tau \rightarrow$

$\text{SilentCycleSegment } C \ \text{side } P \ \tau \ t_0 \ t_1 \rightarrow$

$(\forall n,$

$t_0 \leq n \rightarrow$

$n < t_1 \rightarrow$

$\neg \text{StableWaitOrSigmaEnabledAtCycle } C \ \text{side } P \ \tau \ n) \rightarrow$

$t_1 - t_0 \leq \text{ProcessEpsilonBound } C \ P$

-- Main-document B4 consequence: all consecutive ε -only evolution, including

-- any intra-cycle normalization ε -steps represented inside CycleBucket, is

-- bounded by the single D_P bound before the process either commits a Σ -action

-- or reaches stable wait. In this Lean strategy, that D_P bound is

-- MainB4Bound C P.

unified_epsilon_stutter_bound :

$\forall \text{ side } \tau \ P \ s,$

$\text{InfiniteExecutionOfSide } C \ \text{side } \tau \rightarrow$

$\text{ConsecutiveEpsilonOnlySegment } C \ \text{side } P \ \tau \ s \rightarrow$

$\text{ProcessInternallyAdvancingThroughout } C \ \text{side } P \ \tau \ s \rightarrow$

$\text{NoExternalStallThroughout } C \ \text{side } P \ \tau \ s \rightarrow$

$\text{NoSigmaCommitOrStableWaitBeforeEnd } C \ \text{side } P \ \tau \ s \rightarrow$

$\text{EpsilonStepCount } C \ \text{side } P \ \tau \ s \leq \text{MainB4Bound } C \ P$

$\text{normalizes_or_reaches_sigma_or_wait} :$
 $\forall \text{ side } \tau \text{ P } t_0,$
 $\text{InfiniteExecutionOfSide C side } \tau \rightarrow$
 $\text{ProcessActiveInBucket C side P } (\tau t_0) \rightarrow$
 $\text{BeginsInternalProgressTowardNextObservable C side P } \tau t_0 \rightarrow$
 $\exists n,$
 $t_0 \leq n \wedge$
 $n \leq t_0 + \text{MainB4Bound C P} \wedge$
 $(\text{SigmaCommitsForProcessAt C side P } \tau n \vee$
 $\text{StableWaitOrSigmaEnabledAtCycle C side P } \tau n)$

-- Counts only ε -steps attributed to process P. A global CycleBucket may contain
 -- ε -steps from several processes, so the per-process normalization bound must
 -- be stated over the P-owned slice of the ε -prefix / ε -suffix, not over the
 -- entire global list.

$\text{def ProcessEpsilonStepsIn}$
 $(C : \text{ComparisonInstance})$
 $(\text{side} : \text{Side})$
 $(P : \text{Process})$
 $(\text{steps} : \text{List EpsStep}) : \text{Nat} :=$
 $\text{CountEpsStepsOwnedByProcess C side P steps}$

-- Equality of the non-clock state components that may not change across an
 -- idle quiescent bucket. A bucket may still advance the clock, so B5-style
 -- idle/quiescent reasoning must not assert $\beta.\text{endState} = \beta.\text{startState}$ unless
 -- endState is defined before the tick. However, when the bucket has no
 -- Σ -events and no ε -prefix/ ε -suffix steps, the intra-cycle ε position must not
 -- drift silently.

$\text{def NonClockStateEq}$
 $(C : \text{ComparisonInstance})$
 $(\sigma_0 \sigma_1 : \text{State}) : \text{Prop} :=$
 $\sigma_0.\text{store} = \sigma_1.\text{store} \wedge$
 $\sigma_0.\text{epsPos} = \sigma_1.\text{epsPos}$

-- Finite process universe used to form the main-document B5 maximum.
 -- This is the set of processes in the selected Sys_ref / RTL_B comparison.

$\text{def ProcessUniverse}$
 $(C : \text{ComparisonInstance}) : \text{Finset Process} :=$

ProcessesOfComparison C

```
-- Lean representation of the main-document B5 bound max_P D_P.  
-- Since D_P is represented in this strategy by MainB4Bound C P,  
-- SystemQuiescenceBound C is exactly max_P MainB4Bound C P over the finite  
-- process universe of the selected comparison instance.  
def SystemQuiescenceBound  
(C : ComparisonInstance) : Nat :=  
(ProcessUniverse C).sup (fun P => MainB4Bound C P)
```

```
structure QuiescentNormalization (C : ComparisonInstance) where  
  bucket_prefix_bounded :  
    ∀ side β P,  
    WellFormedCycleBucket C side β →  
    ProcessEpsilonStepsIn C side P β.epsPrefix ≤ IntraCycleNormalizationBound C P  
  bucket_suffix_bounded :  
    ∀ side β P,  
    WellFormedCycleBucket C side β →  
    ProcessEpsilonStepsIn C side P β.epsSuffix ≤ IntraCycleNormalizationBound C P  
  end_quiescent_sound :  
    ∀ side β,  
    WellFormedCycleBucket C side β →  
    β.endQuiescent →  
    QuiescentAt C side β.endState
```

```
-- Starting a bucket at an  $\varepsilon$ -quiescent state only says that the incoming state  
-- has no pending internal normalization. It does not forbid  $\Sigma$ -events in the  
-- bucket, and it does not forbid  $\varepsilon$ -suffix normalization caused by those  $\Sigma$ -events.  
-- The no-further- $\varepsilon$  conclusion is valid only for an idle bucket in which no  
-- observable action is enabled and no external/environmental change wakes the  
-- system.
```

```
idle_quiescent_bucket_has_no_epsilon_steps :  
  ∀ side β,  
  WellFormedCycleBucket C side β →  
  QuiescentAt C side β.startState →  
  β.sigmaEvents =  $\emptyset$  →  
  NoObservableActionEnabledAtState C side β.startState →
```

$\text{NoExternalChangeInBucket } C \text{ side } \beta \rightarrow$
 $\beta.\text{epsPrefix} = [] \wedge$
 $\beta.\text{epsSuffix} = [] \wedge$
 $\text{NonClockStateEq } C \beta.\text{startState } \beta.\text{endState}$

$\text{bounded_system_quiescence} :$
 $\forall \text{ side } \tau t,$
 $\text{InfiniteExecutionOfSide } C \text{ side } \tau \rightarrow$
 $\text{NoObservableActionEnabledAtState } C \text{ side } ((\tau t).\text{startState}) \rightarrow$
 $\text{NoExternalChangeFrom } C \text{ side } \tau t \rightarrow$
 $\exists n,$
 $t \leq n \wedge$
 $n \leq t + \text{SystemQuiescenceBound } C \wedge$
 $\text{QuiescentAt } C \text{ side } ((\tau n).\text{endState}) \wedge$
 $\forall m,$
 $n \leq m \rightarrow$
 $\text{QuiescentAt } C \text{ side } ((\tau m).\text{startState}) \wedge$
 $\text{QuiescentAt } C \text{ side } ((\tau m).\text{endState}) \wedge$
 $((\text{NoExternalChangeFrom } C \text{ side } \tau m \wedge$
 $\text{NoObservableActionEnabledAtState } C \text{ side } ((\tau m).\text{startState})) \rightarrow$
 $(\tau m).\text{sigmaEvents} = \emptyset \wedge$
 $(\tau m).\text{epsPrefix} = [] \wedge$
 $(\tau m).\text{epsSuffix} = [])$

The `bounded_system_quiescence` field is the Lean-level semantic assumption corresponding to B5. It is triggered from a single cycle whose start state has no enabled observable action. If no external/environmental change wakes the system, this assumption says that the system reaches the ε -fixed point within the explicit main-document bound `max_P D_P`.

In this Lean strategy, `D_P` means `MainB4Bound C P`. Therefore the B5 phrase “within `max_P D_P`” is represented by `SystemQuiescenceBound C`, defined as the finite maximum of `MainB4Bound C P` over the process universe of the selected comparison instance. The fact that each `MainB4Bound C P` is below `SystemQuiescenceBound C`, and the least-upper-bound property of `SystemQuiescenceBound C`, are not B5 assumptions; they are Lean lemmas proved from the `Finset.sup` definition of `SystemQuiescenceBound C`. `ProcessEpsilonBound C P` is only an internal cycle-progress component and is not the `D_P` used by B5.

After that point, later buckets remain idle only while the no-external-change condition continues to hold and no observable action becomes enabled again. Thus B5 is a bounded quiescence-closure premise from an idle start state, not a requirement to prove in advance that no observable action is disabled forever from the original cycle.

The field `idle_quiescent_bucket_has_no_epsilon_steps` is deliberately weaker than the invalid statement “QuiescentAt β .startState implies β .epsPrefix = [] and β .epsSuffix = [] and β .endState = β .startState.” A quiescent start state may still be followed by committed Σ -events, and those Σ -events may update state and create ϵ -suffix normalization. Also, a bucket may advance the clock, so the idle fixed-point conclusion uses `NonClockStateEq` rather than literal equality of `startState` and `endState`.

`B4_finiteEpsilonDepth` exposes the main-document B4 consequence through `MainB4Bound`: each process has a single finite `D_P` ϵ -stutter bound before a Σ -action or stable wait, and in this Lean strategy that `D_P` bound is exactly `MainB4Bound C P`. `ProcessEpsilonBound` and `IntraCycleNormalizationBound` remain useful Lean-internal components, but they are not separate replacements for B4 and must not be used as the `D_P` in B5. The formalization must prove that the internal decomposition implies the single unified B4 finite-stutter obligation. `QuiescentNormalization` may still use `IntraCycleNormalizationBound` for bucket well-formedness and ϵ -quiescent cutpoint construction, but those bounds are applied per process by counting only the ϵ -steps owned by that process.

Bundle B-rules:

structure `BRules` (`C` : `ComparisonInstance`) where

`B1_postDeterminism` :

`PostDeterministicSigmaTrace C`

-- B2 is not post-only. It applies to any valid infinite execution of the
 -- selected comparison side when that side participates in a progress/liveness
 -- theorem.

`B2_weakFairness` :

$\forall$ side τ ,

`InfiniteExecutionOfSide C side  $\tau \rightarrow$`

`WeakFairness C side  $\tau$`

-- B3 is likewise side-parametric. The P_push/P_pop obligations are required
 -- for the reference-side Sys_ref execution as well as the post-side RTL_B
 -- execution when the theorem uses those executions for liveness/deadlock
 -- reasoning.

B3_channelProgress :

$\forall$ side τ c,

InfiniteExecutionOfSide C side $\tau \rightarrow$

P_push C side τ c $\wedge$ P_pop C side τ c

B4_finiteEpsilonDepth :

FiniteEpsilonDepth C

B5_quiescentNormalization :

QuiescentNormalization C

B6_idleStutterTimeDivergence :

IdleStutterTimeDivergence C

-- Theorem-scope projections of BRules. TraceBRules deliberately omits B4/B5;
 -- NormalizationBRules contains exactly the finite- ϵ / quiescence premises used
 -- by normalization and cutpoint theorems.

structure TraceBRules (C : ComparisonInstance) where

B1_postDeterminism : PostDeterministicSigmaTrace C

B2_weakFairness :

$\forall$ side τ ,

InfiniteExecutionOfSide C side $\tau \rightarrow$

WeakFairness C side τ

B3_channelProgress :

$\forall$ side τ c,

InfiniteExecutionOfSide C side $\tau \rightarrow$

P_push C side τ c $\wedge$ P_pop C side τ c

B6_idleStutterTimeDivergence : IdleStutterTimeDivergence C

structure NormalizationBRules (C : ComparisonInstance) where

B4_finiteEpsilonDepth : FiniteEpsilonDepth C

B5_quiescentNormalization : QuiescentNormalization C

def BRules.toTraceBRules

(C : ComparisonInstance)

```

(hB : BRules C) : TraceBRules C :=
{
  B1_postDeterminism := hB.B1_postDeterminism
  B2_weakFairness := hB.B2_weakFairness
  B3_channelProgress := hB.B3_channelProgress
  B6_idleStutterTimeDivergence := hB.B6_idleStutterTimeDivergence
}

```

```

def BRules.toNormalizationBRules
(C : ComparisonInstance)
(hB : BRules C) : NormalizationBRules C :=
{
  B4_finiteEpsilonDepth := hB.B4_finiteEpsilonDepth
  B5_quiescentNormalization := hB.B5_quiescentNormalization
}

```

B2/B3 remain temporal assumptions usable by K proofs. B4/B5 also remain assumptions, but they are now constructive assumptions: they provide the finite ε -normalization facts needed to bound `epsPrefix` / `epsSuffix` lengths in `CycleBucket` and to prove bounded ε -chain lemmas such as Lemma I.1.

7. Channel semantics and E4

Define channel state over the selected `Sys_ref` boundary.

The abstract occupancy is cycle-boundary, non-fall-through occupancy. It is indexed by the execution side, the infinite bucketed execution, the channel, and the cycle, not by the full intra-cycle state. It must not be a free bookkeeping function disconnected from the committed trace.

```
capacity : C.Sys_ref.Channel → Nat
```

```
-- Number of  $\Sigma$ -visible Push_c commits on channel c strictly before cycle t.
```

```

def PushCommitsBefore
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(c : C.Sys_ref.Channel)
(t : Cycle) : Nat :=
CountSigmaCommitsBefore C side  $\tau$  t (fun e => IsPushCommitOnChannel C side e c)

```



```

-- Number of  $\Sigma$ -visible Pop_c commits on channel c strictly before cycle t.
def PopCommitsBefore
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(c : C.Sys_ref.Channel)
(t : Cycle) : Nat :=
CountSigmaCommitsBefore C side  $\tau$  t (fun e => IsPopCommitOnChannel C side e c)

-- Initial occupancy at the selected comparison boundary. In the usual reset-empty
-- case this is 0. In accounted-initial-occupancy cases it is supplied explicitly
-- by the comparison instance / elaboration package.
initialOcc :
ComparisonInstance  $\rightarrow$  Side  $\rightarrow$  C.Sys_ref.Channel  $\rightarrow$  Nat

-- Commit-history-derived cycle-boundary occupancy.
-- For B(c)=0 rendezvous channels this value is always 0; rendezvous enablement
-- is governed by same-cycle matching endpoint requests, not stored occupancy.
def occAt
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(c : C.Sys_ref.Channel)
(t : Cycle) : Nat :=
if capacity C c = 0 then
0
else
initialOcc C side c

+ PushCommitsBefore C side  $\tau$  c t

- PopCommitsBefore C side  $\tau$  c t

```

For convenience, one may define a state-indexed abbreviation only by projecting the state to its cycle, and only relative to the execution whose state is being inspected:

```

def occAtState
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(c : C.Sys_ref.Channel)

```

```
(σ : State) : Nat :=
occAt C side τ c σ.cycle
```

This abbreviation is not a second occupancy notion. In particular, occupancy is invariant under ε -steps and under same-cycle Σ -events because it is defined from commits strictly before the cycle boundary:

```
theorem occ_invariant_within_cycle
(C : ComparisonInstance)
(side : Side)
(τ : InfiniteExecutionBucketed C.Sigma)
(c : C.Sys_ref.Channel)
(σ1 σ2 : State) :
σ1.cycle = σ2.cycle →
occAtState C side τ c σ1 = occAtState C side τ c σ2
```

The definition above is the canonical occupancy used by E4, ChannelEnabled, ChannelDisabled, WFG construction, B3 progress reasoning, and drain-closure reasoning. If the concrete Lean development keeps occAt opaque for proof-engineering reasons, it must instead assume the following well-formedness condition and use it everywhere occAt is consumed:

```
structure E4OccupancyTraceWF
(C : ComparisonInstance)
(side : Side)
(τ : InfiniteExecutionBucketed C.Sigma) where
occ_matches_committed_prefix :
∀ c t,
occAt C side τ c t =
if capacity C c = 0 then
0
else
initialOcc C side c
+ PushCommitsBefore C side τ c t
- PopCommitsBefore C side τ c t
initial_occupancy_bounded :
∀ c,
initialOcc C side c ≤ capacity C c
occupancy_bounded :
```

$\forall c\ t,$
 $\text{occAt } C \text{ side } \tau\ c\ t \leq \text{capacity } C\ c$
 $\text{no_underflow} :$
 $\forall c\ t,$
 $\text{PopCommitsBefore } C \text{ side } \tau\ c\ t \leq$
 $\text{initialOcc } C \text{ side } c + \text{PushCommitsBefore } C \text{ side } \tau\ c\ t$
 $\text{rendezvous_has_no_stored_occupancy} :$
 $\forall c\ t,$
 $\text{capacity } C\ c = 0 \rightarrow$
 $\text{occAt } C \text{ side } \tau\ c\ t = 0$

Reset-empty alignment (A10). The main document's formal model is reset-empty: after each reset boundary at which the compared execution begins or restarts, every buffered channel with $B(c) > 0$ has logical occupancy 0 at the chosen Σ boundary, and the Appendix J/K occupancy equations and the per-channel Push/Pop precedence lemma are interpreted relative to that reset-empty initial channel state. Document-aligned theorem instances therefore set $\text{initialOcc } C \text{ side } c = 0$, and $\text{A10_reset_empty_fifo} / \text{ResetEmptyFifoAssumption } C$ entails exactly that equation at each compared reset boundary. A nonzero accounted initialOcc is an extension outside the main document's empty-FIFO model: per v42, preloaded channel contents must instead be modeled explicitly as post-reset producer actions, or supported by a separately stated initialized-state extension whose obligations are then carried by the comparison instance. The initialOcc parameter is kept in occAt only so that such an explicitly flagged extension does not require re-deriving the occupancy infrastructure; it must not be used to smuggle preloaded contents into a theorem instance that claims the documented A10 assumption.

-- Domain for one-cycle issue-time request reasoning.
 -- This domain includes every executed source-level message-operation instance,
 -- including failed PushNB / PopNB polls that produce no committed Σ -event.

```

def BoundaryReqIssueDomainSide
(C : ComparisonInstance)
(side : Side)
(q : DynMsgInstance)
( $\sigma$  : State) : Prop :=
 $\exists P,$ 
 $q \in \text{QExecSide } C \text{ side } P \wedge$ 
 $\text{MessageOperationInstance } q \wedge$ 
 $\text{OwnsMessageEndpoint } P\ q \wedge$ 
 $\text{IssuedAtState } C \text{ side } q\ \sigma$ 

```

-- Domain for persistent operation-attributive request/channel reasoning.

-- This is the domain used by ChannelEnabled, ChannelDisabled, Front, NextFront,
 -- automatic flush, and WFG construction. Failed NB-only polls are not included
 -- merely because they executed.

def BoundaryReqPersistentDomainSide

(C : ComparisonInstance)

(side : Side)

(q : DynMsgInstance)

(σ : State) : Prop :=

$\exists P,$

$q \in Q\Omega\text{Side } C \text{ side } P \wedge$

MessageOperationInstance q $\wedge$

OwnsMessageEndpoint P q

-- General BoundaryReq domain.

-- A request may be meaningful either because q is issuing in the current cycle

-- or because q is a persistent message-operation frontier instance.

def BoundaryReqDomainSide

(C : ComparisonInstance)

(side : Side)

(q : DynMsgInstance)

(σ : State) : Prop :=

BoundaryReqIssueDomainSide C side q $\sigma \vee$

BoundaryReqPersistentDomainSide C side q σ

Executed failed non-blocking polls may belong to Qexec,P for replay and witness purposes while remaining absent from Q Ω ,P and Ω_P . Such polls do acquire the one-cycle BoundaryReq-at-issue obligation when they issue, via BoundaryReqIssueDomainSide. They do not acquire persistent BoundaryReqPersistentDomainSide, ChannelEnabled, ChannelDisabled, Front, NextFront, automatic-flush, or WFG obligations merely because they executed.

BoundaryReqSide :

ComparisonInstance $\rightarrow$ Side $\rightarrow$ DynMsgInstance $\rightarrow$ State $\rightarrow$ Prop

-- ChannelEnabledSide is the v42 E4 availability/eligibility predicate at the

-- chosen abstract boundary. It is not itself a Commit predicate. For B(c)>0 it

-- is BoundaryReq conjoined with the cycle-boundary FIFO availability predicate;

-- for B(c)=0 it is BoundaryReq conjoined with the same-cycle complementary

-- endpoint request predicate. Hidden grant/arbitration/back-pressure must not

-- be added as an extra conjunct here unless it is represented as explicit

-- Sys_B^elab boundary structure.

ChannelEnabledSide :

ComparisonInstance $\rightarrow$ Side $\rightarrow$ DynMsgInstance $\rightarrow$ State $\rightarrow$ Prop

ChannelSelectedSide :
 ComparisonInstance → Side → DynMsgInstance → State → Prop

```
def ChannelSelectionCommits
(C : ComparisonInstance)
(side : Side) : Prop :=
  ∀ τ q σ t,
  StateAtCycleInPrefix C side τ t σ →
  BoundaryReqPersistentDomainSide C side q σ →
  ChannelEnabledSide C side q σ →
  ChannelSelectedSide C side q σ →
  CommitAtCycle C side τ q t
```

```
def ChannelDisabledSide
(C : ComparisonInstance)
(side : Side)
(q : DynMsgInstance)
(σ : State) : Prop :=
  BoundaryReqPersistentDomainSide C side q σ ∧
  BoundaryReqSide C side q σ ∧
  ¬ ChannelEnabledSide C side q σ
```

For post-side automatic-flush and issue/commit formulas, use the following abbreviations:

```
def BoundaryReqIssueDomain
(C : ComparisonInstance)
(q : DynMsgInstance)
(σ : State) : Prop :=
  BoundaryReqIssueDomainSide C .post q σ
```

```
def BoundaryReqPersistentDomain
(C : ComparisonInstance)
(q : DynMsgInstance)
(σ : State) : Prop :=
  BoundaryReqPersistentDomainSide C .post q σ
```

```
def BoundaryReqDomain
(C : ComparisonInstance)
(q : DynMsgInstance)
(σ : State) : Prop :=
  BoundaryReqDomainSide C .post q σ
```

```
def BoundaryReq
```

```

(C : ComparisonInstance)
(q : DynMsgInstance)
( $\sigma$  : State) : Prop :=
BoundaryReqSide C .post q  $\sigma$ 

```

```

def ChannelEnabled
(C : ComparisonInstance)
(q : DynMsgInstance)
( $\sigma$  : State) : Prop :=
ChannelEnabledSide C .post q  $\sigma$ 

```

```

def ChannelDisabled
(C : ComparisonInstance)
(q : DynMsgInstance)
( $\sigma$  : State) : Prop :=
ChannelDisabledSide C .post q  $\sigma$ 

```

BoundaryReqSide, ChannelEnabledSide, and ChannelDisabledSide are the canonical side-indexed predicates. The shorter BoundaryReq / ChannelEnabled / ChannelDisabled forms are post-side abbreviations used only where the section is already explicitly about RTL_B / post-side behavior.

BoundaryReqSide is used under BoundaryReqDomainSide. This includes both one-cycle issue requests, via BoundaryReqIssueDomainSide, and persistent frontier/channel requests, via BoundaryReqPersistentDomainSide.

ChannelEnabledSide and ChannelDisabledSide are used only under BoundaryReqPersistentDomainSide or its post-side abbreviation BoundaryReqPersistentDomain. Thus failed non-blocking polls in $Q_exec, P \setminus \Omega_P$ may satisfy the Appendix C one-cycle request-at-issue rule, but they still do not become message-operation frontier items and do not acquire ChannelEnabled, ChannelDisabled, Front, NextFront, automatic-flush, or WFG obligations merely because they executed.

ChannelEnabled is allowed to inspect the settled boundary-request state σ for persistent message-operation frontier instances, but its capacity/occupancy component must use the commit-history-linked value $occAt\ C\ side\ \tau\ q.channel\ \sigma.cycle$. Thus E4 enablement is evaluated against the beginning-of-cycle abstract occupancy, with no same-cycle fall-through through ε -settling or same-cycle Σ -events.

Capacity cases:

inductive CapacityKind
 | rendezvous -- B(c) = 0
 | buffered -- B(c) > 0

Derived E4 lemmas:

BufferedPushEnabled
 BufferedPopEnabled
 RendezvousEnabled
 FIFOOrderPreservation
 NoDropNoDuplicate
 OccupancyUpdatePush
 OccupancyUpdatePop

The occupancy-update lemmas are derived from the commit-history definition of `occAt`, or from `E4OccupancyTraceWF` if `occAt` is kept opaque. They are not independent facts about a free primitive. In particular, `OccupancyUpdatePush` and `OccupancyUpdatePop` must prove that moving from cycle t to $t+1$ changes `occAt` exactly according to the `Push_c` / `Pop_c` commits in cycle t , subject to the non-fall-through rule and the single-transfer-per-cycle endpoint constraints.

E4 should be parameterized by `Sys_ref`, so in elaborated cases it ranges over explicit elaborated channels, not merely outer channels.

B-faithfulness is a distinct implementation-side obligation, not a consequence of E4 or trace compatibility. It is stated per back-annotated explicit abstract channel as a behavioral refinement of the selected concrete boundary by the explicit E4 abstract channel:

```
structure BFaithful
(C : ComparisonInstance)
(c : Channel) : Prop where
boundary_complete :
  ∀ e,
    RTLVisibleTransferOnChannel C c e →
    CrossesChosenAbstractBoundaryAsCommittedPushOrPop C c e
capacity_upper_bound :
  ∀ τ t,
    RealizedUnpoppedItemsAtChosenBoundary C τ c t ≤ capacity C c
boundary_behavior_exact :
  ∀ τ t action,
```

RealizedBoundaryActionPermitted $C \tau c t \text{ action} \leftrightarrow$
 E4BoundaryActionPermitted $C \tau c t \text{ action}$
 no_hidden_storage_or_credit :
 NoUnmodeledStorageOrCreditBeyondChosenBoundary $C c$
 no_hidden_enablement :
 NoHiddenGrantArbitrationOrBackpressureAtBoundary $C c \vee$
 RepresentedByExplicitSysBElabBoundaryStructure $C c$
 internal_motion_epsilon :
 InternalChannelMovementBetweenBoundaryCommitsIsEpsilon $C c$
 evidence :
 BFaithfulnessEvidence $C c$

The capacity field is intentionally named `capacity_upper_bound`: by itself it proves only that concrete occupancy does not exceed $B(c)$. Exactness is supplied by `boundary_behavior_exact` together with boundary completeness, absence of hidden storage/credit, absence or explicit elaboration of hidden enablement, and Σ -opacity of internal movement. The concrete implementation-assumption bundle carries the full behavioral obligation:

structure ImplementationAssumptions
 (C : ComparisonInstance) where
 e4ChannelRealization : E4ChannelRealizationConformance C
 bFaithful :
 $\forall c,$
 BackAnnotatedExplicitAbstractChannel $C c \rightarrow \text{BFaithful } C c$
 otherNamedImplementationObligations : OtherImplementationAssumptions C

Consequently, `JTraceAssumptions`, `JCutpointAssumptions`, `JSigmaDUTSafetyAssumptions`, `IConstructionAssumptions`, and `KStructuralAssumptions` import B-faithfulness mechanically through `ImplementationAssumptions`. `J.AbsConf` remains separate because it is witness- and selected-cutpoint-dependent.

8. Appendix Q elaboration interface

Appendix Q should be introduced early. In the updated document, elaboration is not a late proof afterthought: in elaborated cases, $\text{Sys_ref} = \text{Sys_B}^{\text{elab}}(E)$, and all references to `B`, `occ`, `BoundaryReq`, `ChannelEnabled/Disabled`, `Front_P`, and `WFG` are interpreted over the explicit channels of the elaborated model.

As in the ordinary E4 semantics, elaborated-channel occupancy is cycle-boundary, non-fall-through occupancy. Thus $\text{occ_E}(d,t)$ and $A_E(c,t)$ are indexed by cycle t . State-indexed forms such as $\text{occ_E}(d,\sigma)$ or $A_E(c,\sigma)$, when used informally at ε -quiescent cutpoints, are only abbreviations for $\text{occ_E}(d,\sigma.\text{cycle})$ and $A_E(c,\sigma.\text{cycle})$.

Chan_outer should be a parameter fixed by the chosen outer Sys_B , not a field chosen internally by the elaboration package.

The channel-level elaboration projection follows Appendix Q's single-valued optional form. A proof-internal elaborated channel is either hidden from the outer DUT/testbench boundary or assigned to one outer channel. Separately, a proof-internal token/event may account for multiple outer Σ_DUT -visible channel effects in bundled or joined cases.

```

structure ElaborationPackage
(Sys_B : System)
(Chan_outer : Type)
( $\Sigma\_DUT$  : Type) where
Sys_elab : System
Chan_elab : Type
finite_chan_elab : Fintype Chan_elab
ElabToken : Type
B_E : Chan_elab  $\rightarrow$  Nat
occ_E : Chan_elab  $\rightarrow$  Cycle  $\rightarrow$  Nat

-- Channel-level projection/accounting support.
--  $\Pi\_elab\ d$  = some  $c$  means that elaborated channel  $d$  belongs to the
-- channel-level fiber for outer channel  $c$ .  $\Pi\_elab\ d$  = none corresponds to
-- the  $\perp$  case in Appendix Q:  $d$  is proof-internal and hidden from the outer
-- DUT/testbench boundary.
Pi_elab : Chan_elab  $\rightarrow$  Option Chan_outer

-- Token/event-level accounting support.
--  $\Pi\_token\_elab\ tok$  is the set of outer channels accounted for by one
-- proof-internal token/event. This may contain multiple outer channels for
-- bundled or joined work items. Token-level multiplicity does not make the
-- channel-level  $\Pi\_elab$  relation multi-valued.
Pi_token_elab : ElabToken  $\rightarrow$  Set Chan_outer

```

tokenOfSigma : Sys_elab.Sigma → Option ElabToken

-- Bucketed outer-projection of the elaborated proof-internal execution.
-- This replaces any flat Trace-level projection. The projection may hide
-- proof-internal staged/bundle/join events, but it must preserve the
-- cycle-bucket structure needed for ε -quiescent cutpoints, same-cycle atomic
-- units, and temporal B2/B3 reasoning.

piSigmaBucket :

CycleBucket Sys_elab.Sigma →

CycleBucket Σ _DUT

piSigmaFinite :

ExecutionBucketed Sys_elab.Sigma →

ExecutionBucketed Σ _DUT

piSigmaInfinite :

InfiniteExecutionBucketed Sys_elab.Sigma →

InfiniteExecutionBucketed Σ _DUT

piSigma_bucket_wf : Prop

piSigma_finite_agrees_with_bucket : Prop

piSigma_infinite_agrees_with_bucket : Prop

A_E : Chan_outer → Cycle → Nat

outer_bucketed_execution_preservation : Prop

occupancy_preservation : Prop

internal_refinement : Prop

no_hidden_cross_channel_disablement : Prop

wfg_visibility : Prop

combinational_wellformedness : Prop

For each outer channel c , the channel-level fiber of the elaboration is the
single-valued optional fiber

$\Pi_{\text{elab}}^{-1}(c) = \{ d : \text{Chan_elab} \mid \text{Pi_elab } d = \text{some } c \}$.

This channel-level optional map matches Appendix Q's Pi_elab :

$\text{Chan_elab}(E) \rightarrow \text{Chan_outer} \text{ union } \{\text{bottom}\}$. The token-level projection Pi_token_elab is a

Lean-side refinement used to make bundled/joined multi-channel accounting explicit when one proof-internal staged/bundle/join token accounts for several outer Sigma_DUT-visible channel effects. It is not an additional named component of Appendix Q.1; the scheduling-rules document folds this accounting into A_E and the corresponding Q.1 accounting clauses. The bucketed projection piSigmaBucket, together with piSigmaFinite and piSigmaInfinite, and the occupancy/accounting map A_E must be consistent with Pi_elab and the token-level refinement. Elaboration projection preserves the cycle-bucket decomposition needed for epsilon-quiescent cutpoints, same-cycle atomic units, R2C fixed-point buckets, and B2/B3 temporal reasoning while hiding proof-internal staged/bundle/join events from the outer Sigma_DUT view.

For convenience, the package may define state-indexed abbreviations $\text{occ_E_atState } d \text{ sigma} := \text{occ_E } d \text{ sigma.cycle}$ and $\text{A_E_atState } c \text{ sigma} := \text{A_E } c \text{ sigma.cycle}$. These abbreviations are invariant within a cycle and are not separate occupancy notions.

Reference choice is dependent and definitionally selects the reference system:

inductive ReferenceKind

| ordinarySysB

| elaborated

structure SelectedElaboration

(outerSysB : System)

(Chan_outer : Type)

(Sigma Sigma_DUT : Alphabet) where

E : ElaborationPackage outerSysB Chan_outer Sigma_DUT

-- The selected elaborated system emits the shared proof-internal alphabet used

-- by the comparison instance.

sigma_schema_agrees : E.Sys_elab.Sigma = Sigma

inductive SysRefChoice

(outerSysB : System)

(Chan_outer : Type)

(Sigma Sigma_DUT : Alphabet)

| ordinary

| elaborated

(selected : SelectedElaboration outerSysB Chan_outer Sigma Sigma_DUT)

namespace SysRefChoice

def kind :

SysRefChoice outerSysB Chan_outer Sigma Sigma_DUT -> ReferenceKind

| .ordinary => .ordinarySysB

| .elaborated _ => .elaborated

def Sys_ref :

SysRefChoice outerSysB Chan_outer Sigma Sigma_DUT -> System

| .ordinary => outerSysB

| .elaborated selected => selected.E.Sys_elab

def elab? :

SysRefChoice outerSysB Chan_outer Sigma Sigma_DUT ->

Option (ElaborationPackage outerSysB Chan_outer Sigma_DUT)

| .ordinary => none

| .elaborated selected => some selected.E

end SysRefChoice

Because ComparisonInstance.C.Sys_ref is defined as C.refChoice.Sys_ref, an elaborated instance cannot name one package in refChoice and a different Sys_ref in theorem statements. ElaborationProjectionWF, BFaithful elaborated boundary chains, KElaborationPrerequisite, sync-boundary gate coverage, and the side-condition audit must all consume the selected value carried by the elaborated constructor. A proposition such as sys_ref_matches_kind : Prop is not an adequate substitute for this dependent construction.

This also handles sync-boundary epoch gates: if proof-relevant withholding exists, it must be represented by explicit gate channels in Sys_B^elab(E), not by hidden gating on an otherwise E4-enabled outer channel. The earlier review correctly flagged this as a point that must be explicit in the Lean strategy.

Locality note. Although ElaborationPackage is presented as a single structure parameterizing the theorem instance, its components (Sys_elab, Chan_elab, Π _elab, Π _token,elab, π_Σ ,E, A_E) are constructed locally per process and per channel boundary. The channel-level projection Π _elab is single-valued and optional, matching Appendix Q's map Chan_elab(E) $\rightarrow$ Chan_outer $\cup$ { $\perp$ }. The token-level projection Π _token,elab is the object allowed to map one

proof-internal staged/bundle/join token to multiple outer Σ_{DUT} -visible channels. Each per-process elaboration is determined by that process's local channel structure (number of inputs, sync-boundary discipline, pipelining configuration) and is independent of the elaboration chosen for any other process. The system-level `ElaborationPackage` is the disjoint composition of these per-process pieces. Treating `E` as a theorem-instance parameter is the right Lean encoding of "the verifier has chosen per-process elaborations and the proof takes them as given"; it is not a claim that the verifier must reason globally to construct `E`.

9. R-rules, E-rules, and modeling contracts

The R-rule bundle should include only the document's actual R-rules plus an optional Lean-side array-rule pointer if desired. It should not include a fictional `R5_syncBoundaryNoCrossing`; sync-boundary crossing is `E5`. The document explicitly labels "Messages cannot cross their immediate synchronization boundary" as `E5`.

However, `R5` itself is a real R-rule: `R5` (Channel capacity semantics; `Sys` vs. `Sys_B`). In the Lean strategy, `R5` should record the chosen source/reference-model capacity convention: raw `Sys` uses rendezvous capacity, while `Sys_B` / `Sys_ref` uses the selected capacity $B(c)$, with $B(c)=0$ interpreted as rendezvous and $B(c)>0$ interpreted as cycle-boundary, non-fall-through bounded FIFO. The concrete enablement and occupancy-update obligations remain the `E4` channel-capacity semantics; `R5` records which source/reference capacity semantics are being used by the selected comparison instance.

The predicates `RendezvousChannel C c` and `BufferedChannel C c` are definitional abbreviations for the zero-capacity and positive-capacity cases. Therefore the substantive `R5` obligation is not the tautological case split $\text{capacity } C\ c = 0 \rightarrow \text{RendezvousChannel } C\ c$ and $0 < \text{capacity } C\ c \rightarrow \text{BufferedChannel } C\ c$. The substantive obligation is that the selected source reference model uses the intended capacity convention for the comparison instance: in the raw `Sys` case this means rendezvous capacity; in `Sys_B` / `Sys_B^elab` this means the selected $B(c)$ capacities, with `E4` supplying the corresponding `ChannelEnabled` / `ChannelDisabled` and occupancy-update semantics.

```

def R2CCombinationalFixedPointRule
(C : ComparisonInstance) : Prop :=
SequentialOnly C ∨
∃ hR2C : R2C_WF C,
R2CEventsObservedOnlyInStutterCanonicalUnits C hR2C ∧
R2CUnitsMatchedByComponentLocalOrdinal C hR2C

```

```

def R5ChannelCapacitySemantics
(C : ComparisonInstance) : Prop :=
SysRefUsesSelectedCapacitySemantics C

```

```

structure RRules (C : ComparisonInstance) where
R1_syncOrder : Prop
R2_signalAnchorSequential : Prop
R2C_combinationalFixedPoint :
R2CCombinationalFixedPointRule C
R3_syncSemantics : Prop
R4_messageIssueOrder : Prop
R5_channelCapacitySemantics :
R5ChannelCapacitySemantics C

```

R4_messageIssueOrder is deliberately named only as an issue-order rule. It records the source-induced no-reverse discipline for message-operation issue. It does not impose a separate commit-order rule; commit timing is constrained by E4 channel availability, FIFO/rendezvous semantics, backpressure, and the chosen Sys_ref / RTL_B comparison instance.

R2 source well-formedness (RULE 1 / RULE 2). In v42, RULE 1 conformance is a source well-formedness obligation for proof-backed flows, not merely a recommended diagnostic. R2_signalAnchorSequential therefore carries, in addition to the anchoring equations, the following obligations: every observable sequential-process signal Read/Write has a unique real bounding synchronization anchor on each dynamic path on which it occurs (with a particular START_P occurrence usable only when the modeling convention includes that reset-epoch start as a real synchronization event in S, and a particular FINISH_P occurrence usable only for an actually terminating process in the same reset epoch); no observable Read/Write event is assigned clock time ∞ ; and if a blocking message-passing operation separates a signal read or write from its corresponding synchronization anchor, the theorem instance is admissible only if the model is rewritten so the signal IO is adjacent to its intended anchor, or an explicit source-level well-formedness discharge

shows that the signal IO has a unique unconditional real anchor on every dynamic path and cannot observe a value inconsistent with that anchor. In the Lean development these facts are part of `C.WellFormed` together with the `read_anchor_defined / write_anchor_defined / reset-epoch-indexed anchor-realized` fields of `DynamicUniverse`; an unresolved tool warning is not by itself a discharge.

E-rules:

```
structure ERules (C : ComparisonInstance) where
  E1_sync_order_preservation : Prop
  E2_signal_anchor_preservation : Prop
  E3_message_order_preservation : Prop
  E4_channel_capacity_semantics : Prop
  E5_sync_boundary_preservation : Prop
```

Modeling contracts live in the single `ModelingContracts` structure defined in Section 4.2. That one structure includes direct-input contracts, array-access mapping rules, and latency-sensitive local-protocol handling. Section 9 does not redeclare it. Concrete theorem bundles carry a `ModelingContracts C` field and install the corresponding local `HasModelingContracts C` instance before using schematic expressions such as `C.contracts.arrays` or `C.contracts.directInputs`.

-- No second `ModelingContracts` declaration appears here.
-- Use the Section 4.2 declaration and its `HasModelingContracts` interface.

This avoids two independent contract types and also avoids the circular declaration pattern formerly created by storing `ModelingContracts C` inside `ComparisonInstance`. The same Section 4.2 bundle is used by R/E-rule theorem instances, Appendix I/J witness construction, Appendix L snooping/WC, direct-input late-sampling proofs, external-array proofs, and cadence-sensitive local-protocol isolation obligations.

10. Array access mapping

External arrays are not merely internal `Store`. The document states that external arrays are mapped onto IO operations by an array access mapping layer, and that array accesses before a synchronization call must commit no later than the sync commit, while accesses

after the sync must commit strictly after it. It also permits transformations such as caching, merging, splitting, and reordering when allowed by the mapping layer.

Define:

inductive MemoryEvent

| read (array : ArrayId) (addr : ArrayAddress) (value : Value)

| write (array : ArrayId) (addr : ArrayAddress) (value : Value)

-- Memory events are carried through the same proof-internal core-IO/ Σ mapping

-- as the other externally visible operations; they are not an unrelated trace.

def MemoryEvent.toCoreIOEvent : MemoryEvent $\rightarrow$ CoreIOEvent

| e $\Rightarrow$ CoreIOEvent.memory e

structure ArrayAccessMapping where

mappedCoreIOEvents : SourceArrayAccess $\rightarrow$ List CoreIOEvent

mappedMemoryEvents : SourceArrayAccess $\rightarrow$ List MemoryEvent

commitTime : SourceArrayAccess $\rightarrow$ Cycle

fixedLatency : SourceArrayAccess $\rightarrow$ Nat

memory_events_are_core_io_projection :

$\forall$ a,

mappedCoreIOEvents a =

(mappedMemoryEvents a).map MemoryEvent.toCoreIOEvent

conflictFreeReorderAllowed :

SourceArrayAccess $\rightarrow$ SourceArrayAccess $\rightarrow$ Prop

transformedEquivalent :

SourceArrayAccess $\rightarrow$ List SourceArrayAccess $\rightarrow$ Prop

Rules:

Array accesses are constrained by the same dynamic synchronization intervals as the ordinary source-order frontier, but SourceArrayAccess values are not themselves members of $\text{prefS}(s)$ or $\text{suffS}(s)$. The array-access mapping layer must therefore provide an explicit bridge from source array accesses to the dynamic frontier/synchronization interval structure.

structure ArraySyncIntervalBridge

(C : ComparisonInstance) where
arrayFrontierItemOf :
SourceArrayAccess → Option FrontierItem

array_item_in_dynamic_universe :
 $\forall a x,$
arrayFrontierItemOf a = some x →
 $\exists P W,$
 $x \in (W.universe P).\Omega$

array_before_sync_matches_prefS :
 $\forall a s x,$
arrayFrontierItemOf a = some x →
ArrayAccessBeforeSync C a s ↔
FrontierItemInPrefS C x s

array_after_sync_matches_suffS :
 $\forall a s x,$
arrayFrontierItemOf a = some x →
ArrayAccessAfterSync C a s ↔
FrontierItemInSuffS C x s

array_commit_cycle_is_mapping_commit_cycle :
 $\forall a,$
ArrayCommitCycle C a = C.contracts.arrays.mapping.commitTime a

-- External/shared array accesses are interpreted through the array access
-- mapping layer, not as ordinary internal Store updates. The mapping layer
-- supplies the externally relevant memory operation, the addressed storage
-- location, the commit cycle at which that storage is accessed, and the
-- declared memory semantics used to compare Sys_ref and RTL_B.

inductive ArrayOpKind
| read
| write

structure ExternalStorageLocation where
storage : ExternalArrayStorage

addr : ArrayAddress

```
def ArrayMappedOpKind
(C : ComparisonInstance)
(a : SourceArrayAccess) : ArrayOpKind :=
...
```

```
def ArrayMappedLocation
(C : ComparisonInstance)
(a : SourceArrayAccess) : ExternalStorageLocation :=
...
```

```
def ArrayWriteValue
(C : ComparisonInstance)
(a : SourceArrayAccess) : Option Value :=
...
```

```
def ArrayReadValue
(C : ComparisonInstance)
(a : SourceArrayAccess) : Option Value :=
...
```

```
def SameExternalStorageLocation
(C : ComparisonInstance)
(a1 a2 : SourceArrayAccess) : Prop :=
ArrayMappedLocation C a1 = ArrayMappedLocation C a2
```

-- The mapping layer may declare that a transformation splits, merges, caches,
-- or otherwise rewrites source array accesses. Such transformations are legal
-- only when the mapped external memory behavior is equivalent under the
-- declared memory semantics.

```
def MappingEquivalentUnderSharedMemorySemantics
(C : ComparisonInstance)
(a : SourceArrayAccess)
(transformed : List SourceArrayAccess) : Prop :=
C.contracts.arrays.mapping.transformedEquivalent a transformed ∧
PreservesExternalMemoryBehavior C a transformed
```

```

-- Serialization order for accesses that touch the same externally visible
-- storage location and are not proven conflict-free. This is a semantic order
-- over mapped memory commits, not a new source-order rule for ordinary core IO.
def SharedStorageSerializedBefore
(C : ComparisonInstance)
(a1 a2 : SourceArrayAccess) : Prop :=
SameExternalStorageLocation C a1 a2 ∧
C.contracts.arrays.mapping.commitTime a1 < C.contracts.arrays.mapping.commitTime a2

def SharedStorageSameCycleCompatible
(C : ComparisonInstance)
(a1 a2 : SourceArrayAccess) : Prop :=
SameExternalStorageLocation C a1 a2 ∧
C.contracts.arrays.mapping.commitTime a1 = C.contracts.arrays.mapping.commitTime a2
∧
DeclaredSameCycleMemoryPolicyAllows C a1 a2

def SharedStorageConflict
(C : ComparisonInstance)
(a1 a2 : SourceArrayAccess) : Prop :=
SameExternalStorageLocation C a1 a2 ∧
¬ C.contracts.arrays.mapping.conflictFreeReorderAllowed a1 a2

def SharedStorageCoherent
(C : ComparisonInstance) : Prop :=
∀ aRead aWrite,
ArrayMappedOpKind C aRead = ArrayOpKind.read →
ArrayMappedOpKind C aWrite = ArrayOpKind.write →
SameExternalStorageLocation C aWrite aRead →
C.contracts.arrays.mapping.commitTime aWrite <
C.contracts.arrays.mapping.commitTime aRead →
NoInterveningWriteToSameLocation C aWrite aRead →
ArrayReadValue C aRead = ArrayWriteValue C aWrite

-- Every definition below uses SourceArrayAccess. ArrayAccess is not a second
-- carrier. A transformation result is a List SourceArrayAccess because splitting,
-- merging, caching, or rewriting need not be one-to-one.

```

```

structure ArrayAccessMappingRules (C : ComparisonInstance) where
mapping : ArrayAccessMapping
syncIntervalBridge :
ArraySyncIntervalBridge C
beforeSyncCommitsNoLater :
 $\forall a\ s,$ 
ArrayAccessBeforeSync C a s  $\rightarrow$ 
C.contracts.arrays.mapping.commitTime a  $\leq$  SyncCommitTime s
afterSyncCommitsAfter :
 $\forall a\ s,$ 
ArrayAccessAfterSync C a s  $\rightarrow$ 
SyncCommitTime s < C.contracts.arrays.mapping.commitTime a
issueConsistentWithFixedLatency :
 $\forall a,$ 
C.contracts.arrays.mapping.commitTime a =
ArrayIssueTime a + C.contracts.arrays.mapping.fixedLatency a

```

```

-- Transformed operations must preserve the memory behavior declared by the
-- array access mapping layer, including any externally visible shared-memory
-- effects.

```

```

transformedOpsRespectMapping :
 $\forall a$  transformed,
TransformedFromList a transformed  $\rightarrow$ 
MappingEquivalentUnderSharedMemorySemantics C a transformed

```

```

-- Reordering of external array/memory accesses is legal only when the mapping
-- layer declares the reordering conflict-free.

```

```

conflictFreeReorderingOnly :
 $\forall a_1\ a_2,$ 
Reordered a1 a2  $\rightarrow$ 
C.contracts.arrays.mapping.conflictFreeReorderAllowed a1 a2

```

```

-- If two mapped accesses touch the same externally visible storage location and
-- are not covered by a declared conflict-free reordering proof, the mapping
-- layer must provide a serialization relation or an explicit same-cycle memory
-- policy. This is the shared-storage rule required by the main scheduling
-- document for external/shared memories.

```

```

sharedStorageConflictsAreSerializedOrPolicyCompatible :

```

$\forall a_1 a_2,$
 $a_1 \neq a_2 \rightarrow$
 SharedStorageConflict C $a_1 a_2 \rightarrow$
 SharedStorageSerializedBefore C $a_1 a_2 \vee$
 SharedStorageSerializedBefore C $a_2 a_1 \vee$
 SharedStorageSameCycleCompatible C $a_1 a_2$

-- Reads observe the value determined by the declared serialization/coherence
 -- semantics of the shared external storage.
 sharedStorageCoherence :
 SharedStorageCoherent C

For early milestones, it is acceptable to assume:

AllExternalArrayAccessesAlreadyMappedToCoreIO C

but the full proof strategy should include array mapping.

11. R2C combinational fixed point

ComparisonInstance now has:
 combModel? : Option CombComposition

R2C should use that field when combinational processes are in scope. In v42, combinational Read/Write events are not emitted unconditionally in every observable cycle bucket. Instead, each explicitly modeled combinational process or combinational network component C_comb has a selected observable slice $Obs(C_comb)$. The composed combinational network still has a unique ε -quiescent fixed-point valuation μ_t in each compared cycle, but a Σ -visible R2C fixed-point signal-observation unit is emitted only at canonical observation occurrences: (1) the first cycle in the compared execution in which the component's observable slice is defined and observed; (2) a later cycle in which the vector of fixed-point values on $Obs(C_comb)$ differs from the most recent emitted unit for that component; (3) a cycle in which an explicitly modeled observer/sampler requires a fresh observation unit; or (4) every relevant cycle for an explicitly cycle-locked R2C component.

The Lean model should therefore preserve four facts:

1. R2C fixed-point values are read from the composed valuation μ_t .

2. The composed valuation is obtained from explicit local combinational process components, rather than from an unexplained monolithic global object.
3. Σ -visible R2C units are stutter-canonicalized per component unless the component is explicitly cycle-locked.
4. Cross-model matching is by component identity and local R2C occurrence ordinal, not by equality of global cycle bucket number or by co-bucketing with unrelated events.

Define:

```
structure R2CObservableLabel where
  proc : Process
  sig : Signal
  direction : SignalDirection -- read or write
```

```
structure LocalCombNetwork where
  proc : Process
  combStepLocal : Store → Store → Prop
  reads : Finset Signal
  writes : Finset Signal
  obsSlice : Finset R2CObservableLabel
  localDeterministic : Prop
```

```
structure CombNetwork where
  combStep : Store → Store → Prop
  deterministic : Prop
```

```
structure CombComposition where
  components :
    Process → Option LocalCombNetwork
```

```
composed :
  CombNetwork
```

```
component_processes_are_combinational :
  ∀ P N,
  components P = some N →
  CombinationalProcess P
```

component_owner_matches_process :

$\forall P N,$

components $P = \text{some } N \rightarrow$

$N.\text{proc} = P$

observable_slice_well_formed :

$\forall P N \text{ label},$

components $P = \text{some } N \rightarrow$

$\text{label} \in N.\text{obsSlice} \rightarrow$

$\text{label}.\text{proc} = P \wedge (\text{label}.\text{sig} \in N.\text{reads} \vee \text{label}.\text{sig} \in N.\text{writes})$

local_step_embeds_in_composed_step :

$\forall P N \sigma \sigma',$

components $P = \text{some } N \rightarrow$

$N.\text{combStepLocal } \sigma \sigma' \rightarrow$

$\text{composed}.\text{combStep } \sigma \sigma'$

composed_step_generated_by_components :

$\forall \sigma \sigma',$

$\text{composed}.\text{combStep } \sigma \sigma' \rightarrow$

$\exists P N,$

components $P = \text{some } N \wedge$

$N.\text{combStepLocal } \sigma \sigma'$

resolved_driver_or_disjoint_write_policy :

$\forall P_1 P_2 N_1 N_2 \text{ sig},$

components $P_1 = \text{some } N_1 \rightarrow$

components $P_2 = \text{some } N_2 \rightarrow$

$\text{sig} \in N_1.\text{writes} \rightarrow$

$\text{sig} \in N_2.\text{writes} \rightarrow$

$P_1 \neq P_2 \rightarrow$

$\text{ResolvedMultiDriverPolicy } \text{composed } \text{sig}$

composed_deterministic_from_components :

$(\forall P N, \text{components } P = \text{some } N \rightarrow N.\text{localDeterministic}) \rightarrow$

$\text{composed}.\text{deterministic}$

The scheduling document's R2C rule requires deterministic ε -settling to a unique

observable fixed point. It does not globally require structural combinational acyclicity for every admissible combinational network. Pure combinational oscillation, ambiguous unresolved multi-driver behavior, intentional combinational-loop behavior outside a unique fixed-point interpretation, or non-convergent ε -behavior are excluded through the unique-fixed-point obligation below unless they are represented by explicit state or synchronization elsewhere.

Structural acyclicity may still be imposed as a separate well-formedness requirement for specific elaboration/coupler constructions in Appendix Q, but it is not a universal requirement of LocalCombNetwork, CombNetwork, or CombComposition itself.

```
def CombQuiescent (N : CombNetwork) ( $\sigma$  : State) : Prop :=
 $\neg \exists \sigma', N.combStep \sigma.store \sigma'.store$ 
```

```
def CombFixedPointAtCycle
(N : CombNetwork)
(t : Cycle)
( $\mu$  : Store) : Prop :=
ReachesByCombSteps N t  $\mu$   $\wedge$ 
IsFixedPoint N  $\mu$ 
```

```
def ComponentParticipatesInCycle
(C : ComparisonInstance)
(M : CombComposition)
(P : Process)
(t : Cycle) : Prop :=
 $\exists N,$ 
M.components P = some N  $\wedge$ 
CombinationalProcessActiveAt C P t
```

```
def R2CUnitVector
(C : ComparisonInstance)
(M : CombComposition)
(P : Process)
(t : Cycle)
( $\mu$  : Store) : Vector Value :=
ValuesOfObservableSlice C M P t  $\mu$ 
```



```

def R2CUnitEmissionCanonical
(C : ComparisonInstance)
(M : CombComposition)
(P : Process)
(t : Cycle) : Prop :=
FirstDefinedR2CObservation C M P t V
R2CUnitVectorChangedSincePreviousEmission C M P t V
ExplicitObserverRequiresFreshR2CUnit C M P t V
CycleLockedR2CComponent C M P

```

Well-formedness:

```

structure R2C_WF (C : ComparisonInstance) where
composition_present :
C.combModel?.isSome

```

uniqueComposedFixedPoint :

```

∀ t M,
C.combModel? = some M →
∃! μ, CombFixedPointAtCycle M.composed t μ

```

component_events_use_composed_fixed_point :

```

∀ P e t μ M,
C.combModel? = some M →
ComponentParticipatesInCycle C M P t →
CombReadWriteEventOfProcess C P e →
e ∈ R2CUnitEventsAt C P t →
CombFixedPointAtCycle M.composed t μ →
EventReadsWritesStore e μ

```

all_comb_events_have_component :

```

∀ P e t M,
C.combModel? = some M →
CombReadWriteEventOfProcess C P e →
e ∈ R2CUnitEventsAt C P t →
∃ N,
M.components P = some N

```

units_emit_only_at_canonical_occurrences :

$\forall P \ t \ M,$

$C.combModel? = some \ M \rightarrow$

$R2CUnitEmitted \ C \ P \ t \rightarrow$

$R2CUnitEmissionCanonical \ C \ M \ P \ t$

stutter_cycles_have_no_sigma_unit :

$\forall P \ t \ M,$

$C.combModel? = some \ M \rightarrow$

$\neg R2CUnitEmissionCanonical \ C \ M \ P \ t \rightarrow$

$\neg R2CUnitEmitted \ C \ P \ t$

component_local_occurrence_matching :

$\forall P \ k \ u_ref \ u_post,$

$KthR2CUnit \ C \ .ref \ P \ k \ u_ref \rightarrow$

$KthR2CUnit \ C \ .post \ P \ k \ u_post \rightarrow$

$R2CUnitValuesAgree \ C \ P \ u_ref \ u_post$

multi_input_well_formed :

$\forall P \ M,$

$C.combModel? = some \ M \rightarrow$

$MultInputR2CComponent \ C \ M \ P \rightarrow$

$SingleExplicitAnchoringDomain \ C \ M \ P \ V$

$AllAdmissibleInterleavingsInduceSameR2CUnitSequence \ C \ M \ P \ V$

$R2CComponentHiddenBehindExplicitSynchronizationOrTransactor \ C \ M \ P$

def R2CEventsObservedOnlyInStutterCanonicalUnits

(C : ComparisonInstance)

(hR2C : R2C_WF C) : Prop :=

$\forall P \ e \ t,$

$CombReadWriteEventOfProcess \ C \ P \ e \rightarrow$

$e \in BucketSigmaEvents \ C \ t \rightarrow$

$\exists M \ \mu \ unit,$

$C.combModel? = some \ M \wedge$

$ComponentParticipatesInCycle \ C \ M \ P \ t \wedge$

$CombFixedPointAtCycle \ M.composed \ t \ \mu \wedge$

$R2CUnitEmitted \ C \ P \ t \wedge$

$e \in unit.events \wedge$

EventReadsWritesStore e μ

```
def R2CUnitsMatchedByComponentLocalOrdinal
(C : ComparisonInstance)
(hR2C : R2C_WF C) : Prop :=
 $\forall$  P k u_ref u_post,
KthR2CUnit C .ref P k u_ref  $\rightarrow$ 
KthR2CUnit C .post P k u_post  $\rightarrow$ 
R2CUnitValuesAgree C P u_ref u_post
```

This formulation keeps the semantic fixed point global where R2C needs it, but makes the proof obligation modular and stutter-canonical: each combinational process contributes a local component and selected observable slice, the composed network used for μ_t is justified by the CombComposition fields, and the emitted observable sequence is compared by component-local occurrence ordinal. Thus R2C fixed-point equality remains a system-level statement over μ_t , while conformance remains checkable process-by-process plus an explicit composition/driver-resolution and multi-input anchoring/confluence obligation.

If combinational processes are out of scope for an early milestone, use:

SequentialOnly C
and postpone R2C_WF.

12. Issue/Commit well-formedness

Issue and Commit should be relational plus existence/uniqueness, not necessarily computable total functions.

The Issue/Commit layer is side-parametric and uses the same bucketed execution carrier as the rest of the strategy. It must not introduce a separate flat Trace type.

```
def SystemOfSide
(C : ComparisonInstance) : Side  $\rightarrow$  System
| Side.ref => C.Sys_ref
| Side.post => C.RTL_B
```

abbrev SidePrefix

(C : ComparisonInstance)
 (side : Side) : Type :=
 Prefix (SystemOfSide C side)

abbrev SideInfiniteExecution
 (C : ComparisonInstance)
 (side : Side) : Type :=
 InfiniteExecutionBucketed C.Sigma

IssueAt :
 (C : ComparisonInstance) →
 (side : Side) →
 SidePrefix C side →
 DynMsgInstance →
 Cycle →
 Prop

CommitAt :
 (C : ComparisonInstance) →
 (side : Side) →
 SidePrefix C side →
 DynMsgInstance →
 Cycle →
 Payload →
 Prop

def CommitAtCycle
 (C : ComparisonInstance)
 (side : Side)
 (τ : SidePrefix C side)
 (q : DynMsgInstance)
 (t : Cycle) : Prop :=
 ∃ v, CommitAt C side τ q t v

Well-formedness:

def BoundaryRequestAtIssue
 (C : ComparisonInstance) : Prop :=
 ∀ side τ q tIssue σ,

```

IssuedInPrefix C side  $\tau$  q  $\rightarrow$ 
IssueAt C side  $\tau$  q tIssue  $\rightarrow$ 
StateAtCycleInPrefix C side  $\tau$  tIssue  $\sigma \rightarrow$ 
BoundaryReqIssueDomainSide C side q  $\sigma \wedge$ 
BoundaryReqSide C side q  $\sigma \wedge$ 
((q.kind = MsgKind.Push  $\vee$  q.kind = MsgKind.PushNB)  $\rightarrow$ 
PayloadAtState C side q  $\sigma$  = PayloadAtIssue C side  $\tau$  q tIssue)

```

```

def BlockingBoundaryRequestPersistence
(C : ComparisonInstance) : Prop :=
 $\forall$  side  $\tau$  q tIssue t  $\sigma$ ,
IssuedInPrefix C side  $\tau$  q  $\rightarrow$ 
IssueAt C side  $\tau$  q tIssue  $\rightarrow$ 
StateAtCycleInPrefix C side  $\tau$  t  $\sigma \rightarrow$ 
tIssue  $\leq$  t  $\rightarrow$ 
 $\neg$  CommitsBefore C side  $\tau$  q t  $\rightarrow$ 
Blocking q  $\rightarrow$ 
BoundaryReqPersistentDomainSide C side q  $\sigma \wedge$ 
BoundaryReqSide C side q  $\sigma \wedge$ 
(q.kind = MsgKind.Push  $\rightarrow$ 
PayloadAtState C side q  $\sigma$  = PayloadAtIssue C side  $\tau$  q tIssue)

```

```

def BoundaryRequestSoundness
(C : ComparisonInstance) : Prop :=
BoundaryRequestAtIssue C  $\wedge$ 
BlockingBoundaryRequestPersistence C

```

-- SameEndpointDirection means the same concrete ready/valid endpoint, not
-- merely the same Push-vs-Pop direction. In particular, two Push operations on
-- different channels are not the same endpoint direction and may be outstanding
-- concurrently.

```

def SameEndpointDirection
(q0 q1 : DynMsgInstance) : Prop :=
q0.channel = q1.channel  $\wedge$ 
DirectionOf q0 = DirectionOf q1

```

```

def StableAttribution
(C : ComparisonInstance) : Prop :=

```

$\forall \text{ side } \tau \ q_0 \ q_1 \ t \ \sigma t \ \sigma_{\text{next}},$
 $\text{SameEndpointDirection } q_0 \ q_1 \rightarrow$
 $q_0 \neq q_1 \rightarrow$
 $\text{Blocking } q_0 \rightarrow$
 $\text{StateAtCycleInPrefix } C \text{ side } \tau \ t \ \sigma t \rightarrow$
 $\text{StateAtCycleInPrefix } C \text{ side } \tau \ (t + 1) \ \sigma_{\text{next}} \rightarrow$
 $\text{BoundaryReqDomainSide } C \text{ side } q_0 \ \sigma t \rightarrow$
 $\text{BoundaryReqSide } C \text{ side } q_0 \ \sigma t \rightarrow$
 $\text{BoundaryReqDomainSide } C \text{ side } q_1 \ \sigma_{\text{next}} \rightarrow$
 $\text{BoundaryReqSide } C \text{ side } q_1 \ \sigma_{\text{next}} \rightarrow$
 $\neg \text{EndpointCommitAt } C \text{ side } \tau \ q_0.\text{endpoint } t \rightarrow$
 $\neg \text{EndpointCommitAt } C \text{ side } \tau \ q_1.\text{endpoint } t \rightarrow$
 False

Equivalently, the first premise may be written as $q_0.\text{endpoint} = q_1.\text{endpoint}$ if the concrete Lean development defines endpoint as the pair $(q.\text{channel}, \text{DirectionOf } q)$. The important point is that *StableAttribution* forbids retagging one still-outstanding request to a different operation on the same physical endpoint; it does not forbid independent blocking pushes or pops on different channels.

BoundaryRequestSoundness has two parts. *BoundaryRequestAtIssue* applies to every issued source-level message-operation instance q , including *PushNB* and *PopNB* instances that fail and therefore produce no committed Σ -event. Such a non-blocking poll must still assert the appropriate endpoint request in its issue cycle. *BlockingBoundaryRequestPersistence* is the additional non-withdrawal rule for blocking *Push/Pop* instances: after issue and before commit, the same dynamic operation instance keeps its request asserted, and a *Push* keeps its payload stable.

StableAttribution applies when the earlier outstanding request q_0 is a blocking *Push/Pop*. In that case, attribution cannot change to any distinct same-endpoint operation before a commit on that endpoint direction, whether the later candidate q_1 is blocking or non-blocking. For non-blocking *PushNB / PopNB* calls, request assertion at the issue cycle does not by itself imply persistence beyond that issue cycle, and continuous assertion of the endpoint request signal does not by itself collapse later executed non-blocking call instances into the same dynamic instance q . Repeated executed NB polls remain distinct source-level

dynamic instances unless the witness explicitly identifies the same still-outstanding request instance.

-- Dynamic source interval between synchronization anchors.
 -- This is witness-specific: only executed dynamic instances in the selected
 -- witness universe are classified. Untaken branches and unexecuted loop
 -- iterations have no SyncIntervalId.

```
structure SyncInterval where
proc : Process
pred? : Option SyncCall
succ? : Option SyncCall
intervalId : Nat
```

```
def SyncIntervalId
(C : ComparisonInstance)
(W : WitnessBundle C)
(q : DynMsgInstance) : Option SyncInterval :=
DynamicSourceIntervalOf C W q
```

```
structure SyncIntervalWF
(C : ComparisonInstance)
(W : WitnessBundle C) where
interval_defined_for_frontier_messages :
 $\forall P q,$ 
 $q \in (W.universe P).Q_{\Omega} \rightarrow$ 
 $\exists I, \text{SyncIntervalId } C W q = \text{some } I$ 
```

```
same_interval_iff_same_anchors :
 $\forall q_0 q_1 I_0 I_1,$ 
SyncIntervalId C W q0 = some I0  $\rightarrow$ 
SyncIntervalId C W q1 = some I1  $\rightarrow$ 
I0 = I1  $\leftrightarrow$ 
I0.proc = I1.proc  $\wedge$ 
I0.pred? = I1.pred?  $\wedge$ 
I0.succ? = I1.succ?  $\wedge$ 
I0.intervalId = I1.intervalId
```

$\text{pref_suff_consistent} :$
 $\forall s q,$
 $q \in \text{prefS } s \vee q \in \text{suffS } s \rightarrow$
 $\exists I, \text{SyncIntervalId } C \ W \ q = \text{some } I$

$\text{def SameSyncInterval}$
 $(C : \text{ComparisonInstance})$
 $(W : \text{WitnessBundle } C)$
 $(q_0 \ q_1 : \text{DynMsgInstance}) : \text{Prop} :=$
 $\exists I,$
 $\text{SyncIntervalId } C \ W \ q_0 = \text{some } I \wedge$
 $\text{SyncIntervalId } C \ W \ q_1 = \text{some } I$

$\text{def SourceIssuesBackToBackAfterCommit}$
 $(C : \text{ComparisonInstance})$
 $(W : \text{WitnessBundle } C)$
 $(\text{side} : \text{Side})$
 $(\tau : \text{SidePrefix } C \ \text{side})$
 $(q_0 \ q_1 : \text{DynMsgInstance})$
 $(t : \text{Cycle}) : \text{Prop} :=$
 $\text{SameEndpointDirection } q_0 \ q_1 \wedge$
 $\text{SameSyncInterval } C \ W \ q_0 \ q_1 \wedge$
 $\text{CommitAtCycle } C \ \text{side } \tau \ q_0 \ t \wedge$
 $\text{NextSourceMessageOpOnEndpoint } C \ q_0 \ q_1 \wedge$
 $\text{IssuedInPrefix } C \ \text{side } \tau \ q_1 \wedge$
 $\text{SourceControlReachesAndIssuesAt } C \ W \ \text{side } \tau \ q_1 \ (t + 1)$

$\text{def BackToBackIssueWF}$
 $(C : \text{ComparisonInstance})$
 $(W : \text{WitnessBundle } C) : \text{Prop} :=$
 $\forall \text{side } \tau \ q_0 \ q_1 \ t,$
 $\text{SourceIssuesBackToBackAfterCommit } C \ W \ \text{side } \tau \ q_0 \ q_1 \ t \rightarrow$
 $\text{EndpointRequestContinuouslyAsserted } C \ \text{side } \tau \ q_0.\text{endpoint } t \ (t + 1) \rightarrow$
 $\text{IssueAt } C \ \text{side } \tau \ q_1 \ (t + 1)$

The SameSyncInterval component of SourceIssuesBackToBackAfterCommit prevents

the back-to-back issuance rule from carrying a continuously asserted endpoint request across an explicit wait(), SyncChannel, ac_sync, or other synchronization boundary. Cross-boundary issue timing is governed instead by SyncBoundaryIssueDiscipline / E5.

The continuous physical endpoint assertion is not what causes q_1 to issue. It only explains why the boundary signal may remain asserted across the q_0/q_1 transition without a visible deassertion bubble. The source-level cause is the separate SourceControlReachesAndIssuesAt premise: q_1 has actually been reached and issued as the next same-endpoint operation in the same synchronization interval. Without that premise, a held valid/ready signal could be incorrectly used to infer issuance of a source operation that control flow has not yet reached.

```
def SyncBoundaryIssueDiscipline
(C : ComparisonInstance) : Prop :=
 $\forall$  side  $\tau$   $q_0$   $q_1$  s tSync  $t_0$   $t_1$ ,
 $q_0 \in \text{prefS } s \rightarrow$ 
 $q_1 \in \text{suffS } s \rightarrow$ 
SyncCommitAt C side  $\tau$  s tSync  $\rightarrow$ 
IssueAt C side  $\tau$   $q_0$   $t_0 \rightarrow$ 
IssueAt C side  $\tau$   $q_1$   $t_1 \rightarrow$ 
 $t_0 \leq \text{tSync} \wedge \text{tSync} < t_1$ 
```

```
def NBDynamicInstanceDiscipline
(C : ComparisonInstance) : Prop :=
 $\forall$  P  $q_0$   $q_1$ ,
IsNB  $q_0 \rightarrow$ 
IsNB  $q_1 \rightarrow$ 
ExecutedInProcess C P  $q_0 \rightarrow$ 
ExecutedInProcess C P  $q_1 \rightarrow$ 
SameSourceCallSiteButDifferentExecution  $q_0$   $q_1 \rightarrow$ 
 $q_0 \neq q_1$ 
```

```
structure IssueCommitWF
(C : ComparisonInstance)
(W : WitnessBundle C) where
```

$\text{issue_exists} :$
 $\forall \text{ side } \tau q,$
 $\text{IssuedInPrefix } C \text{ side } \tau q \rightarrow$
 $\exists t, \text{IssueAt } C \text{ side } \tau q t$
 $\text{issue_unique} :$
 $\forall \text{ side } \tau q t_1 t_2,$
 $\text{IssueAt } C \text{ side } \tau q t_1 \rightarrow$
 $\text{IssueAt } C \text{ side } \tau q t_2 \rightarrow$
 $t_1 = t_2$
 $\text{commit_exists_unique} :$
 $\forall \text{ side } \tau q,$
 $\text{CommitsInPrefix } C \text{ side } \tau q \rightarrow$
 $\exists ! tp : \text{Cycle} \times \text{Payload},$
 $\text{CommitAt } C \text{ side } \tau q tp.1 tp.2$
 $\text{boundary_request_soundness} :$
 $\text{BoundaryRequestSoundness } C$
 $\text{stable_attribution} :$
 $\text{StableAttribution } C$
 $\text{back_to_back_issue} :$
 $\text{BackToBackIssueWF } C W$
 $\text{sync_boundary_discipline} :$
 $\text{SyncBoundaryIssueDiscipline } C$
 $\text{nb_dynamic_instance_discipline} :$
 $\text{NBDynamicInstanceDiscipline } C$

Derived timestamp:
 $\text{noncomputable def IssueTime}$
 $(hWF : \text{IssueCommitWF } C W)$
 $(h : \text{IssuedInPrefix } C \text{ side } \tau q) : \text{Cycle} :=$
 $\text{Classical.choose } (hWF.\text{issue_exists side } \tau q h)$

This preserves Appendix C's use of $\text{Issue}(q)$ as an auxiliary timestamp while avoiding an unnecessary computability requirement.

13. Drain-only blocked-frontier discipline, automatic flush, and sync-boundary policy

Define:

Pending $P \sigma$

Front $P \sigma$

NextFront $P \sigma$

BlockingMsgNextFront $P \sigma$

Done $P \sigma$

Remain $P \sigma$

The generalized drain-only blocked-frontier discipline, and Appendix B AutomaticFlush as its pipelined constructor, must be stated over the same dynamic universe and source-order objects used by Appendix I/J/K. In particular, neither may be an opaque Prop bundle.

Section 13 reasons primarily about FrontierItem values, while Section 7's BoundaryReqSide / ChannelEnabledSide / ChannelDisabledSide predicates are operation-attributive predicates over DynMsgInstance values. Therefore Section 13 uses the following item-level wrappers. These wrappers are not new semantic predicates; they merely read a message-operation frontier item through its owning dynamic message instance.

```
def MsgOfFrontierItemSide
(C : ComparisonInstance)
(side : Side)
(x : FrontierItem) : Option DynMsgInstance :=
MsgInstanceOfFrontierItem C x.proc x

def BoundaryReqItemSide
(C : ComparisonInstance)
(side : Side)
(x : FrontierItem)
( $\sigma$  : State) : Prop :=
 $\exists q,$ 
MsgOfFrontierItemSide C side x = some q  $\wedge$ 
BoundaryReqPersistentDomainSide C side q  $\sigma \wedge$ 
BoundaryReqSide C side q  $\sigma$ 
```

```

def ChannelEnabledItemSide
(C : ComparisonInstance)
(side : Side)
(x : FrontierItem)
( $\sigma$  : State) : Prop :=
 $\exists$  q,
MsgOfFrontierItemSide C side x = some q  $\wedge$ 
ChannelEnabledSide C side q  $\sigma$ 

```

```

def ChannelDisabledItemSide
(C : ComparisonInstance)
(side : Side)
(x : FrontierItem)
( $\sigma$  : State) : Prop :=
 $\exists$  q,
MsgOfFrontierItemSide C side x = some q  $\wedge$ 
ChannelDisabledSide C side q  $\sigma$ 

```

```

def PayloadAtFrontierItemState
(C : ComparisonInstance)
(side : Side)
(x : FrontierItem)
( $\sigma$  : State) : Payload :=
PayloadAtState C side (Classical.choose (MsgOfFrontierItemSide_isSome C side x))  $\sigma$ 

```

```

def FrontierItemIssuedAt
(C : ComparisonInstance)
(side : Side)
( $\tau$  : SidePrefix C side)
(x : FrontierItem)
(t : Cycle) : Prop :=
 $\exists$  q,
MsgOfFrontierItemSide C side x = some q  $\wedge$ 
IssueAt C side  $\tau$  q t

```

```

def FrontierItemCommitsBeforeCycle
(C : ComparisonInstance)
(side : Side)

```

$(\tau : \text{SidePrefix } C \text{ side})$
 $(x : \text{FrontierItem})$
 $(t : \text{Cycle}) : \text{Prop} :=$
 $\exists q,$
 $\text{MsgOfFrontierItemSide } C \text{ side } x = \text{some } q \wedge$
 $\text{CommitsBefore } C \text{ side } \tau q t$

$\text{def FrontierItemCommitsBetween}$
 $(C : \text{ComparisonInstance})$
 $(\text{side} : \text{Side})$
 $(\tau : \text{SidePrefix } C \text{ side})$
 $(x : \text{FrontierItem})$
 $(t_0 t : \text{Cycle}) : \text{Prop} :=$
 $\exists q,$
 $\text{MsgOfFrontierItemSide } C \text{ side } x = \text{some } q \wedge$
 $\text{CommitsBetween } C \text{ side } \tau q t_0 t$

$\text{def FrontierItemCommitsAtCycle}$
 $(C : \text{ComparisonInstance})$
 $(\text{side} : \text{Side})$
 $(\tau : \text{SidePrefix } C \text{ side})$
 $(x : \text{FrontierItem})$
 $(t : \text{Cycle}) : \text{Prop} :=$
 $\exists q,$
 $\text{MsgOfFrontierItemSide } C \text{ side } x = \text{some } q \wedge$
 $\text{CommitAtCycle } C \text{ side } \tau q t$

$\text{def FrontierBlocked}$
 $(C : \text{ComparisonInstance})$
 $(\text{side} : \text{Side})$
 $(P : \text{Process})$
 $(\sigma : \text{State}) : \text{Prop} :=$
 $\text{Front } C \text{ side } P \sigma \neq \emptyset \wedge$
 $\forall x,$
 $x \in \text{Front } C \text{ side } P \sigma \rightarrow$
 $\text{BlockingMessageFrontierItem } x \wedge$
 $\text{BoundaryReqItemSide } C \text{ side } x \sigma \wedge$

ChannelDisabledItemSide C side x σ

def LaterThanBlockedFrontier

(C : ComparisonInstance)

(side : Side)

(P : Process)

(σ : State)

(y : FrontierItem) : Prop :=

$\exists x,$

$x \in \text{Front C side P } \sigma \wedge$

SourceOrderLt C side P x y

def FrontierRemainsBlockedOver

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ : SidePrefix C side)

(t_0 t : Cycle) : Prop :=

$\forall n \sigma n,$

$t_0 \leq n \rightarrow$

$n \leq t \rightarrow$

StateAtCycleInPrefix C side τ n $\sigma n \rightarrow$

FrontierBlocked C side P σn

def NoNewLaterWorkWhileBlocked

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ : SidePrefix C side) : Prop :=

$\forall t_0 t \sigma_0 y,$

StateAtCycleInPrefix C side τ t_0 $\sigma_0 \rightarrow$

FrontierBlocked C side P $\sigma_0 \rightarrow$

FrontierRemainsBlockedOver C side P τ t_0 t $\rightarrow$

LaterThanBlockedFrontier C side P σ_0 y $\rightarrow$

$\neg \text{FrontierItemIssuedAt C side } \tau$ y t

-- Definition R.16 clause (2) is more general than message issue. The dynamic

-- operation carrier ranges over synchronization, signal/R2C, message, array, and

-- other proof-visible operation occurrences represented in Remain/NextFront.

def DynamicOperationLaunchedAt

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ : SidePrefix C side)

(op : FrontierItem)

(t : Cycle) : Prop :=

DynamicOccurrenceLaunchAt C side P τ op t

def NoNewDynamicOperationLaunchWhileBlocked

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ : SidePrefix C side) : Prop :=

$\forall t_0 t \sigma_0 \text{ op},$

StateAtCycleInPrefix C side $\tau t_0 \sigma_0 \rightarrow$

FrontierBlocked C side P $\sigma_0 \rightarrow$

FrontierRemainsBlockedOver C side P $\tau t_0 t \rightarrow$

LaterThanBlockedFrontier C side P $\sigma_0 \text{ op} \rightarrow$

$\neg$ DynamicOperationLaunchedAt C side P $\tau \text{ op } t$

def NoWithdrawalWhileBlocked

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ : SidePrefix C side) : Prop :=

$\forall t_0 t \sigma_0 \sigma t x,$

StateAtCycleInPrefix C side $\tau t_0 \sigma_0 \rightarrow$

StateAtCycleInPrefix C side $\tau t \sigma t \rightarrow$

$t_0 \leq t \rightarrow$

$x \in \text{Front } C \text{ side } P \sigma_0 \rightarrow$

BoundaryReqItemSide C side $x \sigma_0 \rightarrow$

$\neg$ FrontierItemCommitsBetween C side $\tau x t_0 t \rightarrow$

BoundaryReqItemSide C side $x \sigma t \wedge$

(IsPushFrontierItem $x \rightarrow$

PayloadAtFrontierItemState C side $x \sigma t =$

PayloadAtFrontierItemState C side $x \sigma_0$)

-- WFGEdgeCauselItem is the attribution relation used by the concrete WFG
 -- definition: it identifies the frontier item from which a $P \rightarrow Q$ edge is drawn.

def WFGEdgeAttributedToFrontierItem

(C : ComparisonInstance)

(side : Side)

(P : Process)

(x : FrontierItem)

(σ : State) : Prop :=

$\exists Q,$

WFGEdge C side σ P Q $\wedge$

WFGEdgeCauselItem C side σ P Q x

def NonFrontierAssertedRequestNotWFGSource

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ : SidePrefix C side) : Prop :=

$\forall t \sigma x,$

StateAtCycleInPrefix C side $\tau t \sigma \rightarrow$

BlockingMessageFrontierItem x $\rightarrow$

BoundaryReqItemSide C side x $\sigma \rightarrow$

x $\notin$ Front C side P $\sigma \rightarrow$

$\neg$ WFGEdgeAttributedToFrontierItem C side P x σ

Full Remain_P / NextFront_P minimality is deliberately not a field of K.Drain, AutomaticFlush, or DrainOnlyBlockedFrontierDiscipline. Definition R.16 requires only that already-asserted non-frontier requests are not used as WFG sources.

The stronger statement that the selected blocking frontier is minimal in the full Remain_P(σ) order is the conclusion proved by K0_front_nextfront_classification from R2/R3/R4, E1–E5, Issue/Commit attribution, and cutpoint well-formedness.

def WFGInertForProcess

(C : ComparisonInstance)

(side : Side)

(P : Process)

($\sigma_0 \sigma t$: State) : Prop :=

$\forall Q,$

WFGEdge C side σ_0 P Q $\leftrightarrow$ WFGEdge C side σt P Q

```
def NoNewBlockingFrontierCause
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma_0$   $\sigma t$  : State) : Prop :=
 $\forall x$ ,
 $x \in \text{Front C side P } \sigma t \rightarrow$ 
BlockingMessageFrontierItem  $x \rightarrow$ 
ChannelDisabledItemSide C side  $x \sigma t \rightarrow$ 
 $\exists y$ ,
 $y \in \text{Front C side P } \sigma_0 \wedge$ 
BlockingMessageFrontierItem  $y \wedge$ 
SameBlockingCause C side P  $y x$ 
```

```
def WFGInertUnderInternalOnlyBetween
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\tau$  : SidePrefix C side) : Prop :=
 $\forall t_0 t \sigma_0 \sigma t$ ,
StateAtCycleInPrefix C side  $\tau t_0 \sigma_0 \rightarrow$ 
StateAtCycleInPrefix C side  $\tau t \sigma t \rightarrow$ 
 $t_0 \leq t \rightarrow$ 
FrontierBlocked C side P  $\sigma_0 \rightarrow$ 
FrontierRemainsBlockedOver C side P  $\tau t_0 t \rightarrow$ 
InternalOrEpsilonOnlyBetween C side  $\tau t_0 t \rightarrow$ 
WFGInertForProcess C side P  $\sigma_0 \sigma t \wedge$ 
NoNewBlockingFrontierCause C side P  $\sigma_0 \sigma t$ 
```

```
def DrainableDownstreamRequest
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma_0$  : State)
( $x$  : FrontierItem) : Prop :=
AlreadyIssuedBeforeBlockedCutpoint C side P  $\sigma_0 x \wedge$ 
```

DownstreamOfBlockedFrontier C side P $\sigma_0 \times \Lambda$
 BlockingMessageFrontierItem $x \wedge$
 BoundaryReqItemSide C side $x \sigma_0$

def DrainOnlyCommitDiscipline
 (C : ComparisonInstance)
 (side : Side)
 (P : Process)
 (τ : SidePrefix C side) : Prop :=
 $\forall t_0 t \sigma_0 e,$
 StateAtCycleInPrefix C side $\tau t_0 \sigma_0 \rightarrow$
 $t_0 \leq t \rightarrow$
 FrontierBlocked C side P $\sigma_0 \rightarrow$
 FrontierRemainsBlockedOver C side P $\tau t_0 t \rightarrow$
 $e \in \text{SigmaEventsAtCycleInPrefix C side } \tau t \rightarrow$
 EventOwnedByProcess C side P $e \rightarrow$
 ObservableCommitEvent $e \rightarrow$
 $\exists x,$
 DrainableDownstreamRequest C side P $\sigma_0 \times \Lambda$
 EventCommitsFrontierItem C side $x e \wedge$
 $\neg \text{FrontierItemCommitsBeforeCycle C side } \tau x t_0 \wedge$
 FrontierItemCommitsAtCycle C side $\tau x t$

The K.Drain and $K\epsilon$ obligations are separate. DrainOnlyCommitDiscipline permits observable commits of already-issued downstream work while the frontier remains blocked. WFGInertUnderInternalOnlyBetween is the independent A11/ $K\epsilon$ property for intervals containing no observable Σ -commit. Fair drain normalization and Appendix K may consume both premises, but neither is hidden inside the other.

structure AutomaticFlush
 (C : ComparisonInstance)
 (side : Side)
 (P : Process)
 (τ : SidePrefix C side) where
 block_point :
 $\forall t \sigma,$
 StateAtCycleInPrefix C side $\tau t \sigma \rightarrow$
 FrontierBlocked C side P $\sigma \rightarrow$

FlushBlockedPoint C side P τ t σ

no_new_dynamic_launch :

NoNewDynamicOperationLaunchWhileBlocked C side P τ

no_withdrawal :

NoWithdrawalWhileBlocked C side P τ

nonfrontier_not_wfg_source :

NonFrontierAssertedRequestNotWFGSource C side P τ

drain_only_progress :

DrainOnlyCommitDiscipline C side P τ

sync_boundary_policy :

SyncBoundaryPolicy C side P τ

-- Appendix B AutomaticFlush is the pipelined construction of the more general
-- property consumed by Appendix K. K.2b can also encounter liberal same-interval
-- eager issue in ordinary non-pipelined control, so the K layer must not require
-- every relevant blocked interval to be a pipelined interval.

def BlockingIntervallnPrefix

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ : SidePrefix C side) : Prop :=

$\exists t_0 \ t \ \sigma_0,$

StateAtCycleInPrefix C side τ t_0 $\sigma_0 \wedge$

$t_0 \leq t \wedge$

FrontierBlocked C side P $\sigma_0 \wedge$

FrontierRemainsBlockedOver C side P τ t_0 t

-- This predicate records semantic overlap in the selected prefix rather than
-- leaving the classification as an opaque proof placeholder.

def PipelinedControlCanContainOverlappedDynamicWork

(C : ComparisonInstance)

(side : Side)

(P : Process)

```

( $\tau$  : SidePrefix C side) : Prop :=
 $\exists t \sigma op_1 op_2,$ 
StateAtCycleInPrefix C side  $\tau t \sigma \wedge$ 
 $op_1 \neq op_2 \wedge$ 
DynamicOperationActiveAt C side P  $\tau op_1 t \wedge$ 
DynamicOperationActiveAt C side P  $\tau op_2 t$ 

-- Ordinary non-overlapped control supplies the same semantic no-launch property
-- directly; this name is retained as the constructor-facing alias.
def NoNewDynamicOperationLaunchAfterBlockedPoint
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\tau$  : SidePrefix C side) : Prop :=
NoNewDynamicOperationLaunchWhileBlocked C side P  $\tau$ 

def PipelinedBlockingIntervalInPrefix
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\tau$  : SidePrefix C side) : Prop :=
BlockingIntervalInPrefix C side P  $\tau \wedge$ 
PipelinedControlCanContainOverlappedDynamicWork C side P  $\tau$ 

def NonOverlappedBlockingControl
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\tau$  : SidePrefix C side) : Prop :=
BlockingIntervalInPrefix C side P  $\tau \wedge$ 
NoNewDynamicOperationLaunchAfterBlockedPoint C side P  $\tau$ 

-- A stored-or-credited boundary item is one unit of accepted work represented
-- at the chosen explicit abstract channel boundary: a resident FIFO element,
-- rendezvous/request item, skid/register item, or an explicitly accounted credit.
-- The concrete sum/quotient carrier is selected by the boundary realization.
def StoredOrCreditedBoundaryItem
(C : ComparisonInstance)

```

```

(c : Channel) : Type := by
  sorry

def FiniteExplicitBoundaryStorage
  (C : ComparisonInstance) : Prop :=
  ∀ c,
  BackAnnotatedExplicitAbstractChannel C c →
  Finite (StoredOrCreditedBoundaryItem C c)

-- The implementation assumptions already include finite back-annotated B(c),
-- explicit boundary accounting, and B-faithfulness. This theorem packages those
-- facts into the finiteness premise consumed by both drain-only constructors.
theorem finite_explicit_boundary_storage_of_implementation
  (C : ComparisonInstance)
  (hImpl : ImplementationAssumptions C) :
  FiniteExplicitBoundaryStorage C :=
  by
    -- For each selected explicit abstract boundary, use the finite capacity B(c),
    -- the realization/accounting map, and exact B-faithfulness to inject every
    -- stored-or-credited item into a finite slot/credit index.
    sorry

def FiniteAlreadyIssuedDrainableWork
  (C : ComparisonInstance)
  (side : Side)
  (P : Process)
  (τ : SidePrefix C side) : Prop :=
  ∀ t₀ σ₀,
  StateAtCycleInPrefix C side τ t₀ σ₀ →
  FrontierBlocked C side P σ₀ →
  Set.Finite
  {x : FrontierItem |
    DrainableDownstreamRequest C side P σ₀ x ∧
    AlreadyIssuedBeforeBlockedCutpoint C side P σ₀ x}

-- This is the complete seven-clause Lean carrier for Definition R.16 / K.Drain.
-- The field order mirrors the document: blocked point; no new source-later
-- dynamic launch; no withdrawal; non-frontier requests excluded as WFG sources;

```

-- drain-only committed progress; finite non-replenishing drain; and explicit
 -- sync-boundary policy.

structure DrainOnlyBlockedFrontierDiscipline

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ : SidePrefix C side) : Prop where

block_point :

$\forall t \sigma,$

StateAtCycleInPrefix C side $\tau t \sigma \rightarrow$

FrontierBlocked C side P $\sigma \rightarrow$

FlushBlockedPoint C side P $\tau t \sigma$

no_new_dynamic_launch :

NoNewDynamicOperationLaunchWhileBlocked C side P τ

no_withdrawal :

NoWithdrawalWhileBlocked C side P τ

nonfrontier_not_wfg_source :

NonFrontierAssertedRequestNotWFGSource C side P τ

drain_only_progress :

DrainOnlyCommitDiscipline C side P τ

finite_nonreplenishing_drain :

FiniteAlreadyIssuedDrainableWork C side P τ

sync_boundary_policy :

SyncBoundaryPolicy C side P τ

-- FiniteAlreadyIssuedDrainableWork is discharged from finite explicit boundary

-- storage/accounting together with the no-new-launch property. It does not say

-- that a blocked pipeline must be replenished in order to drain.

theorem AutomaticFlush.toDrainOnlyBlockedFrontierDiscipline

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ : SidePrefix C side)

(hAF : AutomaticFlush C side P τ)

(hFinite : FiniteExplicitBoundaryStorage C) :

DrainOnlyBlockedFrontierDiscipline C side P τ :=

by

-- Project the six non-finiteness clauses from hAF; use hFinite and

-- no_new_dynamic_launch for the finite non-replenishing drain set.
sorry

theorem NonOverlappedControl.toDrainOnlyBlockedFrontierDiscipline

(C : ComparisonInstance)

(W : WitnessBundle C)

(side : Side)

(P : Process)

(τ : SidePrefix C side)

(hOrd : NonOverlappedBlockingControl C side P τ)

(hIC : IssueCommitWF C W)

(hSync : SyncBoundaryPolicy C side P τ)

(hFinite : FiniteExplicitBoundaryStorage C) :

DrainOnlyBlockedFrontierDiscipline C side P τ :=

by

-- In ordinary non-overlapped control, later work may already have issued in a
-- legal liberal same-interval pattern before the blocked point, but no new
-- dynamic operation is launched after that point. Persistent request behavior
-- supplies no-withdrawal; WFG construction excludes non-frontier requests; only
-- already-issued downstream work can drain; hSync supplies R.16 clause (7).

sorry

-- The document theorem needs R.16 only on reachable selected intervals used by
-- fair drain normalization, K.2b, or K.1. The relevance predicate prevents the
-- semantic theorem from quantifying over unreachable prefixes or unrelated
-- serial intervals.

def ProofRelevantDrainInterval

(C : ComparisonInstance)

(W : WitnessBundle C)

(side : Side)

(P : Process)

(τ : SidePrefix C side) : Prop :=

BlockingIntervalInPrefix C side P τ $\wedge$

SelectedReachableSidePrefix C W side τ $\wedge$

(OutstandingSourceLaterDynamicWork C side P τ $\vee$

RequiredByFairDrainNormalization C W side P τ $\vee$

RequiredByK2bEagerDeferral C W side P τ)

-- K.Drain is the theorem-scoped condition that every proof-relevant interval has
 -- the complete seven-clause Definition R.16 discipline. $K\epsilon$ remains separate.

```
def KDrain
(C : ComparisonInstance)
(W : WitnessBundle C) : Prop :=
  ∀ side P τ,
  ProofRelevantDrainInterval C W side P τ →
  DrainOnlyBlockedFrontierDiscipline C side P τ
```

-- A tool or flow may choose to certify the stronger universal profile. It is a
 -- sufficient implementation condition, not the minimal semantic theorem premise.

```
def UniversalDrainConformance
(C : ComparisonInstance) : Prop :=
  ∀ side P τ,
  BlockingIntervalInPrefix C side P τ →
  DrainOnlyBlockedFrontierDiscipline C side P τ
```

theorem UniversalDrainConformance.impliesKDrain

```
(C : ComparisonInstance)
(W : WitnessBundle C)
(hUniversal : UniversalDrainConformance C) :
KDrain C W :=
by
  intro side P τ hRelevant
  exact hUniversal side P τ hRelevant.1
```

The sync-boundary policy must explicitly split the ordinary and elaborated cases. In the elaborated case, it must state a universal coverage obligation: it is not enough to exhibit one gate for one synchronization call. Every proof-relevant withholding case in which capacity belongs only to $\text{suff_S}(s)$ and can affect producer-facing availability while s is pending must be represented by an explicit sync-boundary / epoch-gate abstract channel, and the resulting refusal must appear as ordinary ChannelDisabled behavior at that explicit boundary.

```
def SyncPendingAt
(C : ComparisonInstance)
(side : Side)
(P : Process)
(τ : SidePrefix C side)
```



```

(s : SyncCall)
(t : Cycle) : Prop :=
SyncIssuedButNotCommitted C side P  $\tau$  s t

-- qSuffix is the future operation in suff_S(s) whose eventual work consumes
-- capacity belonging to the post-sync epoch. E5 normally prevents qSuffix from
-- issuing while s is pending.
def FutureEpochCapacityUse
(C : ComparisonInstance)
(side : Side)
(P : Process)
(s : SyncCall)
(qSuffix : DynMsgInstance) : Prop :=
qSuffix  $\in$  suffS_msg C side P s  $\wedge$ 
MessageOperationInstance qSuffix  $\wedge$ 
CapacityBelongsOnlyToPostSyncInterval C side P s qSuffix

-- qUpstream is the possibly different producer request currently presented at
-- the producer-facing boundary. It may belong to another process.
def UpstreamProducerRequestForFutureEpoch
(C : ComparisonInstance)
(side : Side)
(P : Process)
(s : SyncCall)
(qSuffix qUpstream : DynMsgInstance)
( $\sigma$  : State) : Prop :=
BoundaryReqPersistentDomainSide C side qUpstream  $\sigma$   $\wedge$ 
BoundaryReqSide C side qUpstream  $\sigma$   $\wedge$ 
ProducerFacingRequest qUpstream  $\wedge$ 
UpstreamTransferWouldSupplyCapacityForSuffix
C side P s qUpstream qSuffix

def RequiresProducerFacingWithholding
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\tau$  : SidePrefix C side)
(s : SyncCall)

```

(qSuffix qUpstream : DynMsgInstance)

(t : Cycle)

(σ : State) : Prop :=

StateAtCycleInPrefix C side τ t σ ∧

SyncPendingAt C side P τ s t ∧

FutureEpochCapacityUse C side P s qSuffix ∧

UpstreamProducerRequestForFutureEpoch

C side P s qSuffix qUpstream σ

def SyncBoundaryGateCorrect

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ : SidePrefix C side)

(selected : SelectedElaboration C.outerSysB C.Chan_outer C.Sigma C.Sigma_DUT)

(gateChannel : Channel)

(s : SyncCall)

(qSuffix qUpstream : DynMsgInstance) : Prop :=

GateChannelOf selected.E gateChannel ∧

GateSeparatesUpstreamRequestFromSuffixCapacity

C side P selected.E gateChannel s qUpstream qSuffix ∧

∀ t σ,

RequiresProducerFacingWithholding

C side P τ s qSuffix qUpstream t σ →

RequestPresentedAtGateBoundary

C side selected.E gateChannel qUpstream σ →

ChannelDisabledSide C side qUpstream σ

structure SyncBoundaryWithholdingCoverage

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ : SidePrefix C side)

(selected : SelectedElaboration C.outerSysB C.Chan_outer C.Sigma C.Sigma_DUT) where
covered :

∀ s qSuffix qUpstream t σ,

RequiresProducerFacingWithholding

C side P τ s qSuffix qUpstream t σ →

$\exists$ gateChannel,
 SyncBoundaryGateCorrect
 C side P τ selected gateChannel s qSuffix qUpstream

inductive SyncBoundaryPolicy
 (C : ComparisonInstance)
 (side : Side)
 (P : Process)
 (τ : SidePrefix C side)
 | no_withholding_in_ordinary_SysB :
 C.refChoice = .ordinary $\rightarrow$
 NoProofRelevantSyncBoundaryWithholding C side P τ $\rightarrow$
 SyncBoundaryPolicy C side P τ
 | explicit_gates_in_selected_elaboration :
 $\forall$ selected,
 C.refChoice = .elaborated selected $\rightarrow$
 SyncBoundaryWithholdingCoverage C side P τ selected $\rightarrow$
 SyncBoundaryPolicy C side P τ

The two dynamic operations are intentionally distinct. qSuffix supplies the suff_S(s) ownership fact; qUpstream supplies the live BoundaryReq and is the operation whose producer-facing availability becomes ChannelDisabled. Requiring one q to play both roles is generally vacuous under E5, because the suffix operation has not yet issued while s is pending. The selected-elaboration equality also prevents a gate from an unrelated package from satisfying the coverage obligation.

In the ordinary Sys_B case, the clause adds no hidden gating: the instance proves that no proof-relevant withholding is needed. Otherwise the theorem instance must select Sys_B^{elab}(E), and every withholding case must appear as ordinary ChannelDisabled behavior for qUpstream at an explicit gate in that exact E.

14. Witness selector and Appendix L

The uploaded document's witness selector is a partial strategy over ε -quiescent source cutpoints and ε -modeled choices whose outcomes may affect later Σ -visible behavior; it must be causally available from the matched RTL prefix and global Σ -causal history.

Avoid circularity by defining it over the partial construction so far:

structure WitnessSelector (C : ComparisonInstance) where

choose :

($\tau_{\text{post_prefix}}$: Prefix C.RTL_B) $\rightarrow$

($\tau_{\text{ref_so_far}}$: Prefix C.Sys_ref) $\rightarrow$

(cp : ChoicePoint $\tau_{\text{ref_so_far}}$) $\rightarrow$

Option ChoiceOutcome

causal :

$\forall \tau_p \tau_r \text{ cp out},$

choose $\tau_p \tau_r \text{ cp} = \text{some out} \rightarrow$

DependsOnlyOnAvailablePrefix $\tau_p \tau_r \text{ cp out}$

consistent_with_post_prefix :

$\forall \tau_p \tau_r \text{ cp out},$

choose $\tau_p \tau_r \text{ cp} = \text{some out} \rightarrow$

ConsistentWithRTLPrefix $\tau_p \tau_r \text{ cp out}$

Non-blocking witness mapping.

Failed PushNB / PopNB polls are executed dynamic instances in Q_{exec} but produce no Σ -event. Therefore the Post $\rightarrow$ Ref construction cannot reconstruct them from the Σ -trace alone. The witness bundle must carry an explicit partial dynamic-instance correspondence for such ε -modeled NB instances.

structure NBInstanceMap (C : ComparisonInstance) where

μ_Q :

DynMsgInstance $\rightarrow$ Option DynMsgInstance

maps_only_executed_nb :

$\forall q_{\text{post}} q_{\text{ref}},$

$\mu_Q q_{\text{post}} = \text{some } q_{\text{ref}} \rightarrow$

ExecutedInPost C $q_{\text{post}} \wedge$

ExecutedInRef C $q_{\text{ref}} \wedge$

IsNB $q_{\text{post}} \wedge \text{IsNB } q_{\text{ref}}$

preserves_owner_and_interface :

$\forall q_{\text{post}} q_{\text{ref}},$

$\mu_Q q_{\text{post}} = \text{some } q_{\text{ref}} \rightarrow$

$q_{\text{post}}.\text{proc} = q_{\text{ref}}.\text{proc} \wedge$

$q_{\text{post}}.\text{channel} = q_{\text{ref}}.\text{channel} \wedge$

$q_{\text{post}}.\text{kind} = q_{\text{ref}}.\text{kind}$

```

preserves_nb_outcome :
  ∀ q_post q_ref,
  μ_Q q_post = some q_ref →
  NBOutcomePost C q_post = NBOutcomeRef C q_ref

covers_non_cadence_control_relevant_failed_polls :
  ∀ q_post,
  ExecutedInPost C q_post →
  IsFailedNB q_post →
  AffectsLaterSigmaOrFrontier C q_post →
  ¬ CadenceSensitiveNBInstance C q_post →
  ∃ q_ref, μ_Q q_post = some q_ref

cadence_sensitive_failed_polls_are_not_unhandled_core :
  ∀ q_post,
  ExecutedInPost C q_post →
  IsFailedNB q_post →
  CadenceSensitiveNBInstance C q_post →
  HandledByLatencySensitiveLocalProtocolContract C q_post

source_order_position_preserved :
  ∀ q_post q_ref,
  μ_Q q_post = some q_ref →
  SourceOrderKeyOf C q_post = SourceOrderKeyOf C q_ref

respects_source_order_slice :
  ∀ P q1_post q2_post q1_ref q2_ref,
  μ_Q q1_post = some q1_ref →
  μ_Q q2_post = some q2_ref →
  SourceOrderLe C P (SourceOrderKeyOf C q1_post) (SourceOrderKeyOf C q2_post) ↔
  SourceOrderLe C P (SourceOrderKeyOf C q1_ref) (SourceOrderKeyOf C q2_ref)

structure WitnessBundle (C : ComparisonInstance) where
  selector : WitnessSelector C
  nbMap : NBInstanceMap C
  universe : ∀ P, DynamicUniverseData C P

structure WitnessBundleWF
  (C : ComparisonInstance)

```

(W : WitnessBundle C) where
 universe_wf : $\forall P, \text{WitnessUniverseWF } C \ W \ P$

Witness compatibility:

```
def WC
(C : ComparisonInstance)
(W : WitnessBundle C) : Prop :=
(∀ constructionPrefix,
WitnessChoicesRespectMatchedPrefix C W.selector constructionPrefix) ∧
WitnessNBMapRespectsMatchedPrefix C W.nbMap ∧
WitnessChoicesAndNBMapAgree C W.selector W.nbMap ∧
LatencySensitiveLocalProtocolWC C W C.contracts.latencySensitiveLocalProtocols
```

Thus WC is the Lean-side packaging of witness selection, the μ_Q matching obligation, and the modeling-contract obligation that cadence-sensitive failed-poll behavior is not left as an unconstrained hidden timing dependence inside the ordinary source core. When E3/E5 reasoning or the Post $\rightarrow$ Ref construction refers to failed ε -modeled non-blocking call instances in Q_{exec}, P , the required μ_Q correspondence is supplied by $W.\text{nbMap}$ as part of WC for non-cadence-sensitive or already boundary-selected cases. If a failed-poll sequence is cadence-sensitive because externally visible behavior depends on the number or cadence of failures, then WC is applied at the chosen observable boundary of the isolated/snoop-aligned local protocol block, not to the raw internal failed-poll cadence.

Snooping assumptions:

```
structure SnoopingWrapperAssumptions
(C : ComparisonInstance)
(W : WitnessBundle C) where
covers_eps_choices_affecting_boundary :
∀ P choice,
EpsChoiceAffectsLaterSigmaOrFrontier C P choice  $\rightarrow$ 
ChoiceCoveredBySnoopingOrWitness C W P choice

does_not_relabel_boundary_sigma :
∀  $\tau$ ,
SnoopedExecution C W  $\tau \rightarrow$ 
BoundarySigmaTracePreserved C W  $\tau$ 
```

selected_choices_prefix_causal :

$\forall$ constructionPrefix,

WitnessChoicesRespectMatchedPrefix C W.selector constructionPrefix

failed_polls_remain_epsilon :

$\forall$ q,

IsFailedNB q $\rightarrow$

$\neg \exists e, \text{FailedPollContributesCommittedSigmaEvent } C \ q \ e$

cadence_sensitive_sites_isolated_or_snoop_aligned :

$\forall$ P loc,

CadenceSensitiveNBPollingSite C P loc $\rightarrow$

LocalProtocolSiteHandledByContract C W C.contracts.latencySensitiveLocalProtocols P loc

Lemma L.2:

theorem L2_snooping_establishes_WC

(C : ComparisonInstance)

(W : WitnessBundle C)

(hSnoop : SnoopingWrapperAssumptions C W) :

WC C W

This theorem does not turn failed PushNB/PopNB polls into committed Σ -events. Failed polls remain ε -steps. For ordinary NB-CF sites, the identity/arbitrary witness suffices. For non-NB-CF but non-cadence-sensitive sites, the witness selects the ε -modeled outcomes needed for the matched prefix. For cadence-sensitive retry/timeout/poll-arbiter sites, the theorem may be used only after the local protocol block has been isolated, snoop-aligned, or given an explicit cycle-accurate model, and the comparison is made at that block's chosen observable boundary.

IdentityOrArbitraryNBWitnessBundle C is the witness bundle used in the NB-CF case. Its selector uses identity choices for ε -modeled non-blocking outcomes that are fixed by the matched prefix, and arbitrary choices for non-blocking outcomes that NB-CF proves cannot affect later Σ -visible control flow or NextFront behavior. Its nbMap maps matched executed NB instances to the corresponding same dynamic source-call instance when such a match is relevant, and is arbitrary on failed NB instances that NB-CF proves irrelevant. It is not a discharge for cadence-sensitive retry/timeout/poll-arbiter sites unless those sites have first been shown not to affect later Σ -visible behavior, or have been handled by the latency-sensitive local-protocol contract.

```

def IdentityOrArbitraryNBWitnessBundle
(C : ComparisonInstance) : WitnessBundle C :=
{
  selector := IdentityOrArbitraryNBSelector C,
  nbMap := IdentityOrArbitraryNBMap C
}

```

Corollary L.3:

The document-level WC predicate is global for the selected comparison instance and now includes the latency-sensitive local-protocol contract. Therefore a theorem returning global WC must not be stated from a single-process NB-CF premise alone.

Use a global theorem when proving WC for the full comparison instance:

```

theorem L3_nb_cf_identity_witness
(C : ComparisonInstance)
(hNBCF :  $\forall P, \text{NBCF } C P$ )
(hLocalProtocols :
  LatencySensitiveLocalProtocolWC
  C
  (IdentityOrArbitraryNBWitnessBundle C)
  C.contracts.latencySensitiveLocalProtocols) :
WC C (IdentityOrArbitraryNBWitnessBundle C)

```

If a per-process theorem is useful, state it as a local witness fact rather than as global WC:

```

theorem L3_nb_cf_identity_witness_local
(C : ComparisonInstance)
(P : Process)
(hNBCF :  $\text{NBCF } C P$ ) :
LocalWCForProcess C (IdentityOrArbitraryNBWitnessBundle C) P

```

The global theorem is then obtained by combining the local theorem over all processes with the latency-sensitive local-protocol contract:

```

theorem L3_nb_cf_identity_witness_from_local
(C : ComparisonInstance)
(hLocal :  $\forall P, \text{LocalWCForProcess } C (\text{IdentityOrArbitraryNBWitnessBundle } C) P$ )
(hLocalProtocols :
  LatencySensitiveLocalProtocolWC

```


C

(IdentityOrArbitraryNBWitnessBundle C)

C.contracts.latencySensitiveLocalProtocols) :

WC C (IdentityOrArbitraryNBWitnessBundle C)

Corollary L.4 (cadence-sensitive polling handled by isolation). The scheduling-rules document names this fact Corollary L.4; the strategy already enforces its content through the LatencySensitiveLocalProtocolContract and the WC definition, but the Lean development should also expose it under its document name:

theorem L4_cadence_sensitive_isolated_or_out_of_scope

(C : ComparisonInstance)

(P : Process)

(loc : SourceLoc)

(hCadence : CadenceSensitiveNBPollingSite C P loc) :

HandledByLatencySensitiveLocalProtocolContractAt

C C.contracts.latencySensitiveLocalProtocols P loc V

OutsideOrdinaryAppendixIJSrcCoreAt C P loc

That is: if NB-CF fails because the source observes the number or cadence of failed PushNB/PopNB polls — retry counters, timeout counters, or arbitration policies whose later Σ -visible behavior depends on how long a channel remained empty or full — then the site is latency-sensitive at that polling site, and it must be isolated, snoop-aligned, or given an explicit cycle-accurate local protocol model under Appendix D / Appendix L before the ordinary Appendix I/J source-core proof is applied. The left disjunct is exactly the contract obligation `cadence_sensitive_nb_sites_handled`; the right disjunct records that an unhandled site removes the process from the scope of the ordinary trace-compatibility theorem at the chosen observable boundary. Lemma L.2 may then be used only at the chosen observable boundary of the handled block, never against the raw internal failed-poll cadence.

Appendix L Note 1: Σ _DUT-safety-inert local timing choices.

The preceding L.2/L.3 facts are WC-producing facts for the ordinary proof-internal C.Sigma theorem. Appendix L Note 1 is different: it is a projected-safety principle for the outer DUT/testbench alphabet C.Sigma_DUT. It says that an internal latency-sensitive interface need not be snooped for the Σ _DUT-safety fragment when its ε -modeled timing choices cannot change which Σ _DUT-visible observable units occur, cannot change their projected values, and can only permute projected units that are causally independent at the DUT boundary.

This is not a replacement for WC in the full C.Sigma theorem, and it is not an Appendix K deadlock/liveness exemption. A Σ_{DUT} -safety-inert interface that is also used in Appendix K reasoning must still satisfy the WFG-inert / $K\epsilon$ obligations, such as A11_Kepsilon_WFGInert.

-- A proof-internal observable unit whose projection is visible at the chosen
-- DUT/testbench boundary.

SigmaDUTVisibleUnit :

ComparisonInstance $\rightarrow$ ObservableUnit C.Sigma $\rightarrow$ Prop

-- Required projected Appendix J order at the DUT boundary.
-- This is the mandated projected $\prec_J$ relation induced by E1–E5, channel
-- semantics, synchronization semantics, atomic-unit constraints,
-- witness-control dependencies that survive projection, and elaboration /
-- σ_{DUT} -projection constraints. It is not an incidental order that merely
-- appears in one admissible schedule.

ProjectedRequiredJOrderDUT :

ComparisonInstance $\rightarrow$

ObservableUnit C.Sigma $\rightarrow$

ObservableUnit C.Sigma $\rightarrow$

Prop

-- Two proof-internal observable units are ordered at the DUT boundary exactly
-- when their projected Σ_{DUT} -visible units are related by the required projected
-- Appendix J order.

def DUTOrdered

(C : ComparisonInstance)

(u v : ObservableUnit C.Sigma) : Prop :=

SigmaDUTVisibleUnit C u $\wedge$

SigmaDUTVisibleUnit C v $\wedge$

ProjectedRequiredJOrderDUT C u v

-- A pair is causally comparable if either order direction is required at the
-- projected DUT boundary.

def DUTCausallyComparable

(C : ComparisonInstance)

(u v : ObservableUnit C.Sigma) : Prop :=

DUTOrdered C u v $\vee$ DUTOrdered C v u

-- The choice may change the projected DUT-visible presence or value label of u.

ChoiceMayAddRemoveOrRelabelSigmaDUTUnit :

ComparisonInstance $\rightarrow$ Process $\rightarrow$ EpsModeledChoice C P $\rightarrow$ ObservableUnit C.Sigma $\rightarrow$

Prop

-- The choice may change the relative order of two projected DUT-visible units.

ChoiceMayReorderSigmaDUTUnits :

ComparisonInstance $\rightarrow$ Process $\rightarrow$ EpsModeledChoice C P $\rightarrow$

ObservableUnit C.Sigma $\rightarrow$ ObservableUnit C.Sigma $\rightarrow$ Prop

-- A single ε -modeled local timing choice is inert for Σ_{DUT} safety when it

-- neither changes projected event presence/value nor reorders any projected

-- pair whose order is causally required at the DUT boundary.

def EpsChoiceIsSigmaDUTSafetyInert

(C : ComparisonInstance)

(P : Process)

(choice : EpsModeledChoice C P) : Prop :=

($\forall$ u,

SigmaDUTVisibleUnit C u $\rightarrow$

$\neg$ ChoiceMayAddRemoveOrRelabelSigmaDUTUnit C P choice u) $\wedge$

($\forall$ u v,

ChoiceMayReorderSigmaDUTUnits C P choice u v $\rightarrow$

$\neg$ DUTCausallyComparable C u v)

-- A latency-sensitive interface is Σ_{DUT} -safety-inert when all of the ε -modeled

-- local timing choices owned by the interface are safety-inert in the sense above.

def SigmaDUTSafetyInertInterface

(C : ComparisonInstance)

(I : LatencySensitiveInterface C) : Prop :=

$\forall$ P choice,

ChoiceOwnedByInterface C I P choice $\rightarrow$

EpsChoiceIsSigmaDUTSafetyInert C P choice

-- A choice has a Σ_{DUT} -safety discharge either by ordinary snooping/witness

-- coverage or by belonging to a certified Σ_{DUT} -safety-inert interface.

def ChoiceHasSigmaDUTSafetyDischarge

(C : ComparisonInstance)

(W : WitnessBundle C)

(P : Process)

(choice : EpsModeledChoice C P) : Prop :=

ChoiceCoveredBySnoopingOrWitness C W P choice $\vee$

$\exists$ I,

ChoiceOwnedByInterface C I P choice $\wedge$

SigmaDUTSafetyInertInterface C I

-- The projected-safety obligation is deliberately weaker than WC. It covers

-- only choices that can matter to the Σ_{DUT} projected safety theorem.

def SigmaDUTSafetyWitnessObligation

(C : ComparisonInstance)

```

(W : WitnessBundle C) : Prop :=
  ∀ P choice,
  EpsChoiceCanAffectSigmaDUTProjection C P choice →
  ChoiceHasSigmaDUTSafetyDischarge C W P choice

```

```

-- Local Appendix L Note 1 corollary: a certified safety-inert interface
-- discharges only its own choices. This theorem is intentionally local; it does
-- not claim that an arbitrary identity witness satisfies the global obligation
-- for every other choice in the design.

```

```

theorem LNote1_sigmaDUT_safety_inert_interface

```

```

(C : ComparisonInstance)

```

```

(W : WitnessBundle C)

```

```

(I : LatencySensitiveInterface C)

```

```

(hI : SigmaDUTSafetyInertInterface C I) :

```

```

  ∀ P choice,

```

```

  ChoiceOwnedByInterface C I P choice →

```

```

  ChoiceHasSigmaDUTSafetyDischarge C W P choice

```

```

-- No global Appendix L Note 1 assembly theorem is stated here. The global obligation
-- SigmaDUTSafetyWitnessObligation C W is intentionally an assumption of
-- JSigmaDUTSafetyAssumptions. Individual certified inert-interface choices are
-- discharged locally by LNote1_sigmaDUT_safety_inert_interface; all other
-- Σ_DUT-relevant choices must be discharged by the ordinary snooping/witness
-- machinery at the point where the global obligation is established.

```

The Appendix L Note 1 projected-safety facts are safety-side counterparts of the WFG-inert $\varepsilon / K\varepsilon$ idea, but they live in Appendix L/J rather than Appendix K. They are used only for the Σ_{DUT} -projected safety theorem below. They do not produce WC C W, do not prove CompatibleSys over the full proof-internal C.Sigma alphabet, and do not discharge A11_Kepsilon_WFGInert for deadlock/liveness proofs.

15. Concrete ObsSigma, Obs-congruence, and deterministic replay

Do not define ObsSigma as an informal least fixed point. Define it as a concrete record of the source-state slices required for replay. The record is P-local: it contains the P-owned state, frontier, pending, attribution, signal-anchor, direct-input, array-mapping, and R2C component-local observation-unit facts needed to determine later P-local Σ -visible behavior and Appendix I-K proof predicates. It does not independently store ChannelEnabled / ChannelDisabled as P-local facts; enablement is obtained from the P-local request/frontier/BoundaryReq facts together with the chosen Sys_ref boundary state and E4. For combinational-process signal IO, ObsSigma records the selected component identity, local R2C occurrence ordinal, observable slice, and fixed-point values for emitted R2C units, not every stutter cycle of the combinational network.

```

structure ObsSigma
(C : ComparisonInstance)
(W : WitnessBundle C)
(P : Process)
( $\sigma$  : State) where
control_pos : ControlPos P
relevant_vars : RelevantVarSlice C P  $\sigma$ 
selected_eps_outcomes : SelectedOutcomeSlice C W.selector P  $\sigma$ 
nb_witness_slice : NBWitnessSlice C W.nbMap P  $\sigma$ 
msg_attribution_slice : MsgAttributionSlice C W P  $\sigma$ 
pending_slice : PendingSlice C P  $\sigma$ 
front_slice : FrontSlice C P  $\sigma$ 
nextfront_slice : NextFrontSlice C P  $\sigma$ 
boundary_req_slice : BoundaryReqSlice C P  $\sigma$ 
signal_anchor_slice : SignalAnchorSlice C P  $\sigma$ 
array_mapping_slice : ArrayMappingSlice C P  $\sigma$ 
direct_input_slice : DirectInputSlice C P  $\sigma$ 
r2c_unit_slice : R2CUnitSlice C P  $\sigma$ 

```

Here `msg_attribution_slice` records the $Q_{\text{exec}}, P / Q_{\Omega}, P$ attribution facts needed for E3/E5, Issue/Commit replay, WFG construction, and BoundaryReq ownership. It is broader than `nb_witness_slice`: `nb_witness_slice` records μ_Q matching for ε -modeled failed NB calls, while `msg_attribution_slice` records the source-level operation attribution of pending and frontier message-operation instances generally. The `r2c_unit_slice` field is the previously missing explicit carrier for the combinational-process facts that the document's Definition I.Obs already requires: for each R2C component owned by P , the selected component identity, the local R2C occurrence ordinal of the most recent emitted unit, the selected observable slice `Obs(C_comb)`, and the fixed-point values of emitted units — not every stutter cycle of the combinational network. The combinational-R2C branch of `IObs_congruence_one_observable_unit` below propagates this agreement at the granularity of a complete atomic observable unit; without the `r2c_unit_slice` field that branch would have no carrier to transport, which is why earlier revisions could only assume it.

Cadence-sensitive failed-poll behavior is not represented by adding failed-poll cadence events to `ObsSigma` or to Σ . Failed PushNB/PopNB polls remain ε -steps with no committed transfer event. If a retry counter, timeout counter, polling arbiter, or other local policy computes externally visible behavior from the number or cadence of failed polls, then that site must already be handled by `C.contracts.latencySensitiveLocalProtocols`. The ordinary

source-core ObsSigma record is then applied at the chosen observable boundary of the isolated/snoop-aligned/explicitly modeled block.

```
def SigmaRelevantEq C W P  $\sigma_1$   $\sigma_2$  :=  
ObsSigma C W P  $\sigma_1$  = ObsSigma C W P  $\sigma_2$ 
```

Two replay modes must be kept distinct. Plain Theorem J.1 advances through canonical immediate post-unit semantic states: the state immediately after the transition, or explicitly atomic transition group, that realizes the next P-local observable unit, before any optional maximal ε -normalization. Corollary J.1 and Appendix K instead compare explicitly ε -normalized cutpoints. Both modes use complete ObservableUnit objects, never individual labels of an atomic rendezvous, paired synchronization, or R2C fixed-point unit. SameObservableUnitImmediateStep states that u is the next P-local unit and that the step stops at the canonical immediate post-unit state. SameObservableUnitCutpointStep additionally carries the finite ε -normalization to the ε -quiescent cutpoint after u .

```
def CanonicalImmediatePostUnitState  
(C : ComparisonInstance)  
(W : WitnessBundle C)  
(P : Process)  
( $\rho$  : ExecutionPrefix C.Sigma)  
(k : Nat) : State :=  
ImmediateStateAfterKthPLocalObservableUnit C W P  $\rho$  k
```

```
theorem IObs_congruence_one_immediate_observable_unit  
(C : ComparisonInstance)  
(W : WitnessBundle C)  
(hB : TraceBRules C)  
(hR : RRules C)  
(hE : ERules C)  
(hIC : IssueCommitWF C W)  
(hContracts : ModelingContracts C)  
(hWC : WC C W)  
(P : Process)  
( $\sigma_1$   $\sigma_2$   $\sigma_1'$   $\sigma_2'$  : State)  
(u : ObservableUnit C.Sigma)  
(hObs : SigmaRelevantEq C W P  $\sigma_1$   $\sigma_2$ )  
(hStep1 : SameObservableUnitImmediateStep C W P  $\sigma_1$  u  $\sigma_1'$ )  
(hStep2 : SameObservableUnitImmediateStep C W P  $\sigma_2$  u  $\sigma_2'$ )
```

(hBoundary : ChosenObservableBoundaryWF C W P hContracts)

(hCadence :

AllCadenceSensitiveNBSitesHandled

C W P hContracts.latencySensitiveLocalProtocols) :

SigmaRelevantEq C W P σ_1' σ_2'

Proof obligations for IObs_congruence_one_immediate_observable_unit should be decomposed by source construct. The theorem consumes only TraceBRules: the next complete observable unit, the fixed stimulus, and W determine the finite transition segment through that unit. It does not invoke maximal ε -closure and does not claim that σ_1' or σ_2' is ε -quiescent.

- deterministic local computation: equal relevant variables and control position determine equal successor ObsSigma fields;
- anchored signal reads/writes: the fixed reactive stimulus and the common next observable unit u determine equal latched Read/Write facts; an ordinary signal event is represented by a singleton unit, while an explicitly atomic signal-observation unit is consumed as a whole;
- combinational R2C units: when u is an R2CFixedPointObservationUnit owned by P, hObs supplies equal prior r2c_unit_slice facts. Together with equal relevant component inputs/state, the fixed stimulus, and R2C_WF, this gives the same unique composed fixed point, canonical-emission decision, component identity, next local occurrence ordinal, selected Obs(C_comb) slice, and fixed-point values. Ordinary R2C stutter cycles emit no unit and are not induction steps;
- blocking Push/Pop and synchronization: the common unit u, together with equal pending/frontier/attribution/BoundaryReq facts, determines equal immediate successor ObsSigma fields; rendezvous and paired synchronization units are consumed atomically rather than replayed as separate labels;
- successful PushNB/PopNB: the singleton committed-transfer unit determines equal data, control, attribution, and frontier updates;
- failed PushNB/PopNB: the failed poll is ε and is not itself an observable-unit induction step. NB-CF or WC fixes every failed outcome that can affect the next unit; cadence-sensitive sites are handled only at the isolated/snoop-aligned/explicit protocol boundary required by hCadence;
- other ε -only choices before the next unit: WC supplies every outcome capable of affecting that unit or later Appendix I–K proof predicates.

The explicitly ε -quiescent variant is obtained by composing the immediate theorem with the constructive B4/B5 normalization supplied by NormalizationBRules:

```

theorem IObs_congruence_one_observable_unit
(C : ComparisonInstance)
(W : WitnessBundle C)
(hB : TraceBRules C)
(hNorm : NormalizationBRules C)
(hR : RRules C)
(hE : ERules C)
(hIC : IssueCommitWF C W)
(hContracts : ModelingContracts C)
(hWC : WC C W)
(P : Process)
( $\sigma_1 \sigma_2 \sigma_1' \sigma_2'$  : State)
(u : ObservableUnit C.Sigma)
(hObs : SigmaRelevantEq C W P  $\sigma_1 \sigma_2$ )
(hStep1 : SameObservableUnitCutpointStep C W P  $\sigma_1 u \sigma_1'$ )
(hStep2 : SameObservableUnitCutpointStep C W P  $\sigma_2 u \sigma_2'$ )
(hBoundary : ChosenObservableBoundaryWF C W P hContracts)
(hCadence :
AllCadenceSensitiveNBSitesHandled
C W P hContracts.latencySensitiveLocalProtocols) :
SigmaRelevantEq C W P  $\sigma_1' \sigma_2'$  :=
by
-- Apply IObs_congruence_one_immediate_observable_unit, then use hNorm to
-- construct the corresponding finite  $\varepsilon$ -normalizations and prove that the
-- normalized ObsSigma slices remain equal.
sorry

```

Deterministic replay therefore has two theorem forms:

```

theorem IDR_immediate_post_unit_deterministic_replay
(C : ComparisonInstance)
(W : WitnessBundle C)
(hB : TraceBRules C)
(hR : RRules C)
(hE : ERules C)
(hIC : IssueCommitWF C W)

```


(hContracts : ModelingContracts C)
 (hWC : WC C W)
 (P : Process)
 ($\rho_1 \rho_2$: ExecutionPrefix C.Sigma)
 (hInit : SameInitialSourceState C W P $\rho_1 \rho_2$)
 (hSameUnits : SamePLocalObservableUnitSequence C P $\rho_1 \rho_2$)
 (hBoundary : ChosenObservableBoundaryWF C W P hContracts)
 (hCadence :
 AllCadenceSensitiveNBSitesHandled
 C W P hContracts.latencySensitiveLocalProtocols) :
 $\forall k,$
 SigmaRelevantEq C W P
 (CanonicalImmediatePostUnitState C W P $\rho_1 k$)
 (CanonicalImmediatePostUnitState C W P $\rho_2 k$)

-- This is the Lean counterpart of scheduling-rules Lemma I.DR'. It is the
 -- replay theorem used by plain J.1 and imports no B4/B5 premise.

theorem IDR_deterministic_replay
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (hB : TraceBRules C)
 (hNorm : NormalizationBRules C)
 (hR : RRules C)
 (hE : ERules C)
 (hIC : IssueCommitWF C W)
 (hContracts : ModelingContracts C)
 (hWC : WC C W)
 (P : Process)
 ($\rho_1 \rho_2$: ExecutionPrefix C.Sigma)
 (hInit : SameInitialSourceState C W P $\rho_1 \rho_2$)
 (hSameUnits : SamePLocalObservableUnitSequence C P $\rho_1 \rho_2$)
 (hBoundary : ChosenObservableBoundaryWF C W P hContracts)
 (hCadence :
 AllCadenceSensitiveNBSitesHandled
 C W P hContracts.latencySensitiveLocalProtocols) :
 $\forall k,$
 SigmaRelevantEq C W P

(EpsQuiescentCutpointAfterObservableUnit C W P ρ_1 k)

(EpsQuiescentCutpointAfterObservableUnit C W P ρ_2 k)

Proof architecture. Prove `IDR_immediate_post_unit_deterministic_replay` by induction on `k` using `IObs_congruence_one_immediate_observable_unit`. The base case follows from `hInit`; the successor case consumes the same complete next P-local observable unit in both executions. Prove `IDR_deterministic_replay` by applying the immediate theorem at each `k` and then the paired finite ε -normalizations supplied by `hNorm`. The proof must not identify a raw immediate terminal state with its ε -normalized successor; it proves equality separately at the two explicitly chosen cutpoint modes.

The same `ObsSigma` carrier is reused in both modes. Plain

`J1_fixed_RTL_time_selected_witness_construction` and `J1_post_to_ref` use

`IDR_immediate_post_unit_deterministic_replay` and maintain canonical immediate post-unit replay states. Lemma `S1`, `J1_cutpoint_correspondence`, Corollary `J.1`, and Appendix `K` use `IDR_deterministic_replay` only after `NormalizationBRules` has produced the relevant ε -quiescent cutpoints. Atomic rendezvous, synchronization, and `R2C` units remain indivisible in both modes. For `RTL_B-side` request-attribution, `BoundaryReq`, pending, endpoint-request, and enablement facts, `J.AbsConf` for the same selected $(\tau_{\text{post}}, \tau_{\text{ref}}, \rho_{\text{post}}, \sigma_{\text{ref}}, W)$ triple remains necessary; neither replay theorem derives those implementation-side facts from the P-local observable-unit sequence alone.

16. Appendix I per-process construction

Bundle the assumptions:

`structure IConstructionAssumptions`

`(C : ComparisonInstance)`

`(W : WitnessBundle C)` where

`wellformed : C.WellFormed`

`bRules : TraceBRules C`

`rRules : RRules C`

`eRules : ERules C`

`issueCommitWF : IssueCommitWF C W`

`witnessUniverseWF : WitnessBundleWF C W`

`implementation : ImplementationAssumptions C`

`witnessCompatible : WC C W`

`contracts : ModelingContracts C`

`environment : ReactiveEnvironmentWF C.env`

initialAndReset : InitialAndResetCorrespondence C W

elaborationProjectionWF :

∀ side τ, ElaborationProjectionWF C side τ

-- B4/B5 are supplied only to I1_bounded_epsilon_closure and explicitly
-- normalized replay/cutpoint theorems. I3, I4, the countably-infinite local
-- branch of I5, and plain J.1 use the trace/admissibility bundle above.

Main theorems:

theorem I1_bounded_epsilon_closure

(C : ComparisonInstance)

(hNorm : NormalizationBRules C) :

...

theorem I2_channel_progress_instance

(C : ComparisonInstance)

(hTrace : TraceBRules C) :

...

theorem I3_finite_extension_step

(C : ComparisonInstance)

(W : WitnessBundle C)

(hI : IConstructionAssumptions C W) :

...

theorem I4_finite_prefix_construction

(C : ComparisonInstance)

(W : WitnessBundle C)

(hI : IConstructionAssumptions C W) :

...

def NoFurtherPLocalObservableUnitsAfter

(C : ComparisonInstance)

(side : Side)

(P : Process)

(τ∞ : InfiniteExecutionBucketed C.Sigma)

(n₀ : Nat) : Prop :=

∀ n,

$n_0 \leq n \rightarrow$

$\text{PLocalObservableUnitsInBucket } C \text{ side } P (\tau^\infty n) = \emptyset$

structure FinitePLocalMatch

(C : ComparisonInstance)

(W : WitnessBundle C)

(P : Process)

(ρ_{post} : InfiniteExecutionBucketed C.Sigma) where

cutoff : Nat

post : AnnotatedPrefix C W .post

ref : AnnotatedPrefix C W .ref

post_prefix_of_execution :

PrefixOfInfiniteExecution C .post post.prefix ρ_{post}

captures_all_post_P_units :

NoFurtherPLocalObservableUnitsAfter C .post P ρ_{post} cutoff

compatible_local_prefix : CompatibleP C W P ref post

-- The finite-P-local-event branch stops here at the per-process layer. The

-- absence of later P-local units does not authorize an arbitrary B6 suffix.

theorem I5_finite_local_prefix_construction

(C : ComparisonInstance)

(W : WitnessBundle C)

(hl : IConstructionAssumptions C W)

(P : Process)

(ρ_{post} : InfiniteExecutionBucketed C.Sigma)

(hPost : PostAdmissible C ρ_{post})

(hFinite : FinitePLocalObservableUnits C .post P ρ_{post}) :

FinitePLocalMatch C W P ρ_{post}

-- This theorem covers only the countably-infinite P-local-unit branch.

theorem I5_countably_infinite_local_limit_construction

(C : ComparisonInstance)

(W : WitnessBundle C)

(hl : IConstructionAssumptions C W)

(P : Process)

(ρ_{post} : InfiniteExecutionBucketed C.Sigma)

(hPost : PostAdmissible C ρ_{post})

(hInf : CountablyInfinitePLocalObservableUnits C .post P ρ_{post}) :

```

 $\exists$  p_ref_local,
InfinitePLocalReferenceTrace C W P p_ref_local  $\wedge$ 
CompatiblePInfinite C W P p_ref_local
(ProjectPLocalExecution C .post P p_post)

```

-- Backward-compatible alias for the countably-infinite branch only. The finite
-- branch is completed system-wide by the two theorems in Section 17.
theorem I5_infinite_limit_construction :=
I5_countably_infinite_local_limit_construction

I3 should use IssueAt plus derived IssueTime, not an unconstrained guessed issue cycle.
For v42 it must also keep the E4/B2 split explicit: matched channel history and B3 can
establish that a pending operation eventually becomes continuously ChannelEnabled; B2
then gives eventual scheduler/arbitrator selection, and the selection-commit soundness
clause gives the actual Commit in the selected cycle. I3 should not collapse
ChannelEnabled into immediate same-cycle Commit.

17. Appendix J system-level correspondence

Define execution scopes:

```

def FairProgressSatisfying
(C : ComparisonInstance)
(side : Side)
( $\tau^\infty$  : InfiniteExecutionBucketed C.Sigma) : Prop :=
InfiniteExecutionOfSide C side  $\tau^\infty$   $\wedge$ 
WeakFairness C side  $\tau^\infty$   $\wedge$ 
 $\forall$  c,
P_push C side  $\tau^\infty$  c  $\wedge$ 
P_pop C side  $\tau^\infty$  c

def PostAdmissible
(C : ComparisonInstance)
( $\tau^\infty$  : InfiniteExecutionBucketed C.Sigma) : Prop :=
InfinitePostExecute C  $\tau^\infty$   $\wedge$ 
FairProgressSatisfying C .post  $\tau^\infty$ 

def RefAdmissible
(C : ComparisonInstance)
( $\tau^\infty$  : InfiniteExecutionBucketed C.Sigma) : Prop :=

```

InfiniteRefExecution $C \tau^\infty \wedge$
FairProgressSatisfying $C .ref \tau^\infty$

def Exec_post ($C : ComparisonInstance$) ($\tau : Prefix C.RTL_B$) : Prop :=
 $\exists \tau^\infty,$
Extends $\tau^\infty \tau \wedge$
PostAdmissible $C \tau^\infty$

def Exec_ref ($C : ComparisonInstance$) ($\tau : Prefix C.Sys_ref$) : Prop :=
 $\exists \tau^\infty,$
Extends $\tau^\infty \tau \wedge$
RefAdmissible $C \tau^\infty$

Here FairProgressSatisfying is side-parametric. It includes the B2/B3 fairness/progress assumptions for the selected execution side: weak fairness and the P_push / P_pop channel-progress obligations. PostAdmissible is the post-side instance of this condition together with InfinitePostExecution; RefAdmissible is the reference-side instance together with InfiniteRefExecution. Thus Exec_post and Exec_ref are defined analogously, and any τ_ref produced or consumed by J.1 / K.1 carries the ref-side B2/B3 progress assumptions needed by the drain-closure and deadlock-preservation arguments.

B6 is the globally fixed-point idle-stutter theorem, not an arbitrary deadlock-cutpoint extension principle. It is the general time-divergence / idle-stutter convention only for a reachable finite prefix/state at which the system has reached a B5/global fixed point: after no further Σ -action or internal ε -microstep is enabled under the same stimulus, the global clock may continue by infinite idle-stutter ticks. Each idle-stutter bucket has no sigmaEvents and performs no ε -prefix or ε -suffix work. It changes no non-clock state; the only permitted state change is the clock/tick bookkeeping associated with the idle cycle.

def ClockOnlyIdleStep
($C : ComparisonInstance$)
(side : Side)
($\sigma_0 \sigma_1 : State$) : Prop :=
NonClockStateEq $C \sigma_1 \sigma_0 \wedge$
 $\sigma_1.cycle = \sigma_0.cycle + 1 \wedge$
 $\sigma_1.epsPos = 0$

def IdleStutterBucket
($C : ComparisonInstance$)

```

(side : Side)
( $\sigma$  : State)
( $\beta$  : CycleBucket C.Sigma) : Prop :=
 $\beta$ .startState = AdvanceClockOnly  $\sigma$   $\beta$ .cycle  $\wedge$ 
 $\beta$ .sigmaEvents =  $\emptyset$   $\wedge$ 
 $\beta$ .epsPrefix = []  $\wedge$ 
 $\beta$ .epsSuffix = []  $\wedge$ 
ClockOnlyIdleStep C side  $\beta$ .startState  $\beta$ .endState  $\wedge$ 
 $\beta$ .endQuiescent

```

```

def FixedPointState
(C : ComparisonInstance)
(side : Side)
( $\sigma$  : State) : Prop :=
EpsQuiescent  $\sigma$   $\wedge$ 
NoObservableActionEnabledAtState C side  $\sigma$   $\wedge$ 
NoInternalEpsilonStepEnabledAtState C side  $\sigma$ 

```

```

theorem B6_extend_reachable_prefix_by_idle_stutter

```

```

(C : ComparisonInstance)
(side : Side)
( $\pi$  : Prefix (SystemOfSide C side))
( $\sigma$  : State)
(hReach : TerminalCutpoint  $\pi$   $\sigma$ )
(hFP : FixedPointState C side  $\sigma$ ) :
 $\exists \tau$  : InfiniteExecutionBucketed C.Sigma,
ExtendsPrefix C side  $\tau$   $\pi$   $\wedge$ 
 $\forall n$ ,
 $n \geq \text{PrefixLength } \pi \rightarrow$ 
IdleStutterBucket C side  $\sigma$  ( $\tau$   $n$ ) :=
by

```

```

-- General B6 theorem: any reachable finite execution prefix/state at a B5/global
-- fixed point is extendable to an infinite execution by appending idle-stutter
-- ticks under the same stimulus. B6 supplies time divergence only. It does not
-- by itself prove the B2/B3 fairness/progress side conditions required for
-- membership in Exec_post or Exec_ref at an arbitrary partial internal-MP
-- deadlock cutpoint.

```

```

sorry

```

Thus any broad prose summary of B6 should be read with this B5/global fixed-point side condition.

The v42 K.1 bridge for a partial internal-MP deadlock is therefore not this B6 idle-stutter theorem. It is the separate fair drain-normalization construction in Section 18: already-issued D-owned non-frontier requests that later become ChannelEnabled may commit as finite drain-only progress; outside-D and environment endpoints continue under the ordinary B2/B3 progress premises; and pure idle-stutter is used only for D-owned internal frontier blockers that remain ChannelDisabled inside the closed WFG component.

Finite-P-local-event completion for Appendix I.5. The per-process finite-prefix theorem is completed only after a global Sys_ref continuation has been selected. Define:

The predicate NoFurtherPLocalObservableUnitsAfter was defined in Section 16.

```
def EmptyPLocalProjectionAfter
  (C : ComparisonInstance)
  (side : Side)
  (P : Process)
  ( $\tau^\infty$  : InfiniteExecutionBucketed C.Sigma)
  ( $n_0$  : Nat) : Prop :=
  NoFurtherPLocalObservableUnitsAfter C side P  $\tau^\infty$   $n_0$ 
```

```
structure RefContinuationWithoutFurtherPUnits
  (C : ComparisonInstance)
  (P : Process)
  ( $\pi_{\text{ref}}$  : Prefix C.Sys_ref)
  ( $n_0$  : Nat) where
   $\tau^\infty$  : InfiniteExecutionBucketed C.Sigma
  extends_prefix : Extends  $\tau^\infty$   $\pi_{\text{ref}}$ 
  admissible : RefAdmissible C  $\tau^\infty$ 
  no_further_P_units :
  NoFurtherPLocalObservableUnitsAfter C .ref P  $\tau^\infty$   $n_0$ 
```

-- Non-fixed-point branch: project the selected admissible global continuation.

-- Other processes and the environment may continue.

theorem l5_finite_local_projection_from_admissible_global_continuation

(C : ComparisonInstance)

(W : WitnessBundle C)

(P : Process)

(π_{ref} : Prefix C.Sys_ref)

(n_0 : Nat)

(hCont : RefContinuationWithoutFurtherPUnits C P π_{ref} n_0) :

$\exists \tau^\infty$,

Extends $\tau^\infty \pi_{\text{ref}} \wedge$

RefAdmissible C $\tau^\infty \wedge$

EmptyPLocalProjectionAfter C .ref P $\tau^\infty n_0 :=$

$\langle \text{hCont}.\tau^\infty,$

hCont.extends_prefix,

hCont.admissible,

hCont.no_further_P_units $\rangle$

-- General fixed-point helper: B6 supplies the idle extension, while the B2/B3

-- obligations are vacuous because no observable action or persistent matching

-- channel request is enabled at or after the fixed point.

theorem fixedpoint_idle_extension_ref_admissible

(C : ComparisonInstance)

(W : WitnessBundle C)

(hB : TraceBRules C)

(π_{ref} : Prefix C.Sys_ref)

(σ_{ref} : State)

(τ^∞ : InfiniteExecutionBucketed C.Sigma)

(hTerm : TerminalCutpoint π_{ref} σ_{ref})

(hFP : FixedPointState C .ref σ_{ref})

(hExt : ExtendsPrefix C .ref $\tau^\infty \pi_{\text{ref}}$)

(hIdle :

$\forall n,$

$n \geq \text{PrefixLength } \pi_{\text{ref}} \rightarrow$

IdleStutterBucket C .ref σ_{ref} ($\tau^\infty n$) :

RefAdmissible C $\tau^\infty :=$

by

-- InfiniteRefExecution follows from the finite prefix and hIdle. Weak fairness

-- and P_push/P_pop hold vacuously on the suffix because hFP and every hIdle

-- bucket contain no enabled or committed observable action and no newly

-- asserted endpoint request.

sorry

-- Fixed-point branch: B6 is applied only to the complete global state.

theorem I5_finite_local_projection_from_global_fixedpoint

(C : ComparisonInstance)

(W : WitnessBundle C)

(hB : TraceBRules C)

(P : Process)

(π_{ref} : Prefix C.Sys_ref)

(σ_{ref} : State)

(n_0 : Nat)

(hTerm : TerminalCutpoint π_{ref} σ_{ref})

(hFP : FixedPointState C .ref σ_{ref})

(hNoLaterPInPrefix : NoLaterPLocalUnitAfterIndex C .ref P π_{ref} n_0) :

$\exists \tau^\infty$,

Extends $\tau^\infty \pi_{\text{ref}} \wedge$

RefAdmissible C $\tau^\infty \wedge$

EmptyPLocalProjectionAfter C .ref P $\tau^\infty n_0 :=$

by

rcases B6_extend_reachable_prefix_by_idle_stutter

C .ref π_{ref} σ_{ref} hTerm hFP with $\langle \tau^\infty, \text{hExt}, \text{hIdle} \rangle$

have hAdm : RefAdmissible C $\tau^\infty :=$

fixedpoint_idle_extension_ref_admissible

C W hB π_{ref} σ_{ref} τ^∞ hTerm hFP hExt hIdle

exact $\langle \tau^\infty, \text{hExt}, \text{hAdm},$

no_further_P_units_in_fixedpoint_idle_extension

C W P π_{ref} σ_{ref} $\tau^\infty n_0$ hNoLaterPInPrefix hFP hExt hIdle $\rangle$

The Section 17 J.1 construction must choose one of these branches for every process with finitely many local observable units. It may not infer a global idle extension merely from the fact that P itself has stopped emitting units.

def BeforeNextResetAffectingProcess

(C : ComparisonInstance)

(side : Side)

(P : Process)

(epoch : ResetEpochId)

(τ : InfiniteExecutionBucketed C.Sigma)

(n : Cycle) : Prop :=

NoResetAffectingProcessSinceEpochStart C side P epoch τ n

theorem finish_process_idle_padding_until_next_reset

(C : ComparisonInstance)

(side : Side)

(P : Process)

(epoch : ResetEpochId)

(π : Prefix (SystemOfSide C side))

(σ : State)

(hReach : TerminalCutpoint π σ)

(hEpoch : CurrentResetEpoch C side P σ = some epoch)

(hFin : ProcessFinishedInEpoch C side P epoch σ)

(hNoMoreP : NoFurtherProcessStepEnabledInEpoch C side P epoch σ) :

$\exists \tau$: InfiniteExecutionBucketed C.Sigma,

ExtendsPrefix C side τ $\pi \wedge$

$\forall n$,

$n \geq \text{PrefixLength } \pi \rightarrow$

BeforeNextResetAffectingProcess C side P epoch τ $n \rightarrow$

ProcessIdleStuttered C side P (τ n) :=

by

-- The FINISH_P occurrence remains a realized reset-epoch-indexed SyncAnchor
-- for R2/E2 signal anchoring. Until the next reset affecting P, the process is
-- padded by process-local idle stutter. This padding creates no new process-owned
-- Σ -events, no new BoundaryReq, no new Front/NextFront item, and no new WFG edge
-- for P. A later reset ends this conclusion, creates a new reset epoch, and may
-- be followed by a new START_P occurrence.

sorry

The FINISH_P corollary is process-local and reset-epoch-bounded: a process that has finished in epoch e is padded by ε /idle-stutter while the global system may continue executing other processes, but only until the next reset that affects P. This padding must not be counted as additional B4/B5 internal progress by P, because it performs no enabled internal ε -step and emits no Σ -event. After a later reset, the earlier FINISH_P occurrence has no anchoring, ordering, or WFG force in the new epoch.

Directed compatibility:

Do not define `CompatibleP` or `CompatibleSys` as equality of bucketed executions, and do not type them over bare `Prefix` pairs. Their carriers must retain the selected witness-relative annotation data needed to interpret dynamic occurrences, `Issue/Commit`, μ_Q , request attribution, and `RequiredJOrder`.

First define the observable units extracted from a bucketed execution. An observable unit may be: a single ordinary Σ -event; an explicitly atomic multi-endpoint unit, such as a rendezvous `Push/Pop` pair or a `SyncChannel/ac_sync` pair; or a component-local R2C fixed-point signal-observation unit emitted under the v42 stutter-canonical R2C convention. Independent same-cycle events are not ordered merely because they share a `CycleBucket`; any strict required order among observable units is imposed only by `RequiredJOrder` / $<_J$. The broader causal-dependency graph $\rightarrow$ may still be recorded for explanatory information-flow reasoning, but it is not the induction order.

```
structure ObservableUnit ( $\Sigma$  : Type) where
  events : Finset  $\Sigma$ 
  atomic : Prop
```

```
-- ObservableUnitWF is the well-formedness predicate for the unit extraction used
-- by Appendix J. It is where non-splitting and grouping obligations are checked:
-- rendezvous Push/Pop pairs, paired SyncChannel/ac_sync endpoints, R2C
-- fixed-point observations, and any other explicitly atomic same-cycle
-- transaction must appear as one ObservableUnit rather than as strict internal
-- RequiredJOrder edges. It also records that every  $\Sigma$ -event in the finite prefix
-- belongs to exactly one extracted unit and that independent same-cycle events
-- are not ordered merely by bucket membership.
```

```
structure ObservableUnitWF
  (C : ComparisonInstance)
  (side : Side)
  ( $\tau$  : Prefix (SystemOfSide C side)) where
  covers_prefix_events :
     $\forall e$ ,
    SigmaEventInPrefix C side  $\tau e \rightarrow$ 
     $\exists u$ ,
     $u \in \text{ObservableUnitsOf } C \text{ side } \tau \wedge$ 
     $e \in u.\text{events}$ 
  events_have_unique_unit :
     $\forall e \ u \ v$ ,
     $u \in \text{ObservableUnitsOf } C \text{ side } \tau \rightarrow$ 
     $v \in \text{ObservableUnitsOf } C \text{ side } \tau \rightarrow$ 
```

$e \in u.\text{events} \rightarrow$
 $e \in v.\text{events} \rightarrow$
 $u = v$
 $\text{atomic_rendezvous_grouped} :$
 $\forall u,$
 $\text{AtomicRendezvousUnit } C \text{ side } \tau u \rightarrow$
 $u \in \text{ObservableUnitsOf } C \text{ side } \tau \wedge u.\text{atomic}$
 $\text{atomic_sync_grouped} :$
 $\forall u,$
 $\text{AtomicSyncUnit } C \text{ side } \tau u \rightarrow$
 $u \in \text{ObservableUnitsOf } C \text{ side } \tau \wedge u.\text{atomic}$
 $\text{r2c_fixed_point_unit_grouped} :$
 $\forall u,$
 $\text{R2CFixedPointObservationUnit } C \text{ side } \tau u \rightarrow$
 $u \in \text{ObservableUnitsOf } C \text{ side } \tau \wedge u.\text{atomic}$
 $\text{atomic_units_not_split} :$
 $\forall u \ e_1 \ e_2,$
 $u \in \text{ObservableUnitsOf } C \text{ side } \tau \rightarrow$
 $u.\text{atomic} \rightarrow$
 $e_1 \in u.\text{events} \rightarrow$
 $e_2 \in u.\text{events} \rightarrow$
 $\text{SameObservableUnit } C \text{ side } \tau$
 $(\text{ObservableUnit.singleton } e_1)$
 $(\text{ObservableUnit.singleton } e_2)$
 $\text{independent_same_cycle_not_ordered_by_bucket} :$
 $\forall u \ v,$
 $u \in \text{ObservableUnitsOf } C \text{ side } \tau \rightarrow$
 $v \in \text{ObservableUnitsOf } C \text{ side } \tau \rightarrow$
 $u \neq v \rightarrow$
 $\text{SameCycleOnly } C \text{ side } \tau u \ v \rightarrow$
 $\neg \text{RequiredJOrderGenerator } C \text{ side } \tau u \ v \wedge$
 $\neg \text{RequiredJOrderGenerator } C \text{ side } \tau v \ u$

ObservableUnitsOf extracts Σ -visible events from the finite bucketed prefix and groups them into comparison units as follows:

- an ordinary committed message, signal, or synchronization event may form a singleton observable unit;
- explicitly atomic multi-endpoint events, such as rendezvous Push/Pop pairs and SyncChannel/ac_sync handshakes, form one atomic observable unit and may not be split by CompatibleP or CompatibleSys;
- R2C combinational observations are emitted as component-local fixed-point signal-observation units according to R2C_WF and are matched by local occurrence ordinal;
- independent same-cycle events are unordered merely by virtue of being in the

same CycleBucket;

- same-cycle events that are merely coincident in one implementation are not forced to remain same-cycle in the other implementation unless they are explicitly atomic or otherwise ordered by RequiredJOrder.

Because τ is a finite prefix, ObservableUnitsOf returns a Finset. If a Set view is needed elsewhere, it should be obtained by coercion from this Finset, rather than by introducing a second observable-unit universe.

```
def ObservableUnitsOf
(C : ComparisonInstance)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side)) :
Finset (ObservableUnit (C.Sigma)) :=
ExtractSigmaUnitsFinset
C side  $\tau$ 
(AtomicRendezvousUnits C side  $\tau$ )
(AtomicSyncUnits C side  $\tau$ )
(R2CFixedPointObservationUnits C side  $\tau$ )
(IndependentSameCycleUnits C side  $\tau$ )
```

```
def ObservableUnitsOfSet
(C : ComparisonInstance)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side)) :
Set (ObservableUnit (C.Sigma)) :=
fun u => u ∈ ObservableUnitsOf C side  $\tau$ 
```

CausalDependency is an information-flow graph. It is useful for explaining that a Pop observes the corresponding Push value, that signal handshakes communicate information, or that a witness-selected ε choice affects later visible behavior. It is not itself the strict partial order used by Theorem J.1, and the Lean development should not assume the unquotiented CausalDependency graph is acyclic. CausalDependency is not part of the core mechanized theorem spine for Theorem J.1, Corollary J.1, Appendix K, or Appendix R. The core proof order is RequiredJOrder / $\leq_J$ over observable units, together with ObservableUnitWF, E1–E5 legality/alignment, witness compatibility, cutpoint correspondence, and the Appendix K WFG/deadlock machinery. If CausalDependency is formalized in Lean, it should be treated as an optional documentation/design-well-formedness layer used to mirror the scheduling-rules document’s explanatory causal-dependency relation, not as a dependency of the main trace-compatibility or liveness-preservation theorems.

```
def CausalDependency
(C : ComparisonInstance)
(side : Side)
```

```

(τ : Prefix (SystemOfSide C side))
(u v : ObservableUnit (C.Sigma)) : Prop :=
Relation.TransGen
(CausalDependencyGenerator C side τ)
u v

```

```

-- Intra-process causal trace/source order is part of CausalDependency, because it
-- records source-level information flow and design intent inside one dynamic
-- process execution. It is intentionally non-temporal: by itself it does not
-- assert strict commit-cycle order, and it is not a RequiredJOrder edge unless
-- some explicit required-order generator below also applies.

```

```

def IntraProcessCausalTraceOrder
(C : ComparisonInstance)
(side : Side)
(τ : Prefix (SystemOfSide C side))
(u v : ObservableUnit (C.Sigma)) : Prop :=
SameOwningProcess C side τ u v ∧
SourceTracePrecedes C side τ u v ∧
CausallyRelevantSourceFlow C side τ u v

```

```

-- Declared shared-storage / memory-serialization dependence is also part of
-- CausalDependency. It records only dependencies supplied by the array-access
-- mapping layer, explicit synchronization, atomic memory semantics, or declared
-- serialization/coherence rules for mapped shared storage. It does not turn a
-- memory information-flow edge into a proof-order edge unless the selected
-- mapping contract also generates an explicit RequiredJOrder edge or atomic unit.

```

```

def DeclaredMemorySerializationDependency
(C : ComparisonInstance)
(side : Side)
(τ : Prefix (SystemOfSide C side))
(u v : ObservableUnit (C.Sigma)) : Prop :=
SharedStorageConflict C side τ u v ∧
(ArrayMappingSerializes C side τ u v V
ExplicitSyncOrdersMemoryAccess C side τ u v V
DeclaredAtomicMemorySemanticsOrder C side τ u v V
DeclaredCoherenceSerializationOrder C side τ u v)

```

```

def CausalDependencyGenerator
(C : ComparisonInstance)
(side : Side)
(τ : Prefix (SystemOfSide C side))
(u v : ObservableUnit (C.Sigma)) : Prop :=
MessageInformationFlow C side τ u v V
SignalInformationFlow C side τ u v V

```

SynchronizationInformationFlow C side $\tau u v v$
 IntraProcessCausalTraceOrder C side $\tau u v v$
 DeclaredMemorySerializationDependency C side $\tau u v v$
 WitnessControlDependency C side $\tau u v v$
 ElaborationInformationFlow C side $\tau u v$

RequiredJOrder is the mandated Appendix J order over observable units. It is narrower than CausalDependency. It contains only strict order constraints that CompatibleP / CompatibleSys must preserve as order. It does not contain mere information-flow dependencies, grouping requirements, or source-lexical order that v42 intentionally treats as non-ordering.

The generator must encode the v42 exclusions explicitly:

- E2 signal anchoring contributes equality/alignment of the corresponding anchor cycle and value label; it never contributes an independent strict RequiredJOrder edge.
- E3 same-interface message operations may impose strict same-interface order. Strict issue/commit attribution is used by the fixed RTL-time witness construction schedule, not as an independent RequiredJOrder / $\prec_J$ generator. In particular, issue-attribution facts may help choose the next replay unit, but they do not add observable order among units that v42 otherwise leaves unordered.
- E3 distinct-interface lexical source order is only a no-reverse legality constraint. It is not a positive strict edge between the distinct-interface commits. In particular, two distinct-interface operations that v42 permits to issue/commit in parallel do not create $u \prec_J v$ merely because one appears earlier in source text.
- Atomic rendezvous, SyncChannel/ac_sync handshakes, and R2C fixed-point observations are represented as single ObservableUnit objects. Their non-splitting constraints are checked by ObservableUnitWF and CompatibleSys; they do not add strict internal edges.
- Witness-control dependencies remain in CausalDependency and WC, where they justify selected ε outcomes. They are not RequiredJOrder generators.
- Elaboration legality is represented by the explicit Sys_B^elab channels, E4 channel-realization semantics, and Π_{elab} / sigmaDUTProjection projection constraints. It is not a separate RequiredJOrder generator unless it is realized as one of the explicit strict channel or synchronization edges below.

```

def E2SignalAnchorAlignmentConstraint
(C : ComparisonInstance)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side))
( $u v$  : ObservableUnit (C.Sigma)) : Prop :=
SignalAnchorCyclesAndLabelsAligned C side  $\tau u v$ 

```



```

def E3MessageStrictRequiredOrder
(C : ComparisonInstance)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side))
( $u\ v$  : ObservableUnit (C.Sigma)) : Prop :=
E3SameInterfaceMessageOrder C side  $\tau\ u\ v$ 

def E3DistinctInterfaceNoReverseLegality
(C : ComparisonInstance)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side)) : Prop :=
 $\forall\ u\ v$ ,
DistinctMessageInterfaces C side  $\tau\ u\ v \rightarrow$ 
SourceLexicallyBefore C side  $\tau\ u\ v \rightarrow$ 
 $\neg$  RequiredReverseCommitOrder C side  $\tau\ v\ u$ 

-- E5 has two logically different parts.
--
-- The pref_S(s) side is non-strict legality:
--  $q \in \text{pref\_S}(s)$  gives  $\text{issue}(q) \leq \text{clk}(s)$ , and for committed blocking  $q$ 
-- the commit cycle of  $q$  is  $\leq \text{clk}(s)$ , with equality allowed.
-- It is therefore not a strict RequiredJOrder generator.
--
-- The suff_S(s) side is the strict post-boundary side:
--  $q \in \text{suff\_S}(s)$  gives  $\text{issue}(q) > \text{clk}(s)$ , and commit cannot precede issue.
-- Thus the synchronization unit for  $s$  is strictly before the observable
-- message unit for  $q$ . This is the only E5 contribution to RequiredJOrder.
def E5PrefSyncBoundaryLegality
(C : ComparisonInstance)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side)) : Prop :=
 $\forall\ s\ q\ u\text{Sync}\ u\text{Msg}$ ,
SyncUnitOf C side  $\tau\ s\ u\text{Sync} \rightarrow$ 
MessageUnitOf C side  $\tau\ q\ u\text{Msg} \rightarrow$ 
 $q \in \text{prefS}\ s \rightarrow$ 
IssuedInPrefix C side  $\tau\ q \rightarrow$ 
IssueTimeOf C side  $\tau\ q \leq \text{CommitCycleOfUnit C side } \tau\ u\text{Sync} \wedge$ 
(Blocking  $q \rightarrow$ 
  CommittedMessageUnit C side  $\tau\ q\ u\text{Msg} \rightarrow$ 
  CommitCycleOfUnit C side  $\tau\ u\text{Msg} \leq \text{CommitCycleOfUnit C side } \tau\ u\text{Sync}$ )

def E5StrictSuffSyncBoundaryRequiredOrder
(C : ComparisonInstance)

```

```

(side : Side)
(τ : Prefix (SystemOfSide C side))
(u v : ObservableUnit (C.Sigma)) : Prop :=
  ∃ s q,
  SyncUnitOf C side τ s u ∧
  MessageUnitOf C side τ q v ∧
  q ∈ suffS s ∧
  IssuedInPrefix C side τ q ∧
  CommitCycleOfUnit C side τ u < CommitCycleOfUnit C side τ v

-- Keep the old name as the RequiredJOrder generator name, but define it to be
-- exactly the strict successor-side E5 relation.
def E5SyncBoundaryRequiredOrder
(C : ComparisonInstance)
(side : Side)
(τ : Prefix (SystemOfSide C side))
(u v : ObservableUnit (C.Sigma)) : Prop :=
  E5StrictSuffSyncBoundaryRequiredOrder C side τ u v

-- The actual generator is intentionally the narrow v42 generator below.
def RequiredJOrderGenerator
(C : ComparisonInstance)
(side : Side)
(τ : Prefix (SystemOfSide C side))
(u v : ObservableUnit (C.Sigma)) : Prop :=
  E1SyncOrder C side τ u v v
  E3MessageStrictRequiredOrder C side τ u v v
  E4ChannelRequiredOrder C side τ u v v
  E5SyncBoundaryRequiredOrder C side τ u v v
  SynchronizationUnitRequiredOrder C side τ u v

structure RequiredJOrderWF
(C : ComparisonInstance)
(side : Side)
(τ : Prefix (SystemOfSide C side)) where
-- This is stated as a generator-level exclusion for proof hygiene. A more
-- minimal formalization could prove the same fact as a consequence of the
-- concrete generator definitions, but keeping it in RequiredJOrderWF is
-- harmless and matches the v42 intent that E2 contributes alignment only, not
-- strict observable order.
e2_alignment_only :
  ∀ u v,
  E2SignalAnchorAlignmentConstraint C side τ u v →

```

$\neg \text{RequiredJOrderGenerator } C \text{ side } \tau \ u \ v \wedge$
 $\neg \text{RequiredJOrderGenerator } C \text{ side } \tau \ v \ u$
 $\text{distinct_interface_no_positive_source_edge} :$
 $\forall u \ v,$
 $\text{DistinctMessageInterfaces } C \text{ side } \tau \ u \ v \rightarrow$
 $\text{SourceLexicallyBefore } C \text{ side } \tau \ u \ v \rightarrow$
 $\neg \text{RequiredJOrderGenerator } C \text{ side } \tau \ u \ v$
 $\text{atomicity_is_grouping_not_order} :$
 $\forall u \ v,$
 $\text{AtomicUnitNonSplittingConstraint } C \text{ side } \tau \ u \ v \rightarrow$
 $\text{SameObservableUnit } C \text{ side } \tau \ u \ v$
 $\text{witness_control_not_order} :$
 $\forall u \ v,$
 $\text{WitnessControlDependency } C \text{ side } \tau \ u \ v \rightarrow$
 $\neg \text{RequiredJOrderGenerator } C \text{ side } \tau \ u \ v$
 $\text{elaboration_not_separate_order} :$
 $\forall u \ v,$
 $\text{ElaborationInformationFlow } C \text{ side } \tau \ u \ v \rightarrow$
 $(\text{RequiredJOrderGenerator } C \text{ side } \tau \ u \ v \rightarrow$
 $\text{E1SyncOrder } C \text{ side } \tau \ u \ v \vee$
 $\text{E3MessageStrictRequiredOrder } C \text{ side } \tau \ u \ v \vee$
 $\text{E4ChannelRequiredOrder } C \text{ side } \tau \ u \ v \vee$
 $\text{E5SyncBoundaryRequiredOrder } C \text{ side } \tau \ u \ v \vee$
 $\text{SynchronizationUnitRequiredOrder } C \text{ side } \tau \ u \ v)$

RequiredJOrderWF is not intended to be a new external tool-flow assumption. It is a definitional sanity theorem about the concrete RequiredJOrderGenerator disjuncts once E1–E5, issue/commit well-formedness, observable-unit grouping, and elaboration well-formedness have been instantiated. The Lean development may keep it as a named structure for proof modularity, but J.1 should obtain it from the following lemma rather than expecting it to appear magically in JTraceAssumptions.

```

theorem RequiredJOrderWF_from_concrete_generators
  (C : ComparisonInstance)
  (W : WitnessBundle C)
  (side : Side)
  (τ : Prefix (SystemOfSide C side))
  (hE : ERules C)
  (hIC : IssueCommitWF C W)
  (hOU : ObservableUnitWF C side τ)
  (hImpl : ImplementationAssumptions C)
  (hElab : ElaborationProjectionWF C side τ) :
  RequiredJOrderWF C side τ :=
  
```

by

- Unfold RequiredJOrderGenerator and discharge each WF field by cases on the
- concrete v42 generator disjunct:
- • E2 contributes alignment only;
- • distinct-interface E3 lexical order is no-reverse legality only;
- • the pref_S(s) side of E5 is non-strict issue/commit legality only and
- contributes no RequiredJOrder edge;
- • the suff_S(s) side of E5 is the only E5 strict RequiredJOrder generator;
- • atomicity is already enforced by ObservableUnitWF grouping;
- • witness-control dependencies remain in WC and CausalDependency only; and
- • elaboration information flow contributes order only when realized through
- an explicit E1/E3-same-interface/E4-strict-channel/E5-suff/synchronization
- generator.

sorry

```
def RequiredJOrder
(C : ComparisonInstance)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side))
(u v : ObservableUnit (C.Sigma)) : Prop :=
Relation.TransGen
(RequiredJOrderGenerator C side  $\tau$ )
u v
```

theorem RequiredJOrder_acyclic_on_observable_units

```
(C : ComparisonInstance)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side))
(hWF : RequiredJOrderWF C side  $\tau$ ) :
Acyclic (RequiredJOrder C side  $\tau$ ) :=
by
```

- Strict RequiredJOrder edges are exactly the v42 strict observable-order
- edges above. Each edge is justified by an explicit synchronization order,
- same-interface message order, FIFO/channel order, rendezvous/channel order,
- the strict successor side of synchronization-boundary separation, or paired
- synchronization/barrier before-after constraints. Atomic multi-endpoint
- transactions are single observable units, E2 signal anchors are alignment
- constraints, distinct-interface lexical order is no-reverse legality rather
- than a positive edge, and witness-control / elaboration information-flow
- dependencies are not in this generator.

--

- Issue/commit attribution is intentionally not part of RequiredJOrder. It is
- used by RTLTimeNextObservableUnit and the fixed RTL-time ideal-chain
- construction schedule to choose a realizable next unit, but that construction

```

-- schedule adds no observable order.
--
-- The acyclicity proof is the v42 Theorem J.1 argument instantiated per fixed
-- legal execution prefix, but with the E5 split made explicit. Assign each
-- observable unit  $\gamma$  its commit cycle  $\text{clk}(\gamma)$  in that prefix, with explicitly
-- atomic same-cycle units carrying the single common commit cycle of their
-- endpoints. Every strict generator edge  $\gamma_1 \prec_J \gamma_2$  implies  $\text{clk}(\gamma_1) < \text{clk}(\gamma_2)$ :
--
-- • E1 synchronization order is strict between successive synchronization units.
-- • E3 same-interface message order is strict for the same-interface required
--   order generator.
-- • E4 channel-order generators are strict for reset-empty, bounded-occupancy,
--   non-fall-through, and explicit rendezvous/channel ordering constraints.
-- • E5 contributes a strict edge only from the synchronization unit  $s$  to a
--   committed message unit for  $q \in \text{suff\_S}(s)$ . This edge is strict because
--    $\text{issue}(q) > \text{clk}(s)$  and  $\text{commit}(q)$  cannot occur before  $\text{issue}(q)$ .
-- • Paired synchronization/barrier before-after constraints are strict by their
--   own generator definitions.
--
-- The  $\text{pref\_S}(s)$  half of E5 is deliberately absent from RequiredJOrder. For
--  $q \in \text{pref\_S}(s)$ , E5 gives  $\text{issue}(q) \leq \text{clk}(s)$ , and for a committed blocking  $q$ 
-- the commit cycle of  $q$  is  $\leq \text{clk}(s)$ , with equality allowed. Those facts are
-- E5 legality / boundary well-formedness facts used by issue/commit,
-- frontier, cutpoint, and compatibility proofs; they are not strict  $\prec_J$  edges
-- and are not used in the acyclicity measure.
--
-- Hence any hypothetical  $\prec_J$  cycle would force
--  $\text{clk}(\gamma_0) < \dots < \text{clk}(\gamma_0)$ , which is impossible. The measure for the actual
-- strict RequiredJOrder graph is commit cycle alone. Issue timestamps are used
-- only to prove that the strict  $\text{suff\_S}(s)$  E5 generator is indeed commit-cycle
-- increasing; they are not a second component of the acyclicity measure.
-- E2 alignment, atomic grouping, distinct-interface no-reverse legality,
--  $\text{pref\_S}(s)$  E5 legality, witness-control dependencies, and elaboration
-- information flow contribute no strict edges, per RequiredJOrderWF.
sorry
17.1 Witness-relative annotated comparison carrier

```

The same visible Prefix pair can satisfy the scheduling obligations under one dynamic-operation matching and fail them under another. The selected witness and annotation maps are therefore part of the semantic carrier, not merely proof provenance. ObservableUnitsOf still extracts units from the underlying bucketed prefix, but compatibility is stated over an AnnotatedPrefix whose fields fix the dynamic universe and the attribution data used by the proof.

```

structure DynamicPrefixUniverse
(C : ComparisonInstance)
(W : WitnessBundle C)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side)) where
qexec : Process  $\rightarrow$  Set DynMsgInstance
omega : Process  $\rightarrow$  Set FrontierItem
qomega : Process  $\rightarrow$  Set DynMsgInstance
qomega_is_projection : QOmegaProjectionWF C W side  $\tau$  qexec omega qomega
belongs_to_witness : DynamicUniverseBelongsToWitness C W side  $\tau$  qexec omega

```

```

structure PrefixAnnotations
(C : ComparisonInstance)
(W : WitnessBundle C)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side)) where
dynamicUniverse : DynamicPrefixUniverse C W side  $\tau$ 
issueCommit : IssueCommitAnnotation C W side  $\tau$ 
occurrenceMatching : OccurrenceMatching C W side  $\tau$ 
requestAttribution : RequestAttribution C W side  $\tau$ 
observableUnitsWF : ObservableUnitWF C side  $\tau$ 
annotationWF :
  PrefixAnnotationWF C W side  $\tau$ 
  dynamicUniverse
  issueCommit
  occurrenceMatching
  requestAttribution

```

```

structure AnnotatedPrefix
(C : ComparisonInstance)
(W : WitnessBundle C)
(side : Side) where
prefix : Prefix (SystemOfSide C side)
annotations : PrefixAnnotations C W side prefix

```

```

def RefAnnotatedPrefix (C : ComparisonInstance) (W : WitnessBundle C) :=
  AnnotatedPrefix C W .ref

```

```

def PostAnnotatedPrefix (C : ComparisonInstance) (W : WitnessBundle C) :=
  AnnotatedPrefix C W .post

```

CompatibleP C W P α_{ref} α_{post} means that the P-local observable units of the two annotated prefixes are related by a value-label-preserving bijection; explicitly atomic units map to explicitly atomic units and are not split; the mapping preserves the applicable P-

local RequiredJOrder constraints; E2 anchor equality/alignment, pref_S issue/commit legality, distinct-interface no-reverse legality, and elaboration legality hold through their dedicated predicates; and independent units need not occupy the same cycle bucket. The relation reads dynamic-occurrence identity, μ_Q , Issue/Commit, and request attribution from α_{ref} .annotations and α_{post} .annotations rather than reconstructing them from Σ alone.

```
def CompatibleP
(C : ComparisonInstance)
(W : WitnessBundle C)
(P : Process)
( $\alpha_{\text{ref}}$  : RefAnnotatedPrefix C W)
( $\alpha_{\text{post}}$  : PostAnnotatedPrefix C W) : Prop :=
ProcessObservableUnitBijection C W P  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$   $\wedge$ 
ProcessRequiredJOrderPreserved C W P  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$   $\wedge$ 
AtomicUnitsPreserved C W P  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$   $\wedge$ 
E2AnchorAlignmentPreserved C W P  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$   $\wedge$ 
E5PrefLegalityPreserved C W P  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$   $\wedge$ 
DistinctInterfaceNoReversePreserved C W P  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$   $\wedge$ 
DynamicOccurrenceAndAttributionPreserved C W P  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$ 
```

```
def CompatibleSys
(C : ComparisonInstance)
(W : WitnessBundle C)
( $\alpha_{\text{ref}}$  : RefAnnotatedPrefix C W)
( $\alpha_{\text{post}}$  : PostAnnotatedPrefix C W) : Prop :=
( $\forall P$ , CompatibleP C W P  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$ )  $\wedge$ 
ChannelsWellFormedAndMatched C W  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$   $\wedge$ 
GlobalRequiredJOrderPreserved C W  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$   $\wedge$ 
GlobalAtomicUnitsPreserved C W  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$   $\wedge$ 
ElaborationProjectionLegality C W  $\alpha_{\text{ref}}$   $\alpha_{\text{post}}$ 
```

-- The notation $\alpha_{\text{ref}} \approx_{\text{sys}} \alpha_{\text{post}}$ abbreviates $\text{CompatibleSys } C \ W \ \alpha_{\text{ref}} \ \alpha_{\text{post}}$.

-- It is directed $\text{Post} \rightarrow \text{Ref}$ compatibility, not symmetric trace equivalence.

Proposition J.0 composes process-local compatibility only after the same witness-relative annotations have been fixed for every process and channel. Its B-rule input is TraceBRules , not the full BRules bundle; pairwise composition does not need B4/B5 normalization. Its conclusion includes $\text{ElaborationProjectionLegality}$ only after pair-specific $\text{ElaborationProjectionWF}$ facts for the selected reference and post prefixes have been supplied.

```
theorem J0_pairwise_composition
(C : ComparisonInstance)
```

```

(W : WitnessBundle C)
(hC : C.WellFormed)
(hB : TraceBRules C)
(hR : RRules C)
(hE : ERules C)
(hIC : IssueCommitWF C W)
(hContracts : ModelingContracts C)
( $\alpha$ _ref : RefAnnotatedPrefix C W)
( $\alpha$ _post : PostAnnotatedPrefix C W)
(hElabRef : ElaborationProjectionWF C .ref  $\alpha$ _ref.prefix)
(hElabPost : ElaborationProjectionWF C .post  $\alpha$ _post.prefix)
(hProc :  $\forall$  P, CompatibleP C W P  $\alpha$ _ref  $\alpha$ _post)
(hChan : ChannelsWellFormedAndMatched C W  $\alpha$ _ref  $\alpha$ _post) :
CompatibleSys C W  $\alpha$ _ref  $\alpha$ _post

```

```

-- ElaborationProjectionLegality in CompatibleSys is derived from hElabRef and
-- hElabPost. In an ordinary non-elaborated instance these premises are discharged
-- by the identity-projection theorem; in an elaborated instance they come from
-- the selected well-formed ElaborationPackage.

```

```

-- Projected variants retain the same annotations and compare only the
-- C.sigmaDUTProjection image.
def CompatiblePDUT
(C : ComparisonInstance)
(W : WitnessBundle C)
(P : Process)
( $\alpha$ _ref : RefAnnotatedPrefix C W)
( $\alpha$ _post : PostAnnotatedPrefix C W) : Prop :=
CompatiblePOnProjection C W P C.Sigma_DUT C.sigmaDUTProjection  $\alpha$ _ref  $\alpha$ _post

```

```

def CompatibleSysDUT
(C : ComparisonInstance)
(W : WitnessBundle C)
( $\alpha$ _ref : RefAnnotatedPrefix C W)
( $\alpha$ _post : PostAnnotatedPrefix C W) : Prop :=
CompatibleSysOnProjection C W C.Sigma_DUT C.sigmaDUTProjection  $\alpha$ _ref  $\alpha$ _post

```

```

theorem J0_pairwise_composition_DUT
(C : ComparisonInstance)
(W : WitnessBundle C)
(hC : C.WellFormed)
(hB : TraceBRules C)
(hR : RRules C)
(hE : ERules C)

```



```

(hlC : IssueCommitWF C W)
(hContracts : ModelingContracts C)
( $\alpha$ _ref : RefAnnotatedPrefix C W)
( $\alpha$ _post : PostAnnotatedPrefix C W)
(hElabRef : ElaborationProjectionWF C .ref  $\alpha$ _ref.prefix)
(hElabPost : ElaborationProjectionWF C .post  $\alpha$ _post.prefix)
(hProc :  $\forall$  P, CompatiblePDUT C W P  $\alpha$ _ref  $\alpha$ _post)
(hChan : ChannelsWellFormedAndMatchedOnProjection
  C W C.Sigma_DUT C.sigmaDUTProjection  $\alpha$ _ref  $\alpha$ _post) :
CompatibleSysDUT C W  $\alpha$ _ref  $\alpha$ _post
17.2 J.1 assumption threading

```

The plain trace theorem imports only TraceBRules. The projection constructor below is part of the architecture: it demonstrates that every Appendix I premise used by J.1 is actually present in JTraceAssumptions, including the reactive environment, initial/reset correspondence, and elaboration projection. Similar explicit constructors are required at every later layer.

```

structure JTraceAssumptions
(C : ComparisonInstance)
(W : WitnessBundle C) where
wellformed : C.WellFormed
bRules : TraceBRules C
rRules : RRules C
eRules : ERules C
issueCommitWF : IssueCommitWF C W
implementation : ImplementationAssumptions C
witnessCompatible : WC C W
witnessUniverseWF : WitnessBundleWF C W
contracts : ModelingContracts C
environment : ReactiveEnvironmentWF C.env
initialAndReset : InitialAndResetCorrespondence C W
elaborationProjectionWF :
   $\forall$  side  $\tau$ , ElaborationProjectionWF C side  $\tau$ 

```

```

def JTraceAssumptions.toIConstructionAssumptions
(C : ComparisonInstance)
(W : WitnessBundle C)
(hJ : JTraceAssumptions C W) : IConstructionAssumptions C W :=
{ wellformed := hJ.wellformed
  bRules := hJ.bRules
  rRules := hJ.rRules
  eRules := hJ.eRules

```

```

issueCommitWF := hJ.issueCommitWF
witnessUniverseWF := hJ.witnessUniverseWF
implementation := hJ.implementation
witnessCompatible := hJ.witnessCompatible
contracts := hJ.contracts
environment := hJ.environment
initialAndReset := hJ.initialAndReset
elaborationProjectionWF := hJ.elaborationProjectionWF }

```

```

structure JSigmaDUTSafetyAssumptions
(C : ComparisonInstance)
(W : WitnessBundle C) where
wellformed : C.WellFormed
bRules : TraceBRules C
rRules : RRules C
eRules : ERules C
issueCommitWF : IssueCommitWF C W
implementation : ImplementationAssumptions C
sigmaDUTSafetyWitness : SigmaDUTSafetyWitnessObligation C W
witnessUniverseWF : WitnessBundleWF C W
contracts : ModelingContracts C
environment : ReactiveEnvironmentWF C.env
initialAndReset : InitialAndResetCorrespondence C W
elaborationProjectionWF :
  ∀ side τ, ElaborationProjectionWF C side τ
17.3 Source enablement and one coherent ideal-chain construction

```

RTLTimeNextObservableUnit remains the fixed construction schedule over the selected post prefix. The former NextJUnitRealizable disjunction was too weak: a frontier or pending fact alone does not prove channel legality, counterpart availability, payload agreement, atomic completion, or a finite source transition. FiniteMatchedUnitExtension is the actual successor obligation.

```

def RTLTimeNextObservableUnit
(C : ComparisonInstance)
(W : WitnessBundle C)
(α_post : PostAnnotatedPrefix C W)
(H : Finset (ObservableUnit C.Sigma))
(u : ObservableUnit C.Sigma) : Prop :=
u ∈ ObservableUnitsOf C .post α_post.prefix ∧
u ∉ H ∧
AllRequiredJPredecessorsIn C .post α_post.prefix H u ∧
RefinesSelectedRTLCommitCycleOrder C W α_post H u ∧

```

WithinCycleWitnessReplayAttributionCompatible C W α_{post} H u $\wedge$
 AtomicUnitAlreadyGrouped C .post α_{post} .prefix u

structure RTLTimeIdealChain
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (α_{post} : PostAnnotatedPrefix C W) where
 N : Nat
 idealAt : Fin (N + 1) $\rightarrow$ Finset (ObservableUnit C.Sigma)
 initial_empty : idealAt 0 = $\emptyset$
 terminal_units : idealAt (Fin.last N) = ObservableUnitsOf C .post α_{post} .prefix
 single_unit_step :
 $\forall k : \text{Fin } N,$
 $\exists u,$
 RTLTimeNextObservableUnit C W α_{post} (idealAt k.castSucc) u $\wedge$
 idealAt k.succ = idealAt k.castSucc $\cup \{u\}$
 rtl_time_prefix_closed :
 $\forall k u v,$
 $v \in \text{idealAt } k \rightarrow$
 RequiredJOrder C .post α_{post} .prefix u v $\rightarrow$
 $u \in \text{idealAt } k$

structure FiniteMatchedUnitExtension
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (α_{post} : PostAnnotatedPrefix C W)
 ($\alpha_{\text{ref_before}}$: RefAnnotatedPrefix C W)
 (u : ObservableUnit C.Sigma) where
 preparation : MatchedEpochPreparation C W α_{post} $\alpha_{\text{ref_before}}$ u
 source_instance_selected :
 CorrectDynamicSourceOccurrence
 C W α_{post} preparation.prepared_ref u
 control_and_history_ready :
 SourceControlBranchLoopAndOrderReady
 C W preparation.prepared_ref u
 source_enabled :
 SourceObservableUnitEnabled C W preparation.prepared_ref u
 channel_and_boundary_legal :
 SourceChannelBoundaryLegality C W preparation.prepared_ref u
 matched_labels_and_values :
 ObservableUnitLabelsAndValuesMatch
 C W α_{post} preparation.prepared_ref u
 atomic_counterparts_available :
 AtomicCounterpartsAvailable C W preparation.prepared_ref u

```

ref_after : RefAnnotatedPrefix C W
proper_unit_extension :
  PrefixProperExtension preparation.prepared_ref.prefix ref_after.prefix
whole_step_extends_prior_prefix :
  PrefixExtends ref_after.prefix  $\alpha$ _ref_before.prefix
finite_segment :
  FiniteEpsilonStarThenObservableUnit
  C W preparation.prepared_ref ref_after u
canonical_immediate_terminal :
  CanonicalImmediateTerminalForUnit C W ref_after u
no_extra_observable_unit :
  AddsExactlyObservableUnit
  C W preparation.prepared_ref ref_after u

```

```

structure ReferenceIdealChain
(C : ComparisonInstance)
(W : WitnessBundle C)
( $\alpha$ _post : PostAnnotatedPrefix C W) where
rtl : RTLTimeIdealChain C W  $\alpha$ _post
refAt :  $\forall k : \text{Fin } (\text{rtl.N} + 1), \text{RefAnnotatedPrefix C W}$ 
base_correspondence :
  ApplicableInitialOrResetEpochStart
  C W  $\alpha$ _post (refAt 0)
nested :
   $\forall k : \text{Fin } \text{rtl.N},$ 
  PrefixExtends (refAt k.succ).prefix (refAt k.castSucc).prefix
realizes_ideal :
   $\forall k,$ 
  ReplayPrefixForIdeal C W  $\alpha$ _post (refAt k) (rtl.idealAt k)
step :
   $\forall k : \text{Fin } \text{rtl.N},$ 
   $\exists (u : \text{ObservableUnit C.Sigma})$ 
  (ext : FiniteMatchedUnitExtension C W  $\alpha$ _post (refAt k.castSucc) u),
  RTLTimeNextObservableUnit C W  $\alpha$ _post (rtl.idealAt k.castSucc) u  $\wedge$ 
  rtl.idealAt k.succ = rtl.idealAt k.castSucc  $\cup \{u\} \wedge$ 
  ext.ref_after = refAt k.succ
invariant :
   $\forall k, \text{JConstructionInvariant C W } \alpha$ _post (refAt k) (rtl.idealAt k)
reset_splices_reestablish_invariant :
   $\forall k : \text{Fin } \text{rtl.N},$ 
  EveryResetCrossedByChainStepReestablishesInvariant
  C W  $\alpha$ _post (refAt k.castSucc) (refAt k.succ)

```

-- T.3.1 is not hidden as a field of the finite ideal chain. The finite chain

```

-- construction and the admissible global-extension construction are separate
-- proof obligations.
theorem reference_ideal_chain_terminal_has_admissible_extension
(C : ComparisonInstance)
(W : WitnessBundle C)
(hJ : JTraceAssumptions C W)
( $\alpha$ _post : PostAnnotatedPrefix C W)
(hExecPost : Exec_post C  $\alpha$ _post.prefix)
(chain : ReferenceIdealChain C W  $\alpha$ _post) :
Exec_ref C (chain.refAt (Fin.last chain.rtl.N)).prefix :=
by
-- Construct an admissible infinite global reference execution extending the
-- finite terminal chain prefix. Split explicitly between continuation inherited
-- or replayed from an existing admissible global execution and the genuine
-- B5/global-fixed-point branch closed by B6. Discharge B2/B3 fairness and
-- progress for every still-enabled global action; do not treat finite semantic
-- prefix realizability as an automatic Exec_ref fact.
sorry

theorem ReferenceIdealChain.compatible_terminal
(C : ComparisonInstance)
(W : WitnessBundle C)
(hJ : JTraceAssumptions C W)
( $\alpha$ _post : PostAnnotatedPrefix C W)
(chain : ReferenceIdealChain C W  $\alpha$ _post) :
CompatibleSys C W (chain.refAt (Fin.last chain.rtl.N))  $\alpha$ _post :=
by
-- Induct over the nested chain. Each successor first discharges its complete
-- MatchedEpochPreparation, using ResetEpochCorrespondence when a dynamic reset
-- lies before the next RTL-time unit, and then applies the ordinary finite
-- matched-unit extension and J0_pairwise_composition, passing
-- hJ.elaborationProjectionWF .ref and .post for the exact selected pair.
sorry

structure SelectedJWitnessEvidence
(C : ComparisonInstance)
(W : WitnessBundle C)
( $\alpha$ _post : PostAnnotatedPrefix C W)
( $\alpha$ _ref : RefAnnotatedPrefix C W) where
chain : ReferenceIdealChain C W  $\alpha$ _post
terminal_eq :  $\alpha$ _ref = chain.refAt (Fin.last chain.rtl.N)

def SelectedJWitness
(C : ComparisonInstance)

```

(W : WitnessBundle C)
 (α_post : PostAnnotatedPrefix C W)
 (α_ref : RefAnnotatedPrefix C W) : Prop :=
 Nonempty (SelectedJWitnessEvidence C W α_post α_ref)
 The fixed-chain theorem now constructs one ReferenceIdealChain. It cannot satisfy
 different ideals with unrelated existential prefixes: every successor extends the previous
 terminal prefix, the finite source transition is explicit, the replay invariant is stored at every
 index, and the selected source witness is definitionally the terminal chain element.

theorem post_prefix_annotation_of_exec
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (hJ : JTraceAssumptions C W)
 (τ_post : Prefix C.RTL_B)
 (hExec : Exec_post C τ_post) :
 ∃ α_post : PostAnnotatedPrefix C W,
 α_post.prefix = τ_post

theorem J1_fixed_RTL_time_selected_witness_construction
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (hJ : JTraceAssumptions C W)
 (α_post : PostAnnotatedPrefix C W)
 (hExec : Exec_post C α_post.prefix) :
 ∃ chain : ReferenceIdealChain C W α_post,
 Exec_ref C (chain.refAt (Fin.last chain.rtl.N)).prefix ∧
 CompatibleSys C W (chain.refAt (Fin.last chain.rtl.N)) α_post :=
 by
 -- Use hJ.toIConstructionAssumptions. Construct one nested ReferenceIdealChain;
 -- at each successor discharge MatchedEpochPreparation and then
 -- FiniteMatchedUnitExtension. A crossed dynamic reset is represented by a
 -- MatchedResetEpochSplice that re-establishes the invariant before replaying
 -- the next observable unit. Then invoke
 -- reference_ideal_chain_terminal_has_admissible_extension to prove Exec_ref
 -- for the terminal prefix. Do not invoke NormalizationBRules.
 sorry

theorem J1_post_to_ref
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (hJ : JTraceAssumptions C W)
 (τ_post : Prefix C.RTL_B)
 (hExec : Exec_post C τ_post) :
 ∃ α_post : PostAnnotatedPrefix C W,

```

 $\exists \alpha_{\text{ref}} : \text{RefAnnotatedPrefix } C \ W,$ 
 $\alpha_{\text{post}}.\text{prefix} = \tau_{\text{post}} \wedge$ 
 $\text{SelectedJWitness } C \ W \ \alpha_{\text{post}} \ \alpha_{\text{ref}} \wedge$ 
 $\text{Exec\_ref } C \ \alpha_{\text{ref}}.\text{prefix} \wedge$ 
 $\text{CompatibleSys } C \ W \ \alpha_{\text{ref}} \ \alpha_{\text{post}}$ 

-- OrderedObservableTrace retains the complete observable units and only the
-- strict Appendix-J required-order relation among them. It does not impose a
-- fake order on independent units that merely share a cycle bucket.
structure OrderedObservableTrace ( $\Sigma : \text{Type}$ ) where
units : Finset (ObservableUnit  $\Sigma$ )
requiredBefore : ObservableUnit  $\Sigma \rightarrow$  ObservableUnit  $\Sigma \rightarrow$  Prop

-- Project one finite prefix to its observable-unit carrier and  $\leq_J$  relation.
def OrderedObservableProjectionOfPrefix
(C : ComparisonInstance)
(side : Side)
( $\tau : \text{Prefix (SystemOfSide } C \ \text{side})$ ) : OrderedObservableTrace C.Sigma :=
{
  units := ObservableUnitsOf C side  $\tau$ 
  requiredBefore := RequiredJOrder C side  $\tau$ 
}

-- Two post prefixes are at the same compared observation cut when they expose
-- the same complete ordered post-side observable projection. Their internal
-- states and  $\varepsilon$ -structure need not be equal.
def SamePostObservationCut
(C : ComparisonInstance)
( $\tau_1 \ \tau_2 : \text{Prefix } C.\text{RTL\_B}$ ) : Prop :=
OrderedObservableProjectionOfPrefix C .post  $\tau_1 =$ 
OrderedObservableProjectionOfPrefix C .post  $\tau_2$ 

-- Side-parametric equality of the complete ordered observable projection.
def SameOrderedSigmaProjection
(C : ComparisonInstance)
(side : Side)
( $\tau_1 \ \tau_2 : \text{Prefix (SystemOfSide } C \ \text{side})$ ) : Prop :=
OrderedObservableProjectionOfPrefix C side  $\tau_1 =$ 
OrderedObservableProjectionOfPrefix C side  $\tau_2$ 

-- This relation is deliberately separate from ordered- $\Sigma$  equality. It holds when
-- the two prefixes differ only by finite B4-bounded  $\varepsilon$ -only normalization/stutter
-- segments around the same observable cut; the concrete construction uses the
-- NormalizationBRules/EpsilonNormalizationSuffix layer.

```

```

def FiniteEpsilonStutterDifference
(C : ComparisonInstance)
(side : Side)
( $\tau_1 \tau_2$  : Prefix (SystemOfSide C side)) : Prop := by
  sorry

def OrderedSigmaProjection
(C : ComparisonInstance)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side)) : OrderedObservableTrace C.Sigma :=
  OrderedObservableProjectionOfPrefix C side  $\tau$ 

def SameComparedObservationHorizon
(C : ComparisonInstance)
( $\tau_1 \tau_2$  : Prefix C.RTL_B) : Prop :=
  SamePostObservationCut C  $\tau_1 \tau_2$ 

-- Observable-projection uniqueness is weaker than stateful witness uniqueness.
-- Under standing B1, any two selected J.1 matches at the same compared observation
-- horizon have the same ordered  $\Sigma$  projection, even though their internal source
-- states,  $\varepsilon$ -structure, or admissible witness choices may differ.
theorem J1_matching_sigma_projection_unique
(C : ComparisonInstance)
(W : WitnessBundle C)
(hJ : JTraceAssumptions C W)
( $\alpha_{\text{post}_1} \alpha_{\text{post}_2}$  : PostAnnotatedPrefix C W)
(hExecPost1 : Exec_post C  $\alpha_{\text{post}_1}$ .prefix)
(hExecPost2 : Exec_post C  $\alpha_{\text{post}_2}$ .prefix)
(hHorizon : SameComparedObservationHorizon C  $\alpha_{\text{post}_1}$ .prefix  $\alpha_{\text{post}_2}$ .prefix)
( $\alpha_{\text{ref}_1} \alpha_{\text{ref}_2}$  : RefAnnotatedPrefix C W)
(hCompat1 : CompatibleSys C W  $\alpha_{\text{ref}_1} \alpha_{\text{post}_1}$ )
(hCompat2 : CompatibleSys C W  $\alpha_{\text{ref}_2} \alpha_{\text{post}_2}$ ) :
  OrderedSigmaProjection C .ref  $\alpha_{\text{ref}_1}$ .prefix =
  OrderedSigmaProjection C .ref  $\alpha_{\text{ref}_2}$ .prefix :=
  by
    -- Use standing B1 from hJ.bRules to identify the post-side ordered projection,
    -- then use hCompat1 and hCompat2. Do not conclude  $\alpha_{\text{ref}_1} = \alpha_{\text{ref}_2}$ .
    sorry

theorem J1_selected_ref_prefix_is_liberal
(C : ComparisonInstance)
(W : WitnessBundle C)
(hJ : JTraceAssumptions C W)
( $\alpha_{\text{post}}$  : PostAnnotatedPrefix C W)

```



```

( $\alpha_{ref}$  : RefAnnotatedPrefix C W)
(hSel : SelectedJWitness C W  $\alpha_{post}$   $\alpha_{ref}$ )
(hExecRef : Exec_ref C  $\alpha_{ref}.prefix$ ) :
LiberalReferencePrefix C  $\alpha_{ref}.prefix$ 

```

-- Appendix L Note 1 projected safety theorem. It retains the annotations but
-- concludes only outer Σ_{DUT} compatibility.

theorem J1_post_to_ref_sigmaDUT_safety

```

(C : ComparisonInstance)
(W : WitnessBundle C)
(hJ : JSigmaDUTSafetyAssumptions C W)
( $\tau_{post}$  : Prefix C.RTL_B)
(hExec : Exec_post C  $\tau_{post}$ ) :
 $\exists \alpha_{post}$  : PostAnnotatedPrefix C W,
 $\exists \alpha_{ref}$  : RefAnnotatedPrefix C W,
 $\alpha_{post}.prefix = \tau_{post} \wedge$ 
Exec_ref C  $\alpha_{ref}.prefix \wedge$ 
CompatibleSysDUT C W  $\alpha_{ref}$   $\alpha_{post}$ 

```

-- Guardrail: do not use J1_post_to_ref_sigmaDUT_safety as a substitute for
-- J1_post_to_ref. The projected theorem concludes only CompatibleSysDUT at
-- the outer C.Sigma_DUT boundary; the full proof-internal theorem requires
-- the JTraceAssumptions/WC path and concludes CompatibleSys over C.Sigma.

-- RTLConsistentSourcePrefix remains a source-side restriction. Standing B1
-- supplies uniqueness of the matching RTL_B Sigma-trace under the fixed
-- stimulus; it does not force arbitrary Sys_ref executions to have RTL_B matches.

theorem J1_ref_to_post_restricted

```

(C : ComparisonInstance)
(W : WitnessBundle C)
(hJ : JTraceAssumptions C W)
( $\alpha_{ref}$  : RefAnnotatedPrefix C W)
(hExec : Exec_ref C  $\alpha_{ref}.prefix$ )
(hConsistent : RTLConsistentSourcePrefix C W  $\alpha_{ref}$ ) :
 $\exists \alpha_{post}$  : PostAnnotatedPrefix C W,
Exec_post C  $\alpha_{post}.prefix \wedge$ 
CompatibleSys C W  $\alpha_{ref}$   $\alpha_{post}$ 

```

Cutpoint abstraction and correspondence:

-- J.Abs is an RTL_B cutpoint-to-source-slice abstraction relation, not a
-- universal bit-level projection from arbitrary RTL registers to source states.

def JAbs

```

(C : ComparisonInstance)
(W : WitnessBundle C)

```

```

( $\rho_{\text{post}}$  : State)
( $\sigma_{\text{ref}}$  : State) : Prop :=
ObservablePrefixAgreement C W  $\rho_{\text{post}}$   $\sigma_{\text{ref}}$   $\wedge$ 
PerProcessSourceFrontierSliceAgreement C W  $\rho_{\text{post}}$   $\sigma_{\text{ref}}$   $\wedge$ 
SourceVisibleValueSliceAgreement C W  $\rho_{\text{post}}$   $\sigma_{\text{ref}}$   $\wedge$ 
AbstractChannelStateSliceAgreement C W  $\rho_{\text{post}}$   $\sigma_{\text{ref}}$   $\wedge$ 
RendezvousEndpointRequestSliceAgreement C W  $\rho_{\text{post}}$   $\sigma_{\text{ref}}$   $\wedge$ 
BlockingRequestAttributionPendingSliceAgreement C W  $\rho_{\text{post}}$   $\sigma_{\text{ref}}$   $\wedge$ 
EnablementSliceAgreement C W  $\rho_{\text{post}}$   $\sigma_{\text{ref}}$ 

```

```

-- The selected J.1 prefixes and their normalized representatives are different
-- carriers. A normalization extension preserves the ordered observable projection
-- but may append a nonempty finite  $\varepsilon$ -suffix.

```

```

structure PostEpsilonNormalizationExtension
(C : ComparisonInstance)
(W : WitnessBundle C)
(base normalized : PostAnnotatedPrefix C W) where
base_terminal : State
normalized_terminal : State
base_exec : Exec_post C base.prefix
normalized_exec : Exec_post C normalized.prefix
base_terminal_cutpoint : TerminalCutpoint base.prefix base_terminal
extends : PrefixExtends normalized.prefix base.prefix
epsilon_suffix :
  EpsilonNormalizationSuffix C .post base.prefix base_terminal normalized_terminal
normalized_terminal_cutpoint :
  TerminalCutpoint normalized.prefix normalized_terminal
same_ordered_sigma :
  SameOrderedSigmaProjection C .post base.prefix normalized.prefix
normalized_quiescent : EpsQuiescent normalized_terminal

```

```

structure RefEpsilonNormalizationExtension
(C : ComparisonInstance)
(W : WitnessBundle C)
(base normalized : RefAnnotatedPrefix C W) where
base_terminal : State
normalized_terminal : State
base_exec : Exec_ref C base.prefix

```

normalized_exec : Exec_ref C normalized.prefix
 base_terminal_cutpoint : TerminalCutpoint base.prefix base_terminal
 extends : PrefixExtends normalized.prefix base.prefix
 epsilon_suffix :
 EpsilonNormalizationSuffix C .ref base.prefix base_terminal normalized_terminal
 normalized_terminal_cutpoint :
 TerminalCutpoint normalized.prefix normalized_terminal
 same_ordered_sigma :
 SameOrderedSigmaProjection C .ref base.prefix normalized.prefix
 normalized_quiescent : EpsQuiescent normalized_terminal

structure SelectedNormalizedPrefixPair
 (C : ComparisonInstance)
 (W : WitnessBundle C) where
 -- Exact pair produced by plain J.1.
 base_post : PostAnnotatedPrefix C W
 selected_ref : RefAnnotatedPrefix C W
 selected : SelectedJWitness C W base_post selected_ref
 compatible : CompatibleSys C W selected_ref base_post
 -- Distinct ε -normalized representatives used by Corollary J.1 and Appendix K.
 normalized_post : PostAnnotatedPrefix C W
 normalized_ref : RefAnnotatedPrefix C W
 post_normalization :
 PostEpsilonNormalizationExtension C W base_post normalized_post
 ref_normalization :
 RefEpsilonNormalizationExtension C W selected_ref normalized_ref
 normalized_pair :
 EpsilonNormalizedPair C normalized_ref.prefix normalized_post.prefix

structure JAbsConf
 (C : ComparisonInstance)
 (W : WitnessBundle C) where
 holds_for_selected_normalized_pair :
 $\forall N : \text{SelectedNormalizedPrefixPair } C \ W,$
 JAbs C W
 N.post_normalization.normalized_terminal
 N.ref_normalization.normalized_terminal
 normalization_preserves_abs_slice :

$\forall \tau_post \rho_0 \rho_post,$
 $EpsilonNormalizationSuffix\ C.\text{post}\ \tau_post\ \rho_0\ \rho_post \rightarrow$
 $SigmaHistoryUnchanged\ C.\text{post}\ \rho_0\ \rho_post \wedge$
 $RequestPendingAttributionSlicePreservedModuloIssueCommitE4Boundary$
 $C\ W\ \rho_0\ \rho_post$

structure JCutpointAssumptions
 (C : ComparisonInstance)
 (W : WitnessBundle C) extends JTraceAssumptions C W where
 normalization : NormalizationBRules C
 jAbsConf : JAbsConf C W

def JCutpointAssumptions.asJTraceAssumptions
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (hJ : JCutpointAssumptions C W) : JTraceAssumptions C W :=
 hJ.toJTraceAssumptions

structure CutpointCorrespondence
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (σ_ref : State)
 (ρ_post : State) where
 j_abs : JAbs C W $\rho_post\ \sigma_ref$
 ref_quiescent : EpsQuiescent σ_ref
 post_quiescent : EpsQuiescent ρ_post
 ref_wellformed : WellFormedCutpoint C .ref σ_ref
 post_wellformed : WellFormedCutpoint C .post ρ_post
 direct_input_anchor_agreement : DirectInputAnchorAgreement C W $\sigma_ref\ \rho_post$
 array_mapping_agreement : ArrayMappingAgreement C W $\sigma_ref\ \rho_post$
 r2c_fixed_point_agreement : R2CFixedPointAgreement C W $\sigma_ref\ \rho_post$

theorem cutpoint_nextfront_agreement
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 ($\sigma_ref\ \rho_post$: State)
 (hCorr : CutpointCorrespondence C W $\sigma_ref\ \rho_post$) :
 NextFrontAgreement C W $\sigma_ref\ \rho_post$:= by sorry

```

theorem cutpoint_rendezvous_endpoint_request_agreement
(C : ComparisonInstance)
(W : WitnessBundle C)
( $\sigma\_ref$   $\rho\_post$  : State)
(hCorr : CutpointCorrespondence C W  $\sigma\_ref$   $\rho\_post$ ) :
RendezvousEndpointRequestSliceAgreement C W  $\rho\_post$   $\sigma\_ref$  := by sorry

```

```

theorem cutpoint_pending_enablement_agreement
(C : ComparisonInstance)
(W : WitnessBundle C)
( $\sigma\_ref$   $\rho\_post$  : State)
(hCorr : CutpointCorrespondence C W  $\sigma\_ref$   $\rho\_post$ ) :
BlockingRequestAttributionPendingSliceAgreement C W  $\rho\_post$   $\sigma\_ref$   $\wedge$ 
EnablementSliceAgreement C W  $\rho\_post$   $\sigma\_ref$  := by sorry

```

```

-- Construct the normalization link only after plain J.1 has selected its exact
-- post/reference pair. Neither selected prefix is overwritten by its extension.

```

```

theorem normalize_selected_J1_pair
(C : ComparisonInstance)
(W : WitnessBundle C)
(hJ : JCutpointAssumptions C W)
( $\alpha\_post$  : PostAnnotatedPrefix C W)
( $\alpha\_ref$  : RefAnnotatedPrefix C W)
(hExecPost : Exec_post C  $\alpha\_post$ .prefix)
(hExecRef : Exec_ref C  $\alpha\_ref$ .prefix)
(hSel : SelectedJWitness C W  $\alpha\_post$   $\alpha\_ref$ )
(hCompat : CompatibleSys C W  $\alpha\_ref$   $\alpha\_post$ ) :
 $\exists$  N : SelectedNormalizedPrefixPair C W,
N.base_post =  $\alpha\_post$   $\wedge$ 
N.selected_ref =  $\alpha\_ref$  :=
by
-- Normalize both reachable selected prefixes with B4/B5, construct the distinct
-- annotated extensions, preserve the ordered  $\Sigma$  projection, and retain the exact
-- selected-witness and compatibility facts on the unnormalized pair.
sorry

```

```

theorem J1_cutpoint_correspondence

```

```

(C : ComparisonInstance)
(W : WitnessBundle C)
(hJ : JCutpointAssumptions C W)
(N : SelectedNormalizedPrefixPair C W) :
CutpointCorrespondence C W
  N.ref_normalization.normalized_terminal
  N.post_normalization.normalized_terminal :=
by
-- All reachability evidence is carried by N. Populate j_abs from
-- hJ.jAbsConf.holds_for_selected_normalized_pair N and populate quiescence and
-- cutpoint well-formedness from the two explicit normalization-extension objects.
sorry

```

```

structure SelectedMatchedCutpoint
(C : ComparisonInstance)
(W : WitnessBundle C) where
normalization : SelectedNormalizedPrefixPair C W
-- K.2b consumes the normalized liberal reference prefix.
ref_liberal : LiberalReferencePrefix C normalization.normalized_ref.prefix
correspondence :
  CutpointCorrespondence C W
  normalization.ref_normalization.normalized_terminal
  normalization.post_normalization.normalized_terminal

```

```

def SelectedMatchedCutpoint.τ_post
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) : Prefix C.RTL_B :=
M.normalization.base_post.prefix

```

```

def SelectedMatchedCutpoint.τ_ref
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) : Prefix C.Sys_ref :=
M.normalization.selected_ref.prefix

```

```

def SelectedMatchedCutpoint.tbar_post
(C : ComparisonInstance)

```

```

(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) : Prefix C.RTL_B :=
M.normalization.normalized_post.prefix

def SelectedMatchedCutpoint.tbar_ref
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) : Prefix C.Sys_ref :=
M.normalization.normalized_ref.prefix

def SelectedMatchedCutpoint.p_post
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) : State :=
M.normalization.post_normalization.normalized_terminal

def SelectedMatchedCutpoint.σ_ref
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) : State :=
M.normalization.ref_normalization.normalized_terminal

def SelectedMatchedCutpoint.selected
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) :
SelectedJWitness C W M.normalization.base_post M.normalization.selected_ref :=
M.normalization.selected

def SelectedMatchedCutpoint.compatible
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) :
CompatibleSys C W M.normalization.selected_ref M.normalization.base_post :=
M.normalization.compatible

def SelectedMatchedCutpoint.post_exec
(C : ComparisonInstance)

```

```

(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) : Exec_post C M.τ_post :=
M.normalization.post_normalization.base_exec

def SelectedMatchedCutpoint.ref_exec
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) : Exec_ref C M.τ_ref :=
M.normalization.ref_normalization.base_exec

def SelectedMatchedCutpoint.normalized_post_exec
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) : Exec_post C M.tbar_post :=
M.normalization.post_normalization.normalized_exec

def SelectedMatchedCutpoint.normalized_ref_exec
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) : Exec_ref C M.tbar_ref :=
M.normalization.ref_normalization.normalized_exec

def SelectedMatchedCutpoint.normalized
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) :
EpsilonNormalizedPair C M.tbar_ref M.tbar_post :=
M.normalization.normalized_pair

def SelectedMatchedCutpoint.post_terminal
(C : ComparisonInstance)
(W : WitnessBundle C)
(M : SelectedMatchedCutpoint C W) : TerminalCutpoint M.tbar_post M.p_post :=
M.normalization.post_normalization.normalized_terminal_cutpoint

def SelectedMatchedCutpoint.ref_terminal
(C : ComparisonInstance)
(W : WitnessBundle C)

```



```
(M : SelectedMatchedCutpoint C W) : TerminalCutpoint M.τbar_ref M.σ_ref :=
M.normalization.ref_normalization.normalized_terminal_cutpoint
```

```
theorem J1_selected_matched_cutpoint
```

```
(C : ComparisonInstance)
```

```
(W : WitnessBundle C)
```

```
(hJ : JCutpointAssumptions C W)
```

```
(τ_post : Prefix C.RTL_B)
```

```
(hExecPost : Exec_post C τ_post) :
```

```
∃ M : SelectedMatchedCutpoint C W,
```

```
M.τ_post = τ_post :=
```

```
by
```

```
-- Apply J1_post_to_ref once to obtain the exact annotated selected pair. Apply
```

```
-- normalize_selected_J1_pair to construct distinct reachable normalized
```

```
-- representatives, then apply J1_cutpoint_correspondence to that same carrier.
```

```
-- The equality names the original requested post prefix, not its ε-extension.
```

```
sorry
```

The cutpoint theorem consumes one reachable selected-normalization object. It does not reconstruct Issue/Commit, dynamic occurrence, request-attribution, or execution facts from a visible prefix, and it cannot accidentally apply J.AbsConf to a different witness, an unreachable state pair, or an ε-extended prefix masquerading as the original J.1 selection.

```
def EquivalentUpToFiniteEpsilonStutter
```

```
(C : ComparisonInstance)
```

```
(side : Side)
```

```
(τ1 τ2 : Prefix (SystemOfSide C side)) : Prop :=
```

```
SameOrderedSigmaProjection C side τ1 τ2 ∧
```

```
FiniteEpsilonStutterDifference C side τ1 τ2
```

```
theorem J1_normalized_projection_unique_up_to_finite_epsilon
```

```
(C : ComparisonInstance)
```

```
(W : WitnessBundle C)
```

```
(hJ : JCutpointAssumptions C W)
```

```
(M1 M2 : SelectedMatchedCutpoint C W)
```

```
(hHorizon : SameComparedObservationHorizon C M1.τ_post M2.τ_post) :
```

```
EquivalentUpToFiniteEpsilonStutter
```

```
C.ref M1.τbar_ref M2.τbar_ref :=
```

by

- Apply `J1_matching_sigma_projection_unique` to the exact selected prefixes,
- then use the two explicit reference normalization extensions and `B4/B5` to
- bound the remaining ε -only differences between `tbar_ref` representatives.
- This still does not assert equality of source states.

sorry

Uniqueness scope note. The first theorem is the plain `J.1/B1` conclusion: the matching ordered observable projection is unique at a fixed comparison horizon. The second is a cutpoint-level refinement that additionally consumes `B4/B5` and concludes uniqueness only for the distinct normalized representatives, up to finite ε -stutter. Neither theorem makes the selected stateful `Sys_ref` witness unique.

The proof of `J1_cutpoint_correspondence` now receives one `SelectedNormalizedPrefixPair N`. The carrier contains the exact `J.1`-selected `base_post/selected_ref` pair, its `SelectedJWitness` and `CompatibleSys` evidence, the separate `normalized_post/normalized_ref` extensions, and `Exec_post/Exec_ref` evidence for both the base and normalized prefixes. The theorem therefore populates `CutpointCorrespondence.j_abs` from `hJ.jAbsConf.holds_for_selected_normalized_pair N` without inventing reachability premises or conflating an original prefix with its ε -extension. Quiescence and terminal-cutpoint facts come from `N.post_normalization` and `N.ref_normalization`; side-specific execution well-formedness comes from the embedded normalized execution evidence together with `hJ.wellformed`. Plain `J.1` continues to use `IDR_immediate_post_unit_deterministic_replay` on the exact selected pair. Only after `normalize_selected_J1_pair` constructs the two extensions does `IDR_deterministic_replay` apply at the normalized cutpoints. `J.AbsConf` supplies the `RTL_B`-side request, pending, attribution, endpoint-request, and enablement slices for that same carrier; these are not inferred from the `P`-local Σ -label sequence alone. Legacy agreement facts, when needed, should be obtained through projection lemmas from `correspondence.j_abs` rather than through duplicate fields.

18. Appendix K WFG and liveness

Define WFG over the selected proof-internal `Sys_ref / RTL_B` boundaries, not over the outer `Sigma_DUT` projection. In ordinary cases this distinction may collapse to the identity projection. In elaborated cases, however, `Sys_ref = Sys_B^elab(E)`, and the WFG must see the explicit staged/bundle/join/epoch-gate abstract channels introduced by `E` even when `sigmaDUTProjection /  $\Pi_{elab}$` hides their events from the outer `DUT/testbench`-visible alphabet.

```

def FrontierItemChannelSide
(C : ComparisonInstance)
(side : Side)
(x : FrontierItem)
(c : Channel) : Prop :=
 $\exists q,$ 
MsgOfFrontierItemSide C side x = some q  $\wedge$ 
q.channel = c

```

```

def ComplementOwnerItemSide
(C : ComparisonInstance)
(side : Side)
(x : FrontierItem)
(Q : Process) : Prop :=
 $\exists q,$ 
MsgOfFrontierItemSide C side x = some q  $\wedge$ 
ComplementOwner C side q = Q

```

```

def InternalChannelOfFrontierItem
(C : ComparisonInstance)
(side : Side)
(x : FrontierItem) : Prop :=
 $\exists q,$ 
MsgOfFrontierItemSide C side x = some q  $\wedge$ 
InternalChannel C side q.channel

```

```

def WFGProofInternalBoundaryItem
(C : ComparisonInstance)
(side : Side)
(x : FrontierItem) : Prop :=
 $\exists q,$ 
MsgOfFrontierItemSide C side x = some q  $\wedge$ 
q.channel  $\in$  ProofInternalChannels C side  $\wedge$ 
ChannelRepresentedInSigma C.Sigma q.channel  $\wedge$ 
 $\neg$  ChannelVisibilityDefinedOnlyBySigmaDUT C q.channel

```

```

def WFG

```

```

(C : ComparisonInstance)
(side : Side)
( $\sigma$  : State) : Digraph Process :=
{ edge := WFGEdge C side  $\sigma$  }

```

```

def WFGEdge
(C : ComparisonInstance)
(side : Side)
( $\sigma$  : State)
(P Q : Process) : Prop :=
 $\exists x$ ,
x  $\in$  Front C side P  $\sigma$   $\wedge$ 
BlockingMessageFrontierItem x  $\wedge$ 
BoundaryReqItemSide C side x  $\sigma$   $\wedge$ 
ChannelDisabledItemSide C side x  $\sigma$   $\wedge$ 
ComplementOwnerItemSide C side x Q  $\wedge$ 
InternalChannelOfFrontierItem C side x  $\wedge$ 
WFGProofInternalBoundaryItem C side x

```

```

theorem WFG_ignores_sigmaDUTProjection
(C : ComparisonInstance)
(side : Side)
( $\sigma$  : State)
(P Q : Process) :
WFGEdge C side  $\sigma$  P Q  $\leftrightarrow$ 
WFGEdgeComputedOverProofInternalSigma C side  $\sigma$  P Q :=
by
-- WFG, Front, BoundaryReqItemSide, ChannelEnabledItemSide, and
-- ChannelDisabledItemSide are interpreted over C.Sigma and the chosen
-- Sys_ref / RTL_B abstract channels through the owning dynamic message
-- instance of each message-operation frontier item. sigmaDUTProjection is used
-- only after the proof-internal comparison when projecting to the outer
-- DUT/testbench-visible observation boundary.
sorry

```

Edges are generated from Front_P(σ), not arbitrary non-frontier asserted requests. This intentionally makes the WFG sparser than a graph over all asserted boundary requests: only pending blocking frontier items generate wait-for edges.

For message-operation frontier items, `BoundaryReqItemSide` and `ChannelDisabledItemSide` recover the owning `DynMsgInstance` and then apply the operation-attributive `BoundaryReqSide` / `ChannelDisabledSide` predicates. Failed non-blocking polls and already-asserted non-frontier requests may affect state or occupancy through other rules, but they do not themselves create WFG dependency edges.

The Lean development must include the Appendix K.2.2 / Appendix R.1 structural classification theorem as an explicit milestone. $\text{Front_P}(\sigma)$ is not an independent primitive for liveness proofs: at ε -quiescent cutpoints it is derived from the blocking message-operation slice of `NextFront_P` together with the operation-attributive `BoundaryReq` predicate.

-- Message-operation frontier items whose owning dynamic message instance lies
 -- in the Q_Ω slice for process P. This is a predicate on `FrontierItem`, not a
 -- direct membership test in Q_Ω . The document's Q_Ω, P is a set of
 -- `DynMsgInstance`, while Ω_msg, P / `NextFront_P` are `FrontierItem`-level objects.
 -- Therefore this predicate relates the two universes through
 -- `MsgInstanceOfFrontierItem` / `MsgOfFrontierItemSide` rather than by literal set
 -- intersection.

```
def MessageFrontierItemInQOmega
(C : ComparisonInstance)
(W : WitnessBundle C)
(side : Side)
(P : Process)
(x : FrontierItem) : Prop :=
   $\exists q,$ 
  MsgOfFrontierItemSide C side x = some q  $\wedge$ 
  q  $\in$  QOmegaOfProcess C W P
```

Q_Ω membership is therefore invariant as the proof moves among cutpoints of the selected witness. Only `NextFront_P`, `BoundaryReq`, `ChannelEnabled`, and `ChannelDisabled` are state dependent. The side-indexed `MsgOfFrontierItemSide` projection must identify the same witness-owned dynamic operation instance on corresponding reference and post cutpoints; that agreement is supplied by the selected witness and `J.Abs/J.AbsConf`, not by constructing separate side-specific Q_Ω sets.

```

def BlockingMsgNextFront
(C : ComparisonInstance)
(W : WitnessBundle C)
(side : Side)
(P : Process)
( $\sigma$  : State) : Set FrontierItem :=
{ x |
x  $\in$  NextFront C side P  $\sigma$   $\wedge$ 
MessageFrontierItemInQOmega C W side P x  $\wedge$ 
BlockingMessageFrontierItem x  $\wedge$ 
MessageKindOfFrontierItem C side x  $\in$  {MsgKind.Push, MsgKind.Pop} }

```

This preserves the document's distinction between $\Omega_{\text{msg},P}$ / NextFront_P , whose elements are FrontierItem objects, and $Q_{\Omega,P}$, whose elements are DynMsgInstance objects. A message frontier item belongs to the blocking NextFront slice only when its owning dynamic message instance is defined and lies in $Q_{\Omega,P}$.

```

def FrontFromNext
(C : ComparisonInstance)
(W : WitnessBundle C)
(side : Side)
(P : Process)
( $\sigma$  : State) : Set FrontierItem :=
{ x |
x  $\in$  BlockingMsgNextFront C W side P  $\sigma$   $\wedge$ 
BoundaryReqItemSide C side x  $\sigma$  }

```

```

theorem K0_front_nextfront_classification
(C : ComparisonInstance)
(W : WitnessBundle C)
(side : Side)
(P : Process)
( $\sigma$  : State)
(hK : KStructuralAssumptions C W)
(hQ : EpsQuiescent  $\sigma$ )
(hWF : WellFormedCutpoint C side  $\sigma$ ) :
Front C side P  $\sigma$  =

```

FrontFromNext C W side P $\sigma :=$

by

-- Appendix K.2.2 / Appendix R.1 proof target, aligned with the v42 expanded
-- proof of Lemma K.0. The forward direction must prove that a pending blocking
-- frontier item is not merely minimal among pending blocking items, but minimal
-- in the full $\text{Remain_P}(\sigma) / \text{NextFront_P}$ sense. The proof therefore splits on
-- any hypothetical $y <_P x$:
-- 1. y cannot be a preceding synchronization item, because the current
-- interval definition plus R3/E1 puts it in $\text{Done_P}(\sigma)$;
-- 2. y cannot be a preceding anchored sequential signal item, because R2/E2
-- realizes it at $\text{pred_S}/\text{succ_S}$, nor a preceding combinational item whose
-- emitted R2C unit has already been realized;
-- 3. y cannot be an earlier same-interval message item, because R4/E3 gives
-- issue order; if it committed it is Done, if it is pending blocking it
-- contradicts Front-minimality, if it is successful NB it is Done, and if
-- it is failed NB it is not an Ω_P frontier item;
-- 4. y cannot come from an earlier source interval, because the closing sync
-- has committed and R3/E5 forces earlier blocking pref_S work to have
-- committed no later than that sync.
-- The reverse direction uses Appendix C stable attribution: an asserted
-- BoundaryReq for $x \in \text{BlockingMsgNextFront}$ is attributed to its owning q_x and
-- cannot be retagged to another same-endpoint operation without an intervening
-- commit. Younger already-asserted requests may exist as non-frontier protocol
-- state under Appendix B no-withdrawal/drain-only rules, but they do not enter
-- $\text{Front_P}(\sigma)$ while an earlier pending blocker is frontier-minimal.

sorry

theorem K0_front_correspondence_from_cutpoint

(C : ComparisonInstance)

(W : WitnessBundle C)

(P : Process)

($\sigma_ref \ \rho_post$: State)

(hK : KStructuralAssumptions C W)

(hCorr : CutpointCorrespondence C W $\sigma_ref \ \rho_post$) :

Front C .ref P σ_ref = Front C .post P ρ_post :=

by

-- Apply K0_front_nextfront_classification on both sides using hK, with
-- hCorr.ref_quiescent / hCorr.post_quiescent supplying the ε -quiescence

-- hypotheses and hCorr.ref_wellformed / hCorr.post_wellformed supplying the
 -- WellFormedCutpoint hypotheses. Then use projection lemmas from hCorr.j_abs,
 -- such as cutpoint_nextfront_agreement and
 -- cutpoint_pending_enablement_agreement, for NextFront agreement and
 -- operation-attributive BoundaryReqItemSide agreement on the common
 -- witness-fixed BlockingMsgNextFront slice. The document-numbered A1-A4 and A6-A11
 facts are accessed
 -- through hK.docNoA5; the Appendix C Issue/Commit attribution premise is accessed
 -- through hK.issue_commit_wf.
 sorry

This theorem is intentionally stated for BlockingMsgNextFront_P, not for the
 broader MsgNextFront_P. Successful non-blocking message-operation frontier items
 may contribute committed Push_c / Pop_c Σ -labels, and failed non-blocking polls
 are ε -steps, but neither creates a pending blocking WFG cause.

```
def ProcessStalledAtFront
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma$  : State) : Prop :=
FrontierBlocked C side P  $\sigma$ 
```

```
theorem process_stalled_at_front_iff_frontier_blocked
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma$  : State) :
ProcessStalledAtFront C side P  $\sigma$   $\leftrightarrow$ 
FrontierBlocked C side P  $\sigma$  :=
lff.rfl
```

-- ProcessStalledAtFront is now a document-facing alias, not a second
 -- independently maintained conjunction. The Section 13 progress machinery,
 -- the WFG construction, the canonical closed/drain-closed component, and the
 -- Appendix K deadlock predicates therefore consume exactly the same
 -- FrontierBlocked definition. This eliminates the former proof seam in which
 -- equivalence of two textually duplicated bodies was only promised by a comment.


```

def WFGClosed
(C : ComparisonInstance)
(side : Side)
(D : Finset Process)
( $\sigma$  : State) : Prop :=
 $\forall P Q,$ 
 $P \in D \rightarrow$ 
WFGEdge C side  $\sigma$  P Q  $\rightarrow$ 
 $Q \in D$ 

def WFGTerminalSCC
(C : ComparisonInstance)
(side : Side)
(D : Finset Process)
( $\sigma$  : State) : Prop :=
D.Nonempty  $\wedge$ 
StronglyConnectedSubgraph (WFG C side  $\sigma$ ) D  $\wedge$ 
WFGClosed C side D  $\sigma$ 

-- A drain escape is not necessarily a WFG edge leaving D. It is an enabled
-- non-frontier blocking transfer/drain that would discharge already-issued
-- downstream work for some process in D. Such a drain may be intra-process or
-- may not appear as a graph edge at all, because WFG edges are generated only
-- from current minimal blocking frontier items Front_P( $\sigma$ ).
--
-- Important: “non-frontier” here means “not in the current minimal pending
-- blocking frontier Front_P( $\sigma$ ).” It does not mean that the operation instance is
-- absent from the broader dynamic/message-frontier universe  $\Omega_{\text{msg},P} / Q_{\Omega,P}$ .
-- Conversely, it should not be characterized as a member of the current
-- BlockingMsgNextFront_P( $\sigma$ ) slice together with BoundaryReq, because K.0 proves
-- that, at  $\varepsilon$ -quiescent well-formed cutpoints, those items are exactly Front_P( $\sigma$ ).
--
-- Downstream automatic-flush work is already-issued downstream protocol state:
-- its owning message-operation frontier item may still be part of the broader
-- witness universe, may still have BoundaryReq asserted, and may commit as
-- drain-only progress, but its item is not currently minimal in Front_P( $\sigma$ ).
def OwnsCurrentFrontierMsgInstance
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma$  : State)
(q : DynMsgInstance) : Prop :=

```

```

 $\exists x,$ 
 $x \in \text{Front } C \text{ side } P \ \sigma \wedge$ 
 $\text{MsgOfFrontierItemSide } C \text{ side } x = \text{some } q$ 

def NonFrontierBlockingDrainCandidate
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma$  : State)
(q : DynMsgInstance) : Prop :=
 $\exists x,$ 
DrainableDownstreamRequest C side P  $\sigma$  x  $\wedge$ 
 $\text{MsgOfFrontierItemSide } C \text{ side } x = \text{some } q \wedge$ 
 $x \notin \text{Front } C \text{ side } P \ \sigma$ 

def EnabledNonFrontierBlockingDrainForProcess
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma$  : State) : Prop :=
 $\exists q,$ 
OwnsMessageEndpoint P q  $\wedge$ 
Blocking q  $\wedge$ 
NonFrontierBlockingDrainCandidate C side P  $\sigma$  q  $\wedge$ 
BoundaryReqPersistentDomainSide C side q  $\sigma$   $\wedge$ 
BoundaryReqSide C side q  $\sigma$   $\wedge$ 
ChannelEnabledSide C side q  $\sigma$ 
def NoEnabledNonFrontierBlockingDrain
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma$  : State) : Prop :=
 $\neg \text{EnabledNonFrontierBlockingDrainForProcess } C \text{ side } P \ \sigma$ 

def DrainClosed
(C : ComparisonInstance)
(side : Side)
(D : Finset Process)
( $\sigma$  : State) : Prop :=
 $\forall P,$ 
 $P \in D \rightarrow$ 
NoEnabledNonFrontierBlockingDrain C side P  $\sigma$ 

```

The K.1 SCC contradiction should use this definition. The closed SCC supplies the internal frontier WFG cycle; DrainClosed separately rules out currently enabled non-frontier blocking drains for processes in D. In v42, however, DrainClosed is not misread as saying that outside-D or environment activity can never make such a non-frontier request ChannelEnabled later. The K.1 Exec bridge therefore uses fair drain-normalization: if a D-owned already-issued non-frontier request later becomes continuously serviceable under B2/B3, allow it to commit as drain-only progress, and use the finite-drain argument to reach a later ε -quiescent drain-closed cutpoint when needed.

```
def HasOutgoingWFGEdgeInD
```

```
(C : ComparisonInstance)
```

```
(side : Side)
```

```
(D : Finset Process)
```

```
(P : Process)
```

```
( $\sigma$  : State) : Prop :=
```

```
 $\exists Q,$ 
```

```
 $Q \in D \wedge$ 
```

```
WFGEdge C side  $\sigma$  P Q
```

```
def ClosedDrainClosedWFGComponent
```

```
(C : ComparisonInstance)
```

```
(side : Side)
```

```
(D : Finset Process)
```

```
( $\sigma$  : State) : Prop :=
```

```
WFGTerminalSCC C side D  $\sigma \wedge$ 
```

```
DrainClosed C side D  $\sigma \wedge$ 
```

```
 $(\forall P, P \in D \rightarrow \text{FrontierBlocked } C \text{ side } P \sigma) \wedge$ 
```

```
 $(\forall P, P \in D \rightarrow \text{HasOutgoingWFGEdgeInD } C \text{ side } D P \sigma)$ 
```

```
def ExistsClosedDrainClosedWFGCycle
```

```
(C : ComparisonInstance)
```

```
(side : Side)
```

```
( $\sigma$  : State) : Prop :=
```

```
 $\exists D, \text{ClosedDrainClosedWFGComponent } C \text{ side } D \sigma$ 
```

```
def InternalMPDeadlockShapeAt
```

```
(C : ComparisonInstance)
```

```
(side : Side)
```

```

( $\sigma$  : State) : Prop :=
  EpsQuiescent  $\sigma$   $\wedge$ 
  ExistsClosedDrainClosedWFGCycle C side  $\sigma$ 

```

-- Public document-level deadlock predicate: the shape above together with
 -- reachability in the selected comparison side.

```

def ReachableInternalMPDeadlockAt
  (C : ComparisonInstance)
  (side : Side)
  ( $\sigma$  : State) : Prop :=
  Reachable (SystemOfSide C side)  $\sigma$   $\wedge$ 
  InternalMPDeadlockShapeAt C side  $\sigma$ 

```

The K.1 SCC contradiction uses InternalMPDeadlockShapeAt after reachability has been supplied by an execution prefix or by a separate hypothesis.

ClosedDrainClosedWFGComponent is the single canonical conjunction: the terminal SCC supplies the cycle of internal message-passing dependencies; DrainClosed rules out escape by downstream draining; FrontierBlocked ensures every process in D is stalled on its current blocking frontier, not merely carrying a non-frontier asserted request; and HasOutgoingWFGEdgeInD supplies the non-vacuity/internal-cause condition. Every process in the candidate deadlock set must have at least one WFG edge, generated from an internal proof-visible message-passing frontier item, to another process in the same set. This excludes the spurious singleton case where a process is merely waiting on an external environment or testbench channel and therefore has no internal WFG cause.

ProcessStalledAtFront is definitionally equal to FrontierBlocked, so older document-facing statements that use the former name reduce to this same conjunction without an additional semantic lemma. ExistsClosedDrainClosedWFGCycle, InternalMPDeadlockShapeAt, the per-process participation predicate, and the terminated-peer corollaries all quantify over ClosedDrainClosedWFGComponent directly.

ReachableInternalMPDeadlockAt is the public document-level predicate. The factorization prevents a theorem from accidentally treating an arbitrary state with the deadlock graph shape as a reachable deadlock. Finite-prefix and execution-scoped theorems may use InternalMPDeadlockShapeAt because their terminal-prefix premise already supplies reachability; Appendix K and Appendix R facade statements should expose the reachable wrapper where the document speaks about a reachable deadlock.

Terminated-process convention (v42 K.1). The scheduling-rules document additionally fixes a terminated-process convention: at an examined cutpoint, a process that is blocked solely on channels whose unique complementary endpoint process has terminated in its current reset epoch, with no intervening reset of that peer, is not counted as participating in an internal message-passing deadlock. Such a configuration is a protocol/termination modeling issue, not a K.1 deadlock. In this strategy that convention is not an extra assumption — it is a corollary of `finish_process_idle_padding_until_next_reset` together with the closed-SCC deadlock definition above, and Lean should prove it explicitly:

```
def ProcessParticipatesInInternalMPDeadlockShapeAt
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma$  : State) : Prop :=
EpsQuiescent  $\sigma$   $\wedge$ 
 $\exists D$ ,
 $P \in D \wedge$ 
ClosedDrainClosedWFGComponent C side D  $\sigma$ 
```

```
theorem process_participates_iff_deadlock_component_member
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma$  : State) :
ProcessParticipatesInInternalMPDeadlockShapeAt C side P  $\sigma \leftrightarrow$ 
EpsQuiescent  $\sigma \wedge$ 
 $\exists D, P \in D \wedge$  ClosedDrainClosedWFGComponent C side D  $\sigma :=$ 
Iff.rfl
```

```
theorem terminated_peer_waiting_not_in_deadlock_component
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma$  : State)
(hP2P : PointToPointChannels C)
(hWait : BlockedOnlyOnPeersTerminatedInCurrentEpoch C side P  $\sigma$ ) :
 $\neg \exists D$ ,
 $P \in D \wedge$  ClosedDrainClosedWFGComponent C side D  $\sigma :=$ 
by
```

```
-- A peer terminated in its current reset epoch, with no later reset before  $\sigma$ ,
-- contributes no active complementary endpoint request at  $\sigma$ . Under point-to-point
-- ownership, a process waiting only on such peers cannot satisfy the outgoing-edge
-- and terminal-SCC conjuncts of the canonical internal-deadlock component.
```

-- A later reset creates a new epoch and is outside this theorem premise.
sorry

theorem terminated_peer_waiting_not_internal_MP_deadlock
(C : ComparisonInstance)
(side : Side)
(P : Process)
(σ : State)
(hP2P : PointToPointChannels C)
(hWait : BlockedOnlyOnPeersTerminatedInCurrentEpoch C side P σ) :
 $\neg$ ProcessParticipatesInInternalMPDeadlockShapeAt C side P σ :=
by
intro hPart
exact terminated_peer_waiting_not_in_deadlock_component
C side P σ hP2P hWait hPart.2

Proof sketch. By A6 point-to-point channels, each channel in P's blocking frontier has a unique complementary endpoint process Q. The epoch-aware hWait premise says that, at the examined cutpoint σ , every such Q has a FINISH_Q occurrence in Q's current reset epoch and no reset affecting Q has occurred after that finish. By finish_process_idle_padding_until_next_reset, Q is padded with idle ε -steps from that FINISH_Q occurrence through σ : its Front is empty, it asserts no BoundaryReq, it commits no message operations, and it generates no blocking WFG cause at σ . Hence Q fails FrontierBlocked — its frontier is empty, violating the first conjunct — so Q cannot belong to a ClosedDrainClosedWFGComponent at σ . But every WFG edge out of P generated by P's blocking frontier targets such a Q. Therefore, for any D containing P, either P's outgoing WFG edges leave D, violating the HasOutgoingWFGEdgeInD/terminal-SCC conditions, or D would have to contain a terminated Q, violating the FrontierBlocked condition for Q. Either way no canonical closed drain-closed WFG component containing P exists at σ . A later reset of Q creates a new epoch and invalidates the hWait premise rather than being suppressed by this result. The per-process corollary follows directly by unfolding ProcessParticipatesInInternalMPDeadlockShapeAt; no unproved ProcessStalledAtFront/FrontierBlocked conversion is needed.

Scope warning (v42 K.1). This convention is a scoping rule for the deadlock definition only. It does not verify, and must not be read as verifying, intended termination behavior: explicit EOF, done, shutdown, or cancellation protocols, and any design intent that a consumer should observe a producer's termination, must be modeled explicitly as message-passing or signal protocol in Sys_ref and RTL_B and proved through the ordinary I/J/K obligations, or verified outside Appendix K. A process left waiting forever on a terminated peer is excluded from the K.1 deadlock alarm precisely because the document treats it as a protocol-level

modeling defect, not as an internal message-passing deadlock; the exclusion must therefore never be used to argue that such a configuration is acceptable.

K assumptions should keep the document-numbered Appendix K assumptions visible, but the Lean proof must also separate the structural K premises from A5. Lemma K.2b is the bridge used before A5 is invoked, so K.2b and the K.0/K.2 structural lemmas must not take an assumption bundle that already contains A5. The DockStructuralAssumptions structure below therefore stores A1 and A6–A11. The document-numbered A2–A4 names are derived projections of the B4, B2, and B3 fields already carried by A1’s BRules component, rather than duplicate assumption fields. A5 is carried only by the final KAssumptions wrapper used for K.1.

The v45 Appendix K update changes A5 substantially. A5 is no longer a scheduler-robust liveness premise over all source scheduling choices. It is the conservative serial-blocking source-liveness premise: for A5 only, a process may not issue a later same-interval blocking Push/Pop until every earlier same-interval blocking Push/Pop of that process has committed. The ordinary Appendix J witness, Corollary J.1 cutpoint correspondence, and RTL_B implementation may still contain overlapped / liberal issue. Lemma K.2b is the bridge from such a witness-relative overlapped Sys_ref deadlock to a serial-source deadlock that contradicts A5.

The J.1-to-K.2b path is finite-prefix throughout and uses Prefix C.Sys_ref as its semantic carrier throughout. Serial blocking is a restriction of the ordinary liberal finite source-prefix semantics, not a replacement for the R4/E3/E5/no-reverse discipline. Infinite executions remain the carrier for B2/B3 fairness and admissibility, but J.1 produces a semantic finite reference prefix, every K.2b one-step reduction constructs another semantic finite reference prefix, and A5 consumes the resulting finite conservative-serial prefix. ExecutionBucketed C.Sigma appears in K.2b only through RefPrefixBuckets as a derived measure/stream representation; no arbitrary edited bucket list is treated as realizable without constructing the corresponding Prefix C.Sys_ref.

```
def A11_Kepsilon_WFGInert
(C : ComparisonInstance) : Prop :=
  ∀ side P τ t0 t σ0 σ t,
  StateAtCycleInPrefix C side τ t0 σ0 →
  StateAtCycleInPrefix C side τ t σ t →
  t0 ≤ t →
  InternalOrEpsilonOnlyBetween C side τ t0 t →
  FrontierBlocked C side P σ0 →
  FrontierRemainsBlockedOver C side P τ t0 t →
  WFGInertForProcess C side P σ0 σ t ∧
  NoNewBlockingFrontierCause C side P σ0 σ t
```

```

def BarrierWellFormed
(C : ComparisonInstance)
(W : WitnessBundle C) : Prop :=
  ∀ b,
  SyncChannellInstance C b →
  DesignatedParticipantsReachSameBarrierCount C W b ∧
  NoPermanentBarrierBypass C W b

```

```

inductive SourceBlockingIssuePolicy
| liberal
| serial

```

```

-- Every K.2b theorem below reduces semantic reference-system prefixes. The
-- bucket list is a derived observation of an already-realized Prefix C.Sys_ref;
-- it is never an independent carrier from which the proof later assumes a
-- semantic prefix can be reconstructed.

```

```

def RefPrefixBuckets
(C : ComparisonInstance)
(p_ref : Prefix C.Sys_ref) : ExecutionBucketed C.Sigma :=
  ObservableBucketsOfSystemPrefix C .ref p_ref

```

```

-- Completion of an earlier blocking operation before a later same-interval issue.
-- In the document-aligned serial policy, serial means the earlier blocking
-- operation has committed before the later operation issues; no same-cycle
-- commit-then-issue exception is assumed. If a future document revision adds a
-- precise intra-bucket commit-before-issue convention, represent it as a
-- separately named optional extension theorem, not as part of A5 by default.

```

```

def EarlierBlockingOpCompletedBeforeLaterIssue
(C : ComparisonInstance)
(side : Side)
(τ : ExecutionBucketed C.Sigma)
(qEarlier qLater : DynMsgInstance)
(tlIssueLater : Cycle) : Prop :=
  CommitsBefore C side τ qEarlier tlIssueLater

```

```

-- qLater is an eager issue if it is issued before some source-earlier blocking
-- Push/Pop in the same dynamic interval has completed in the sense above. Eager
-- issue is permitted by the liberal witness policy, but forbidden by the serial A5
-- policy.

```

```

def EagerBlockingIssueAt
(C : ComparisonInstance)
(side : Side)
(τ : ExecutionBucketed C.Sigma)

```



```

(qLater qEarlier : DynMsgInstance)
(t : Cycle) : Prop :=
Blocking qLater  $\wedge$  Blocking qEarlier  $\wedge$ 
SameOwningProcess qEarlier qLater  $\wedge$ 
SameDynamicSourceInterval C side  $\tau$  qEarlier qLater  $\wedge$ 
SourceOrderBefore C side  $\tau$  qEarlier qLater  $\wedge$ 
IssueAtCycle C side  $\tau$  qLater t  $\wedge$ 
 $\neg$  EarlierBlockingOpCompletedBeforeLaterIssue C side  $\tau$  qEarlier qLater t

```

```

structure EagerBlockingIssueEvent
(C : ComparisonInstance)
(side : Side)
(p : ExecutionBucketed C.Sigma) where
qLater : DynMsgInstance
qEarlier : DynMsgInstance
tIssue : Cycle
eager : EagerBlockingIssueAt C side p qLater qEarlier tIssue

```

```

-- The concrete implementation can choose any finite representation of the eager
-- issue events of the finite prefix p, provided membership is equivalent to the
-- EagerBlockingIssueAt witness above. This Finset is the measure used by K.2b.

```

```

def EagerBlockingIssueEvents
(C : ComparisonInstance)
(side : Side)
(p : ExecutionBucketed C.Sigma) : Finset (EagerBlockingIssueEvent C side p) :=
AllEagerBlockingIssueEventsInFinitePrefix C side p

```

```

-- IssuedBlockingEndpointSourceOrderStream is ordered by source/endpoint order,
-- not by issue time. Deferring one eager issue may move an event later in time,
-- so an issue-time-ordered stream would not be prefix-monotone. The prefix theorem
-- below is about the source-order endpoint stream: deferral may withhold a suffix
-- of the endpoint stream, but it never drops an earlier endpoint operation while
-- retaining a later one.

```

```

def IssuedBlockingEndpointSourceOrderStream
(C : ComparisonInstance)
(side : Side)
(p : ExecutionBucketed C.Sigma)
(c : Channel)
(dir : EndpointDirection) : List DynMsgInstance :=
BlockingEndpointInstancesIssuedInPrefixOrderedBySourceEndpointOrder C side p c dir

```

```

-- DefersOneEagerIssue is the run-surgery relation used by the one-step reduction.
-- It identifies a particular eager issue event in p, removes or defers that
-- operation to its serial slot, preserves all retained earlier committed transfers

```

-- and their values, and changes no endpoint stream except by suppressing a
 -- source-order suffix on explicit WFG-relevant point-to-point channels. The relation
 -- must also record that any new terminal cutpoint σ' used after the edit is reached
 -- by finite ε /drain normalization from the edited prefix.

```
def DefersOneEagerIssue
(C : ComparisonInstance)
(side : Side)
( $\rho \rho'$  : ExecutionBucketed C.Sigma) : Prop :=
 $\exists$  ev : EagerBlockingIssueEvent C side  $\rho$ ,
RemovesOrDelaysExactlyEagerIssue C side  $\rho \rho'$  ev  $\wedge$ 
RetainedCommittedTransferPrefix C side  $\rho' \rho \wedge$ 
 $\forall$  c dir,
ChannelCanContributeWFGEdge C c  $\rightarrow$ 
IsPrefix
(IssuedBlockingEndpointSourceOrderStream C side  $\rho' c$  dir)
(IssuedBlockingEndpointSourceOrderStream C side  $\rho c$  dir)
```

```
def ConservativeSerialBlockingRun
(C : ComparisonInstance)
(side : Side)
( $\tau$  : ExecutionBucketed C.Sigma) : Prop :=
 $\forall$  qLater qEarlier t,
EagerBlockingIssueAt C side  $\tau$  qLater qEarlier t  $\rightarrow$  False
```

```
def LiberalBlockingRun
(C : ComparisonInstance)
(side : Side)
( $\tau$  : ExecutionBucketed C.Sigma) : Prop :=
IssueOrderConformsToR4E3E5 C side  $\tau \wedge$ 
NoReverseDistinctInterfaceIssue C side  $\tau$ 
```

-- Prefix predicate for all finite prefixes of an infinite execution. The policy
 -- restrictions must hold on every finite prefix, equivalently on the infinite run;
 -- they are not checked only on one arbitrarily chosen prefix.

```
def LiberalReferencePrefix
(C : ComparisonInstance)
( $\rho_{\text{ref}}$  : Prefix C.Sys_ref) : Prop :=
Exec_ref C  $\rho_{\text{ref}}$   $\wedge$ 
IssueOrderConformsToR4E3E5 C .ref  $\rho_{\text{ref}}$   $\wedge$ 
NoReverseDistinctInterfaceIssue C .ref  $\rho_{\text{ref}}$ 
```

```
def ConservativeSerialReferencePrefix
(C : ComparisonInstance)
( $\rho_{\text{ref}}$  : Prefix C.Sys_ref) : Prop :=
```

```

LiberalReferencePrefix C p_ref ∧
NoEagerBlockingIssue C .ref p_ref

```

```

theorem conservative_serial_reference_prefix_implies_liberal
(C : ComparisonInstance)
(p_ref : Prefix C.Sys_ref) :
ConservativeSerialReferencePrefix C p_ref →
LiberalReferencePrefix C p_ref :=
by
intro h
exact h.1

```

-- A5 is stated directly over finite admissible source prefixes.

```

def InternalMPDeadlockFreeOnSerialPrefixesRef
(C : ComparisonInstance) : Prop :=
∀ p_ref σ_ref,
ConservativeSerialReferencePrefix C p_ref →
TerminalCutpoint p_ref σ_ref →
¬ InternalMPDeadlockShapeAt C .ref σ_ref

```

-- Convenience alias only; it adds no infinite-execution domain.

```

def InternalMPDeadlockFreeOnSerialExecRef
(C : ComparisonInstance) : Prop :=
InternalMPDeadlockFreeOnSerialPrefixesRef C

```

-- Finite measure used by Lemma K.2b. For a finite prefix p, EagerIssueCount is
-- the number of dynamic blocking issue events in p that satisfy
-- EagerBlockingIssueAt. The count is finite because p is finite.

```

def EagerIssueCount
(C : ComparisonInstance)
(side : Side)
(p : ExecutionBucketed C.Sigma) : Nat :=
Finset.card (EagerBlockingIssueEvents C side p)

```

theorem zero_eager_iff_serial_blocking_prefix

```

(C : ComparisonInstance)
(side : Side)
(p : ExecutionBucketed C.Sigma) :
EagerIssueCount C side p = 0 ↔
ConservativeSerialBlockingRun C side p :=
by
-- Unfold EagerIssueCount, EagerBlockingIssueEvents, and
-- ConservativeSerialBlockingRun.
sorry

```

-- Semantic-prefix wrappers for the bucket-level eager-issue machinery. Because
 -- both arguments are Prefix C.Sys_ref, DefersOneEagerIssueRefPrefix includes the
 -- realizability fact that the edited bucket projection comes from an actual
 -- admissible reference-system prefix.

```
def RefEagerIssueCount
(C : ComparisonInstance)
(p_ref : Prefix C.Sys_ref) : Nat :=
EagerIssueCount C .ref (RefPrefixBuckets C p_ref)
```

```
def RefEndpointIssuedBlockingStreamPrefix
(C : ComparisonInstance)
(p_ref' p_ref : Prefix C.Sys_ref)
(c : Channel)
(dir : EndpointDirection) : Prop :=
IsPrefix
(IssuedBlockingEndpointSourceOrderStream
  C .ref (RefPrefixBuckets C p_ref') c dir)
(IssuedBlockingEndpointSourceOrderStream
  C .ref (RefPrefixBuckets C p_ref) c dir)
```

```
structure DefersOneEagerIssueRefPrefix
(C : ComparisonInstance)
(p_ref p_ref' : Prefix C.Sys_ref) : Prop where
-- This field is the explicit semantic/admissibility obligation. It requires an
-- infinite RefAdmissible extension of the edited finite prefix, including the
-- applicable B2/B3 fairness and progress premises.
output_exec_ref : Exec_ref C p_ref'
output_issue_order : IssueOrderConformsToR4E3E5 C .ref p_ref'
output_no_reverse : NoReverseDistinctInterfaceIssue C .ref p_ref'
output_drain_only :
  ∀ P,
  BlockingIntervallnPrefix C .ref P p_ref' →
  DrainOnlyBlockedFrontierDiscipline C .ref P p_ref'
bucket_deferral :
  DefersOneEagerIssue
  C .ref
  (RefPrefixBuckets C p_ref)
  (RefPrefixBuckets C p_ref')
```

```
theorem DefersOneEagerIssueRefPrefix.output_liberal
(C : ComparisonInstance)
(p_ref p_ref' : Prefix C.Sys_ref)
(h : DefersOneEagerIssueRefPrefix C p_ref p_ref') :
```

```

LiberalReferencePrefix C p_ref' :=
<h.output_exec_ref, h.output_issue_order, h.output_no_reverse>
theorem zero_ref_eager_iff_serial_reference_prefix
(C : ComparisonInstance)
(p_ref : Prefix C.Sys_ref)
(hLiberal : LiberalReferencePrefix C p_ref) :
RefEagerIssueCount C p_ref = 0 ↔
ConservativeSerialReferencePrefix C p_ref :=
by
-- Unfold the semantic wrappers, use zero_eager_iff_serial_blocking_prefix on
-- RefPrefixBuckets C p_ref, and identify ConservativeSerialBlockingRun of that
-- projection with NoEagerBlockingIssue C .ref p_ref.
sorry

structure DockStructuralAssumptions
(C : ComparisonInstance)
(W : WitnessBundle C) where
-- A1. R-rules and B-rules.
-- This is the document-numbered Appendix K assumption that the selected
-- Sys_ref / RTL_B comparison satisfies the semantic rule bundles needed by K.
-- E-rules are still required by the imported Appendix J compatibility theorem,
-- but they are not renamed as a document-numbered K assumption here.
A1_R_and_B_rule_conformance :
RRules C ∧ BRules C

-- A2–A4 are not stored a second time. Their document-facing names are
-- defined below as projections from A1_R_and_B_rule_conformance.2, preventing
-- drift between the Appendix K aliases and the underlying B4/B2/B3 obligations.

-- A5 is deliberately not included in this structural bundle. A5 is the
-- source-level serial liveness input consumed only by the final K.1 contradiction,
-- after K.2b has converted the witness-relative / overlapped reference deadlock
-- into a serial-source deadlock. This prevents K.2b from depending, even
-- accidentally, on the premise it is meant to help contradict.

-- A6. Point-to-point channels.
-- This is the document-numbered Appendix K A6 assumption. It records the
-- ordinary point-to-point channel premise. The stronger Lean-side requirement
-- that every WFG-relevant explicit abstract boundary, including introduced
-- Sys_B^elab staged, bundle, join, or epoch-gate boundary channels, be
-- point-to-point is carried by KElaborationPrerequisite below, not by A7.
A6_point_to_point_channels :
PointToPointChannels C

```

```

-- A7. Determinism / witness selection / cutpoint abstraction.
-- RTL_B determinism is the B1 part used by Appendix K; WC records the selected
-- Sys_ref witness supplied by the Appendix I/J/L witness machinery; and
-- J.AbsConf records that the same selected RTL_B cutpoint, Sys_ref cutpoint,
-- and witness discharge the source-slice abstraction used by Corollary J.1 and
-- Appendix K.
--
-- Keep A7 aligned with the scheduling-rules document: A7 has these three
-- conjuncts only. Point-to-point/WFG-boundary obligations belong to A6 and to
-- the Lean-side KElaborationPrerequisite.
A7_determinism_witness_selection_cutpoint_abstraction :
  PostDeterministicSigmaTrace C  $\wedge$ 
  WC C W  $\wedge$ 
  JAbsConf C W

-- A8. Barrier well-formedness (SyncChannels).
-- Barrier stalls caused by mismatched or conditional participation are outside
-- Appendix K's internal message-passing deadlock analysis.
A8_barrier_well_formedness :
  BarrierWellFormed C W

-- A9. Single-transfer-per-cycle endpoint constraint.
A9_single_transfer_per_cycle :
  SingleTransferPerCycleEndpoints C

-- A10. Reset-empty FIFO assumption.
A10_reset_empty_fifo :
  ResetEmptyFifoAssumption C

-- A11. WFG-inert  $\varepsilon$  /  $K\varepsilon$ .
A11_Kepsilon_WFG_inert :
  A11_Kepsilon_WFGInert C

-- Document-numbered A2–A4 aliases, derived from the B-rule bundle already
-- carried by A1. These are definitions/projections, not additional assumptions.
def DockStructuralAssumptions.A2_finite_pipeline_depth_B4
  (C : ComparisonInstance)
  (W : WitnessBundle C)
  (h : DockStructuralAssumptions C W) :
  FiniteEpsilonDepth C :=
  h.A1_R_and_B_rule_conformance.2.B4_finiteEpsilonDepth

def DockStructuralAssumptions.A3_weak_fairness_B2
  (C : ComparisonInstance)

```

```

(W : WitnessBundle C)
(h : DockStructuralAssumptions C W) :
  ∀ side τ,
  InfiniteExecutionOfSide C side τ →
  WeakFairness C side τ :=
  h.A1_R_and_B_rule_conformance.2.B2_weakFairness

def DockStructuralAssumptions.A4_channel_progress_B3
(C : ComparisonInstance)
(W : WitnessBundle C)
(h : DockStructuralAssumptions C W) :
  ∀ side τ c,
  InfiniteExecutionOfSide C side τ →
  P_push C side τ c ∧ P_pop C side τ c :=
  h.A1_R_and_B_rule_conformance.2.B3_channelProgress

-- Lean-side prerequisite for applying Appendix K to the chosen proof-internal
-- Sys_ref boundary structure. This is not a new document-numbered A-assumption.
-- It refines the document's A6 point-to-point premise for the explicit abstract
-- boundaries over which WFG, BoundaryReq, ChannelEnabled, and ChannelDisabled
-- are actually evaluated.
def KElaborationPrerequisite
(C : ComparisonInstance) : Prop :=
  ∀ c,
  ChannelCanContributeWFGEdge C c →
  ChosenAbstractBoundaryOfSysRef C c ∧
  PointToPointAtWFGEvaluationBoundary C c ∧
  NoHiddenSyncBoundaryGatingAtOrdinaryBufferedBoundary C c ∧
  (RequiresSyncBoundaryWithholding C c →
    C.refChoice.kind = ReferenceKind.elaborated)
Lean-side additions:
structure KStructuralAssumptions
(C : ComparisonInstance)
(W : WitnessBundle C) where
docNoA5 : DockStructuralAssumptions C W

-- Imported Appendix J compatibility uses E1-E5. Keep this as a Lean-side
-- theorem prerequisite rather than relabeling it as one of Appendix K's
-- document-numbered A1-A11 assumptions.
e_rules_for_imported_J :
ERules C

-- Appendix C Issue/Commit well-formedness supplies persistent blocking-request
-- attribution, including the ordinary-boundary proof used by

```

-- AutomaticFlush.no_withdrawal. It is not one of Appendix K's numbered A1-A11
-- assumptions; the current document's A6 is point-to-point channel structure.

issue_commit_wf :

IssueCommitWF C W

witness_universe_wf :

WitnessBundleWF C W

-- Appendix K consumes a generalized drain-only blocked-frontier discipline.
-- Pipelined intervals obtain it from Appendix B AutomaticFlush; ordinary
-- non-overlapped control obtains it without pretending that every liberal eager
-- same-interval issue came from loop pipelining.

blocking_interval_mode_complete :

$\forall$ side P τ ,

ProofRelevantDrainInterval C W side P $\tau \rightarrow$

PipelinedBlockingIntervalInPrefix C side P $\tau \vee$

NonOverlappedBlockingControl C side P τ

automatic_flush :

$\forall$ side P τ ,

ProofRelevantDrainInterval C W side P $\tau \rightarrow$

PipelinedBlockingIntervalInPrefix C side P $\tau \rightarrow$

AutomaticFlush C side P τ

-- Ordinary non-overlapped intervals must also discharge the sync-boundary
-- clause of R.16: either no proof-relevant withholding exists in ordinary Sys_B,
-- or the selected Sys_B^{elab} package represents every such case explicitly.

ordinary_sync_boundary_policy :

$\forall$ side P τ ,

ProofRelevantDrainInterval C W side P $\tau \rightarrow$

NonOverlappedBlockingControl C side P $\tau \rightarrow$

SyncBoundaryPolicy C side P τ

-- The K-stack invokes Appendix J / Corollary J.1 to obtain corresponding
-- Sys_ref and RTL_B cutpoints from a post-side deadlock cutpoint. Therefore the
-- non-K-numbered prerequisites required by JTraceAssumptions and
-- JCutpointAssumptions must also be available to the K theorem instance.

wellformed_for_imported_J :

C.WellFormed

implementation_for_imported_J :

ImplementationAssumptions C

environment_for_imported_J :

ReactiveEnvironmentWF C.env

elaboration_projection_wf_for_imported_J :

$\forall$ side τ , ElaborationProjectionWF C side τ

initial_and_reset_for_imported_J :
InitialAndResetCorrespondence C W

contracts : ModelingContracts C
r2c? : Option (R2C_WF C)
elaboration_prerequisite :
KElaborationPrerequisite C

-- Final Appendix K theorem bundle. This is the only K bundle that carries A5.
-- Structural lemmas, including K.2b, take KStructuralAssumptions instead.

structure KAssumptions
(C : ComparisonInstance)
(W : WitnessBundle C) where
structural : KStructuralAssumptions C W
A5_source_liveness :
InternalMPDeadlockFreeOnSerialPrefixesRef C

-- Derived constructor used whenever a structural K proof imports Appendix J's
-- post-to-reference or cutpoint-correspondence theorem.

def KStructuralAssumptions.toJTraceAssumptions
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W) : JTraceAssumptions C W :=
{
 wellformed := hK.wellformed_for_imported_J
 bRules := BRules.toTraceBRules C hK.docNoA5.A1_R_and_B_rule_conformance.2
 rRules := hK.docNoA5.A1_R_and_B_rule_conformance.1
 eRules := hK.e_rules_for_imported_J
 issueCommitWF := hK.issue_commit_wf
 witnessUniverseWF := hK.witness_universe_wf
 implementation := hK.implementation_for_imported_J
 witnessCompatible :=
 hK.docNoA5.A7_determinism_witness_selection_cutpoint_abstraction.2.1
 contracts := hK.contracts
 environment := hK.environment_for_imported_J
 elaborationProjectionWF := hK.elaboration_projection_wf_for_imported_J
 initialAndReset := hK.initial_and_reset_for_imported_J
}

def KAssumptions.toJTraceAssumptions
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KAssumptions C W) : JTraceAssumptions C W :=
KStructuralAssumptions.toJTraceAssumptions C W hK.structural

```

-- Cutpoint transfer imports B4/B5 and J.AbsConf in addition to the trace bundle.
def KStructuralAssumptions.toJCutpointAssumptions
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W) : JCutpointAssumptions C W :=
{
toJTraceAssumptions := KStructuralAssumptions.toJTraceAssumptions C W hK
normalization :=
  BRules.toNormalizationBRules C hK.docNoA5.A1_R_and_B_rule_conformance.2
jAbsConf :=
  hK.docNoA5.A7_determinism_witness_selection_cutpoint_abstraction.2.2
}

```

```

def KAssumptions.toJCutpointAssumptions
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KAssumptions C W) : JCutpointAssumptions C W :=
KStructuralAssumptions.toJCutpointAssumptions C W hK.structural

```

This avoids fictional A12/A13 labels and also avoids silently renumbering the document's A1-A11 assumptions inside Lean. Structural K proofs use hK.docNoA5 for document-numbered facts other than A5, use drainOnlyDisciplineFor to classify each proof-relevant reachable blocked interval and KStructuralAssumptions.kDrain to discharge the theorem-scoped K.Drain audit condition and obtain either the Appendix B automatic-flush construction or the ordinary non-overlapped construction, use KStructuralAssumptions.toJTraceAssumptions for plain J.1, and use KStructuralAssumptions.toJCutpointAssumptions for normalized cutpoint transfer. The final K.1 theorem uses KAssumptions C W and accesses A5 as hK.A5_source_liveness. The JTrace projection carries the explicit initial/reset correspondence used by the reset-aware J.1 construction. The JCutpoint projection packages B4/B5 and J.AbsConf for the same reachable selected annotated-prefix/cutpoint/witness triple. A7 remains exactly the scheduling document's three-conjunct assumption; point-to-point WFG-boundary structure is projected from KElaborationPrerequisite.

```

def KStructuralAssumptions.jAbsConf
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W) : JAbsConf C W :=
hK.docNoA5.A7_determinism_witness_selection_cutpoint_abstraction.2.2

```

```

def KAssumptions.jAbsConf
(C : ComparisonInstance)

```

```

(W : WitnessBundle C)
(hK : KAssumptions C W) : JAbsConf C W :=
KStructuralAssumptions.jAbsConf C W hK.structural

```

```

def KStructuralAssumptions.drainOnlyDisciplineFor
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
(side : Side)
(P : Process)
(τ : SidePrefix C side)
(hRelevant : ProofRelevantDrainInterval C W side P τ) :
DrainOnlyBlockedFrontierDiscipline C side P τ :=
by
cases hK.blocking_interval_mode_complete side P τ hRelevant with
| inl hPipe =>
exact AutomaticFlush.toDrainOnlyBlockedFrontierDiscipline
C side P τ
(hK.automatic_flush side P τ hRelevant hPipe)
(finite_explicit_boundary_storage_of_implementation
C hK.implementation_for_imported_J)
| inr hOrd =>
exact NonOverlappedControl.toDrainOnlyBlockedFrontierDiscipline
C W side P τ hOrd hK.issue_commit_wf
(hK.ordinary_sync_boundary_policy side P τ hRelevant hOrd)
(finite_explicit_boundary_storage_of_implementation
C hK.implementation_for_imported_J)

```

```

def KAssumptions.drainOnlyDisciplineFor
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KAssumptions C W)
(side : Side)
(P : Process)
(τ : SidePrefix C side)
(hRelevant : ProofRelevantDrainInterval C W side P τ) :
DrainOnlyBlockedFrontierDiscipline C side P τ :=
KStructuralAssumptions.drainOnlyDisciplineFor
C W hK.structural side P τ hRelevant

```

-- The theorem-scoped K.Drain audit condition is derived from the same complete
-- per-interval projection consumed by fair drain normalization, K.2b, and K.1.

```

def KStructuralAssumptions.kDrain
(C : ComparisonInstance)

```

```

(W : WitnessBundle C)
(hK : KStructuralAssumptions C W) :
KDrain C W :=
by
  intro side P  $\tau$  hRelevant
  exact hK.drainOnlyDisciplineFor side P  $\tau$  hRelevant

```

```

def KAssumptions.kDrain
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KAssumptions C W) :
KDrain C W :=
KStructuralAssumptions.kDrain C W hK.structural

```

```

def KStructuralAssumptions.pointToPointAtWFGBoundary
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W) :
 $\forall$  c, ChannelCanContributeWFGEEdge C c  $\rightarrow$ 
PointToPointAtWFGEvaluationBoundary C c :=
fun c hc => (hK.elaboration_prerequisite c hc).2.1

```

```

def KAssumptions.pointToPointAtWFGBoundary
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KAssumptions C W) :
 $\forall$  c, ChannelCanContributeWFGEEdge C c  $\rightarrow$ 
PointToPointAtWFGEvaluationBoundary C c :=
KStructuralAssumptions.pointToPointAtWFGBoundary C W hK.structural

```

Standing B1 import for Appendix K. Since Appendix K uses the same main theorem stack as Appendix J, KStructuralAssumptions C W imports the standing RTL_B B1 field from the B-rule / JTraceAssumptions layer. K.1 uses B4/B5 only where normalization, fair drain normalization, epsilon-quiescent endpoint reasoning, or finite-stutter closure is required. The theorem-scoped drainOnlyDisciplineFor projection is the complete Definition R.16/K.Drain object: it supplies the blocked point, no new source-later dynamic launch, no withdrawal, non-frontier WFG-source exclusion, drain-only committed progress, finite non-replenishing drain, and sync-boundary policy. automatic_flush remains the pipelined constructor of that property, ordinary non-overlapped control supplies the other constructor, and A11/K ϵ remains an independent premise.

K lemmas:

The single canonical K.0 interface is K0_front_nextfront_classification above, together with

K0_front_correspondence_from_cutpoint. No second placeholder
K0_frontier_blocking_classification declaration is retained.

theorem K0a_rendezvous_endpoint_request_agreement

(C : ComparisonInstance)

(W : WitnessBundle C)

(hCorr : CutpointCorrespondence C W σ_{ref} ρ_{post})

(c : Channel)

(hB0 : capacity c = 0) :

Req_prod c σ_{ref} = Req_prod c ρ_{post} $\wedge$

Req_cons c σ_{ref} = Req_cons c ρ_{post}

K0a is not proved from matched observable history alone. For $B(c)=0$ rendezvous, ChannelEnabled depends on the raw complementary endpoint request in the same ε -quiescent cycle, and a raw request may persist as protocol state while not being the current frontier item. The proof therefore projects the required raw producer-side and consumer-side endpoint-request equality from hCorr.j_abs, using the J.AbsConf discharge for the same selected $(\tau_{\text{post}}, \tau_{\text{ref}}, \rho_{\text{post}}, \sigma_{\text{ref}}, W)$ triple.

-- Definitional unfolding lemma.

-- This is useful in proofs, but it is not the substantive Appendix K.1

-- channel-case characterization.

theorem InternalMPDeadlockShapeAt_unfold_at_quiescent

(C : ComparisonInstance)

(W : WitnessBundle C)

(hK : KStructuralAssumptions C W)

(σ : State)

(hQ : EpsQuiescent σ) :

InternalMPDeadlockShapeAt C .post $\sigma \leftrightarrow$

ExistsClosedDrainClosedWFGCycle C .post $\sigma :=$

by

-- Unfold InternalMPDeadlockShapeAt and use hQ for the ε -quiescence conjunct.

sorry

-- Channel-side cause exposed by Appendix K.1.

-- These cases are derived from ChannelDisabledItemSide plus E4 at the chosen

-- abstract Sys_ref / RTL_B boundary. For buffered channels, disabled Push means

-- no space and disabled Pop means no data. For rendezvous channels, disabled
 -- Push/Pop means the unique complementary endpoint request is absent in the
 -- same cycle.

inductive K1BlockedChannelCause

(C : ComparisonInstance)

(side : Side)

(σ : State)

(x : FrontierItem)

(q : DynMsgInstance)

(c : Channel) : Prop

| buffered_push_full :

q.channel = c $\rightarrow$

IsPushKind q.kind $\rightarrow$

0 < capacity C c $\rightarrow$

occAt c σ .cycle = capacity C c $\rightarrow$

K1BlockedChannelCause C side σ x q c

| buffered_pop_empty :

q.channel = c $\rightarrow$

IsPopKind q.kind $\rightarrow$

0 < capacity C c $\rightarrow$

occAt c σ .cycle = 0 $\rightarrow$

K1BlockedChannelCause C side σ x q c

| rendezvous_push_missing_pop :

q.channel = c $\rightarrow$

IsPushKind q.kind $\rightarrow$

capacity C c = 0 $\rightarrow$

$\neg$ ComplementaryPopRequestAt C side c σ $\rightarrow$

K1BlockedChannelCause C side σ x q c

| rendezvous_pop_missing_push :

q.channel = c $\rightarrow$

IsPopKind q.kind $\rightarrow$

capacity C c = 0 $\rightarrow$

$\neg$ ComplementaryPushRequestAt C side c σ $\rightarrow$

K1BlockedChannelCause C side σ x q c

-- Substantive Appendix K.1 characterization.

-- This is the theorem corresponding to the main document's Lemma K.1:

-- every process in the deadlocked SCC is blocked at its current blocking

```

-- message frontier, and every such disabled blocking channel operation has one
-- of the E4 channel-side causes below.
theorem K1_deadlock_characterization
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
( $\sigma$  : State)
(hReach : Reachable C.RTL_B  $\sigma$ )
(hDead : InternalMPDeadlockShapeAt C .post  $\sigma$ ) :
 $\exists$  D,
WFGTerminalSCC C .post D  $\sigma \wedge$ 
DrainClosed C .post D  $\sigma \wedge$ 
 $\forall$  P,
P  $\in$  D  $\rightarrow$ 
ProcessStalledAtFront C .post P  $\sigma \wedge$ 
 $\forall$  x q c,
x  $\in$  Front C .post P  $\sigma \rightarrow$ 
BlockingMessageFrontierItem x  $\rightarrow$ 
MsgOfFrontierItemSide C .post x = some q  $\rightarrow$ 
q.channel = c  $\rightarrow$ 
BoundaryReqItemSide C .post x  $\sigma \rightarrow$ 
ChannelDisabledItemSide C .post x  $\sigma \rightarrow$ 
K1BlockedChannelCause C .post  $\sigma$  x q c :=
by
-- Proof outline:
-- 1. Unfold InternalMPDeadlockShapeAt using
-- InternalMPDeadlockShapeAt_unfold_at_quiescent to obtain an  $\varepsilon$ -quiescent
-- closed drain-closed WFG SCC D.
-- 2. Use ProcessStalledAtFront to show each P  $\in$  D is stalled on its current
-- blocking message frontier, not on an internal  $\varepsilon$ -step.
-- 3. Use MsgOfFrontierItemSide to recover the owning dynamic message instance q.
-- 4. Use ChannelDisabledItemSide, hK.docNoA5.A1_R_and_B_rule_conformance, and
-- hK.e_rules_for_imported_J / E4 channel semantics at  $\sigma$ .cycle:
-- • disabled buffered Push_c implies  $\text{occAt } c \ \sigma.\text{cycle} = \text{capacity } C \ c$ ;
-- • disabled buffered Pop_c implies  $\text{occAt } c \ \sigma.\text{cycle} = 0$ ;
-- • disabled rendezvous Push_c implies the complementary Pop request is
-- absent in  $\sigma$ ;
-- • disabled rendezvous Pop_c implies the complementary Push request is

```

-- absent in σ .
 -- 5. Use `hK.elaboration_prerequisite` to ensure that `c` is the chosen explicit
 -- abstract boundary of the `Sys_ref` / `RTL_B` comparison after any required
 -- `Sys_B^elab` elaboration. Synchronization-boundary ramp-down / epoch-gating
 -- is therefore represented at explicit abstract channels, not as a hidden
 -- exception to E4 on an otherwise ordinary buffered channel.
 sorry

```
theorem K2_dependency_refinement
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
( $\sigma_{\text{ref}}$   $\rho_{\text{post}}$  : State)
(hCorr : CutpointCorrespondence C W  $\sigma_{\text{ref}}$   $\rho_{\text{post}}$ ) :
WFG C .ref  $\sigma_{\text{ref}}$  = WFG C .post  $\rho_{\text{post}}$ 
```

```
theorem K2a_drain_closure_transfer
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
(D : Finset Process)
( $\sigma_{\text{ref}}$   $\rho_{\text{post}}$  : State)
(hCorr : CutpointCorrespondence C W  $\sigma_{\text{ref}}$   $\rho_{\text{post}}$ )
(hDrainPost : DrainClosed C .post D  $\rho_{\text{post}}$ ) :
DrainClosed C .ref D  $\sigma_{\text{ref}}$ 
```

`K2_dependency_refinement` and `K2a_drain_closure_transfer` intentionally do not take separate `hQref` / `hQpost` or `hWFref` / `hWFpost` parameters. Their cutpoint quiescence facts are part of `hCorr`, via `hCorr.ref_quiescent` and `hCorr.post_quiescent`, and their cutpoint well-formedness facts are part of `hCorr`, via `hCorr.ref_wellformed` and `hCorr.post_wellformed`. Thus these theorems can invoke `K0_front_correspondence_from_cutpoint` without any extra side hypotheses.

K.2b serial-source dominance. The scheduling-rules v45 proof no longer tries to rebuild a fresh serial replay toward the same frontier. Lean should formalize K.2b as a well-founded reduction on the number of eager blocking issues in a finite liberal semantic reference prefix. Every theorem that creates or edits a run quantifies over `Prefix C.Sys_ref`. `RefPrefixBuckets` is used only to compute finite eager-event measures and endpoint streams from those semantic prefixes; the proof may not first construct an arbitrary `ExecutionBucketed` value and then assume that it lifts to a legal `Sys_ref` prefix.


```

structure DrainClosedDeadlockingRefPrefix
(C : ComparisonInstance)
(p_ref : Prefix C.Sys_ref)
( $\sigma$  : State) where
liberal : LiberalReferencePrefix C p_ref
terminal : TerminalCutpoint p_ref  $\sigma$ 
deadlocked : InternalMPDeadlockShapeAt C.ref  $\sigma$ 
component :
 $\exists$  D, ClosedDrainClosedWFGComponent C.ref D  $\sigma$ 

```

```

-- Prefix-suppression is not arbitrary deletion. It may remove or defer only a
-- suffix/order-ideal of each endpoint's issued blocking-operation stream. This
-- prefix property is the reason occupancy domination is true; naive "fewer
-- Pushes and fewer Pops" is not monotone.
theorem K2b_endpoint_prefix_suppression
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
(p_ref p_ref' : Prefix C.Sys_ref)
(hDef : DefersOneEagerIssueRefPrefix C p_ref p_ref') :
 $\forall$  c dir,
ChannelCanContributeWFGEdge C c  $\rightarrow$ 
RefEndpointIssuedBlockingStreamPrefix C p_ref' p_ref c dir :=
by
-- Project both semantic prefixes through RefPrefixBuckets, then use A6
-- point-to-point ownership, same-endpoint source order, E3 no-reverse
-- issue/commit order and hDef.bucket_deferral. The hDef.output_exec_ref field
-- is the separately constructed admissibility evidence for p_ref'; this theorem
-- projects the endpoint-prefix consequence and does not reconstruct Exec_ref.
sorry

```

```

theorem K2b_joint_value_control_consistency
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
(p_ref p_ref' : Prefix C.Sys_ref)
(hPref :  $\forall$  c dir,
  RefEndpointIssuedBlockingStreamPrefix C p_ref' p_ref c dir)
(hRetained : RetainedCommittedTransferPrefix
  C.ref
  (RefPrefixBuckets C p_ref')
  (RefPrefixBuckets C p_ref)) :
JointSourceReplayAgreementUpToRetainedCommits

```

```

C W
(RefPrefixBuckets C  $\rho_{ref}$ )
(RefPrefixBuckets C  $\rho_{ref}$ ) :=
by
-- This is a simultaneous source/dataflow theorem, not merely a per-process
-- deterministic replay lemma. Fixed stimulus, fixed NB-CF / witness-selected NB
-- outcomes, matched retained Push payloads, and matched retained Pop-return
-- values imply the same operation identities, guards, payloads, and branches up
-- to the retained commit prefix.
sorry

theorem K2b_occupancy_disabledness_domination
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
( $\rho_{ref}$   $\rho_{ref}'$  : Prefix C.Sys_ref)
( $\sigma$   $\sigma'$  : State)
(x : FrontierItem)
(hPrefix :  $\forall c \text{ dir},$ 
  ChannelCanContributeWFGEEdge C c  $\rightarrow$ 
  RefEndpointIssuedBlockingStreamPrefix C  $\rho_{ref}'$   $\rho_{ref}$  c dir)
(hFront : CorrespondingBlockedFrontierItem
  C .ref
  (RefPrefixBuckets C  $\rho_{ref}$ )
  (RefPrefixBuckets C  $\rho_{ref}'$ )
   $\sigma$   $\sigma'$  x)
(hDisabled : ChannelDisabledItemSide C .ref x  $\sigma$ ) :
ChannelDisabledItemSide C .ref x  $\sigma'$  :=
by
-- Evaluate on the explicit abstract channel where ChannelDisabled is defined.
-- For a blocked Pop, endpoint-prefix suppression preserves the relevant empty
-- condition; for a blocked Push, it preserves the relevant full condition; for
-- rendezvous, suppressing eager issue cannot create the missing unique
-- complementary endpoint request. Sys_B^elab cases lift only when the introduced
-- boundary channel is point-to-point at the WFG evaluation boundary.
sorry

-- This monotonicity fact prevents the eager-reduction proof from asserting,
-- without proof, that serialization cannot dissolve the deadlock. The two run
-- arguments are semantic Sys_ref prefixes; their bucket projections expose the
-- occupancy edit, while admissibility and fixed-stimulus facts remain attached
-- to the prefixes themselves.
theorem K2b_no_third_party_enable_under_eager_deferral
(C : ComparisonInstance)

```

```

(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
(ρ_ref ρ_ref' : Prefix C.Sys_ref)
(σ σ' : State)
(D : Finset Process)
(hDef : DefersOneEagerIssueRefPrefix C ρ_ref ρ_ref')
(hPrefix : ∀ c dir,
  ChannelCanContributeWFGEdge C c →
  RefEndpointIssuedBlockingStreamPrefix C ρ_ref' ρ_ref c dir)
(hPTP : ∀ c,
  ChannelCanContributeWFGEdge C c →
  PointToPointAtWFGEvaluationBoundary C c)
(hComponent : ClosedDrainClosedWFGComponent C .ref D σ) :
NoThirdPartyEnablementOrOutgoingEscape
  C .ref
  (RefPrefixBuckets C ρ_ref)
  (RefPrefixBuckets C ρ_ref')
  σ σ' D :=
by
-- Split on whether the deferred eager operation is still pending at σ or
-- committed in ρ_ref. Pending eager requests are non-frontier protocol state
-- and are inert under Kε. If it committed, point-to-point plus endpoint-prefix
-- occupancy domination shows that withholding the early commit cannot enable a
-- third process or create an outgoing escape from the canonical component.
sorry

```

```

theorem fair_drain_normalize_closed_component
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
(side : Side)
(ρ : Prefix (SystemOfSide C side))
(σ : State)
(D : Finset Process)
(hTerm : TerminalCutpoint ρ σ)
(hClosed : ClosedWFGComponent C side D σ)
(hBlocked : ∀ P, P ∈ D → FrontierBlocked C side P σ)
(hDrainOnly :
  ∀ P, P ∈ D →
  DrainOnlyBlockedFrontierDiscipline C side P ρ)
(hNoEscape : NoThirdPartyEnablementOrOutgoingEscapeAt C side ρ σ D) :
∃ ρ' σ',
  ExtendsFinitePrefix C side ρ' ρ ∧
  TerminalCutpoint ρ' σ' ∧

```

ReachableByFiniteDrainNormalization C side D $\sigma \sigma' \wedge$
 EpsQuiescent $\sigma' \wedge$
 DrainClosed C side D $\sigma' \wedge$
 $\exists D', \text{ClosedDrainClosedWFGComponent } C \text{ side } D' \sigma' :=$
 by
 -- This is the side-parametric finite fair drain-normalization construction
 -- referenced by scheduling-rules K.2b and by the “Exec_post applicability at a
 -- deadlock cutpoint” part of Theorem K.1. hDrainOnly is the load-bearing carrier:
 -- it applies both to pipelined automatic-flush intervals and to ordinary
 -- non-overlapped liberal intervals. It allows only already-issued non-frontier
 -- drain commits, interleaves fair outside-D/environment progress, and issues no
 -- new work across the blocked frontier.
 sorry

 -- Once disabledness and no-third-party-enable are known, invoke
 -- fair_drain_normalize_closed_component; the edited semantic prefix then has a
 -- canonical terminal closed drain-closed WFG component.
 theorem K2b_terminal_scc_persists_after_eager_deferral
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (hK : KStructuralAssumptions C W)
 (p_ref p_ref' : Prefix C.Sys_ref)
 ($\sigma \sigma'$: State)
 (D : Finset Process)
 (hDef : DefersOneEagerIssueRefPrefix C p_ref p_ref')
 (hOld : ClosedDrainClosedWFGComponent C .ref D σ)
 (hNoEscape : NoThirdPartyEnablementOrOutgoingEscape
 C .ref
 (RefPrefixBuckets C p_ref)
 (RefPrefixBuckets C p_ref')
 $\sigma \sigma' D$)
 (hBlocked' : $\forall P, P \in D \rightarrow \text{FrontierBlocked } C \text{ .ref } P \sigma'$)
 (hDrain' : DrainClosed C .ref D σ') :
 $\exists D', \text{ClosedDrainClosedWFGComponent } C \text{ .ref } D' \sigma' :=$
 by
 -- Follow WFG edges from blocked processes in D at σ' . Point-to-point ensures the
 -- complementary owner is unique and no third-party arbitration can be newly
 -- enabled by removing the eager operation. Finiteness of the process set yields
 -- a nonempty terminal SCC. Drain closure rules out downstream drain escapes.
 sorry

 theorem K2b_construct_admissible_eager_deferral
 (C : ComparisonInstance)
 (W : WitnessBundle C)

```

(hK : KStructuralAssumptions C W)
(ρ_ref : Prefix C.Sys_ref)
(σ : State)
(hDead : DrainClosedDeadlockingRefPrefix C ρ_ref σ)
(hEager : 0 < RefEagerIssueCount C ρ_ref) :
∃ ρ_ref',
DefersOneEagerIssueRefPrefix C ρ_ref ρ_ref' :=
by
-- This is the explicit Exec_ref-preservation obligation. The construction also
-- fills hDef.output_drain_only for every blocked interval of the edited semantic
-- prefix, using the pipelined AutomaticFlush constructor or the ordinary
-- non-overlapped-control constructor as applicable. Choose the eager issue
-- to defer, construct the edited finite Sys_ref prefix under the same fixed
-- stimulus/witness/capacity/boundary instance, and then construct an infinite
-- RefAdmissible extension of that edited prefix. The extension proof must
-- re-establish the applicable B2/B3 fairness and progress conditions; it is not
-- obtained merely by editing RefPrefixBuckets or by asserting that the output is
-- a semantic Prefix C.Sys_ref. The output_issue_order and output_no_reverse
-- fields establish the remaining LiberalReferencePrefix conjuncts.
sorry

```

```

theorem K2b_one_step_eager_reduction
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
(ρ_ref : Prefix C.Sys_ref)
(σ : State)
(hDead : DrainClosedDeadlockingRefPrefix C ρ_ref σ)
(hEager : 0 < RefEagerIssueCount C ρ_ref) :
∃ ρ_ref' σ',
DefersOneEagerIssueRefPrefix C ρ_ref ρ_ref' ∧
DrainClosedDeadlockingRefPrefix C ρ_ref' σ' ∧
RefEagerIssueCount C ρ_ref' < RefEagerIssueCount C ρ_ref :=
by
-- First project the generalized DrainOnlyBlockedFrontierDiscipline from hK for
-- the input reference prefix, then invoke K2b_construct_admissible_eager_deferral
-- and use hDef.output_drain_only for the edited prefix. This is where the
-- edited finite prefix and its fair/progress-satisfying RefAdmissible extension
-- are constructed. Then use the record's bucket_deferral field. Endpoint-prefix
-- suppression and occupancy-disabledness domination preserve the
-- relevant blockers; no-third-party-enable and fair_drain_normalize_closed_component,
-- supplied with the generalized drain-only evidence, recover
-- a canonical component at an ε-quiescent drain-closed cutpoint.
sorry

```

```

theorem K2b_minimal_eager_deadlock_is_serial
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
(ρ_ref : Prefix C.Sys_ref)
(σ : State)
(hDead : DrainClosedDeadlockingRefPrefix C ρ_ref σ)
(hMin : ∀ ρ_ref' σ',
  DrainClosedDeadlockingRefPrefix C ρ_ref' σ' →
  RefEagerIssueCount C ρ_ref ≤ RefEagerIssueCount C ρ_ref') :
ConservativeSerialReferencePrefix C ρ_ref :=
by
-- If RefEagerIssueCount is positive, K2b_one_step_eager_reduction constructs an
-- actual semantic prefix with a smaller count, contradicting minimality. Then
-- zero_ref_eager_iff_serial_reference_prefix and hDead.liberal yield seriality.
sorry

```

```

theorem K2b_serial_source_dominance_for_internal_deadlock
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
(ρ_ref : Prefix C.Sys_ref)
(σ_ref : State)
(hLiberal : LiberalReferencePrefix C ρ_ref)
(hTerm : TerminalCutpoint ρ_ref σ_ref)
(hDead : InternalMPDeadlockShapeAt C .ref σ_ref) :
∃ ρ_ser σ_ser,
ConservativeSerialReferencePrefix C ρ_ser ∧
TerminalCutpoint ρ_ser σ_ser ∧
InternalMPDeadlockShapeAt C .ref σ_ser :=
by
-- Package the selected prefix as DrainClosedDeadlockingRefPrefix, extracting its
-- canonical component from hDead. Among all semantic finite liberal deadlocking
-- prefixes reachable under the same fixed stimulus, witness choices, capacities,
-- and explicit boundaries, choose one with minimal RefEagerIssueCount. The
-- one-step theorem produces another Prefix C.Sys_ref whenever the count is
-- positive, so the minimum is zero. The zero-count theorem yields a
-- ConservativeSerialReferencePrefix, which is returned directly to A5.
sorry

```

Implementation note. K.2b is now the single proof bridge that replaces the former scheduler-robust A5 quantification. Its public theorem and every proof-producing reduction lemma use Prefix C.Sys_ref. Bucketed execution values

are derived only by RefPrefixBuckets to compute eager counts, endpoint streams, and local occupancy facts. Consequently the zero-eager reduction result already is the semantic p_{ser} returned to A5; no InfiniteExecutionBucketed substitution or hidden realizability step is allowed. Exec_ref membership is nevertheless a real proof obligation: K2b_construct_admissible_eager_deferral must construct the RefAdmissible infinite extension for every edited finite prefix, and the later K.2b lemmas consume that explicit evidence rather than assuming it automatically. Exec membership for K.1 should distinguish three cases.

First, for a globally fixed-point cutpoint, B6 idle-stutter may be used to build an infinite execution extension:

```
theorem fixedpoint_cutpoint_in_Exec_post
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KStructuralAssumptions C W)
( $\rho_{\text{post}}$  : State)
(hReach : Reachable C.RTL_B  $\rho_{\text{post}}$ )
(hFP : FixedPointState C .post  $\rho_{\text{post}}$ )
(hIdle : hK.docNoA5.A1_R_and_B_rule_conformance.2.B6_idleStutterTimeDivergence) :
 $\exists \tau_{\text{post}}$ ,
TerminalCutpoint  $\tau_{\text{post}}$   $\rho_{\text{post}}$   $\wedge$ 
Exec_post C  $\tau_{\text{post}}$ 
```

Second, a partial internal-MP deadlock is not justified by B6 idle-stutter alone, because processes outside D or environment endpoints may still have enabled observable actions. The v42 proof supplies the missing bridge by fair drain-normalization:

```
def DrainNormalizationStep
(C : ComparisonInstance)
(side : Side)
(D : Finset Process)
( $\sigma \sigma'$  : State) : Prop :=
 $\exists P q$ ,
 $P \in D \wedge$ 
OwnsMessageEndpoint P q  $\wedge$ 
Blocking q  $\wedge$ 
NonFrontierBlockingDrainCandidate C side P  $\sigma$  q  $\wedge$ 
BoundaryReqSide C side q  $\sigma \wedge$ 
ChannelEnabledSide C side q  $\sigma \wedge$ 
CommitDrainOnly C side q  $\sigma \sigma' \wedge$ 
NoNewWorkCrossesDisabledInternalFrontier C side D  $\sigma \sigma'$ 
```

Thus drain-normalization is intentionally stated over already-issued downstream blocking requests, not over the current `BlockingMsgNextFront_P( $\sigma$ )` slice. At ε -quiescent well-formed cutpoints, K.0 identifies `Front_P( $\sigma$ )` with the `BoundaryReq-enabled` blocking `NextFront` slice; drain-only progress is for already-issued downstream requests outside that current minimal frontier, while `NoNewWorkCrossesDisabledInternalFrontier` prevents such commits from advancing new source work past the blocked frontier.

```
def DrainNormalizedDeadlockCutpoint
```

```
(C : ComparisonInstance)
```

```
(side : Side)
```

```
(D : Finset Process)
```

```
( $\sigma_0 \sigma'$  : State) : Prop :=
```

```
ReachableByFiniteDrainNormalization C side D  $\sigma_0 \sigma'$   $\wedge$ 
```

```
EpsQuiescent  $\sigma'$   $\wedge$ 
```

```
DrainClosed C side D  $\sigma'$   $\wedge$ 
```

```
 $\forall P,$ 
```

```
P  $\in$  D  $\rightarrow$  ProcessStalledAtFront C side P  $\sigma'$ 
```

```
theorem deadlock_cutpoint_in_Exec_post_via_fair_drain_normalization
```

```
(C : ComparisonInstance)
```

```
(W : WitnessBundle C)
```

```
(hK : KStructuralAssumptions C W)
```

```
( $\rho_{\text{post}}$  : State)
```

```
(hReach : Reachable C.RTL_B  $\rho_{\text{post}}$ )
```

```
(hDead : InternalMPDeadlockShapeAt C .post  $\rho_{\text{post}}$ ) :
```

```
 $\exists \tau_{\text{post}} \rho_{\text{post}}' D,$ 
```

```
TerminalCutpoint  $\tau_{\text{post}} \rho_{\text{post}}' \wedge$ 
```

```
Exec_post C  $\tau_{\text{post}} \wedge$ 
```

```
DrainNormalizedDeadlockCutpoint C .post D  $\rho_{\text{post}} \rho_{\text{post}}' \wedge$ 
```

```
InternalMPDeadlockShapeAt C .post  $\rho_{\text{post}}' :=$ 
```

```
by
```

```
-- Use K1_deadlock_characterization to obtain the closed SCC D. B3 cannot force
```

```
-- progress of D-owned internal frontier blockers without contradicting
```

```
-- disabled-frontier, drain-closure, or WFG closure; rendezvous blockers are
```

```
-- handled by absence of the same-cycle complementary endpoint request.
```

```
-- Environment-facing frontier items are not WFG edges and are handled as
```


- ordinary fair continuation, not as WFG-closure contradictions. For
- already-issued D-owned non-frontier requests, drain-closure only says they are
- not ChannelEnabled at the current cutpoint; if outside-D or environment
- progress later makes one continuously serviceable, allow it to commit as
- drain-only progress. The generalized drain-only blocked-frontier discipline (constructed from automatic flush for pipelined intervals or ordinary non-overlapped control otherwise), finite explicit capacities, finite
- ε -depth, and no new work across the disabled internal frontier imply only
- finitely many such D-owned drain commits before a new ε -quiescent
- drain-closed cutpoint is reached, unless a frontier item commits and the
- proof uses the corresponding reachable cutpoint according to the v45 case
- split. Interleave this finite drain-normalization with fair outside-D and
- environment continuation to obtain Exec_post.

sorry

Third, the main K theorem can be exposed in both execution-scoped and reachable forms. The execution-scoped post-side form is convenient internally; the reachable form is what Appendix K states after applying the fair drain-normalization bridge. The source-side liveness premise used to contradict a reference deadlock is now serial, not liberal/witness-relative.

```
def InternalMPDeadlockFreeOnExecPost
(C : ComparisonInstance) : Prop :=
  ∀ τ_post p_post,
  Exec_post C τ_post →
  TerminalCutpoint τ_post p_post →
  ¬ InternalMPDeadlockShapeAt C .post p_post
```

```
theorem K1_liveness_preservation_on_exec
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KAssumptions C W) :
  InternalMPDeadlockFreeOnExecPost C
```

```
def InternalMPDeadlockFreeReachablePost
(C : ComparisonInstance) : Prop :=
  ∀ p_post,
  ¬ ReachableInternalMPDeadlockAt C .post p_post
```

Main K theorem:

theorem K1_liveness_preservation

(C : ComparisonInstance)

(W : WitnessBundle C)

(hK : KAssumptions C W) :

InternalMPDeadlockFreeReachablePost C

Proof dependency note. Prove the reachable post-side conclusion by contradiction.

Assume `ReachableInternalMPDeadlockAt C .post p_post` and apply

`deadlock_cutpoint_in_Exec_post_via_fair_drain_normalization` to obtain one specific

`Exec_post` prefix τ_{post} ending at the ε -quiescent drain-normalized cutpoint ρ_{post} !

Import Appendix J with `hJCut := KAssumptions.toJCutpointAssumptions C W hK` and invoke

`J1_selected_matched_cutpoint` on that exact τ_{post} . The B4/B5 normalization premises

and the selected `J.AbsConf` evidence are packaged in `hJCut` rather than supplied to the

plain trace theorem. Destruct the resulting package `M : SelectedMatchedCutpoint C W` and

retain the equality $M.\tau_{\text{post}} = \tau_{\text{post}}$.

All subsequent facts are projections from `M`. The exact `J.1` pair is named by `M. $\tau_{\text{post}}$` ,

`M. $\tau_{\text{ref}}$` , `M.selected`, `M.post_exec`, `M.ref_exec`, and `M.compatible`. The distinct normalized

representatives are named by `M. $\tau_{\text{bar\_post}}$` , `M. $\tau_{\text{bar\_ref}}$` , `M.normalized_post_exec`,

`M.normalized_ref_exec`, `M.normalized`, `M.post_terminal`, `M.ref_terminal`, `M.ref_liberal`, and

`M.correspondence`. Consequently `J.AbsConf`, the selected witness, and the cutpoint

transfer cannot silently refer to a different post prefix, reference prefix, or normalized pair.

Use `K2_dependency_refinement`, `K2a_drain_closure_transfer`, and

`K0a_rendezvous_endpoint_request_agreement` with `M.correspondence` to transfer the

closed drain-closed internal WFG deadlock from `M. $\rho_{\text{post}}$` to `M. $\sigma_{\text{ref}}$` .

Apply `K2b_serial_source_dominance_for_internal_deadlock` to `M. $\tau_{\text{bar\_ref}}$` , `M. $\sigma_{\text{ref}}$` ,

`M.ref_liberal`, `M.ref_terminal`, and the transferred deadlock. It returns ρ_{ser} and σ_{ser}

satisfying `ConservativeSerialReferencePrefix C  $\rho_{\text{ser}}$` , `TerminalCutpoint  $\rho_{\text{ser}}$   $\sigma_{\text{ser}}$` , and

`InternalMPDeadlockShapeAt C .ref  $\sigma_{\text{ser}}$` . These three facts contradict

`hK.A5_source_liveness` directly. The proof never converts the selected finite `J.1` prefix into

an `InfiniteExecutionBucketed` argument. No additional fair-extension assumption is

introduced at the `K.1` contradiction step: `K.2b` returns a `ConservativeSerialReferencePrefix`

whose `LiberalReferencePrefix` component already contains `Exec_ref`, and

`K2b_construct_admissible_eager_deferral` is the explicit theorem that constructs and

proves the required `RefAdmissible` extension after each eager deferral.

This is intentionally one-way. Do not state:

InternalMPDeadlockFreeOnSerialPrefixesRef C <->

InternalMPDeadlockFreeReachablePost C

A5 separation and K.1 deduplication. A5 is the finite-prefix predicate

InternalMPDeadlockFreeOnSerialPrefixesRef C, matching the scheduling-rules formulation over reachable cutpoints of admissible conservative-serial source prefixes. It is deliberately not part of KStructuralAssumptions, because K.2b and the structural K.0/K.2 lemmas are used to reach the serial-source deadlocking prefix before A5 is invoked. The final KAssumptions wrapper adds exactly one A5 field, hK.A5_source_liveness.

Consequently K1_liveness_preservation and K1_liveness_preservation_on_exec no longer take a second explicit source-liveness premise; they consume hK.A5_source_liveness at the final contradiction step. InternalMPDeadlockFreeOnSerialExecRef remains only a definitional convenience alias for the finite-prefix A5 predicate; it introduces no second execution domain. This removes duplicate assumptions and prevents circular dependence of K.2b on A5. It does not erase the admissibility obligation: because ConservativeSerialReferencePrefix contains LiberalReferencePrefix and therefore Exec_ref, the fair/progress-satisfying extension of each edited prefix must be constructed explicitly inside K2b_construct_admissible_eager_deferral before the serial prefix can be returned to A5.

19. Appendix R facade

Appendix R should be formalized last as aliases/wrappers over earlier theorems.

Use Lean names that avoid the doc's R.2 collision:

theorem R_lemma1_front_nextfront_classification := ...

theorem R_theorem1_prefix_matching := ...

theorem R_corollary1_cutpoint_correspondence := ...

theorem R_corollary1a_rendezvous_endpoint_request_agreement := ...

theorem R_lemma2_dependency_refinement := K2_dependency_refinement

theorem R_theorem2_closed_cycle_and_one_way_drain_transfer := ...

Appendix R uses per-type numbering. The Lean identifiers include the item type and therefore remain unambiguous: the closed-cycle/drain-transfer result is the scheduling document's Theorem R.2 and is named

R_theorem2_closed_cycle_and_one_way_drain_transfer. Do not renumber it to R.3 merely to avoid cross-type reuse of the numeral R.2.

Do not call cutpoint correspondence R1_cutpoint_correspondence; in the doc structure, it corresponds to a corollary, not Theorem R.1.

Appendix R cutpoint correspondence should expose the v42 naming: the post-side terminal cutpoint is ρ_{post} , the relation $\sigma_{\text{ref}} \equiv \rho_{\text{post}}$ is the compact notation for the $J.\text{Abs}/J.\text{AbsConf}$ source-slice correspondence $\rho_{\text{post}} \Downarrow_{\{ \text{Abs}, J, w \}} \sigma_{\text{ref}}$, and it is not equality of arbitrary RTL microstate.

20. What remains as hypotheses

These remain theorem parameters or design/tool obligations:

1. R-rules R1, R2, R2C, R3, R4, R5.
2. E-rules E1-E5, with the v42 RequiredJOrder restriction made explicit: E2 is alignment only, distinct-interface E3 source order is no-reverse legality rather than a positive strict edge, atomic multi-endpoint operations are single observable units rather than internal order edges, and E5 contributes strict RequiredJOrder only on its suff_S side. The pref_S side of E5 remains a non-strict issue/commit legality and sync-boundary well-formedness obligation: $q \in \text{pref_S}(s)$ gives $\text{issue}(q) \leq \text{clk}(s)$, and for committed blocking q the commit cycle is $\leq \text{clk}(s)$, with equality allowed. The sanity package RequiredJOrderWF is not a new external assumption; it should be proved inside Lean as RequiredJOrderWF_from_concrete_generators once the concrete E1-E5, IssueCommitWF, ObservableUnitWF, R2C unit, and elaboration/projection definitions are instantiated.
- 2a. Before substantive proofs, a signature-skeleton file shall type-check the final carriers and theorem interfaces for plain J.1, selected normalization/cutpoint transfer, and K.1. A theorem consumer whose premises cannot construct its carrier is a blocking architecture defect, not a proof hole.
3. B-rule semantic obligations are theorem-scoped rather than imported through one undifferentiated J bundle. TraceBRules carries B1, B2, B3, and B6 for the plain Appendix J trace theorem and its admissible execution construction. NormalizationBRules carries B4 and B5 and is imported only by JCutpointAssumptions and by Appendix K, where ε -normalization, ε -quiescent endpoint reasoning, or finite-stutter closure is actually used. The full BRules bundle remains the document-numbered Appendix K A1 component. B1 is the standing RTL_B deterministic- Σ -trace assumption PostDeterministicSigmaTrace C for the fixed comparison instance; it is not a source-side determinism premise. B2/B3 are encoded temporally, and B5 is limited to genuine quiescence closure under the no-external-change qualification, including bounded_system_quiescence. B2 must encode the split in which ChannelEnabled means eligibility, selection is the fairness event, and the

selected enabled action commits in the selected cycle. B3 remains the channel-progress obligation that can discharge persistent full/empty/rendezvous-blocked premises before a pending transfer becomes ChannelEnabled. The max and least-upper-bound facts for SystemQuiescenceBound C are Lean lemmas proved from Finset.sup, not hypotheses. The Appendix K A2, A3, and A4 names are projections from the B4, B2, and B3 fields of A1's full BRules component. Appendix K A5 remains the separate finite-prefix serial-source liveness premise InternalMPDeadlockFreeOnSerialPrefixesRef C; the witness-relative reference deadlock is connected to it only by K2b_serial_source_dominance_for_internal_deadlock.

4. Well-formedness of the selected comparison instance C.WellFormed.

4a. InitialAndResetCorrespondence C W, including process-control, source-visible-value, explicit channel-state, request/synchronization-state, anchor/direct-input, reset-empty occupancy, and no-stale-state facts at startup and at every relevant dynamic-reset epoch. This is the explicit Appendix T.2.1 induction base, not a hidden component of C.WellFormed.

5. ImplementationAssumptions C, including E4 channel-realization conformance and the mechanically carried per-channel field $\forall c$, BackAnnotatedExplicitAbstractChannel C $c \rightarrow$ BFaithful C c , plus the other named implementation obligations such as I.A0.

6. Appendix C Issue/Commit well-formedness and request-attribution discipline IssueCommitWF C W.

7. Reactive-environment well-formedness ReactiveEnvironmentWF C.env: the semantic carrier may retain cycle and empty-bucket information, but ordinary Appendix I/J theorem instances must be cross-cycle Σ -causal, so equal prior global Σ -visible histories produce equal responses. Raw cycle/cadence dependence not represented in Σ must be routed through the Appendix D/L latency-sensitive mechanisms.

8. Witness compatibility WC C W for the full proof-internal C.Sigma theorem, obtained from SideConditionDischargeRecord.wc through WCDischarge.proof. Lemma L.2 snooping, Corollary L.3 global NB-CF plus its identity-witness and local-protocol premises, or a complete mixed per-site partition may provide the proof-producing provenance, but none replaces the actual WC premise.

8a. For the Appendix L Note 1 Σ _DUT-projected safety theorem only, the weaker SigmaDUTSafetyWitnessObligation C W may replace WC. This obligation is discharged choice-by-choice: each Σ _DUT-relevant ε -modeled local timing choice is either covered by ordinary snooping/witness machinery or belongs to a certified SigmaDUTSafetyInertInterface. This does not discharge full WC and does not discharge Appendix K WFG-inertness / K_e obligations.

8b. J.AbsConf C W for the selected RTL_B implementation, abstract boundary, witness, and normalized cutpoints. The theorem-scoped audit makes this a RequiredDischarge for j1Cutpoint and k1Liveness scopes and permits inapplicability only for a j1Trace-only scope. This is a tool/RTL-flow compliance obligation alongside E4 channel-realization conformance and B-faithfulness; it is not proved by deterministic replay alone.

8c. B-faithfulness of each back-annotated capacity B(c), as a named tool/RTL-flow compliance hypothesis BFaithful C c per explicit abstract channel: (i) boundary completeness — every Σ -visible transfer of c in RTL_B crosses the agreed abstract boundary as a committed Push_c or Pop_c, with no hidden side entrance or exit; (ii) exact

boundary behavior — the concrete boundary permits exactly the Push/Pop actions permitted by E4, with occupancy bounded by $B(c)$, no hidden storage or credit, and no hidden grant, arbitration, or backpressure condition unless that condition is represented by explicit $\text{Sys_B}^{\text{elab}}$ boundary structure; (iii) ε -opacity of internal movement — internal movement of data within the realized channel between commit points is ε and contributes no Σ -labels and no WFG causes; and (iv) an evidence obligation — each $B(c)$ is supported by a trusted back-annotation algorithm, a per-channel certificate, a structural RTL check, an interface monitor, or a formal flow, recorded in the theorem-scoped `SideConditionDischargeRecord C W` scope of Section 1.1 and carried by `ImplementationAssumptions.bFaithful` into $I/J/K$ theorem instances. Like 8b, this is an obligation on the tool/RTL flow and the chosen comparison instance; it is not derived by E4, Proposition J.0, or Theorem J.1, all of which assume it.

9. Direct-input contracts.

10. Array access mapping rules.

11. R2C fixed-point well-formedness $\text{R2C_WF } C$ when combinational-process signal IO is in scope, including selected `Obs(C_comb)` slices, stutter-canonical unit emission, component-local occurrence-ordinal matching, and either a single anchoring domain or a confluence proof for multi-input R2C components; cycle-locked R2C components must be declared explicitly.

12. $K.\text{Drain}$ / Definition R.16 for every proof-relevant blocked interval, with Appendix B `AutomaticFlush` as the pipelined constructor and ordinary non-overlapped control as the other constructor: (i) the selected point is a fully disabled blocking frontier; (ii) no new source-later dynamic operation is launched after that blocked point; (iii) already asserted blocking requests are not withdrawn before commit and Push payloads remain stable; (iv) an already asserted non-frontier request is not used as a WFG source; (v) any observable commit owned by the blocked process is attributable only to already-issued downstream drainable work; (vi) the already-issued drain set is finite and non-replenishing; and (vii) proof-relevant sync-boundary capacity withdrawal is absent in ordinary Sys_B or represented through explicit $\text{Sys_B}^{\text{elab}}$ gate structure. Full `Remain/NextFront` frontier minimality is not assumed here; `K0_front_nextfront_classification` proves it.

13. Elaboration package E, when $\text{Sys_ref} = \text{Sys_B}^{\text{elab}}(E)$.

14. K elaboration prerequisite `KElaborationPrerequisite C`: every WFG-relevant channel is represented at the chosen abstract Sys_ref / RTL_B comparison boundary; hidden sync-boundary gating is not allowed at ordinary buffered boundaries; and sync-boundary withholding that affects WFG reasoning is represented through explicit $\text{Sys_B}^{\text{elab}}$ channels.

15. Appendix K structural assumptions not already covered above: point-to-point channels, point-to-point WFG evaluation boundaries for any introduced $\text{Sys_B}^{\text{elab}}$ channels, `SyncChannel/barrier` well-formedness, single-transfer-per-cycle endpoint constraints, reset-empty FIFO/accounted initial occupancy, and the independent WFG-inert $\varepsilon / K\varepsilon$ premise. $K\varepsilon$ is not folded into $K.\text{Drain}$.

16. Source-level internal message-passing deadlock freedom for K.1 under A5's conservative serial-blocking source semantics:
`InternalMPDeadlockFreeOnSerialPrefixesRef C`.

This premise quantifies over finite admissible serial prefixes that are also ordinary liberal/R4/E3/E5/no-reverse reference prefixes; it is not a scheduler-robust liveness premise over all source scheduling choices and it is not derived from local R-rules or from RequiredJOrderWF.

K2b_serial_source_dominance_for_internal_deadlock

is the bridge from the witness-relative liberal reference deadlock to this serial A5 premise.

17. Fair drain-normalization for K.1 Exec_post applicability: a reachable partial internal-MP deadlock cutpoint is not extended by pure idle-stutter unless it is a global fixed point; the proof must construct a fair continuation that allows finite D-owned non-frontier drain-only commits when B2/B3 later makes them continuously serviceable, while preserving the disabled internal frontier used for the WFG contradiction.

18. The side-condition discharge record of Section 1.1. This is audit bookkeeping, not an additional semantic assumption. Its WC entry proves the actual WC C W premise and records whether that proof came from NB-CF, Appendix L snooping, direct witness construction, or complete mixed per-site coverage; NB-CF and snooping are provenance routes rather than separate audited theorem propositions. The record separately covers Sys_B^{elab}(E), per-channel B-faithfulness, J.AbsConf, K.Drain, and K ϵ according to theorem scope; K.Drain and K ϵ are independently required for k1Liveness. K.Drain is typed as KDrain C W over ProofRelevantDrainInterval; UniversalDrainConformance is only a sufficient stronger certificate profile. Each constructor carries evidence of the declared method; inapplicability requires a NotRequired proof, and required per-channel B-faithfulness has no inapplicable constructor. Every theorem instance claimed against the document should be accompanied by a fully populated SideConditionDischargeRecord.

21. What should be proved inside Lean

Lean should prove:

- bucketed execution infrastructure and the cycle-indexed ReactiveEnvironment carrier, together with ReactiveEnvironmentWF, SamePriorSigmaVisibleHistory, and reactive_environment_replay_from_same_sigma_history, proving that ordinary Appendix I/J replay is cross-cycle Σ -causal and insensitive to elapsed empty buckets not represented in Σ ;
- PrefixAnnotations / AnnotatedPrefix well-formedness and construction from admissible prefixes, with compatibility typed only over the witness-relative annotated carriers;
- InitialComparisonCorrespondence and ResetEpochCorrespondence projection lemmas, including matched reset occurrences, two-sided reset-empty occupancy, finite MatchedResetEpochSplice construction, fresh post-reset dynamic identities, and re-establishment of JConstructionInvariant;
- WCDischarge.proof and theorem-scope discharge facts, including a blocking-core

direct/identity witness construction available before the B1 trace theorem, plus independent K.Drain and K ϵ scope enforcement for Appendix K;

- finite ObservableUnitsOf extraction and atomic grouping/non-splitting facts;
- temporal lemmas for AlwaysFrom / EventuallyFrom;
- BRules.toTraceBRules and BRules.toNormalizationBRules;
- explicit assumption constructors JTraceAssumptions.toConstructionAssumptions, JCutpointAssumptions.asJTraceAssumptions, KStructuralAssumptions.toJTraceAssumptions, and KStructuralAssumptions.toJCutpointAssumptions;
- dependency-audit theorems showing that J0_pairwise_composition, J0_pairwise_composition_DUT, J1_fixed_RTL_time_selected_witness_construction, and J1_post_to_ref consume TraceBRules but no NormalizationBRules/B4/B5 premise;
- source enablement and actual finite extension lemmas for every unit class: anchored signals, R2C units, buffered Push/Pop, rendezvous, synchronization, and successful non-blocking operations;
- one coherent reset-aware ReferenceIdealChain core whose prefixes are nested, whose successor starts at the prior terminal state or an explicitly matched reset restart, and whose terminal annotated prefix is the selected J.1 witness, without hiding Exec_ref as a structure field;
- ReferenceIdealChain.compatible_terminal derived from the chain invariant and J.0, rather than carried as an independent SelectedJWitnessEvidence field;
- reference_ideal_chain_terminal_has_admissible_extension as the named T.3.1 construction, with the Appendix I finite-local continuation split between an admissible global continuation and the genuine global-fixed-point/B6 branch and explicit B2/B3 fairness/progress discharge;
- PostEpsilonNormalizationExtension, RefEpsilonNormalizationExtension, SelectedNormalizedPrefixPair, and normalize_selected_J1_pair, followed by J.Abs/J.AbsConf cutpoint transfer for that same reachable carrier; the exact selected prefixes and their normalized representatives must remain distinct, and all base/normalized Exec_post and Exec_ref evidence must be supplied explicitly;
- ordinary-boundary B-faithfulness construction/certificate interfaces in Milestone A and elaborated-boundary B-faithfulness in Milestone H;
- DirectInputBucketWF construction from DirectInputContract, with late-sample value equality derived from the permitted-update exclusion rather than assumed;
- elaboration, array, R2C, and Appendix L obligations already listed above;
- the complete seven-clause DrainOnlyBlockedFrontierDiscipline, ProofRelevantDrainInterval, theorem-scoped KDrain C W, UniversalDrainConformance.impliesKDrain, AutomaticFlush and ordinary non-overlapped

constructors, `KStructuralAssumptions.drainOnlyDisciplineFor` and `kDrain`, fair drain normalization using separate `K.Drain` and `Kε` premises, `K.2b` eager-count reduction with explicit `RefAdmissible` and `K.Drain` preservation, and the one-way Appendix K liveness result;

- Appendix R aliases;
- the non-circular `ComparisonInstance / HasModelingContracts` signature, definitionally selected `Sys_ref`, and dependent `ElaborationAuditObligation`, with type-checking tests that reject an unrelated elaboration package or an inapplicable elaboration audit for an elaborated instance;
- sync-boundary gate soundness with distinct `qSuffix` and `qUpstream` carriers, proving that `E5` prevents premature suffix issue while the actual asserted upstream request is disabled at the selected explicit gate;
- `A6` point-to-point complementary-owner existence/uniqueness for every WFG-contributing channel, plus `KElaborationPrerequisite` for originally shared/arbitrated resources.

Optional non-core formalization. The Lean development may additionally prove `CausalDependencyGenerator` coverage, showing that the informational relation mirrors the scheduling-rules causal-dependency definition: message information flow, signal information flow, synchronization information flow, intra-process causal trace/source order, declared memory-serialization/shared-storage dependence, witness-control dependence, and elaboration information flow. This optional coverage theorem is not a premise of Theorem J.1, Corollary J.1, Appendix K, or Appendix R. Those theorem instances depend on `RequiredJOrder / <_J`, `ObservableUnitWF`, `E1–E5`, witness/cutpoint correspondence, and the relevant progress/liveness assumptions, not on acyclicity or completeness of the broader `CausalDependency` graph.

21.1 Appendix T coverage matrix

The matrix below is normative for the Lean architecture: every scheduling-specific obligation named by Appendix T has an explicit carrier, intermediate construction theorem or assumption, consuming theorem, and milestone. A blank row, a carrier whose premises cannot be supplied by its consumer, or a merely informal row should be treated as a proof-plan defect.

Appendix T obligation	Lean carrier / proof object	Primary consumer	Milestone
T.2.1 Initial/reset	<code>InitialComparisonCorrespondence</code> with two-sided reset-empty occupancy;	<code>JTrace/I</code> base case; every reset-aware J.1 successor	A0/B1

Appendix T obligation	Lean carrier / proof object	Primary consumer	Milestone
correspondence	ResetEpochCorrespondence; MatchedResetEpochSplice; MatchedEpochPreparation		
T.2.2 Next-unit source enablement	FiniteMatchedUnitExtension.source_instance_selected/ source_enabled/channel_and_boundary_legal	J.1 successor	A/B1
T.2.3 Independence and atomic units	ObservableUnitWF; AtomicUnitsPreserved; RequiredJOrder exclusions	CompatibleP/J.0/J.1	A
T.2.4 Fixed RTL-time ideal chain	RTLTimeIdealChain; reset-aware finite ReferenceIdealChain core; reference_ideal_chain_terminal_h as_admissible_extension	J1_fixed_RTL_time_selected_witness_construction	B1
T.2.5 WC	WC C W; blocking direct/identity construction; WCDischarge.proof; NB-CF/snooping/mixed routes	Appendices I/J	A/B1 for blocki ng; C/D for NB/sn ooping
T.2.6 Deterministic replay	ObsSigma; IObs congruence; IDR immediate/normalized	J.1; cutpoint transfer	B1/B2
T.2.7 E4 occupancy/boundary behavior	ERules; ChannelEnabled/Disabled; explicit boundaries	I3/J.0/K	A/H
T.2.8 B-faithfulness	BFaithful; ordinary-boundary constructor/certificate; ImplementationAssumptions; required discharge; elaborated- chain proof	All concrete theorem instances	A for ordinar y Sys_B; H for elabor ation
T.2.9 Local-to-system composition	CompatibleP/CompatibleSys over AnnotatedPrefix; pair-specific ElaborationProjectionWF premises; J.0	J.1	A/B1

Appendix T obligation	Lean carrier / proof object	Primary consumer	Milestone
T.2.10 J.Abs/J.Abs Conf	JAbs; Post/RefEpsilonNormalizationExtension; SelectedNormalizedPrefixPair with base and normalized execution evidence; reachable-carrier JAbsConf; CutpointCorrespondence	Corollary J.1; K	B2
T.2.11 Finite-local continuation	RefContinuationWithoutFurtherPUnits; fixed-point branch	I.5/J.1	B1
T.2.12 Finite ϵ -normalization	NormalizationBRules; I1; normalize_selected_J1_pair; distinct selected and normalized annotated prefixes; normalized replay	J cutpoint; K	B2
T.3.1 Admissible global executions	Exec_ref/post with TraceBRules.B2/B3; finite ReferenceIdealChain core; named reference_ideal_chain_terminal_has_admissible_extension; NormalizationBRules.B5 only for normalized cutpoint/K branches	J.1; cutpoint transfer; K	B1 for J.1 and admissible terminal extension; B2/B3 for normalized/K uses
T.3.2 Drain normalization/closure	ProofRelevantDrainInterval; theorem-scoped KDrain C W; UniversalDrainConformance sufficient profile; complete seven-clause DrainOnlyBlockedFrontierDiscipline; AutomaticFlush and ordinary	K.2b/K.1 and theorem-scoped K.Drain audit	B3

Appendix T obligation	Lean carrier / proof object	Primary consumer	Milestone
	non-overlapped constructors; fair_drain_normalize_closed_component; DrainNormalizedDeadlockCutpoint; A11/K ϵ carried separately		
T.3.3 K.2b eager reduction	Generalized drain-only evidence on selected/edited prefixes; DefersOneEagerIssueRefPrefix.output_drain_only; admissible deferral; eager-count measure	K.2b/K.1	B3
T.3.4 WFG/serial deadlock	A6_point_to_point_channels; PointToPointAtWFGEvaluationBoundary; ComplementOwner with existence/uniqueness; KElaborationPrerequisite for originally shared/arbitrated resources; ClosedDrainClosedWFGComponent; ConservativeSerialReferencePrefix; A5	K.0/K.2/K.2b/K.1	B3/H

22. Development milestones

Milestone A0 — Typed carrier, selected-reference spine, and reset base case

Add: an early type-checking signature skeleton for the J.1, J.1-cutpoint, and K.1 interfaces; a non-circular ComparisonInstance that contains semantic comparison data only; the dependent SysRefChoice and derived C.Sys_ref; the single ModelingContracts C declaration plus HasModelingContracts C interface; theorem-specific assumption bundles rather than C.assumptions; DynamicPrefixUniverse, PrefixAnnotations, AnnotatedPrefix, InitialComparisonCorrespondence, ResetEpochCorrespondence, InitialAndResetCorrespondence, MatchedResetEpochSplice, MatchedEpochPreparation, the cycle-indexed ReactiveEnvironment carrier, SamePriorSigmaVisibleHistory, ReactiveEnvironmentWF, construction of post annotations from Exec_post, and explicit

assumption-projection constructors. Establish that compatibility cannot be formed from bare Prefix values; an elaborated reference cannot diverge from its selected package; an elaborated audit cannot be marked inapplicable; every document-aligned initial/reset epoch supplies explicit two-sided reset-empty occupancy and no-stale-state facts; and a finite reset splice re-establishes the construction invariant with fresh dynamic-operation identities.

Targets:

signature_skeleton_J1_J1cutpoint_K1_typechecks
ComparisonInstance_nonrecursive_typechecks
SysRefChoice_selected_package_definitionally_determines_Sys_ref
ElaborationAuditObligation_rejects_inapplicable_elaborated_instance
ElaborationAuditObligation_ordinary_requires_no_elaboration_proof
HasModelingContracts_installed_from_theorem_bundle
AnnotatedPrefix_of_Exec_post_blocking
InitialComparisonCorrespondence_blocking
ResetEpochCorrespondence_blocking
ResetEpochCorrespondence_reset_empty_blocking
MatchedResetEpochSplice_blocking
MatchedEpochPreparation_complete_blocking
JConstructionInvariant_reestablished_after_reset_blocking
ReactiveEnvironmentWF_blocking
reactive_environment_replay_from_same_sigma_history_blocking
initial_correspondence_implies_same_source_state
JTraceAssumptions_to_IConstructionAssumptions
JCutpointAssumptions_as_JTraceAssumptions
J0_and_plain_J1_no_NormalizationBRules_dependency

Milestone A — Finite sequential, blocking-only, non-elaborated core

Scope:

Sys_ref = Sys_B;
sequential processes only;
blocking Push/Pop only;
no NB calls, snooping, arrays, direct-input late sampling, or R2C;
finite bucketed prefixes and finite bucketed execution fragments;
witness-relative annotated prefix carriers from Milestone A0;
a direct/identity blocking WC construction and WCDischarge object;
R1-R5 and E1-E5 infrastructure for the sequential/blocking subset;

E4 channel-capacity legality for rendezvous and buffered channels;
ordinary-boundary BFaithful construction/certificate interface and required discharge;
an explicit StoredOrCreditedBoundaryItem carrier and FiniteExplicitBoundaryStorage
construction from implementation assumptions;
Issue/Commit existence, uniqueness, dynamic-occurrence identity, and request
attribution;
ObservableUnitWF and same-cycle independence/atomic grouping;
FiniteMatchedUnitExtension for signals, buffered transfers, rendezvous, and
synchronization;
pairwise composition over TraceBRules only.

Targets:

WC_blocking_identity_or_direct_witness
WCDischarge_blocking
E4_channel_capacity_lemmas_blocking
BFaithful_ordinary_boundary_blocking
BFaithful_from_backannotation_certificate_blocking
ImplementationAssumptions_blocking
SideConditionDischargeRecord_bfaithfulness_blocking
StoredOrCreditedBoundaryItem_blocking
finite_explicit_boundary_storage_of_implementation_blocking
IssueCommitWF_blocking
ObservableUnitWF_blocking
FiniteMatchedUnitExtension_blocking
J0_pairwise_composition_blocking
J0_pairwise_composition_DUT_blocking
J0_elaboration_projection_legality_from_pair_wf_blocking

Milestone A deliberately does not target the temporal J.1 theorem, Corollary J.1, or
Theorem K.1.

Milestone B — Blocking temporal, cutpoint, and liveness stack

Milestone B1 — Temporal trace construction without B4/B5

Add: infinite bucketed executions; AlwaysFrom and EventuallyFrom; B2/B3/B6 admissible
execution machinery; Appendix I finite/infinite continuation; TraceBRules and
JTraceAssumptions; consumption of the blocking WC construction from Milestone A;
cross-cycle same- Σ -history environment replay from Milestone A0; immediate

deterministic replay; OrderedObservableTrace, OrderedObservableProjectionOfPrefix, SamePostObservationCut, and SameOrderedSigmaProjection; RTLTimeIdealChain; one coherent reset-aware finite ReferenceIdealChain core; and the named `reference_ideal_chain_terminal_has_admissible_extension` theorem. Every successor must provide a complete MatchedEpochPreparation followed by a FiniteMatchedUnitExtension. When a reset lies before the next RTL-time unit, the preparation supplies a MatchedResetEpochSplice, reset-empty restart, fresh dynamic universe, and re-established construction invariant. Every `refAt(k+1)` extends `refAt(k)`, and SelectedJWitness identifies the terminal chain element. `Exec_ref` is derived afterward by the named admissible-global-extension theorem rather than stored as a field of the finite chain. CompatibleSys at the terminal prefix is proved from the chain invariant and `J0_pairwise_composition`, not stored as an independent evidence field. Plain J.1 imports no NormalizationBRules.

Targets:

RequiredJOrderWF_from_concrete_generators_blocking
IObs_congruence_one_immediate_observable_unit_blocking
reactive_environment_replay_from_same_sigma_history_blocking
IDR_immediate_post_unit_deterministic_replay_blocking
I5_countably_infinite_local_limit_construction_blocking
I5_finite_local_projection_from_admissible_global_continuation_blocking
I5_finite_local_projection_from_global_fixedpoint_blocking
RTLTimeIdealChain_single_unit_blocking
ReferenceIdealChain_nested_blocking
ReferenceIdealChain_step_finite_matched_extension_blocking
ReferenceIdealChain_reset_epoch_splice_blocking
ReferenceIdealChain_compatible_terminal_blocking
reference_ideal_chain_terminal_has_admissible_extension_blocking
J1_fixed_RTL_time_selected_witness_construction_blocking
J1_post_to_ref_blocking
J1_post_to_ref_across_reset_epochs_blocking
OrderedObservableTrace_blocking
OrderedObservableProjectionOfPrefix_blocking
SamePostObservationCut_blocking
SameOrderedSigmaProjection_blocking
J1_matching_sigma_projection_unique_blocking
J1_selected_ref_prefix_is_liberal_blocking
B3_from_B2_E4_ordinary_boundary_blocking

Milestone B2 — ϵ -normalization and selected cutpoint correspondence

Add: constructive B4/B5 normalization; NormalizationBRules;
IObs_congruence_one_observable_unit as the normalized replay step;
IDR_deterministic_replay at ϵ -quiescent cutpoints; JCutpointAssumptions;
PostEpsilonNormalizationExtension and RefEpsilonNormalizationExtension;
SelectedNormalizedPrefixPair, which retains the exact J.1-selected pair and separately carries reachable normalized representatives; normalize_selected_J1_pair; the J.AbsConf implementation interface and theorem-scoped discharge; the redesigned SelectedMatchedCutpoint accessors $\tau_{\text{post}}/\tau_{\text{ref}}$ versus $\tau_{\text{bar_post}}/\tau_{\text{bar_ref}}$;
FiniteEpsilonStutterDifference and EquivalentUpToFiniteEpsilonStutter; and fixed-point-only Exec_post extension. J.AbsConf is an implementation obligation consumed by the cutpoint theorem, not a theorem to be proved from J.1 replay. Every cutpoint consumer receives explicit base and normalized Exec_post/Exec_ref evidence through one carrier.

Targets:

I1_bounded_epsilon_closure_blocking
IObs_congruence_one_observable_unit_blocking
IDR_deterministic_replay_blocking
PostEpsilonNormalizationExtension_blocking
RefEpsilonNormalizationExtension_blocking
SelectedNormalizedPrefixPair_blocking
normalize_selected_J1_pair_blocking
J1_cutpoint_correspondence_blocking
J1_selected_matched_cutpoint_blocking
SelectedMatchedCutpoint_exact_vs_normalized_accessors_blocking
FiniteEpsilonStutterDifference_blocking
EquivalentUpToFiniteEpsilonStutter_blocking
J1_normalized_projection_unique_up_to_finite_epsilon_blocking
fixedpoint_cutpoint_in_Exec_post_blocking

Milestone B3 — WFG, drain-only normalization, K.2b, and K.1

Add: Appendix B AutomaticFlush as the pipelined constructor; semantic
PipelinedControlCanContainOverlappedDynamicWork;
NoNewDynamicOperationLaunchWhileBlocked and its ordinary-control alias
NoNewDynamicOperationLaunchAfterBlockedPoint; NonOverlappedBlockingControl for
ordinary liberal intervals; NonFrontierAssertedRequestNotWFGSource; the complete

seven-clause DrainOnlyBlockedFrontierDiscipline, ProofRelevantDrainInterval, theorem-scoped KDrain C W, and the stronger sufficient UniversalDrainConformance profile; KStructuralAssumptions.blocking_interval_mode_complete, automatic_flush, ordinary_sync_boundary_policy, drainOnlyDisciplineFor, and kDrain; the theorem-scoped SideConditionDischargeRecord.kdrain field; separate $K\epsilon/A11$ evidence; A6 point-to-point ownership at the WFG boundary, ComplementOwner existence/uniqueness, and KElaborationPrerequisite for originally shared/arbitrated resources; canonical closed drain-closed WFG components; fair_drain_normalize_closed_component with explicit per-process K.Drain and $K\epsilon$ evidence; ConservativeSerialReferencePrefix; finite-prefix InternalMPDeadlockFreeOnSerialPrefixesRef and its convenience alias; the DefersOneEagerIssueRefPrefix.output_drain_only field; K2b_construct_admissible_eager_deferral with explicit Exec_ref and complete K.Drain preservation; eager-count reduction; and InternalMPDeadlockFreeOnExecPost. Full Remain/NextFront minimality remains a K.0 theorem target rather than an implementation-side drain assumption.

Targets:

PipelinedControlCanContainOverlappedDynamicWork_blocking
 NoNewDynamicOperationLaunchWhileBlocked_blocking
 NoNewDynamicOperationLaunchAfterBlockedPoint_blocking
 NonFrontierAssertedRequestNotWFGSource_blocking
 AutomaticFlush_to_drain_only_blocked_frontier_blocking
 NonOverlappedControl_to_drain_only_blocked_frontier_blocking
 ProofRelevantDrainInterval_blocking
 KDrain_theorem_scoped_R16_carrier_blocking
 UniversalDrainConformance_implies_KDrain_blocking
 KStructuralAssumptions_drain_only_projection_blocking
 KStructuralAssumptions_kDrain_blocking
 SideConditionDischargeRecord_kdrain_scope_blocking
 AutomaticFlush_no_withdrawal_from_IssueCommitWF_blocking
 SyncBoundaryGateCorrect_distinct_suffix_and_upstream_request_blocking
 SyncBoundaryWithholdingCoverage_uses_selected_elaboration_blocking
 A6_point_to_point_channels_blocking
 ComplementOwner_exists_unique_blocking
 KElaborationPrerequisite_shared_resource_elaboration_blocking
 K0_front_nextfront_classification_without_frontier_minimality_assumption_blocking
 InternalMPDeadlockFreeOnSerialPrefixesRef_blocking
 fair_drain_normalize_closed_component_with_separate_KDrain_Kepsilon_blocking

K2b_construct_admissible_eager_deferral_blocking
K2b_edited_prefix_preserves_KDrain_blocking
K2b_endpoint_prefix_suppression_blocking
K2b_no_third_party_enable_blocking
K2b_terminal_scc_persists_blocking
K2b_serial_source_dominance_blocking
K1_liveness_preservation_blocking
terminated_peer_not_internal_MP_deadlock_blocking

Milestone C — Non-blocking under NB-CF

Add:

Qexec;

failed NB polls as ε ;

successful NB transfers as Σ ;

Corollary L.3.

Target:

J1_post_to_ref_nb_cf

Milestone D — Witness selector, snooping, and Appendix L Note 1 projected safety

Add:

partial-prefix WitnessSelector;

Lemma L.2;

SigmaDUTSafetyInertInterface;

SigmaDUTSafetyWitnessObligation;

CompatiblePDUT and CompatibleSysDUT;

projected pairwise composition J0_pairwise_composition_DUT;

Appendix L Note 1 projected-safety branch / J1_post_to_ref_sigmaDUT_safety for interfaces whose ε -modeled local timing choices cannot add/remove/relabel Σ _DUT-visible units and can only commute DUT-causally-independent projected units.

Targets:

J1_post_to_ref_with_snooping

J1_post_to_ref_sigmaDUT_safety

Milestone E — Direct inputs

Add:

- DirectInputContract;
- derived lateSamplingPreservesLogicalAnchor;
- DirectInputBucketWF carrying DirectInputContract; bucket placement for logical anchor vs physical late sample; and derivation of late-sample value agreement from NoPermittedSyncControlledUpdateInAnchorWindow.

Targets:

DirectInputBucketWF_from_contract

DirectInputBucketWF_late_sample_value_matches_anchor

J1_post_to_ref_direct_inputs

Milestone F — External arrays

Add:

- One unified SourceArrayAccess carrier, ArrayAddress, MemoryEvent, and MemoryEvent.toCoreIOEvent integration into the proof-internal core-IO/ Σ mapping;
- ArrayAccessMapping and ArrayAccessMappingRules, including list-valued transformed operations and fixed-latency issue/commit consistency;
- ArraySyncIntervalBridge, sync-relative commit rules, transformation equivalence, conflict-free reorder certificates, shared-storage serialization/same-cycle policy, read coherence, and FiniteMatchedUnitExtension support for mapped external-array units.

Target:

ArraySyncIntervalBridge_sound

array_transformation_preserves_external_memory_behavior

array_reordering_requires_conflict_free_certificate

shared_storage_conflicts_serialized_or_policy_compatible

shared_storage_read_coherence

FiniteMatchedUnitExtension_external_array

J1_post_to_ref_external_arrays

Milestone G — R2C

Add:

- combModel?;
- LocalCombNetwork;
- CombNetwork;
- CombComposition;
- R2C_WF;
- selected observable slices Obs(C_comb);
- R2C fixed-point signal-observation units;
- stutter-canonical unit emission and explicit cycle-locked component mode;
- component-local occurrence-ordinal matching;
- multi-input anchoring-domain / interleaving-confluence obligations;
- per-process combinational component obligations;
- composed fixed-point observation semantics.

Target:

J1_post_to_ref_with_R2C_v42_units

Milestone H — Elaboration

Add:

- ElaborationPackage Sys_B Chan_outer Sigma_DUT, WellFormedElaborationPackage, SelectedElaboration, and the dependent SysRefChoice whose elaborated constructor definitionally selects E.Sys_elab;
- Π_{elab} and token-level bundled/joined accounting;
- bucket-preserving π_{Σ}, E projections for finite and infinite executions;
- A_E occupancy/accounting preservation and B-faithful boundary-chain realization;
- explicit-channel interpretation of B, occ, BoundaryReq, ChannelEnabled, Front, and WFG; dependent ElaborationAuditObligation for the exact selected package; projection preservation of observable units, atomicity, RequiredJOrder, and E4; complete sync-epoch-gate coverage with distinct qSuffix and qUpstream carriers; KElaborationPrerequisite; and elaborated initial/reset correspondence.

Targets:

ElaborationProjectionWF_of_package

SelectedElaboration_defines_Sys_ref

ElaborationAuditObligation_selected_package

elaboration_projection_preserves_observable_units
elaboration_projection_preserves_atomic_units
elaboration_projection_preserves_required_order
elaboration_occupancy_accounting_sound
finite_explicit_boundary_storage_of_elaboration
elaboration_E4_boundary_sound
sync_epoch_gate_elaboration_complete
sync_epoch_gate_disables_upstream_request_not_suffix_operation
KElaborationPrerequisite_of_wellformed_package
InitialAndResetCorrespondence_elaborated
BFaithful_elaborated_boundary_chain
J1_post_to_ref_elab
K1_liveness_preservation_elab

Milestone I — Appendix R facade

Add compact R theorem wrappers after the full proof stack is available.

23. Expected difficulty

The hardest pieces remain:

1. Bucketed execution, witness-relative annotation carriers, cycle-indexed reactive stimulus, and coherent reset-aware ideal-chain induction.
2. Temporal B2/B3 over infinite executions.
3. Concrete ObsSigma and I.DR.
4. Appendix J source-enablement proofs, blocking WC construction, and the nested ReferenceIdealChain construction with explicit dynamic-reset splices.
5. Direct-input late-sampling contracts.
6. External array access mapping.
7. R2C fixed-point semantics.
8. Appendix Q elaboration obligations.
9. Appendix K serial-A5 liveness/deadlock preservation, including the generalized drain-only blocked-frontier discipline for both pipelined automatic-flush and ordinary non-overlapped intervals, K.2b eager-count reduction / serial-source

dominance, explicit construction of RefAdmissible extensions and drain-only evidence after eager deferral, and fixed-point-only Exec_post extension.

A polished full formalization is plausibly in the 18k–30k Lean line range if direct inputs, arrays, R2C, temporal fairness/progress, non-blocking witness selection, snooping, automatic flush, Issue/Commit attribution, A11/K ϵ , and elaboration are all included. A core sequential/blocking/non-elaborated proof would be substantially smaller, plausibly 6k–10k lines; an intermediate blocking-plus-temporal-WFG proof would likely fall around 10k–16k lines.